

BVAR Scenario Tool

User Manual and Methodological Documentation¹

Mátyás Farkas

International Monetary Fund

June 18, 2026

Source code: <https://github.com/matyasfarkas/BVARScenarioTool>

¹The views expressed herein are those of the author and should not be attributed to the International Monetary Fund, its Executive Board, or its management. This tool is provided “as is” without warranty of any kind, express or implied. The author and the IMF disclaim all liability for any errors, omissions, or for any loss or damage arising from its use. Users are solely responsible for any decisions made based on the output of this tool.

BVAR Scenario Tool

User Manual and Methodological Documentation

Mátyás Farkas
International Monetary Fund

June 18, 2026

Abstract

This document describes the BVAR Scenario Tool, a browser-based application for conditional forecasting, scenario analysis, and policy counterfactual evaluation. The tool combines a large Bayesian vector autoregression (BVAR) estimated with hierarchical Minnesota priors ([Giannone et al., 2015](#)) with structural impulse responses drawn from the Macroeconomic Model Data Base (MMB; [Wieland et al., 2012, 2016](#)). Users condition the BVAR forecast on arbitrary variable paths via a Durbin–Koopman simulation smoother ([Durbin and Koopman, 2002, 2012](#)), and evaluate monetary and fiscal policy counterfactuals using DSGE-identified impulse response functions. We describe the methodology employed, the implementation of the conditional forecasting algorithms, and the three-layer architecture that separates the unconditional baseline, the user’s scenario, and the policy counterfactual. All computation runs client-side in the browser; the tool requires no server infrastructure and can be deployed as a static website.

Keywords: BVARs, Conditional Forecasting, Scenario Analysis, Kalman Filter, Durbin–Koopman Smoother, Policy Counterfactuals, DSGE Models, MMB

JEL classification: C32, C53, E37, E52, E58

*The views in this paper are solely the responsibility of the author and should not be interpreted as reflecting the views of the International Monetary Fund or any other person associated with the IMF. Email: mfarkas@imf.org.

Contents

1	Introduction	3
1.1	Notation	4
2	BVAR Estimation	6
2.1	Reduced-Form VAR	6
2.2	Data Transformations	6
2.3	Reporting Units	6
2.4	Hierarchical Minnesota Prior	7
2.5	Per-variable own-lag prior mean: random walk vs. stationary	7
3	Companion-Form State Space	8
3.1	State Vector	9
3.2	Transition Equation	9
3.3	Observation Equation	9
3.4	State-Noise Covariance	10
4	Kalman Filter and Smoother	10
4.1	Conditioning via Partial Observations	10
4.2	Forward Pass: Kalman Filter	10
4.3	Backward Pass: Durbin–Koopman Smoother	11
4.4	Conditional Forecast Computation	12
4.5	Delta-Based Display	13
4.6	Uncertainty Quantification	13
5	Policy Counterfactuals	14
5.1	DSGE Impulse Response Functions	14
5.2	Monetary Policy Shock Inversion	14
5.3	Three Conditioning Modes	15
5.4	Optimal Policy: Barnichon–Mesters	16
6	Global VAR (GVAR) and Global Commons	16
6.1	Architecture: an endogenous US and 32 partner blocs	17
6.2	Trade-weighted star variables	18
6.3	Trade weights	19
6.4	Coupling at solve time, not estimation	19
6.5	Two-tier solve: live preview vs. the full fixed point	22
6.6	Global commons: US-determined common shocks	22
6.7	Relation to the fiscal and counterfactual layers	23

7	Fiscal Blocks: Estimated Drivers and Debt Accounting	24
7.1	Estimated Fiscal Drivers	24
7.2	The Budget-Identity Recursion	26
7.3	Active Levers	28
7.4	Debt-Targeting Inversion	29
7.5	Display Layout	29
7.6	Data Treatment: the U.K. Index-Linked-Gilt Spike	30
8	Long-Run Anchoring: Prior vs Entropic Tilt	30
8.1	In-Prior Long-Run Prior (PLR)	31
8.2	After-Estimation Entropic Tilt	32
8.3	When to Use Which	33
9	Economy Coverage, Data Sources, and Estimation	34
9.1	The GVAR System: 33 Integrated Blocs	34
9.2	Standalone Single-Country Blocs	36
9.3	Data Sources	36
9.4	COVID-as-Missing Estimation	37
9.5	Common-Vintage Harmonization to 2025Q4	40
9.6	Predictive Uncertainty Bands	40
9.7	US Specification	41
9.8	Variable Transformations, Priors, and Financial Tags by Economy	43
10	Per-Country Data Sources and Transformations	73
10.1	Per-Economy Data Documentation: the Six Core Blocs	74
10.2	United States (US)	75
10.3	Euro Area (EA)	78
10.4	Japan (JP)	80
10.5	United Kingdom (UK)	82
10.6	Canada (CA)	83
10.7	China (CN)	85
10.8	Euro-core single-country BVARs (AT, BE, DE, FR, NL)	87
10.9	Austria (AT)	88
10.10	Belgium (BE)	91
10.11	Germany (DE)	94
10.12	France (FR)	97
10.13	Netherlands (NL)	99
10.14	Euro-area periphery: data conventions (ES, IT, IE, PT, GR)	102
10.15	Spain (ES)	103
10.16	Italy (IT)	105

10.17Ireland (IE)	108
10.18Portugal (PT)	110
10.19Greece (GR)	113
10.20Euro-area small members: data conventions	115
10.21Estonia (EE)	116
10.22Finland (FI)	118
10.23Luxembourg (LU)	120
10.24Latvia (LV)	122
10.25Lithuania (LT)	125
10.26Slovakia (SK)	127
10.27Slovenia (SI)	129
10.28EU non-euro and hard-gap small blocs: data conventions	131
10.29Czechia (CZ)	132
10.30Hungary (HU)	134
10.31Poland (PL)	135
10.32Costa Rica (CR)	137
10.33Iceland (IS)	139
10.34Advanced-Economy Blocs: Per-Variable Data Provenance	141
10.35Australia (AU)	142
10.36Switzerland (CH)	143
10.37Denmark (DK)	144
10.38Norway (NO)	146
10.39Sweden (SE)	147
10.40New Zealand (NZ)	149
10.41Korea (KR)	150
10.42Israel (IL)	152
10.43Emerging-market blocs I: BR, MX, CL, CO, IN, ID	154
10.44Brazil (BR)	154
10.45Mexico (MX)	156
10.46Chile (CL)	157
10.47Colombia (CO)	159
10.48India (IN)	160
10.49Indonesia (ID)	162
10.50Emerging markets II: data conventions (TR, ZA, SA, AR, RU, TH, MY, TW)	164
10.51Türkiye (TR)	164
10.52South Africa (ZA)	166
10.53Saudi Arabia (SA)	167
10.54Argentina (AR)	169
10.55Russia (RU)	170

10.56	Thailand (TH)	172
10.57	Malaysia (MY)	173
10.58	Taiwan, Province of China (TW)	175
11	User Interface Guide	177
11.1	Top Bar	177
11.2	Chart Panel	177
11.3	Control Panel (Sidebar)	178
11.3.1	Scenario Section	178
11.3.2	Policy Counterfactual Section	178
11.4	Export	178
11.4.1	Multi-Scenario Support	179
12	Technical Notes	179
12.1	Client-Side Computation	179
12.2	Performance Optimizations	179
12.3	Interactive GVAR Architecture: keeping the heavy solve off the drag path	180
12.4	Numerical Conventions	181
12.5	Identifiability and Weak Exogeneity	181
13	Reproducibility and Replication	182
13.1	Repository and Folder Map	183
13.2	Software Dependencies	183
13.3	End-to-End Replication Recipe	184
13.4	Model-Signature Versioning	185
14	Empirical Results	185
14.1	In-sample diagnostics	185
14.2	Out-of-sample forecast horse race	187
14.2.1	Model set	187
14.2.2	Scheme: recursive expanding window	188
14.2.3	Scoring and inference	189
14.2.4	Results	190
14.3	Endogenous-US spillover analysis	191
15	Discussion	194
15.1	Design Philosophy	194
15.2	Limitations	195
15.3	Known issues flagged for fix	196
15.4	Directions for Future Work	197

1 Introduction

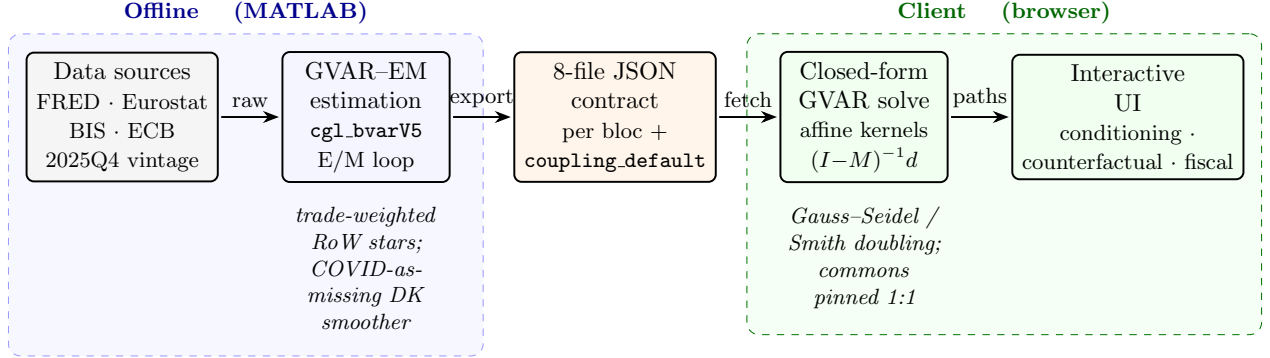


Figure 1: End-to-end system architecture of the EM-Bayesian Global VAR platform. Harmonized macro-financial data (FRED, Eurostat, BIS, ECB; common 2025Q4 vintage) feed the *offline* MATLAB GVAR-EM estimator: per-bloc Bayesian VARs (`cgl.bvarV5`) are re-fit in an EM loop that, at each sweep, rebuilds the trade-weighted rest-of-world “star” variables (ROWDEM/ROWINF/ROWFIN) from partners’ Durbin–Koopman-completed panels and re-estimates every bloc with the 2020Q2–Q4 *macro* columns masked (the foreign-rate star ROWFIN and all domestic financials kept observed; COVID-as-missing), iterating to $\max|\Delta\text{star}| < 0.03$ (cf. §6). Each converged bloc is serialized to the eight-file JSON contract (`spec`, `companion_mean`, `companion_draws`, `baseline`, `baseline_levels`, `historical`, `historical_levels`, `transformations`; plus `fiscal_meta` and the precomputed `coupling_default.json.gz` coupling artifact). In the *client* (browser), the cross-country system is solved in closed form: the scenario delta obeys the affine fixed point $x = Mx + d$, evaluated by Gauss–Seidel sweeps or directly via $(I - M)^{-1}d$ (Smith doubling), with global commons (US WTI oil, US 10Y, US BAA, US equity) pinned 1:1 onto every RoW bloc. The interactive UI exposes conditioning, counterfactuals, and the stock-flow fiscal block (§7).

Conditional forecasting with Bayesian vector autoregressions (BVARs) has become a standard tool in central bank forecasting and policy analysis. The seminal contributions of Doan et al. (1984) and Waggoner and Zha (1999) established the methodological foundations, while Bańbura et al. (2015) extended these techniques to large cross-sections using state-space methods. The BVAR Scenario Tool described in this document builds on this literature to provide an interactive, browser-based platform that makes conditional forecasting and policy counterfactual analysis accessible without specialized software.

The tool implements a three-layer analytical framework:

- (i) **Baseline forecast:** An unconditional forecast from a large BVAR estimated on quarterly macroeconomic and financial data, following the hierarchical prior framework of Giannone et al. (2015).
- (ii) **Conditional scenario:** The user conditions any subset of variables on desired paths; a Durbin–Koopman smoother (Durbin and Koopman, 2002, 2012) propagates these constraints to all remaining variables, respecting the full covariance structure of the VAR.

- (iii) **Policy counterfactual:** An alternative monetary or fiscal policy path is overlaid on the scenario using impulse response functions from a DSGE model drawn from the Macroeconomic Model Data Base (MMB; [Wieland et al., 2012](#)), and the resulting macroeconomic responses are re-conditioned through the BVAR to ensure cross-variable consistency.

The tool is designed for economists at central banks and international institutions who need to construct and communicate conditional projections and policy scenarios. It runs entirely in the browser, requiring no server-side computation, and can be deployed as a static website on any hosting platform.

This document is organized as follows. Section 2 describes the BVAR estimation methodology. Section 3 derives the companion-form state-space representation. Section 4 presents the Kalman filter and Durbin–Koopman smoother used for conditional forecasting. Section 5 details the policy counterfactual framework. Section 9 documents the variable coverage. Section 11 provides a guide to the user interface. Section 12 discusses technical implementation details. Section 15 concludes with a discussion of limitations and directions for future development.

1.1 Notation

Table 1 collects the symbols used throughout the manual, grouped by the layer in which they appear (BVAR estimation, the companion-form state space and smoother, the GVAR coupling, the fiscal recursion, and the long-run tilt). A plain-language *Glossary* of the named concepts—BVAR, GVAR, the trade-weighted stars, COVID-as-missing, the stock-flow adjustment, the EM loop, the endogenous dominant unit, the entropic tilt, and the evaluation statistics (CRPS, Giacomini–White, ...)—is provided in Appendix A.

Table 1: Notation used across the BVAR, GVAR, fiscal, and tilt layers.

Symbol	Meaning
<i>BVAR estimation (Sec. 2)</i>	
y_t	$n \times 1$ vector of endogenous variables of a bloc at quarter t (in estimation space).
n	number of variables in a bloc ($n = 41$ for the endogenous-US bloc).
p	VAR lag order ($p = 3$ throughout).
B_ℓ	$n \times n$ reduced-form autoregressive coefficient matrix on lag ℓ .
ε_t	reduced-form innovation, $\varepsilon_t \sim \mathcal{N}(0, \Sigma)$.
Σ	$n \times n$ innovation covariance; posterior-mean $\hat{\Sigma}$.
$\beta, \hat{\beta}$	stacked VAR coefficients; posterior mean.
λ	overall Minnesota/GLP shrinkage (tightness) hyperparameter.
μ	sum-of-coefficients prior hyperparameter (Giannone et al., 2015).
ψ	residual-variance scale of the prior (per variable).
δ_i	prior mean on variable i 's own first lag: $\delta_i = 1$ (random walk, RW) or $\delta_i = 0$ (stationary white noise, WN).
pos	index set of the WN variables; sets $\text{diagb}(\text{pos}) = 0$ (Sec. 2.5).

continued on next page

Table 1 continued from previous page

Symbol	Meaning
D	number of retained posterior draws $\{(\hat{\beta}^{(d)}, \hat{\Sigma}^{(d)})\}$.
<i>State space $\mathcal{E}$ smoother (Sec. 3–4)</i>	
α_t	companion-form state vector (stacked current and lagged y).
T, Z, Q, H	transition, observation, state-noise, and measurement-noise matrices.
$Z_t^{(o)}$	rows of Z corresponding to the <i>observed</i> variables at t (partial observation).
$\alpha_{t t-1}, P_{t t-1}$	one-step predicted state mean and covariance.
v_t, F_t, K_t	innovation, innovation variance, Kalman gain.
r_t, L_t	Durbin–Koopman backward recursion vector and transition.
$\alpha_{t H}$	smoothed (full-interval) state.
$\mathcal{W}$	COVID-as-missing window (the fixed three quarters 2020Q2–Q4).
<i>GVAR coupling (Sec. 6)</i>	
c, j	bloc (economy) indices; c receiver, j partner.
w_{cj}	trade weight of partner j in bloc c ’s stars, $\sum_j w_{cj} = 1$.
ROWDEM, ROWINF, ROWFIN	row-weighted foreign demand, inflation, and financial “star” variables.
$s_{c,v,t}$	scenario pin-delta of bloc c ’s star v at horizon t .
$g\Delta_{p,t}$	deviation of US-determined global column p (oil, US10Y, USBAA, USeq).
x	stacked deviation state of the coupling fixed point.
M	solve-time coupling operator; $x = Mx + d$.
d	exogenous forcing (own-scenario pins) of the fixed point.
$\rho(M)$	spectral radius of M (multiplier strength; $< 1 \Rightarrow$ contractive).
<i>Fiscal recursion (Sec. 7)</i>	
d_t	gross public debt as a percent of GDP.
$rev_t, pexp_t$	revenue and primary expenditure (% of GDP); FREV_GDP, FPEXP_GDP.
i_t^{eff}	effective interest rate (EFFI, % p.a.).
sfa_t	stock-flow adjustment (SFA, % of GDP).
g_t^n	nominal GDP growth.
pb_t	primary balance = $rev_t - pexp_t$.
<i>Long-run tilt $\mathcal{E}$ evaluation (Sec. 8, 14)</i>	
$\bar{y}^{\text{LR}}$	user-specified long-run anchor (H-step moment of the tilted density).
θ	entropic-tilt multiplier (dual solution).
ESS	effective sample size of the tilted importance weights.
h	forecast horizon (quarters); $h \in \{1, 2, 4, 8\}$ in the horse-race.
t_0	forecast origin in the recursive out-of-sample scheme.
d_t (loss)	loss differential between two models at t (overloads d_t ; context disambiguates).
CRPS	continuous ranked probability score (primary density metric).

2 BVAR Estimation

2.1 Reduced-Form VAR

Consider an n -variable VAR(p) in reduced form:

$$y_t = c + \sum_{\ell=1}^p B_{\ell} y_{t-\ell} + u_t, \quad u_t \sim \mathcal{N}(0, \Sigma), \quad (1)$$

where $y_t \in \mathbb{R}^n$ is the vector of observables, $c \in \mathbb{R}^n$ is a vector of intercepts, $B_1, \dots, B_p \in \mathbb{R}^{n \times n}$ are the autoregressive coefficient matrices, and $\Sigma \in \mathbb{R}^{n \times n}$ is the positive-definite innovation covariance matrix. In the US specification, $n = 41$ and $p = 3$, yielding a coefficient matrix $\hat{\beta} \in \mathbb{R}^{124 \times 41}$ (with row 0 containing the intercept). The 41 US variables include the four US-determined *global commons* (GLB_OIL, GLB_USLR, GLB_USBAA, GLB_USEQ) that the GVAR layer of Section 6 carries into every other bloc, the four-variable fiscal block (Section 7), and the three trade-weighted ROW*_US star variables through which the US, now an endogenous GVAR bloc, receives Rest-of-World feedback.

2.2 Data Transformations

Variables enter the VAR in one of two forms:

- **Log-transformed (log)**: the variable is modeled in log levels. Growth rates displayed to the user are quarter-over-quarter annualized:

$$g_t^{\text{q/q ann.}} = 400 \times (\ln Y_t - \ln Y_{t-1}). \quad (2)$$

Year-over-year growth rates are computed as:

$$g_t^{\text{y/y}} = 100 \times (\ln Y_t - \ln Y_{t-4}). \quad (3)$$

- **Linear (lin)**: the variable enters in levels (e.g., interest rates in percent, unemployment rate). No growth-rate conversion is applied.

2.3 Reporting Units

The display unit is chosen per variable for interpretability, independently of the estimation transformation:

- **National-accounts / activity flows** (GDP, consumption, investment, trade volumes) are shown as growth (q/q annualized or y/y).
- **Prices** (CPI/HICP/deflators) are shown as inflation.
- **Rates and ratios** (policy and market rates, unemployment, fiscal ratios) are shown as levels in percent / percentage points. Unemployment-rate uncertainty bands are floored at 0 (a rate

cannot be negative); policy and market rates are *not* floored, because negative nominal rates do occur.

- **Index / FX / equity / commodity series** (the trade-weighted dollar FXTWM, US equity GLB_USEQ, house prices HP, oil PZTEXP/GLB_OIL, the non-fuel commodity index GSNE, and the other blocs' NEER_EA, NEER_UK, NEER_CA, REER_CN, EQ_CN) are reported as the *index level* (e.g. 2010 = 100) rather than as q/q growth: a trade-weighted exchange rate or equity index q/q-annualized is not an informative number, whereas its level is. These series remain `log` for estimation and are conditioned in the stored $100 \ln(\text{index})$ space; only the display is exponentiated back to the level, $\exp(\text{stored}/100)$, with a level-space uncertainty fan.

2.4 Hierarchical Minnesota Prior

The model is estimated following [Giannone et al. \(2015\)](#), who cast the Minnesota prior ([Litterman, 1986](#); [Doan et al., 1984](#)) in a hierarchical framework with data-driven hyperparameter selection. The prior specification includes:

- (a) The **Minnesota shrinkage** prior on B_ℓ , centered on the random-walk or white-noise prior for each variable (depending on the variable's stationarity properties). The tightness parameter λ controls the overall degree of shrinkage toward the prior mean.
- (b) The **sum-of-coefficients** prior ([Doan et al., 1984](#)), which shrinks toward the belief that a unit increase in every variable's level produces a unit increase in its forecast, imposing a form of cointegration.
- (c) The **single-unit-root** prior, which constrains the system to have at most one common stochastic trend.

The hyperparameters governing the tightness of each prior component are treated as additional parameters and selected by maximizing the marginal likelihood, as advocated by [Giannone et al. \(2015\)](#).

Posterior draws are obtained via Markov chain Monte Carlo (MCMC). The estimator runs a long chain with burn-in, retaining the posterior mean $\hat{\beta}$ and $\hat{\Sigma}$ for the deterministic conditional forecasting engine. For the browser the chain is *thinned* to $D = 500$ retained draws $\{(\hat{\beta}^{(d)}, \hat{\Sigma}^{(d)})\}_{d=1}^D$, the count shipped in every bloc's `companion_draws.json`, used for uncertainty quantification via the simulation smoother (Section 4.6). The thinned export keeps the per-bloc draw artifact small enough to load on demand (the US file is ≈ 34 MB at $D = 500$, $n = 41$, $p = 3$).

2.5 Per-variable own-lag prior mean: random walk vs. stationary

The Minnesota/GLP prior is centered, for each equation, on a univariate model in which the variable's own first lag carries a prior mean of δ_i and every other coefficient a prior mean of zero ([Litterman, 1986](#); [Doan et al., 1984](#); [Giannone et al., 2015](#)). The *default* choice $\delta_i = 1$ encodes a

random-walk (RW) prior belief—“the best forecast of the level is the current level”—which is appropriate for variables that are (near-)integrated in the estimation space (log real activity, log price levels, nominal yields). For a variable that is genuinely *stationary*—a mean-reverting ratio or rate that fluctuates around a constant level—the random-walk center is the wrong null: it shrinks the dynamics toward a unit root and lets the variable *drift* at long horizons rather than return to its historical mean. For such variables the tool sets the own-lag prior mean to $\delta_i = 0$, a stationary white-noise (WN) prior whose null model is “the best forecast is the unconditional mean (the equation constant), not the last observation.”

The mechanism (pos/stationary). The choice is implemented exactly once, at the point where the prior mean of the coefficient matrix is built. In BVAR/MFBVAR/GLPreplicationWeb/bvarGLPmf.m the own-first-lag prior block is the diagonal matrix `diag(diagb)` with `diagb = 1n` by default; the line

$$\text{diagb}(\text{pos}) = 0$$

zeroes the own-lag prior mean for every variable index in the vector `pos`, so those equations are shrunk toward white noise instead of a random walk. The index set is threaded into the sampler as the third positional argument `stationary` of BVAR/MFBVAR/cgl.bvarV5.m (internally `pos = stationary`), which forwards it as the ‘pos’ option to `bvarGLPmf`. The estimators build the set directly from the per-variable prior tag in each bloc’s specification,

$$\text{pos} = \text{find}(\text{prior} == \text{'WN'}),$$

so a variable opts in to the stationary prior simply by carrying `prior:"WN"` in its spec (see `pipeline/export_bvar_gvar24_loop.m` and `pipeline/export_bvar_euro_fsap.m`). For the 44 non-fiscal blocs the set is empty and the call is byte-identical to the historical all-RW behaviour.

Prior, not constraint. Crucially $\delta_i = 0$ sets only the prior *mean*; it does not *impose* stationarity. The posterior own-lag coefficient is the usual GLP balance of prior and likelihood, governed by the estimated tightness λ (itself selected by the marginal-likelihood maximization, Section 2.4). A genuinely persistent series whose data favour a near-unit root will still produce a large posterior own-lag coefficient despite the white-noise center; the WN prior simply removes the *built-in* drift that the random-walk center would otherwise inject at the long end. This is the mechanism that makes the fiscal GDP-ratios mean-revert (Section 7.1).

3 Companion-Form State Space

To apply the Kalman filter, the $\text{VAR}(p)$ in (1) is cast in companion form as a first-order state-space model (see, e.g., Durbin and Koopman, 2012, ch. 3).

3.1 State Vector

Define the np -dimensional state vector:

$$\alpha_t = \begin{pmatrix} y_t \\ y_{t-1} \\ \vdots \\ y_{t-p+1} \end{pmatrix} \in \mathbb{R}^{np}. \quad (4)$$

3.2 Transition Equation

The state evolves according to:

$$\alpha_t = T \alpha_{t-1} + d + R \eta_t, \quad \eta_t \sim \mathcal{N}(0, \Sigma), \quad (5)$$

where the $np \times np$ transition matrix T , the np -dimensional intercept vector d , and the $np \times n$ selection matrix R are:

$$T = \begin{pmatrix} B'_1 & B'_2 & \cdots & B'_{p-1} & B'_p \\ I_n & 0 & \cdots & 0 & 0 \\ 0 & I_n & \cdots & 0 & 0 \\ \vdots & & \ddots & & \vdots \\ 0 & 0 & \cdots & I_n & 0 \end{pmatrix}, \quad d = \begin{pmatrix} c \\ 0 \\ \vdots \\ 0 \end{pmatrix}, \quad R = \begin{pmatrix} I_n \\ 0 \\ \vdots \\ 0 \end{pmatrix}. \quad (6)$$

Remark 1. The B'_ℓ blocks in T are transposed relative to (1) because the coefficient matrix $\hat{\beta}$ is stored as $(1 + np) \times n$, with row 0 containing the intercept c and rows $1, \dots, np$ containing the stacked lag coefficients. Specifically, $T(i, j) = \hat{\beta}(j + 1, i)$ for $0 \leq i < n$ and $0 \leq j < np$.

3.3 Observation Equation

The observation equation extracts the contemporaneous variables:

$$y_t = Z \alpha_t + \varepsilon_t, \quad \varepsilon_t \sim \mathcal{N}(0, H), \quad (7)$$

where $Z = (I_n \mid 0_{n \times n(p-1)})$ and $H = \sigma_\varepsilon^2 I_n$ with $\sigma_\varepsilon^2 = 10^{-12}$ (a negligible measurement error included for numerical stability of the Kalman filter recursions).

3.4 State-Noise Covariance

The $np \times np$ state-noise covariance matrix is:

$$Q = R \Sigma R' = \begin{pmatrix} \Sigma & 0 & \cdots & 0 \\ 0 & 0 & \cdots & 0 \\ \vdots & & \ddots & \\ 0 & 0 & \cdots & 0 \end{pmatrix}. \quad (8)$$

4 Kalman Filter and Smoother

4.1 Conditioning via Partial Observations

The key insight underlying conditional forecasting in a state-space framework, following [Bańbura et al. \(2015\)](#), is that conditioning on variable i at horizon h is equivalent to observing $y_{T+h,i}$ in the Kalman filter. For each forecast period $t = T + 1, \dots, T + H$, the observation vector is partitioned into:

- **Conditioned variables:** $y_t^{(o)}$, with known target values specified by the user.
- **Free variables:** $y_t^{(m)}$, treated as missing observations (coded as NaN).

The Kalman filter processes only the observed subset at each time step, adapting Z_t , H_t , and the innovation v_t accordingly.

4.2 Forward Pass: Kalman Filter

Algorithm: Kalman Filter with Partial Observations. **Input:** T, Z, Q, H, d ; initial state α_0 , covariance P_0 ; partial observations $\{y_t^{(o)}\}_{t=1}^H$.

For $t = 1, \dots, H$:

$$\alpha_{t|t-1} = T \alpha_{t-1|t-1} + d \quad (\text{state prediction}) \quad (9)$$

$$P_{t|t-1} = T P_{t-1|t-1} T' + Q \quad (\text{covariance prediction}) \quad (10)$$

If observations exist at time t , let $Z_t^{(o)}$ be the rows of Z for observed variables, then:

$$v_t = y_t^{(o)} - Z_t^{(o)} \alpha_{t|t-1} \quad (\text{innovation}) \quad (11)$$

$$F_t = Z_t^{(o)} P_{t|t-1} (Z_t^{(o)})' + H_t^{(o)} \quad (\text{innovation variance}) \quad (12)$$

$$K_t = P_{t|t-1} (Z_t^{(o)})' F_t^{-1} \quad (\text{Kalman gain}) \quad (13)$$

$$\alpha_{t|t} = \alpha_{t|t-1} + K_t v_t \quad (\text{state update}) \quad (14)$$

$$P_{t|t} = P_{t|t-1} - K_t Z_t^{(o)} P_{t|t-1} \quad (\text{covariance update}) \quad (15)$$

If no observations exist at time t : $\alpha_{t|t} = \alpha_{t|t-1}$ and $P_{t|t} = P_{t|t-1}$.

The matrix inversion F_t^{-1} is computed via LU decomposition with partial pivoting; singularity is flagged when the pivot falls below 10^{-15} .

4.3 Backward Pass: Durbin–Koopman Smoother

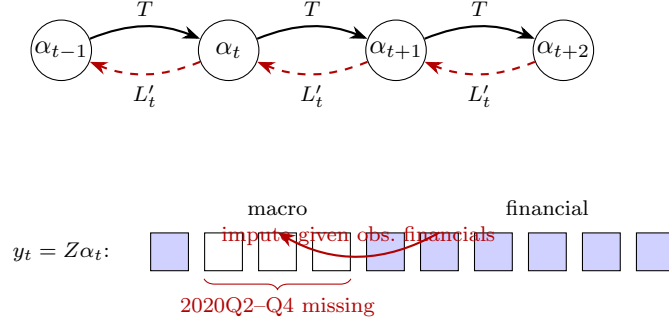


Figure 2: Companion-form state space with the Durbin–Koopman filter/smoothen under the COVID-as-missing treatment. States α_t (Eq. (4)) evolve forward through the transition matrix T (solid, Kalman filter, Eq. (9)); the backward smoother propagates $r_{t-1} = (Z_t^{(o)})' F_t^{-1} v_t + L'_t r_t$ (dashed, Eq. (17)). In the observation row $y_t = Z \alpha_t$ (Eq. (7)), over the fixed window 2020Q2–Q4 (three quarters) the *macro* cells (real activity, prices, and the macro stars ROWDEM/ROWINF) are set missing (hollow) while the *financial* cells (rates, yields, equity, house prices, and the foreign-rate star ROWFIN) remain observed (filled). Because the filter processes only the observed rows $Z_t^{(o)}$ of each partially-observed vector, the macro holes are imputed conditional on the observed financials.

The smoother refines the filtered estimates using all future information, yielding $\alpha_{t|H}$ for all t (Durbin and Koopman, 2002, 2012).

Algorithm: Durbin–Koopman Fixed-Interval Smoother. **Input:** Filtered quantities $\{\alpha_{t|t-1}, P_{t|t-1}, v_t, Z_t^{(o)}\}$. Initialize $r_H = 0$. For $t = H, H-1, \dots, 1$:

If observations exist at time t :

$$L_t = T' (I - P_{t|t-1} (Z_t^{(o)})' F_t^{-1} Z_t^{(o)}), \quad (16)$$

$$r_{t-1} = (Z_t^{(o)})' F_t^{-1} v_t + L'_t r_t. \quad (17)$$

Otherwise: $r_{t-1} = T' r_t$.

The smoothed state is:

$$\alpha_{t|H} = \alpha_{t|t-1} + P_{t|t-1} r_{t-1}. \quad (18)$$

Simulation smoother (draws, not just means). For uncertainty quantification and for the COVID-as-missing data augmentation (Section 9.4) the tool needs *draws* of the missing/conditioned states, not only their smoothed mean. The Durbin and Koopman (2002) simulation smoother obtains a draw $\tilde{\alpha}$ from the conditional distribution $p(\alpha | y^{(o)})$ by the elegant “mean-correction” device: simulate an unconditional path, smooth it, and add the smoothed *discrepancy* back to the

unconditional draw. Algorithm 1 states the form used inside each `cgl_bvarV5` sweep, with partial (ragged + COVID-masked) observations handled by processing only the observed rows $Z_t^{(o)}$ at each date.

Algorithm 1 Durbin–Koopman simulation smoother with partial observations (one draw inside the data-augmentation Gibbs sweep).

Input: system matrices T, Z, Q, H ; partial observations $\{y_t^{(o)}\}_{t=1}^H$ (COVID-macro and ragged rows absent); current draw of (β, Σ) .

Output: one draw $\tilde{\alpha}_{1:H}$ from $p(\alpha_{1:H} \mid y^{(o)}, \beta, \Sigma)$, hence imputed values for the missing y cells.

- 1: $\triangleright$ (1) Forward simulation of an unconditional path.
 - 2: Draw α_0^+ from the initial state law; **for** $t = 1, \dots, H$: draw η_t, ε_t and set $\alpha_t^+ = T\alpha_{t-1}^+ + \eta_t$, $y_t^+ = Z\alpha_t^+ + \varepsilon_t$.
 - 3: Form the synthetic partial observations $y_t^{+(o)}$ on the *same* observed pattern as the data (drop the cells that are NaN in $y_t^{(o)}$).
 - 4: $\triangleright$ (2) Kalman filter (Algorithm of Sec. 4.2) on the artificial series $y_t^{(o)} - y_t^{+(o)}$.
 - 5: **for** $t = 1, \dots, H$: run the prediction/update recursions on $Z_t^{(o)}$, storing $\{\alpha_{t|t-1}, P_{t|t-1}, v_t, F_t\}$.
 $\triangleright$ at fully-missing t , no update
 - 6: $\triangleright$ (3) Backward Durbin–Koopman smoothing recursion.
 - 7: Set $r_H = 0$; **for** $t = H, \dots, 1$:
 - 8: **if** observed at t : $L_t = T'(I - P_{t|t-1}(Z_t^{(o)})'F_t^{-1}Z_t^{(o)})$, $r_{t-1} = (Z_t^{(o)})'F_t^{-1}v_t + L_t'r_t$;
 - 9: **else**: $r_{t-1} = T'r_t$.
 - 10: smoothed discrepancy $\hat{\alpha}_t = \alpha_{t|t-1} + P_{t|t-1}r_{t-1}$.
 - 11: $\triangleright$ (4) Mean-correction: add the discrepancy back to the unconditional draw.
 - 12: $\tilde{\alpha}_t \leftarrow \alpha_t^+ + \hat{\alpha}_t$; read the imputed $\tilde{y}_t = Z\tilde{\alpha}_t$ at the missing cells.
-

Averaging $\tilde{\alpha}$ across Gibbs sweeps recovers the posterior-mean completed panel $\hat{X}_c$ that Algorithm 3 feeds back into the stars; retaining the per-draw paths yields the predictive draw cloud `Dforecast` used for the uncertainty bands (Section 9.6) and for density scoring (Section 14.2).

4.4 Conditional Forecast Computation

The conditional forecast is computed as follows:

1. Initialize the state from the last p historical observations: $\alpha_0 = (y'_T, y'_{T-1}, \dots, y'_{T-p+1})'$.
2. Construct the conditioning matrix $Y^* \in \mathbb{R}^{H \times n}$, where $Y_{t,j}^*$ is the target value for variable j at horizon t if conditioned, and NaN otherwise.
3. Run the Kalman filter (Algorithm 4.2) and smoother (Algorithm 4.3) with partial observations given by the non-NaN entries of Y^* .
4. Extract the observation block (first n elements) from each smoothed state $\alpha_{t|H}$.

4.5 Delta-Based Display

To ensure that zero user deviations produce exactly the stored baseline forecast, the tool computes both an unconditional smoother pass (with all entries of Y^* set to NaN) and a conditional pass. The displayed forecast is:

$$\hat{y}_{T+h}^{\text{display}} = \bar{y}_{T+h}^{\text{baseline}} + \underbrace{(\hat{y}_{T+h}^{\text{cond}} - \hat{y}_{T+h}^{\text{uncond}})}_{\text{smoother delta}}, \quad (19)$$

where $\bar{y}^{\text{baseline}}$ is the pre-computed baseline forecast (posterior mean). This construction guarantees that the displayed forecast matches the stored baseline exactly when no conditioning constraints are imposed, avoiding small numerical discrepancies between the smoother’s unconditional output and the stored baseline. The unconditional smoother result is cached and reused across conditioning changes.

For log-transformed variables, the smoother operates in log-level space, and the delta is scaled to annualized growth-rate space:

$$\Delta g_h = 4 \times \Delta \ell_h, \quad \Delta \ell_h = \ell_h^{\text{cond}} - \ell_h^{\text{uncond}}, \quad (20)$$

where ℓ_h denotes the log-level forecast at horizon h .

4.6 Uncertainty Quantification

Forecast uncertainty bands are computed via the Durbin–Koopman *simulation smoother* (Durbin and Koopman, 2002), which generates draws from the posterior predictive distribution conditional on the user’s constraints.

For each posterior draw $d = 1, \dots, D$:

1. **Forward simulation:** Draw shocks $\eta_t^+ \sim \mathcal{N}(0, I_n)$ and compute a simulated path using the d -th posterior draw of the VAR coefficients:

$$\alpha_t^+ = T^{(d)} \alpha_{t-1}^+ + d^{(d)} + R L^{(d)} \eta_t^+, \quad (21)$$

where $L^{(d)}$ is the lower Cholesky factor of $\Sigma^{(d)}$. The simulated observations are $y_t^+ = Z \alpha_t^+$.

2. **Deviation computation:** At conditioned entries, compute $y_t^* = Y_t^{\text{target}} - y_t^+$; at free entries, set $y_t^* = \text{NaN}$.
3. **Smoother on deviations:** Run the Kalman smoother on $\{y_t^*\}$ with initial state $\alpha_0^* = 0$ and small initial covariance $P_0^* = 10^{-7} I_{np}$, producing adjustment paths $\hat{\alpha}_t^*$.
4. **Combine:** The conditional draw is $\tilde{\alpha}_t = \hat{\alpha}_t^* + \alpha_t^+$.

The 16th and 84th percentiles across D draws yield the reported 68% credible bands. The tool uses $D = 50$ in “Quick” mode and $D = 200$ in “Full” mode. The simulation smoother correctly

integrates over both parameter uncertainty (through the posterior draws) and forecast uncertainty (through the simulated shocks).

5 Policy Counterfactuals

The counterfactual layer evaluates how macroeconomic outcomes change under an alternative policy stance. It operates *on top of* the conditional scenario described in Section 4: starting from the DK-smoothed scenario paths, the user specifies deviations in the policy rate or government spending, and the tool traces the implied effects through a DSGE model’s impulse response functions before re-conditioning the full system through the BVAR.

5.1 DSGE Impulse Response Functions

Impulse response functions (IRFs) are drawn from the Macroeconomic Model Data Base (MMB; [Wieland et al., 2012, 2016](#)). The MMB provides a library of pre-solved DSGE models with standardized variable names and multiple policy rule specifications. The tool includes IRFs from 169 models under 9 monetary policy rules:

Rule	Reference
Taylor	Taylor (1993)
SW	Smets and Wouters (2007)
CEE	Christiano et al. (2005)
GR	Gerdesmeier & Roffia (2004)
LWW	Levin, Wieland & Williams (2003)
OW08	Orphanides & Wieland (2008)
OW13	Orphanides & Wieland (2013)
Coenen	Coenen et al. (2012)
CMR	Christiano, Motto & Rostagno (2014)

Each IRF file provides the dynamic response of key macroeconomic variables—inflation, output gap, interest rate, consumption, investment, and others—to a one-standard-deviation monetary policy shock under the specified rule. Fiscal policy IRFs are available for 60 models that include a fiscal sector.

5.2 Monetary Policy Shock Inversion

Given a user-specified interest-rate adjustment path $\Delta r = (\Delta r_1, \dots, \Delta r_H)'$, the tool solves for the structural monetary policy shock sequence $\varepsilon = (\varepsilon_1, \dots, \varepsilon_H)'$ via the lower-triangular Toeplitz

system:

$$\underbrace{\begin{pmatrix} h_r(0) & 0 & \cdots & 0 \\ h_r(1) & h_r(0) & \cdots & 0 \\ \vdots & & \ddots & \vdots \\ h_r(H-1) & h_r(H-2) & \cdots & h_r(0) \end{pmatrix}}_{\mathcal{H}_r \in \mathbb{R}^{H \times H}} \varepsilon = \Delta r, \quad (22)$$

where $h_r(s)$ is the impulse response of the interest rate at horizon s . Since $\mathcal{H}_r$ is lower-triangular, the system is solved by forward substitution:

$$\varepsilon_t = \frac{\Delta r_t - \sum_{j=0}^{t-1} h_r(t-j) \varepsilon_j}{h_r(0)}, \quad t = 0, 1, \dots, H-1. \quad (23)$$

The implied responses of inflation and output are:

$$\Delta \pi_t^{\text{IRF}} = \sum_{j=0}^t h_\pi(t-j) \varepsilon_j, \quad \Delta y_t^{\text{IRF}} = \sum_{j=0}^t h_y(t-j) \varepsilon_j. \quad (24)$$

An analogous procedure applies to fiscal policy shocks, using the government-spending IRFs (h_g , h_π^g , h_y^g , h_r^g) from the DSGE models in the MMB that include a fiscal sector.

5.3 Three Conditioning Modes

After computing the DSGE-implied deltas, the counterfactual is re-conditioned through the BVAR to enforce cross-variable consistency. Three modes govern what is held fixed versus free in the second-stage DK smoother pass:

- (a) **Hold-targets:** All user-conditioned variables are pinned at their scenario values; the interest rate is pinned at the scenario rate plus the adjustment. Remaining variables are free and determined by the DK smoother:

$$Y_t^{\text{cf}}(j) = \hat{y}_{T+t,j}^{\text{cond}}, \quad \text{for } j \in \mathcal{C}, \quad (25)$$

$$Y_t^{\text{cf}}(r) = \hat{y}_{T+t,r}^{\text{cond}} + \Delta r_t, \quad (26)$$

$$Y_t^{\text{cf}}(j) = \text{NaN}, \quad \text{otherwise.} \quad (27)$$

- (b) **Let-adjust:** Only the policy instrument(s) are conditioned; all other variables, including inflation and output, are free and adjust endogenously through the BVAR covariance structure.
- (c) **IRF-consistent** (default): The interest rate is pinned at the scenario rate plus the adjustment, and inflation and output are pinned at the scenario values plus the DSGE-implied deltas

from (24):

$$Y_t^{\text{cf}}(\pi) = \hat{y}_{T+t,\pi}^{\text{cond}} + \Delta\pi_t^{\text{IRF}}, \quad (28)$$

$$Y_t^{\text{cf}}(y) = \hat{y}_{T+t,y}^{\text{cond}} + \Delta y_t^{\text{IRF}}, \quad (29)$$

$$Y_t^{\text{cf}}(r) = \hat{y}_{T+t,r}^{\text{cond}} + \Delta r_t. \quad (30)$$

The remaining variables are determined by the DK smoother, producing responses consistent with both the DSGE-identified monetary transmission and the BVAR covariance structure.

In all modes, the counterfactual display uses the same delta construction as (19):

$$\hat{y}_{T+h}^{\text{cf,display}} = \hat{y}_{T+h}^{\text{scen,display}} + f(\hat{y}_{T+h}^{\text{cf,smooth}} - \hat{y}_{T+h}^{\text{scen,smooth}}), \quad (31)$$

where $f(\cdot)$ applies the $\times 4$ scaling from (20) for log-transformed variables.

5.4 Optimal Policy: Barnichon–Mesters

Following Barnichon and Mesters (2023), the tool solves for the optimal monetary policy shock sequence that minimizes a quadratic loss function over deviations from target:

$$\mathcal{L} = \sum_{t=1}^H \left[\lambda_\pi (\pi_t - \pi^*)^2 + \lambda_y (y_t - y^*)^2 \right], \quad (32)$$

subject to the linear mapping from shocks to outcomes given by the DSGE impulse responses.

Define the Toeplitz matrices $\mathcal{H}_\pi, \mathcal{H}_y \in \mathbb{R}^{H \times H}$ analogously to (22) using the inflation and output-gap IRFs. Stacking the weighted system:

$$\underbrace{\begin{pmatrix} \sqrt{\lambda_\pi} \mathcal{H}_\pi \\ \sqrt{\lambda_y} \mathcal{H}_y \end{pmatrix}}_{\tilde{\mathcal{H}} \in \mathbb{R}^{2H \times H}} \varepsilon = \underbrace{\begin{pmatrix} \sqrt{\lambda_\pi} \Delta\pi^{\text{target}} \\ \sqrt{\lambda_y} \Delta y^{\text{target}} \end{pmatrix}}_{\tilde{b} \in \mathbb{R}^{2H}}, \quad (33)$$

the optimal shock sequence is obtained via Tikhonov-regularized least squares:

$$\varepsilon^* = (\tilde{\mathcal{H}}' \tilde{\mathcal{H}} + \alpha I_H)^{-1} \tilde{\mathcal{H}}' \tilde{b}, \quad (34)$$

with regularization parameter $\alpha = 10^{-6}$. The implied policy rate path is $\Delta r_t^* = \sum_{j=0}^t h_r(t-j) \varepsilon_j^*$, which is then passed to the IRF-consistent conditioning mode (Section 5.3c) to obtain the full counterfactual.

The user controls the relative weight on price stability versus full employment via a slider that sets $\lambda_\pi / (\lambda_\pi + \lambda_y)$.

6 Global VAR (GVAR) and Global Commons

The single-country engine of Sections 2–5 generalizes to a multi-country setting in which a conditioning imposed on one economy spills over to all others. The tool implements a *Global VAR* (GVAR) layer in the tradition of Pesaran et al. (2004) and Dees et al. (2007), comprising the United States together with the euro area as a single aggregate bloc and 31 further Rest-of-World (RoW) blocs—33 coupled blocs in all—that span approximately 90% of global GDP. The decisive implementation choice is that *cross-country coupling is imposed at solve time, not at estimation time*: each bloc is an independently estimated single-country BVAR, and the spillovers enter only through the scenario *deviation*, which propagates as a linear fixed point. Two further channels distinguish the current version: the United States is now a *fully endogenous* coupled bloc—it drives the rest of the world *and* receives feedback from it, rather than acting as a frozen anchor—and four US-determined *global commons* (oil, the long US rate, US corporate credit, and US equity) are carried by every bloc.

6.1 Architecture: an endogenous US and 32 partner blocs

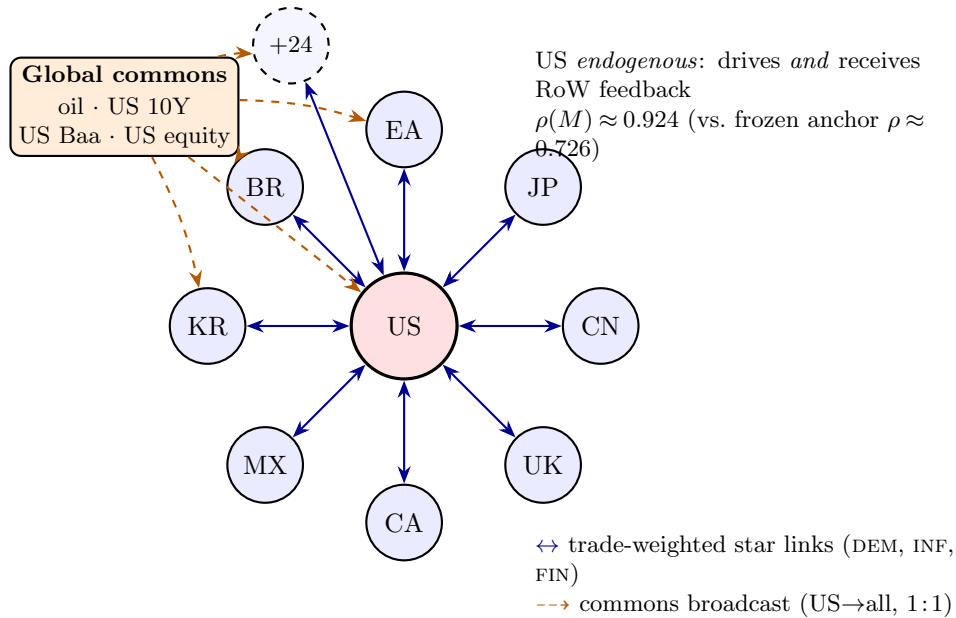


Figure 3: GVAR bloc topology. A central, dominant United States bloc is coupled to a ring of 32 Rest-of-World partner blocs (eight labelled—EA, JP, UK, CA, CN, MX, KR, BR—plus 24 further blocs) through *bidirectional*, trade-weighted “RoW star” links carrying foreign demand (DEM), inflation (INF), and the short rate (FIN). A small global-commons node (oil, the long US rate, US corporate credit, US equity) broadcasts *one-directionally* and identically (1:1) to every bloc. The US is now a *fully endogenous* coupled bloc rather than the classical frozen anchor: it feeds the rest of the world *and* receives feedback, raising the coupling operator’s spectral radius from $\rho(M) \approx 0.726$ (frozen-US, RoW-only) to $\rho(M) \approx 0.924$ at the full 33-bloc endogenous-US configuration (the value stored in the shipped `coupling.default.json.gz`; it moves slightly with each refit), both contractive so the fixed point $x = Mx + d$ is well defined (Sections 6.1–6.4).

The bloc set is $\{\text{US, EA, JP, UK, CA, CN, CH, KR, MX, AU, TR, SE, PL, NO, DK, IN, BR, RU, ID, SA, AR, ZA, T}\}$ (33 blocs: the US, the euro-area aggregate EA, and 31 further blocs). The euro area enters as a *single aggregate* bloc rather than as its individual members: a fully disaggregated euro GVAR is non-stationary ($\rho(M) \approx 1.089 > 1$), and aggregating the euro area into one bloc restores stationarity. The 17 euro-area member states are therefore retained only as *standalone* single-country models outside the coupled system (Section 9.2), not as GVAR blocs.

The US is now endogenous, not a frozen anchor. In earlier versions the US was treated as a *closed anchor*, exogenous in the Dees–PSW sense (Dees et al., 2007): it carried no foreign variables, was solved once, and never iterated in the cross-country fixed point. The current US model instead carries its own three trade-weighted RoW *star* variables (ROWDEM_US, ROWINF_US, ROWFIN_US) built from a US trade-weights row, so the US is a *fully coupled* bloc with $n = 41$ (its domestic and fiscal variables, the global commons of Section 6.6 as its own endogenous variables, and the three US star variables). It both drives every partner—through its weight in their trade-weighted aggregates and through the global commons—and receives partner feedback through its own stars. The US is the first entry of the coupled set and iterates inside the cross-country fixed point alongside everyone else; there is no longer a privileged frozen bloc. This RoW→US feedback is the qualitative gain of the endogenous treatment, and it is also what makes the complete solve more expensive (Section 6.5).

The topology is **not hardcoded**: it self-configures from the deployed model specifications. The US is treated as endogenous if and only if its model carries the three ROW*_US star variables *and* it has a row in the trade-weights file; absent either, the engine falls back to the legacy frozen-anchor solve, which remains byte-identical. More generally, a bloc is an active GVAR member if and only if its `spec.json` carries variables tagged `block: 'row'` *and* it has a row in the trade-weights file (`weights_24bloc.json`; the filename is historical—the file now lists all 33 coupled blocs, the US row included). At load, the engine derives each bloc’s GDP, price, and short-rate columns from its specification (mirroring the estimator’s column-resolution order) and reads its RoW-star indices, so that bloc indices are never assumed by position. Adding an economy to the GVAR therefore requires only shipping its estimated model with the appropriate variable tags and a weights row.

6.2 Trade-weighted star variables

Every coupled bloc—now including the United States—consists of its domestic VAR augmented by three trade-weighted *star* variables that summarize the foreign environment, following the weak-exogeneity device of Pesaran et al. (2004):

- ROWDEM: foreign demand, a trade-weighted index of partner real-GDP growth;
- ROWINF: foreign inflation, the analogous index on partner price levels;
- ROWFIN: foreign financial conditions, a trade-weighted average of partner short-term policy rates.

Let $\mathcal{W}_c \subseteq \{\text{partners of } c\}$ collect the partners that contribute to a given star at quarter t , with w_{cj} the (renormalized) trade weight of partner j in receiver c . For the activity and price stars, the partner level is read in $100 \ln$ space and the star is the annualized growth index

$$\text{ROWDEM}_{c,t} = 4 \sum_{j \in \mathcal{W}_c} \tilde{w}_{cj} (\ell_{j,t}^{\text{gdp}} - \ell_{j,t-1}^{\text{gdp}}), \quad \text{ROWINF}_{c,t} = 4 \sum_{j \in \mathcal{W}_c} \tilde{w}_{cj} (\ell_{j,t}^{\text{price}} - \ell_{j,t-1}^{\text{price}}), \quad (35)$$

where the scaling $4 \Delta(100 \ln \cdot)$ is the quarter-over-quarter annualized convention of (2), and the financial star is the level aggregate

$$\text{ROWFIN}_{c,t} = \sum_{j \in \mathcal{W}_c} \tilde{w}_{cj} r_{j,t}. \quad (36)$$

The weights are *per-component* renormalized over the partners that are present and eligible at each t , $\tilde{w}_{cj} = w_{cj} (\sum_k w_{ck}) / (\sum_{k \in \mathcal{W}_c} w_{ck})$, so that the mass of any absent or excluded partner is reabsorbed and the active weights still sum to one. Three eligibility rules mirror the estimator exactly: rate-less blocs (Switzerland, Norway) carry no own short rate and are dropped from partners' ROWFIN (but retained in ROWDEM/ROWINF); the hyperinflation bloc (Argentina) is dropped from partners' ROWINF and ROWFIN (retained in ROWDEM); and China's GDP is reported in *nominal* terms, so its activity contribution is deflated by *subtraction* in log space, $\ell_{\text{CN}}^{\text{gdp}} - \ell_{\text{CN}}^{\text{price}} = 100 \ln(\text{NGDP}/P)$, rather than by division.

6.3 Trade weights

The bilateral weights are constructed from IMF International Merchandise Trade Statistics (IMTS) and shipped as `weights_24bloc.json` (a historical filename: the file now holds the full roster), which lists the 33 blocs together with 33 weight rows—one per bloc, the US included, now that the US is endogenous—each summing to one (zero diagonal). At load, the live weight matrix W is renormalized over the *active* bloc set, so a deployment that activates a subset of the candidate blocs remains internally consistent.

6.4 Coupling at solve time, not estimation

Because each bloc is estimated independently, there is no joint, block-exogenous GVAR system; the cross-country linkage is imposed when the scenario is solved. The key observation is that, with coefficients fixed at their posterior mean $(\hat{\beta}, \hat{\Sigma})$, the conditional-forecast operator of Section 4 is *affine* in the conditioning values (the Kalman filter and Durbin–Koopman smoother are linear in the pinned data). The *baseline* forecast of each bloc is therefore data-driven and computed exactly once, via an all-`NaN` (unconditional) smoother pass; only the scenario *deviation* couples across blocs.

Because the US is now endogenous it is no longer held outside the fixed point, and the compact deviation state is correspondingly *augmented*. It collects (i) the star pin-deltas $s_{c,v,t}$ of *every* coupled bloc c —the US included—for star variable $v \in \{\text{DEM}, \text{INF}, \text{FIN}\}$ at horizon t , and (ii) the per-period

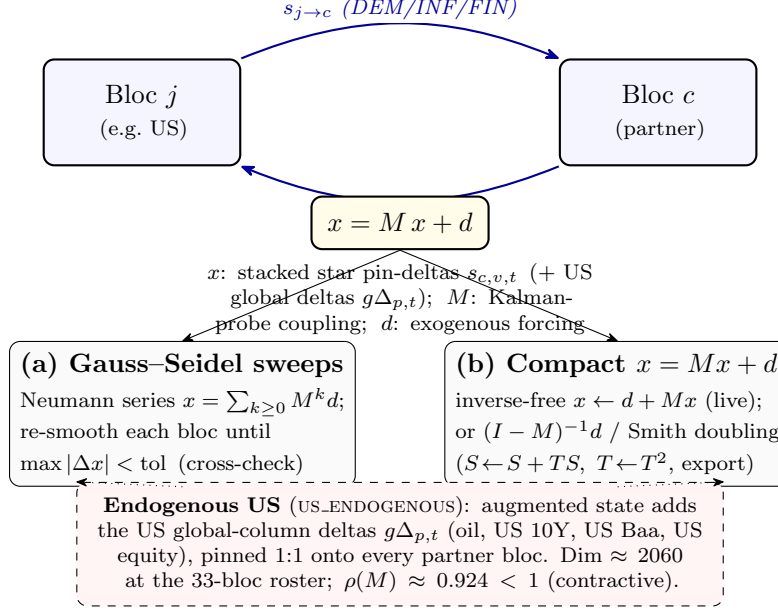


Figure 4: Solve-time cross-country coupling as the affine fixed point $x = Mx + d$. Two coupled blocs exchange trade-weighted star pin-deltas (DEM/INF/FIN), forming the feedback cycle; the deviation state x stacks these pins (plus the US global-column deltas under an endogenous US). The fixed point admits two equivalent solutions—(a) Gauss-Seidel/Neumann sweeps through the Durbin-Koopman smoother, used as the cross-check, and (b) the compact M -state recursion, solved inverse-free by the Neumann fixed-point $x \leftarrow d + Mx$ on the production path (the explicit $(I - M)^{-1}d$ / Smith doubling are reserved for the small frozen-anchor case and the export-time precompute, since the in-browser $O(\text{dim}^3)$ inverse is prohibitive at $\text{dim} \approx 2060$). With the US endogenous the augmented state has dimension ≈ 2060 and spectral radius $\rho(M) \approx 0.924 < 1$, so the iteration contracts and the two routes agree to $\sim 10^{-9}$.

deviation of each US global column $g\Delta_{p,t}$, which under the frozen-anchor design was a constant and is now itself part of the unknown. Stacking these into a vector x , the deviation system is the linear fixed point

$$x = Mx + d, \quad (37)$$

where d is the exogenous forcing (each bloc's own scenario pins moving its linked series at zero partner and global delta) and M now carries three coupling families: star $\leftarrow$ star (receiver c 's star responds to a partner j 's free-variable path—with the US participating both as partner and as receiver), the US-global definition rows (a global column's deviation as the US's own response to its star pins), and star $\leftarrow$ global (a partner's star responds to its own global pin). The US-determined globals therefore live *inside* M rather than in d , which is precisely the new RoW $\rightarrow$ US feedback loop. The augmented state has dimension ≈ 2060 at the full 33-bloc roster.

The fixed point (37) is solved two equivalent ways:

- (a) **Gauss-Seidel / Neumann sweeps** (the cross-check) sum the Neumann series $x = \sum_{k \geq 0} M^k d$ by re-conditioning each bloc in turn through the Durbin-Koopman smoother (Durbin and Koopman, 2002, 2012) until the maximum level change falls below tolerance.

(b) **Direct closed form**

$$x = (I - M)^{-1}d, \quad (38)$$

after which one final conditional-forecast pass per bloc reconstructs the full set of free variables consistent with the converged pin-deltas.

At the full 33-bloc endogenous-US configuration the spectral radius is $\rho(M) \approx 0.924 < 1$, so the Neumann series converges and $(I - M)$ is nonsingular; the two solvers agree to $\sim 10^{-9}$. (The legacy frozen-anchor RoW-only operator had a smaller radius, $\rho(M) \approx 0.726$; closing the RoW→US loop raises the coupling but leaves the system comfortably contractive.)

Algorithm 2 states both solve paths (`app/src/lib/compute/gvarSolve.ts`). The coupling operator M is never formed by writing down a joint VAR; each of its columns is *probed* by perturbing one pinned cell and reading the affine conditional-forecast response of every linked bloc through the Durbin–Koopman smoother—so M encodes the same per-bloc Kalman operator that produces the baseline, restricted to the deviation.

A practical consequence drives the deployment. The operator M depends on the estimated models and on the conditioning *structure* (which cells are pinned) but *not* on the pinned *values*, so it can be assembled once and reused for every value change during interactive dragging. Its dense inverse is, however, expensive: at the augmented endogenous dimension (≈ 2060) the naive in-browser inversion would take tens of minutes (the cubic inverse is of order three-quarters of an hour), making a cold build infeasible even in a Web Worker. The tool therefore **ships a precomputed coupling artifact** (`coupling_default.json.gz`) containing M , $(I - M)^{-1}$, and $\rho(M)$ for the default (empty) pin structure, with the inverse computed offline by a high-performance BLAS routine. The artifact is large (the dense 2060×2060 inverse), so it is shipped *gzipped* (about 30 MB compressed) and decompressed in the browser via the native `DecompressionStream` API at load. This artifact is mandatory, not an optimization: hydrating it reduces the first complete solve from the tens-of-minutes cold build to a few seconds. A model-signature gate guards against using a stale inverse after any re-estimation; a genuinely novel pin structure misses the cache and falls back to an inverse-free Neumann solve (the in-browser cubic inverse is never built live).

6.5 Two-tier solve: live preview vs. the full fixed point

The endogenous US makes the *complete* cross-country solve materially more expensive than the legacy frozen-anchor solve, because placing or moving a US pin re-probes the (large) US bloc, the heavy step. To keep the interface reactive the tool exposes **two solve tiers**:

- **Direct preview** (the default while dragging). A fast, first-order approximation that propagates the scenario through the pre-warmed local per-bloc elasticities—one cross-country hop, without the full multiplier feedback—so the propagated paths update live as the user drags a pin. The per-bloc response kernels are cached by model signature and own-pin structure, so a value-only re-drag (same pinned cells) reuses them and re-solves cheaply; a *structure* change

Algorithm 2 Solve-time coupling fixed point $x = Mx + d$ (Gauss–Seidel/Neumann and direct $(I - M)^{-1}$; `gvarSolve.ts`).

Input: per-bloc posterior-mean models $\{(\hat{\beta}_c, \hat{\Sigma}_c)\}$; scenario pins; trade weights W ; `US_ENDOGENOUS` flag; tolerance `TOL`.

Output: converged deviation state x and per-bloc scenario deltas `delta[c]`; sweep count; $\rho(M)$.

- 1: $\triangleright$ **Forcing.** Build d : each bloc’s own scenario pins, evaluated at *zero* partner-star and global deltas (one DK conditional-forecast pass per pinned bloc).
 - 2: $\triangleright$ **Coupling families in M** (endogenous-US augmented state). `star` $\leftarrow$ `star` (a receiver’s star responds to a partner’s free path; the US is both partner and receiver); US-global rows ($g\Delta$ as the US’s own response to its star pins); `star` $\leftarrow$ `global` (a partner star responds to its own global pin).
 - 3: **Path A — Gauss–Seidel / Neumann sweeps (inverse-free).**
 - 4: $x \leftarrow d$
 - 5: **repeat** (sweep): **for each** coupled bloc c (US included **if** `US_ENDOGENOUS`): re-condition c through the DK smoother on the current partner-star and global deltas $\Rightarrow$ update x ’s block for c .
 - 6: $\triangleright$ this sums the Neumann series $x = \sum_{k \geq 0} M^k d$ term by term.
 - 7: **until** $\max |\Delta x| < \text{TOL}$; record ρ_{emp} from the geometric residual decay.
 - 8: **Path B — direct closed form (cached).**
 - 9: assemble M by probing columns; **if** a precomputed $(I - M)^{-1}$ matches the model signature & pin structure, hydrate it; **else** fall back to Path A (the dense in-browser inverse is never built live).
 - 10: $x \leftarrow (I - M)^{-1}d$; one final per-bloc DK pass reconstructs all free variables consistent with the converged pin-deltas.
 - 11: $\triangleright$ **Cross-check.** The two paths agree to the verified tolerance ($\max |\Delta| < 10^{-4}$ across all blocs in the closed-form vs. iterative regression tests, with the iterative reference itself converged to 10^{-7}), confirming $\rho(M) < 1$; report $\rho(M)$ via power iteration on M (in practice on M^2 , see Sec. 6.5).
 - 12: **return** x , `delta[c]`, sweep count, $\rho(M)$.
-

(pinning a new cell) re-probes the US bloc and is the slower direct step (a few seconds), during which a small “computing...” pill is shown so the wait does not look like a freeze.

- **Full GVAR** (on demand). The complete fixed point of (37)—all feedback loops, including $\text{RoW} \rightarrow \text{US}$ —computed when the user clicks the **Solve full GVAR** button. This consumes the hydrated $(I - M)^{-1}$ artifact and takes a few seconds.

A badge above the charts reports which tier produced the current result—an amber “GVAR DIRECT preview” versus a blue “GVAR full solve”—together with the sweep count, the spectral radius ρ , and the solve time. When a DIRECT preview is on screen, the **Solve full GVAR** button is offered to upgrade it to the complete fixed point. Drag-release dispatch and debouncing keep the preview channel from firing mid-drag, and the heavy coupling artifact is stripped from the worker messages on the direct path (it is only needed for the full solve). For the legacy frozen-anchor

topology—or any deployment where the US is not endogenous—both tiers resolve to the same already-fast cached closed form, so the two-tier distinction is invisible there.

6.6 Global commons: US-determined common shocks

The headline addition of version 2.0 is a set of four *global common* variables that are determined by the United States but carried by *every* bloc:

- GLB_OIL — the spot oil price (WTI);
- GLB_USLR — the US 10-year Treasury yield;
- GLB_USBAA — the US Baa corporate-bond yield (a credit-spread proxy);
- GLB_USEQ — US equity prices.

In the US bloc these enter as ordinary *endogenous* variables; in each partner bloc the same four identifiers (matched one-to-one by *name*, never by position) are carried as weakly-exogenous commons that the solver *pins*.

The pinning is by *deviation*, not by level. Each partner bloc c sets each global column to its own baseline plus the US deviation of the matching global,

$$\widehat{\text{GLB}}_{c,t} = \bar{\text{GLB}}_{c,t} + (\text{GLB}_{\text{US},t}^{\text{scen}} - \bar{\text{GLB}}_{\text{US},t}), \quad (39)$$

where $\bar{(\cdot)}$ denotes the bloc’s own unconditional baseline. This is a load-bearing design choice. Each RoW bloc estimated its *own* baseline path for the globals as weakly-exogenous regressors, and those baseline levels differ from the US’s by tens of 100ln units (for example, of order +17, +70, and +28 for the euro area, Japan, and China on oil). Pinning the US *level* would inject that entire baseline gap as a spurious deviation the instant a bloc is selected, even under a zero scenario. Pinning the US *deviation* on top of each bloc’s own baseline instead yields exactly zero spillover at a zero scenario and an *identical* common shock under conditioning—precisely mirroring the star convention RoW + partner-delta of (35)–(36). A US oil pin therefore propagates the same oil deviation to all 32 partner blocs (verified to within 10^{-9}), while each bloc’s domestic variables respond through its *own* estimated coefficients.

Mechanically, with the US now endogenous the US global *deviation* is itself an unknown of the fixed point rather than a frozen constant. The global columns therefore enter the *augmented* coupling operator M of Section 6.4: a US-global definition row writes each global’s deviation as the US’s own response to its star pins, and a star $\leftarrow$ global block routes each partner’s global pin back into its stars. This is exactly the channel by which a partner shock now feeds back to the US. Indirect effects—a common shock moving a bloc’s macroeconomy, which moves its stars, which spill to other blocs—flow through the same operator and are not double-counted. A bloc that carries no globals reduces the pin to a no-op, leaving the legacy globals-free solve byte-identical.

On the user interface the globals are *conditionable* on the US screen (where they are endogenous) and *read-only*, carried over from the US path, on every other bloc.

6.7 Relation to the fiscal and counterfactual layers

The GVAR layer composes with the single-country machinery rather than replacing it. The displayed scenario line for an active GVAR bloc is the cross-country propagated path of this section; for a standalone (non-GVAR) economy it is the single-country conditional forecast of Section 4. The US derived fiscal block (the debt-to-GDP accounting identity, whose empirical motivation is the fiscal reaction function of [Bohn, 1998](#)) and the DSGE policy counterfactual—including the [Barnichon and Mesters \(2023\)](#) optimal-policy problem—continue to operate on the US bloc’s propagated path. Conditional forecasts throughout use the large-cross-section state-space method of [Bańbura et al. \(2015\)](#) with the hierarchical prior of [Giannone et al. \(2015\)](#), applied independently to each bloc and stitched together only through the linear deviation system (37).

7 Fiscal Blocks: Estimated Drivers and Debt Accounting

Six of the economies in the tool—the United States, the Euro Area, Japan, the United Kingdom, Canada, and China—carry a dedicated *fiscal block*. The block follows the stock-flow-consistent division of labour shared by the U.S. Congressional Budget Office’s conditional-BVAR budget mechanics and the IMF Financial Programming (FPP) / debt-sustainability frameworks: the BVAR is used as a *macro-fiscal scenario generator* that jointly forecasts the macroeconomy together with a small number of fiscal *flow drivers*, while the public-debt stock and the headline reporting variables (primary and overall balance, interest, debt-to-GDP, gross financing needs) are *derived outside the VAR* by the government budget-constraint identity. No reduced-form equation is ever allowed to forecast the debt stock freely, because an unconstrained debt VAR is explosive: the debt-dynamics identity has a near-unit autoregressive root $(1 + i)/(1 + g^n)$, and small parameter errors compound into divergent debt paths. Imposing the identity by construction is exactly the discipline that the CBO and FPP/DSA frameworks demand, and it is the structure implemented in `app/src/lib/compute/fiscalAccounting.ts`.

7.1 Estimated Fiscal Drivers

The BVAR jointly estimates the macro block of Section 9 together with *four* fiscal flow drivers, all entering as linear (`lin`) variables:

- **Revenue** (`FREV_GDP`): general-government receipts as a percentage of GDP, an annual ratio.
- **Primary expenditure** (`FPEXP_GDP`): non-interest outlays as a percentage of GDP, an annual ratio.
- **Effective interest rate** (`EFFI`, % p.a.): the average rate the government actually pays on its outstanding stock, constructed as net interest divided by the lagged gross debt and annualized. Because the stock turns over only as maturing securities are refinanced, this effective rate lags market yields and is therefore distinct from—and estimated separately from—the policy rate and the market yields already in the macro block.

- **Stock-flow adjustment (SFA, % of GDP):** the deficit–debt discrepancy described below.

The effective rate `EFFI` is the estimated interest driver of the deployed block. (The parametrisation has changed over time: an early version carried $\Delta(\text{debt}/\text{GDP})$ directly, and a later one carried debt service to GDP `DSVC`; the current block reverts to the effective rate.) The interest bill is then formed inside the accounting recursion as $\text{int}_t = i_t^{\text{eff}} d_{t-1}/100$.

The stock-flow adjustment as an estimated variable. The novelty of the current block is that the stock-flow adjustment (SFA) is now an *estimated* BVAR variable rather than a constant plug. It is the exact residual of the tool’s own debt recursion—the part of the observed change in gross debt that the flow side (deficit plus the interest–growth “snowball”) does *not* explain. Economically this seam arises because the headline deficit is measured on a federal-flows perimeter while the debt stock is gross debt (so intragovernmental holdings, valuation effects, and other below-the-line “other means of financing” operations move the two apart). It is constructed so that it closes the recursion to machine precision on history, and the pandemic quarters are set to missing (NaN) so the COVID-as-missing Gibbs sampler imputes them (Section 9.4) rather than letting the 2020 swing contaminate the estimate. Carrying SFA as a forecast variable means the debt-to-GDP *projection* now propagates its own SFA path forward (with dynamics and uncertainty bands) instead of defaulting the discrepancy to zero, which materially affects the high-debt blocs whose historical SFA is large and persistent. The historical SFA means (in % of GDP), which each bloc also ships as its `sfaDefault` fallback, are: US 1.75, EA 0.33, JP 2.23, UK 3.15, CA 3.19, CN 4.02.

Because all four drivers are ordinary BVAR variables, they inherit the full estimation machinery of Sections 2–4: the hierarchical Minnesota prior (Giannone et al., 2015), the COVID-as-missing data-augmentation Gibbs treatment, the companion-form Durbin–Koopman conditional smoother (Durbin and Koopman, 2002, 2012), and the Bańbura et al. (2015) conditioning-via-partial-observations construction. In the five Global-VAR fiscal blocs (EA, JP, UK, CA, CN)—and now in the endogenous US bloc itself—the fiscal drivers are part of the domestic block and propagate to the other economies through the shared Rest-of-World stars, exactly as the macro variables do (Pesaran et al., 2004; Dees et al., 2007).

A stationary prior for the fiscal ratios. The three GDP-ratio drivers—revenue `FREV_GDP`, primary expenditure `FPEXP_GDP`, and the stock-flow adjustment `SFA`—are mean-reverting shares that fluctuate around a roughly constant in-sample level; they are *not* integrated. Carrying them under the default random-walk own-lag prior ($\delta = 1$) is therefore the wrong null: it injects a built-in drift that lets the projected ratio wander away from its historical level at the long end of the horizon. These three drivers are accordingly tagged `prior: "WN"` in every fiscal bloc’s specification and estimated with the *stationary* white-noise own-lag prior ($\delta = 0$) of Section 2.5: the estimator collects them via `pos = find(prior == 'WN')` and passes the set as the `stationary` argument of `cgl_bvarV5`, so `diagb(pos) = 0` centers their own-lag dynamics on mean reversion to the equation constant. The effective interest rate `EFFI`, by contrast, *retains* the random-walk prior: it is a slow-

moving stock-average rate (net interest over lagged debt) whose level is genuinely persistent—it turns over only as maturing securities refinance—so a near-unit own-lag is the right prior null for it.

The empirical effect is exactly the intended one. Under the random-walk prior the U.S. revenue ratio was pinned near its end-of-sample value ($\sim 19\%$) and drifted from there; under the white-noise prior it now *reverts* to its $\sim 17.4\%$ in-sample mean over the projection, and the expenditure ratio and SFA likewise return to their historical means rather than carrying the last observation forward indefinitely. We emphasise (per Section 2.5) that this is a shift of the prior *center*, not a hard stationarity constraint: the posterior own-lag coefficient is still the GLP prior–likelihood balance at the estimated tightness λ , so a fiscal ratio whose data genuinely trend will still be allowed to do so—the WN prior only removes the mechanical drift that the random-walk center would otherwise build in. Because `pos` is empty for the 44 non-fiscal blocs, this change is a no-op (byte-identical output) everywhere except the six fiscal blocs.

Each fiscal bloc ships a `fiscal_meta.json` file that pins the column layout used by the accounting recursion. It carries the integer indices `idx = {gdp, deflator, revenue, primaryExp, effRate, sfa}` into the level-space forecast matrix—the first two locate the real-GDP and deflator log-levels that supply nominal growth, the next two locate the revenue and primary-expenditure drivers, `effRate` locates the estimated effective-rate column, and `sfa` locates the estimated stock-flow-adjustment column—together with the seed debt ratio d_0 and a fallback `sfaDefault` (e.g. the U.S. block has $d_0 = 122.57$, `idx.effRate` = 35, `idx.sfa` = 36, and `sfaDefault` = 1.75; the U.K. block has $d_0 = 96.24$, `idx.effRate` = 18, `idx.sfa` = 19). The presence of `idx.sfa` is the switch that selects the per-quarter estimated SFA: when it is set, `computeFiscalBlock` reads the forecast SFA off that column so the debt projection carries the discrepancy forward with its own dynamics and bands; when it is unset, the recursion falls back to the constant `sfaDefault` (or zero). A legacy block that instead carried an `idx.debtService` column would read that column directly as the period interest flow and back out the effective rate; the current effective-rate blocks do not.

7.2 The Budget-Identity Recursion

Let d_t denote the debt-to-GDP ratio (in percent), i_t^{eff} the effective rate (percent per annum), and g_t^n nominal GDP growth (percent per quarter). All flows below are annual ratios in percent of GDP. Nominal growth is read directly off the level-space forecast as the sum of the real-GDP and deflator log-differences, in the $100 \cdot \ln$ space of Section 2.3:

$$g_t^n = (\ell_t^{\text{gdp}} - \ell_{t-1}^{\text{gdp}}) + (\ell_t^{\text{defl}} - \ell_{t-1}^{\text{defl}}), \quad (40)$$

where ℓ is the stored $100 \cdot \ln$ level and the right-hand side is already in percent per quarter (it is not annualized by the factor of four used for display in (2)). The primary balance is the revenue–expenditure gap (positive = surplus),

$$pb_t = rev_t - primexp_t, \quad (41)$$

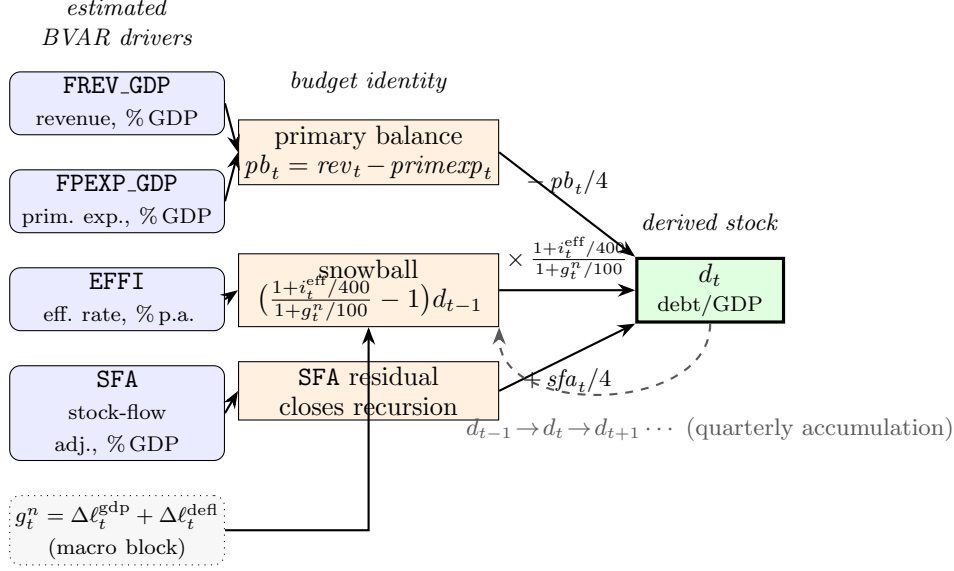


Figure 5: Fiscal stock-flow schematic. The four estimated BVAR drivers (FREV_GDP, FPEXP_GDP, EFFI, SFA; Section 7.1) feed the budget identity: revenue and primary expenditure form the primary balance pb_t (41); the effective rate EFFI and nominal growth g_t^n (40) form the interest-growth *snowball* (46); and the estimated SFA enters as the residual that closes the recursion. These flows accumulate into the derived debt-to-GDP stock d_t by the quarterized roll-forward $d_t = d_{t-1} (1+i_t^{\text{eff}}/400)/(1+g_t^n/100) - pb_t/4 + sfa_t/4$ (44), with the lagged stock d_{t-1} fed back each quarter (dashed). Debt is *derived*, never estimated (`computeFiscalBlock`, Section 7.2).

the interest bill (debt service) is the estimated effective rate applied to the lagged debt stock,

$$int_t = i_t^{\text{eff}} \frac{d_{t-1}}{100}, \quad (42)$$

i.e. the BVAR forecast of the effective rate EFFI drives the interest flow directly, with the lagged debt d_{t-1} converting the rate into a percent-of-GDP flow. (A legacy debt-service block would instead supply int_t as the estimated DSVC ratio and back out the effective rate as $i_t^{\text{eff}} = 100 int_t/d_{t-1}$.) The overall balance and deficit follow as

$$ob_t = pb_t - int_t, \quad def_t = -ob_t = int_t - pb_t. \quad (43)$$

The debt stock is then rolled forward one quarter by the FPP/DSA debt-dynamics equation,

$$d_t = d_{t-1} \frac{1 + i_t^{\text{eff}}/400}{1 + g_t^n/100} - \frac{pb_t}{4} + \frac{sfa_t}{4}, \quad d_{-1} \equiv d_0, \quad (44)$$

seeded at d_0 . The rate enters quarterized as $i_t^{\text{eff}}/400$ and growth as $g_t^n/100$, while the annual flow ratios pb_t and sfa_t are scaled by 1/4 so that one quarter's worth of the flow accrues per step; sfa_t is the stock-flow adjustment (the deficit–debt discrepancy of Section 7.1). In the current blocs sfa_t is the BVAR's *forecast* of the estimated SFA column (read off the level-space path quarter by quarter),

so the debt projection carries the discrepancy forward with its own dynamics and bands; absent an estimated SFA column the recursion falls back to a constant default. Gross financing needs add amortization to the deficit,

$$gfn_t = def_t + amort_t. \quad (45)$$

The recursion in (41)–(45) is the function `computeFiscalBlock` (with the lever-augmented variant in `runFiscalRecursion`); the exact quarterly convention is pinned by a historical back-cast test that reconstructs observed debt-to-GDP from the realized drivers.

Snowball decomposition. Subtracting d_{t-1} from (44) isolates the *automatic* debt dynamics from the discretionary primary balance. The change in the debt ratio is

$$\Delta d_t = \underbrace{\left(\frac{1 + i_t^{\text{eff}}/400}{1 + g_t^n/100} - 1 \right) d_{t-1}}_{\text{snowball (interest-growth)}} - \frac{pb_t}{4} + \frac{sfa_t}{4} \approx \frac{i_t^{\text{eff}} - g_t^n}{1 + g_t^n} d_{t-1} - pb_t + sfa_t, \quad (46)$$

where the right-hand approximation is the familiar Fiscal-Monitor ($i - g$) form (here in annualized flow units). The snowball term is positive—debt rises mechanically—whenever the effective rate exceeds nominal growth; a primary surplus and a favourable growth-interest differential are the two forces that stabilize the ratio. The tool returns the per-quarter snowball contribution $((1 + i_t^{\text{eff}}/400)/(1 + g_t^n/100) - 1)d_{t-1}$ alongside the derived balances so the user can read off the source of any debt drift.

Remark 2. Because the debt stock is *not* a BVAR variable, the estimated effective rate $\hat{i}_t^{\text{eff}}$ and primary balance $\widehat{pb}_t$ are debt-agnostic baselines: the recursion uses *lagged* debt d_{t-1} throughout, so it stays explicit (no within-period fixed point) and the active levers of Section 7.3 layer on without double-counting. This is the design that lets a single forward sweep produce a consistent debt path.

7.3 Active Levers

The accounting layer exposes three user-editable levers with literature-based defaults, governed by a master toggle (`feedbackOn`, default on). They modify the estimated baselines in terms of the lagged debt gap $d_{t-1} - d^*$, where the anchor d^* defaults to d_0 .

Bohn fiscal-reaction function (γ). Following [Bohn \(1998\)](#), the primary balance is allowed to respond to the lagged debt gap, so that fiscal policy consolidates as debt rises above its anchor:

$$pb_t = (rev_t - prime_{exp_t}) + \gamma (d_{t-1} - d^*), \quad \gamma \geq 0. \quad (47)$$

The default $\gamma = 0.03$ (three basis points of primary surplus per percentage point of excess debt) is the canonical [Bohn \(1998\)](#) estimate and the threshold above which a no-Ponzi debt process is mean-reverting. Setting $\gamma = 0$ disables the reaction.

Risk-premium channel (β). The effective rate is allowed to rise with debt above the anchor, capturing a sovereign risk premium:

$$i_t^{\text{eff}} = \widehat{i}_t^{\text{eff}} + \beta \max(0, d_{t-1} - d^*), \quad \beta \geq 0, \quad (48)$$

where the premium bites only *above* the anchor (the $\max(0, \cdot)$ rectifier). The default is $\beta = 0$: the magnitude of the debt-premium elasticity is empirically contested, so the lever is shipped inactive and left to the user. When active, β feeds back into both the interest bill (42) and the snowball term (46), producing the nonlinear debt-spiral dynamics that motivate the FPP/DSA realism literature.

Stock-flow-adjustment closure. The SFA term sfa_t in (44) is now *estimated* as its own BVAR variable (Section 7.1), so the default debt projection already carries the forecast SFA path forward with dynamics and bands rather than defaulting the discrepancy to zero. It can still be overridden—set to zero for pure flow accounting, held at the historical average, or used as a scenario lever to reconcile a target debt path with the flow drivers.

Both feedback channels act on the lagged gap and through the lagged debt already entering (42)–(44), so they are added cleanly to the debt-agnostic baselines without re-estimation and without double-counting.

7.4 Debt-Targeting Inversion

The recursion can also be run backwards. Given a desired debt-to-GDP path $\{d_t^{\text{target}}\}$, the effective rate, nominal growth, and SFA, the annual primary balance required at each quarter is obtained by solving (44) for pb_t along the target trajectory:

$$pb_t = 4 \left[d_{t-1}^{\text{target}} \frac{1 + i_t^{\text{eff}}/400}{1 + g_t^n/100} - d_t^{\text{target}} \right] + sfa_t, \quad (49)$$

with $d_{-1}^{\text{target}} \equiv d_0$. This is the IMF DDT/DSA “required primary balance / adjustment to the debt anchor” (`requiredPrimaryBalance`). The caller then conditions the BVAR on this primary-balance path through the linear constraint $rev_t - \text{primexp}_t = pb_t$, propagated by the DK smoother of Section 4. Because the effective rate and growth themselves respond to the primary balance through the BVAR covariance structure, an exact hit is obtained by iterating invert $\rightarrow$ condition $\rightarrow$ re-read $(i, g) \rightarrow$ re-invert; one or two passes converge.

7.5 Display Layout

The estimated and derived fiscal objects are presented in different chart rows, reflecting their different epistemic status. The four *estimated* fiscal drivers are shown as inputs alongside the macro block: revenue and primary expenditure as % of GDP, and the two new charts

- **Effective interest rate** (EFFI, % p.a.): the average rate the government pays on its outstanding stock. It is plotted as a level in percent per annum next to the policy rate and the market yields. Read it as the slow-moving cost of the debt: it lags the policy rate and the 10-year yield, because only maturing securities reprice, so a rate hike shows up here gradually rather than at once. This is the rate that enters the interest bill (42) and the snowball (46).
- **Stock-flow adjustment** (SFA, % of GDP): the deficit–debt discrepancy. It is plotted as a level in percent of GDP. A positive value means gross debt rose by more than the deficit plus interest–growth dynamics imply (debt-increasing other-means-of-financing operations); a value near zero means the flow accounting closes cleanly. Its history is informative—it is large and persistent for the high-debt blocs (Table values of order +2 to +4% of GDP for JP, UK, CA, CN)—and, crucially, its *forecast* now feeds the debt projection, so a persistently positive SFA pushes the projected debt path *above* what the deficit alone would deliver.

A further row holds the *derived* reporting variables produced by the accounting recursion, in the order

- (i) **Primary Balance** (pb_t , % of GDP), which is itself conditionable: the user can drag a primary-balance path and the tool maps it back to the $rev - primexp$ linear combination;
- (ii) **Δ Debt-to-GDP** (Δd_t , percentage points per quarter), the per-quarter debt change of (46), which is not directly conditionable because it is a pure identity output;
- (iii) **Debt-to-GDP** (d_t , % of GDP), the rolled-forward stock, which can be targeted via the inversion of Section 7.4. Note that this projection now *carries the forecast SFA forward* (with dynamics and bands): a bloc with a large positive estimated SFA will show a debt path that drifts up faster than its primary balance and snowball alone explain, which is the intended, honest behaviour.

Separating the estimated drivers (effective rate, SFA, revenue, expenditure) from the derived balances and debt makes the stock-flow-consistent structure visible on screen: what the data and the prior speak to is shown as an input, and what the budget identity implies is shown as an output.

7.6 Data Treatment: the U.K. Index-Linked-Gilt Spike

Interest series constructed from accrued interest can contain mechanical, non-recurring spikes that are not informative about the underlying refinancing cost. In the United Kingdom, the large stock of *index-linked gilts* produced an interest spike in 2008–09: the inflation uplift on the principal of index-linked debt is recorded as accrued interest, so the 2008 inflation spike (and its subsequent unwinding) translated into an outsized swing in the measured effective rate EFFI that would otherwise contaminate the BVAR’s autoregressive dynamics. These quarters are handled with the same machinery used for the COVID-19 observations: the affected effective-rate entries are set to missing (NaN) and imputed by the Kalman filter inside the data-augmentation Gibbs sampler, rather than

being allowed to enter the likelihood as if they were ordinary observations. This keeps the estimated effective-rate process—and hence the derived interest bill and debt path—free of a one-off accounting artefact while preserving the genuine information in the surrounding quarters.

8 Long-Run Anchoring: Prior vs Entropic Tilt

The forecasts of Section 4 inherit their long-run behaviour entirely from the estimated VAR. Because the real-activity and price blocks are estimated on 100 ln levels of $I(1)$ series, the companion matrix (6) carries roots on or near the unit circle, and the implied long-horizon central path can drift away from any value the user regards as a credible steady state (e.g. a 2% inflation objective, a neutral policy rate, or a balanced great ratio). The tool exposes a *long-run anchor* control that lets the user pin the long-run forecast of a chosen linear combination $h'y$ to a target. We support two conceptually distinct mechanisms, summarized in Table 2: an *in-prior* long-run prior applied *before* estimation, and an *after-estimation* entropic tilt of the predictive density. They answer different questions, and the distinction is methodologically important.

8.1 In-Prior Long-Run Prior (PLR)

The first mechanism is the “prior for the long run” (PLR) of Giannone et al. (2019), which augments the hierarchical Minnesota prior of Section 2.4 with additional dummy observations that shrink a small set of linear combinations Hy toward stationarity (no-cointegration / random-walk behaviour). Let $H \in \mathbb{R}^{q \times n}$ collect q long-run combinations and let y_0 be the prior pre-sample mean. Following the authors’ construction, H is rotated into a full-rank basis $\bar{H} = (H'; \text{null}(H)')'$ with inverse $\bar{H}^{-1}$, and for each active long-run row j a single dummy observation

$$y_j^d = \frac{(y_0 \bar{H}')_j}{\phi_j} (\bar{H}^{-1})'_{.j}, \quad x_j^d = \left(0, \frac{(y_0 \bar{H}')_j}{\phi_j} \mathbf{1}_p \otimes (\bar{H}^{-1})'_{.j}\right), \quad (50)$$

is appended *to the regression system* (both the Gibbs draw and the marginal-likelihood normalizing constant T_d), with ϕ_j controlling the tightness of the j -th long-run restriction. Crucially, (50) combines the *equations* through the columns of $\bar{H}^{-1}$; combining the regressors instead produces a dummy whose residual is identically zero at the sum-of-coefficients mode and hence carries no tightness information. The tightness hyperparameters ϕ_j are held fixed so that the Metropolis–Hastings search over the Minnesota λ and the sum-of-coefficients μ remains two-dimensional.

The PLR is a *proper Bayesian prior*: it modifies the posterior and requires re-estimation. It is therefore computed offline as part of the data-export pipeline, not in the browser.

Remark 3 (What the PLR actually does in the US sample). A US head-to-head ($n = 36$, $p = 3$, COVID treated as missing per Section 4.6, $q = 13$ long-run combinations comprising six great ratios, three nominal anchors, and four rate/credit spreads) shows that the PLR is a *separate dummy block* rather than a change to the Minnesota tightness: the maximizing λ is essentially unchanged (0.563 baseline vs 0.562 with the PLR at either $\phi = 1.0$ or $\phi = 0.1$). Because the sum-of-coefficients-style

dummy in (50) shrinks toward a *random walk* (no cointegration), and the US sample treats most of these ratios as genuinely $I(1)$, the PLR does *not* pull the long-run central forecast toward a historical mean: the mean absolute drift of the $H=40$ forecast from a recent historical anchor moves from 8.49 (baseline) to 8.41 at $\phi = 1.0$ (-1%), and is marginally *worse* at $\phi = 0.1$. The visible effect is on the *bands*, not the path: the GDP-growth 68% band tightens from 4.24 to 3.95 percentage points ($\approx 7\%$) at $\phi = 1.0$. The PLR is thus the appropriate instrument for principled band tightening and a soft no-cointegration restriction; it is *not* a device for pinning a long-run combination to a chosen number.

8.2 After-Estimation Entropic Tilt

The second mechanism is the user-facing one, exposed in the interface as the **Long-run priors** panel (Section 11). It does *not* re-estimate the model. Instead it takes the *existing* posterior draws $\{(\hat{\beta}^{(d)}, \hat{\Sigma}^{(d)})\}_{d=1}^D$ of Section 4.6 and applies an exponential (entropic) tilt — a relative-entropy reweighting in the spirit of Robertson et al. (2005) — so that the *weighted* mean of each long-run moment matches the user’s target. The result is a vector of per-draw importance weights that the band loop applies through weighted quantiles; the underlying draws are never re-solved.

The long-run moment. For draw d the relevant long-run quantity is taken to be the *long-horizon mean forecast*, not the algebraic steady state:

$$g^{(d)} = h' \mathbb{E}[y_{T+H} \mid \hat{\beta}^{(d)}], \quad (51)$$

obtained by iterating the deterministic companion recursion (5) forward H steps with the innovations switched off, exactly as in the mean iteration underlying (21). The default horizon is $H = 40$ (≈ 10 years quarterly), matching the convention of Giannone et al. (2019). We deliberately do *not* use the closed-form steady state $y^* = (I - \sum_{\ell} B_{\ell})^{-1}c$, $g = h'y^*$. For the near-unit-root level VARs the matrix $(I - \sum_{\ell} B_{\ell})$ is near-singular, so y^* is a meaningless quantity of order 10^5 : e.g. for the US GDPH – CH log spread the steady-state evaluation returns $\approx -2.1 \times 10^4$, whereas the $H=40$ forecast moment is the finite $\approx +35$ that matches the chart. The steady-state evaluation is retained only as an opt-in for genuinely stationary combinations.

For a single user variable the moment is stated in the chart’s own display units (Section 2.3): for a `lin` variable (rates, unemployment) the target is the forecast *level* $e_j' \mathbb{E}[y_{T+H}]$; for a `log` variable it is the annualized growth $4(\mathbb{E}[y_{T+H,j}] - \mathbb{E}[y_{T+H-1,j}])$, consistent with (20), so the user enters the number they read off the axis.

KL-minimizing weights. Collecting q restrictions with targets $\bar{g}_r$ and tightness scales s_r , define the centered, scaled moments $\tilde{g}_r^{(d)} = (g_r^{(d)} - \bar{g}_r)/s_r$. We seek weights $w_d \propto \exp(\gamma' \tilde{g}^{(d)})$ that satisfy the moment conditions $\sum_d w_d g_r^{(d)} = \bar{g}_r$ for all r while minimizing the Kullback–Leibler divergence (Kullback and Leibler, 1951) from the uniform weights $1/D$. This is the classical exponential-tilting / minimum-relative-entropy projection: the tilted measure is the I -projection of the empirical

predictive measure onto the affine set of distributions satisfying the long-run moment constraints. The Lagrange dual is the smooth, convex objective

$$F(\gamma) = \log \left(\frac{1}{D} \sum_{d=1}^D \exp(\gamma' \tilde{g}^{(d)}) \right), \quad \nabla F(\gamma) = \mathbb{E}_w[\tilde{g}], \quad \nabla^2 F(\gamma) = \text{Cov}_w(\tilde{g}), \quad (52)$$

whose stationary point $\nabla F(\gamma^*) = 0$ enforces the constraints. We solve (52) by Newton iteration with a backtracking (Armijo) line search and a 10^{-10} ridge on the Hessian, computing F via a numerically stable log-mean-exp; the weights match the target mean exactly regardless of the scales s_r , which serve only to balance restrictions of different magnitudes (each set to the moment’s own posterior standard deviation). The normalized weights w_d are applied as a weighted reduction of the draw cloud: they move the *unconditional central path* (the displayed baseline becomes the weighted mean of the draws, not the unweighted posterior-mean path) and the band quantiles together, on *both* the baseline fan (the precomputed band of Section 4.6 is replaced by the in-browser tilted band) and the conditional/scenario fan, so the entire predictive density—central path included—is pulled toward the anchor.

Effective sample size and degeneracy. The tilt is a pseudo-posterior reweighting, and its reliability is governed by the effective sample size

$$\text{ESS} = \frac{(\sum_d w_d)^2}{\sum_d w_d^2} = \frac{1}{\sum_d w_d^2} \in [1, D], \quad (53)$$

the harmonic measure of how many draws carry non-negligible mass after reweighting. As the target is pushed farther from the prior mean $D^{-1} \sum_d g^{(d)}$, the weights concentrate and ESS falls. The interface surfaces ESS/ D as a color-coded badge (green above 40%, amber between 10% and 40%, red below 10%) so the user can see when an anchor is straining the support of the draws.

Remark 4 (Per-constraint vs joint tilt). In the US comparison of Remark 3, with $D = 500$ draws a *per-constraint* tilt — anchoring one combination at a time — hits every target exactly with a healthy effective sample size (median ESS $\approx 75.9\%$ of D). The *joint* 13-constraint tilt, by contrast, is degenerate: ESS $\rightarrow 1$, because the joint anchor target lies outside the 13-dimensional convex hull of only 500 draws even though each marginal target lies inside its own one-dimensional hull. The practical consequence, reflected in the interface, is that the entropic tilt is well-behaved for one or a few long-run restrictions but should not be over-constrained: the panel defaults to a single anchor and warns when several are enabled simultaneously.

8.3 When to Use Which

The two mechanisms are complementary. The in-prior PLR is the gold standard — a proper offline posterior — but its data-decides property means that, at the tightnesses examined, it leaves the US central long-run forecast essentially unchanged and acts mainly on the bands. The entropic tilt

is in-browser, requires no re-estimation, and *forces* the weighted long-run forecast onto an explicit target, but it is a reweighting of a finite draw cloud and degrades (falling ESS) if over-constrained. As a rule of thumb: to *pin* a long-run combination to a number, use the entropic tilt (or, offline, a tighter targeted prior); for principled band tightening and a soft no-cointegration restriction, use the PLR.

Table 2: Long-run anchoring: in-prior PLR versus after-estimation entropic tilt.

	In-prior PLR	Entropic tilt
Source	Giannone et al. (2019)	Robertson et al. (2005)
Acts on	prior (dummy obs.)	posterior draws (weights)
Re-estimation	yes (offline)	no (in-browser)
Object	proper posterior	pseudo-posterior
Moment	$H y$ stationarity	$h' \mathbb{E}[y_{T+H}]$
Main effect	tightens bands	shifts central path
Failure mode	none (proper)	low ESS if over-constrained

9 Economy Coverage, Data Sources, and Estimation

The tool ships estimated BVARs for roughly fifty economies. Thirty-three of them are integrated into a single *Global VAR* (GVAR) system ([Pesaran et al., 2004](#); [Dees et al., 2007](#)) covering approximately 90% of world GDP; the remaining blocs—including the 17 euro-area member states, which are deliberately kept out of the coupled system because a disaggregated euro GVAR is non-stationary—are standalone single-country models that share the same estimation machinery but are not wired into the cross-country fixed point. This section documents the roster, the underlying data, the COVID-as-missing estimator, the common-vintage harmonization, and the construction of the predictive bands. Throughout, variables enter the VAR in the $100 \ln(\cdot)$ space for `log` series and in levels for `lin` series (Section 2.3); annualized quarter-over-quarter growth and inflation are reported as $4 \times \Delta(100 \ln)$, consistent with (2) and (20).

9.1 The GVAR System: 33 Integrated Blocs

The GVAR comprises the United States—now a fully endogenous coupled bloc, not a frozen anchor (Section 6.1)—the euro area as a single aggregate bloc, plus 31 further Rest-of-World (RoW) blocs:

US; EA (euro-area aggregate), JP, UK, CA, CN (the core majors); CH, KR, MX, AU, TR, SE, PL, NO, DK, IN, BR, RU, ID, SA, AR, ZA, TH, MY; and the nine further non-euro economies CL, CO, CR, CZ, HU, IL, IS, NZ, TW added in the OECD expansion.

The euro area enters as one aggregate bloc by necessity: a fully disaggregated euro GVAR is non-convergent ($\rho(M) \approx 1.089 > 1$), and treating the euro area as a single bloc restores stationarity.

The 17 euro-area members are consequently *standalone* models (Section 9.2), not coupled GVAR blocs. Each bloc—the US included—is an independently estimated single-country BVAR in which the foreign sector is summarized not by a raw foreign-variable block but by three trade-weighted RoW “star” variables, following the GVAR tradition of Pesaran et al. (2004) and Dees et al. (2007):

- ROWDEM — foreign demand: the trade-weighted average of partners’ real-GDP growth;
- ROWINF — foreign inflation: the trade-weighted average of partners’ price inflation;
- ROWFIN — foreign financial conditions: the trade-weighted average of partners’ short rates.

For bloc c with partner set $\mathcal{P}_c$ and weights w_{cj} , the demand and inflation stars are built in annualized growth space from the partners’ stored 100 ln levels,

$$\text{ROWDEM}_{c,t} = 4 \times \frac{\sum_{j \in \mathcal{P}_c} w_{cj} (x_{j,t}^g - x_{j,t-1}^g)}{\sum_{j \in \mathcal{P}_c} w_{cj} \mathbb{I}\{x_{j,t}^g, x_{j,t-1}^g \text{ observed}\}}, \quad (54)$$

where $x_{j,t}^g$ is partner j ’s real-activity level (analogously for ROWINF from prices, and ROWFIN from the level of the short rate). Weights are *renormalized per component* over the partners actually observed in each quarter, so a partner that is missing a given series simply drops out of the corresponding star and the remaining weights are rescaled. Two implementation details matter for the econometrician: China’s reported GDP is nominal, so its real-activity level is formed by subtracting the price level in 100 ln space, $x_{\text{CN}}^g = x_{\text{CN}}^{\text{NGDP}} - x_{\text{CN}}^P$ (a log-space deflation); and Argentina’s 2023–24 hyperinflation episode is excluded from the ROWINF and ROWFIN aggregates of its partners (its genuine real demand is retained in ROWDEM), with renormalization absorbing the dropped weight.

Trade weights. Bilateral weights w_{cj} are constructed from goods-trade flows in the IMF International Trade in Goods statistics (IMTS, the renamed bilateral DOTS), queried from the SDMX 2.1 endpoint api.imf.org over the 2021–2023 average of exports (XG_FOB_USD) plus imports (MG_CIF_USD). The Euro Area is treated as a single counterpart (IMTS aggregate code G163). The weight matrix is row-normalized over the 33 blocs with a zero diagonal; now that the US is endogenous it carries its own trade-weights row, so all 33 blocs have a row.

Rate-optional blocs. The short rate is *optional*: a GDP series and a price series are required of every bloc, but two blocs (Switzerland and Norway) carry no domestic short rate. Their rate column is NaN throughout, they are excluded from their partners’ ROWFIN aggregate, and the ROWFIN weights of any bloc having them as a partner are renormalized accordingly. They still contribute to ROWDEM and ROWINF.

Cross-country coupling. As in Section 9’s original treatment, the baseline forecast of each bloc is computed independently; cross-country coupling enters only through the scenario layer. A shock conditioned in one bloc perturbs its star variables, which propagate to its trade partners, whose

own stars then feed back. Stacking the per-bloc deviations Δ yields the linear fixed point

$$\Delta = M \Delta + d, \quad (55)$$

where M encodes the trade-weighted star couplings together with the four US-determined global commons (the spot oil price, the US 10-year yield, the US Baa corporate yield, and US equity; Section 6.6) and d collects the user’s direct conditioning forcing. With the US endogenous the deviation state is *augmented* to carry the US global-column deviations as unknowns (Section 6.4), so the global commons close the RoW→US feedback loop inside M rather than entering d as a frozen forcing. The system is solved at display time both by a direct $(I - M)^{-1}d$ evaluation and by Neumann sweeps (summing $\sum_k M^k d$); the two agree to $\sim 10^{-9}$. The coupling is contractive, with spectral radius $\rho(M) \approx 0.924$, so the series converges. Because the full 2060×2060 $(I - M)^{-1}$ inverse is expensive in-browser (a cold build is on the order of three-quarters of an hour), it is precomputed offline and shipped as a *gzipped* coupling cache (`coupling_default.json.gz`, about 30 MB compressed), decompressed in the browser via the native `DecompressionStream` API. The interactive interface uses a two-tier solve—a fast direct preview on drag and an explicit “Solve full GVAR” button for the complete fixed point—to keep the endogenous-US solve reactive (Section 6.5).

9.2 Standalone Single-Country Blocs

Beyond the 33 GVAR blocs, the tool ships additional single-country BVARs estimated by the same procedure but not coupled into (55). Together with the GVAR members these complete coverage of all 38 OECD members and add several large non-OECD emerging markets. The standalone remainder is dominated by the 17 *euro-area member states* (DE, FR, IT, ES, NL, AT, BE, FI, GR, IE, LU, PT, SI, SK, EE, LT, LV), which are kept out of the coupled GVAR *by construction*: the euro area is represented in the GVAR by the aggregate **EA** bloc, because a disaggregated euro GVAR is non-stationary ($\rho(M) \approx 1.089 > 1$) whereas aggregating the euro area restores convergence. Each euro member is nonetheless estimated as a full single-country model—its domestic block augmented by the four global commons and by trade-weighted foreign *star* variables—but in *separation* (with the GVAR iteration count fixed to `MAXIT` = 1, so its foreign stars are read off data rather than from a cross-country solve) and it carries no trade-weights row. The country selector groups the roster into two sections, a *GVAR* group and an *Other countries* group; a bloc is GVAR-linked, and hence placed in the *GVAR* group, if and only if it has a trade-weights row and its specification carries the three `ROW*` star variables.

9.3 Data Sources

No single provider supplies current quarterly real GDP for the full roster, so the data are assembled bloc-by-bloc:

- **FRED.** The US block ($n = 41$, Table 3) and many other series are pulled from the St. Louis Fed’s FRED. The China bloc is built entirely from verified real FRED series, with quarterly

real activity anchored by industrial production (PRCNT001CNQ661S) since clean quarterly *real* GDP is not available on FRED for China (only annual constant-price GDP and nominal quarterly GDP that ends early).

- **Eurostat.** The IMF’s new thematic SDMX API exposes no per-country quarterly chain-volume GDP (its national-accounts flow carries current-price values only, for a limited country set), so the 22 EU/EFTA OECD economies are sourced from Eurostat (`namq_10_gdp` chain-linked volumes, `prc_hicp_midx` HICP, `une_rt_q` unemployment, `irt_*` interest rates, `ert_eff_ic_q` NEER), which is current to 2025Q4–2026Q1.
- **National banks and OECD QNA.** The non-EU OECD members (e.g. Chile, Korea, Australia, New Zealand, Mexico, Türkiye) draw real GDP and national accounts from the OECD Quarterly National Accounts, short rates from national-bank series (e.g. RBA/RBNZ), and CPI, unemployment, and NEER from the IMF flows.

Real versus nominal GDP is handled per bloc: where a chain-volume real series exists it is used directly; China’s nominal GDP is deflated in log space as above; and thin or volatile samples are estimated with deliberately wide, honest predictive bands rather than spuriously tight ones.

9.4 COVID-as-Missing Estimation

The 2020 pandemic quarters are an extreme outlier that, left untreated, would contaminate the autoregressive estimates and the innovation covariance. Rather than scaling them out, the tool treats the pandemic window as *missing data* and imputes it within the estimation itself, via a data-augmentation Gibbs sampler that nests a Durbin–Koopman simulation smoother (Durbin and Koopman, 2002, 2012). Each Gibbs sweep alternates:

- (i) a Durbin–Koopman simulation-smoother draw of the missing pandemic rows given the current coefficients, treating those rows as NaN observations in the companion-form state space of Section 3;
- (ii) a Metropolis–Hastings update of the hierarchical-prior hyperparameters on the *completed-data* marginal likelihood, so the overall tightness λ and the sum-of-coefficients parameter μ (Giannone et al., 2015) are selected by empirical Bayes without any fixed-tightness restriction; and
- (iii) a draw of the VAR coefficients (β, Σ) from the completed data.

The pandemic history is thus *preserved* (the surrounding quarters remain in the sample and inform the imputation) while the outlier itself does not drive the dynamics.

Window selection. The COVID window self-calibrates to each bloc’s pandemic trough rather than being hard-coded. It is the *argmin* of the quarter-on-quarter GDP change *restricted to the*

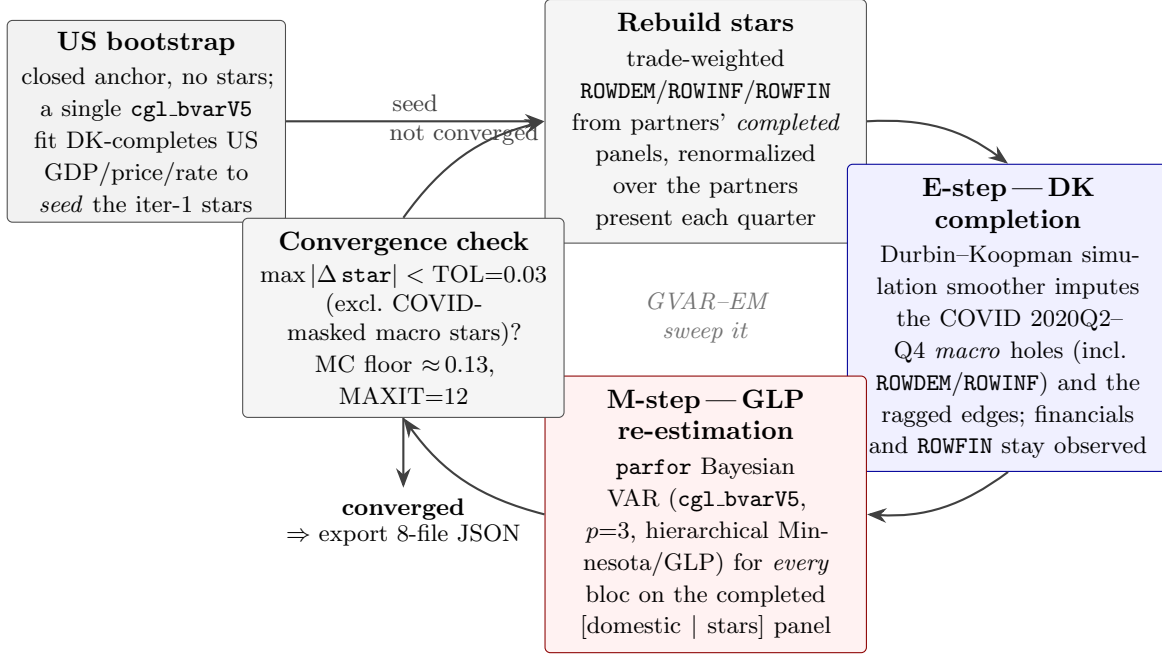


Figure 6: The GVAR-EM estimation loop under pervasive missingness (Giannone et al., 2019; Durbin and Koopman, 2002). A closed US bootstrap (no stars) is estimated once to seed the first trade-weighted foreign variables. Each sweep then (i) rebuilds the ROWDEM/ROWINF/ROWFIN stars from partners’ completed panels, renormalized over the partners present each quarter, and (ii) re-estimates every bloc by hierarchical-prior Bayesian VAR (Gibbs, run in `parfor`). Within each bloc’s single `cgl_bvarV5` pass the **E-step**—a Durbin–Koopman simulation-smoother completion that imputes the COVID macro holes (the fixed 2020Q2–Q4 window, with financial series and ROWFIN held observed) and the ragged, unbalanced-panel edges—and the **M-step**—the posterior re-estimation of the VAR coefficients on the completed [domestic | stars] panel—are carried out jointly by the same sampler. The sweep repeats until the foreign components stop moving, $\max |\Delta \text{star}| < \text{TOL} = 0.03$ (evaluated after the second sweep and excluding the COVID-masked macro stars, whose cross-iteration churn is Monte-Carlo noise with floor ≈ 0.13), capped at $\text{MAXIT} = 12$ iterations.

window 2019Q4–2021Q1:

$$k^* = \arg \min_{t \in \{2019Q4, \dots, 2021Q1\}} (x_t^g - x_{t-1}^g), \quad \mathcal{W} = \{k^* - 1, k^*, k^* + 1, k^* + 2\}, \quad (56)$$

and the four rows of $\mathcal{W}$ (intersected with the sample) are NaN’d across *all* variables before estimation. Restricting the search to 2019Q4–2021Q1 is deliberate: an unrestricted global argmin misfires for economies whose deepest contraction is a different crisis (for example Korea’s 2008Q4 Global-Financial-Crisis drop exceeds its 2020Q2 pandemic drop).

Fixed window and macro-only masking in the production GVAR loop. The auto-detected window above is the legacy single-bloc convention; the production GVAR estimator (`pipeline/export_bvar_gvar2`) instead uses a *fixed* three-quarter window $\mathcal{W} = \{2020Q2, 2020Q3, 2020Q4\}$ for every bloc, which avoids the $\sim$ seven-quarter spans the unrestricted trough search could produce. Equally impor-

tant, the production loop masks *only the macro columns*: over $\mathcal{W}$ the domestic macro variables (`isFinancial = false`) and the two macro stars `ROWDEM/ROWINF` are set `NaN`, while the domestic financial series (rates, yields, equity, house prices, corporate spreads) and the financial star `ROWFIN` are *kept observed*. The Durbin–Koopman smoother then imputes the pandemic macro holes *conditional on* the observed financials, which carry genuine signal through 2020—the masking is thus “macro-masked / financials-observed,” the same convention the out-of-sample evaluation re-applies in-sample at every origin (Section 14.2).

Algorithm 3 states the full estimation loop. It is the operational form of the cycle sketched in Figure 6: a no-stars US bootstrap seeds the first foreign variables, after which each sweep rebuilds every bloc’s trade-weighted stars from partners’ completed panels (the *E-step* regressor), re-estimates all blocs in parallel under macro masking (the per-bloc `cgl_bvarV5` call, whose own Durbin–Koopman completion is the within-bloc *E-step* and whose Gibbs re-estimation is the *M-step*), and recomputes the stars from the freshly completed data, iterating to a fixed point in the completed stars.

Algorithm 3 EM-style GVAR estimation loop (Jacobi star re-convergence; `export_bvar_gvar24_loop.m`).

Input: per-bloc panels $\{X_c\}$ ($X_c = [\text{domestic} \mid \text{stars}]$); trade weights W ; lags $p = 3$; WN/stationary index sets pos_c ; tolerance $\text{TOL} = 0.03$; cap $\text{MAXIT} = 12$; fixed COVID window $\mathcal{W} = \{2020\text{Q2}..\text{Q4}\}$.

Output: per-bloc posterior means $(\hat{\beta}_c, \hat{\Sigma}_c)$, completed panels, predictive draws `Dforecastc`, exported 8-file JSON.

```

1: ▷ Bootstrap (seed the first stars).
2: Mask the US macro columns over  $\mathcal{W}$  and fit the closed, no-stars US bloc with cgl_bvarV5;
   the posterior-mean DK completion gives the initial US real-GDP/price/rate panel (this seeds
   partners’ iteration-1 stars).
3: for  $it = 1, \dots, \text{MAXIT}$  do ▷ Jacobi sweep
4:   ▷ Build all stars from the FROZEN start-of-sweep completed panels.
5:   for each re-estimated bloc  $c$ :  $\text{star}_c \leftarrow$  trade-weighted ROWDEM/ROWINF/ROWFIN from part-
     ners’ completed panels (renormalize over partners present each quarter).
6:   parfor each bloc  $c$  do ▷ independent given the frozen stars
7:      $x_c \leftarrow [\text{domestic}_c \mid \text{star}_c]$ 
8:     over  $\mathcal{W}$ , set NaN the macro columns of  $x_c$  and ROWDEM,ROWINF (keep financials, ROWFIN)
9:      $(\text{Xcond}_c, \beta_c, \Sigma_c, \text{Dforecast}_c) \leftarrow \text{cgl\_bvarV5}(x_c, p, \text{pos}_c, \dots)$  ▷ E-step DK completion +
     M-step Gibbs
10:    completed panel  $\hat{X}_c \leftarrow \text{mean}_{\text{draws}} \text{Xcond}_c$ 
11:  end parfor
12:   $mc \leftarrow \max_c \max |\text{star}_c^{(it)} - \text{star}_c^{(it-1)}|$  ▷ exclude COVID-masked ROWDEM/ROWINF cells
13:  if  $it > 1$  and  $mc < \text{TOL}$  then break ▷ converged
14: end for
15: Export each bloc’s  $(\hat{\beta}_c, \hat{\Sigma}_c)$ , completed history, stars, and predictive draws to the 8-file JSON
    contract.
```

The sweep is a *Jacobi* update—all stars are rebuilt from the common start-of-sweep state *before*

the parallel re-estimation—so the parallel blocs are mutually independent given the current stars and the sweep has the same fixed point as a sequential (Gauss–Seidel) pass, at a wall-clock set by the slowest single bloc (the $n = 41$ US bloc) rather than the sum. The convergence metric excludes the COVID-masked macro-star cells, whose cross-sweep churn is pure imputation Monte-Carlo noise (floor ≈ 0.13); the financial star `ROWFIN`, kept observed through the pandemic, stays in the metric. Each `cg1_bvarV5` call is itself the data-augmentation Gibbs sampler of Algorithm 1—its DK completion is the bloc-level E-step and its hyperparameter/coefficient draws are the M-step—so the whole loop is a block-coordinate EM over (foreign stars, completed-data states, VAR coefficients).

9.5 Common-Vintage Harmonization to 2025Q4

So that the first forecast (nowcast) quarter is uniform across the tool, every bloc’s domestic series is extended to a common 2025Q4 endpoint, making 2026Q1 the first projected quarter everywhere. For blocs whose quarterly data end short of 2025Q4:

- **Real-activity and price series** (GDP, consumption, investment, government, exports, imports, CPI/HICP, all `log`) are benchmarked to IMF World Economic Outlook annual projections (NGDP_RPCH real growth, PCPIPCH inflation) via temporal disaggregation: a within-year quarterly profile is laid down at the WEO-implied annual growth and then proportional-Denton corrected so that each fully appended year’s four-quarter average reproduces the WEO annual level (already-observed quarters in a partial year are held fixed). The realized annual growth of each appended year matches WEO to within 0.2pp.
- **Financial series with no WEO analogue** (NEER, REER, equity indices, and all `lin` rates — short rates, 3-month rates, 10-year yields, unemployment) are carried forward by *hold-last* and flagged as such.

The original raw vintages are left untouched; the extended panels are written separately and consumed by the estimation drivers.

9.6 Predictive Uncertainty Bands

For every bloc the exported baseline carries *predictive* bands, not parameter-only bands. The COVID-missing Gibbs sampler emits a forecast density that integrates over three sources of uncertainty simultaneously — posterior parameter uncertainty, future innovation uncertainty, and missing-data (imputation) uncertainty. The reported quantiles are the 5th, 16th, 84th, and 95th percentiles of this predictive density (with the 50th retained in level space),

$$(q_{05}, q_{16}, q_{84}, q_{95})_{T+h} = \text{Quantile}_{\{0.05, 0.16, 0.84, 0.95\}} \{ \tilde{y}_{T+h}^{(d)} \}_{d=1}^D, \quad (57)$$

yielding the 68% band (q_{16}, q_{84}) shaded in the charts and the 90% band (q_{05}, q_{95}) . Crucially, the central path $\bar{y}^{\text{baseline}}$ remains the *deterministic* forecast from the posterior-mean coefficients (so the unconditional smoother of Section 4.5 reproduces it exactly, and zero user deviations return the

stored baseline); only the bands are predictive. This is the designed fix for the previously over-tight bands that resulted from spreading posterior-mean paths under a tight prior — wide bands now reflect genuine macro volatility (e.g. CN, JP, UK), not the COVID outlier. On-the-fly conditional bands computed in the browser use the same predictive construction, drawing innovations through the simulation smoother of Section 4.6.

9.7 US Specification

The US is the largest bloc, with $n = 41$ quarterly variables and $p = 3$ lags. Its full variable list is given in Table 3; the four-variable fiscal block (FREV_GDP, FPEXP_GDP, EFFI, SFA) supports the fiscal-reaction counterfactuals in the spirit of Bohn (1998), and the conditional-forecast and optimal-policy machinery of Bańbura et al. (2015) and Barnichon and Mesters (2023) applies uniformly across all blocs. The count of 41 comprises the domestic and fiscal variables, the four global commons, and the three trade-weighted ROW*_US star variables that make the US a fully coupled GVAR bloc (Section 6.1).

Table 3: Variable coverage for the US BVAR model. “Trans.” denotes the data transformation: `log` = log levels (growth rates displayed), `lin` = levels. “Prior” denotes the prior mean: RW = random walk, WN = white noise.

ID	Description	Trans.	Prior
GDPH	Real Gross Domestic Product	log	RW
CH	Real Personal Consumption Expenditures	log	RW
FRH	Real Private Residential Fixed Investment (nom/deflator)	log	RW
FNH	Real Private Nonresidential Fixed Investment (nom/deflator)	log	RW
XH	Real Exports of Goods & Services	log	RW
MH	Real Imports of Goods & Services	log	RW
GH	Real Government Consumption Expenditures & Gross Investment	log	RW
JGDP	Gross Domestic Product: Chain Price Index	log	RW
JC	Personal Consumption Expenditures: Chain Price Index	log	RW
JCXFE	PCE less Food & Energy: Chain Price Index	log	RW
UI	CPI-U: All Items	log	RW
UIXFDG	CPI-U: All Items Less Food & Energy	log	RW

ID	Description	Trans.	Prior
LXBC	Business Sector: Compensation Per Hour (nominal)	log	RW
LANAGRA	All Employees: Total Nonfarm	log	RW
LR	Civilian Unemployment Rate: 16 yr +	lin	RW
IP	Industrial Production Index	log	RW
CUMFG	Capacity Utilization: Manufacturing [SIC]	lin	RW
HST	Housing Starts	log	RW
YPDH	Real Disposable Personal Income	log	RW
CSENT	University of Michigan: Consumer Sentiment	lin	RW
POLRATE	Effective Federal Funds Rate (policy rate)	lin	RW
FCM2	2-Year Treasury Note Yield at Constant Maturity	lin	RW
FCM5	5-Year Treasury Note Yield at Constant Maturity	lin	RW
GLB-USLR	US 10-Year Treasury Yield (global common)	lin	RW
FAAA	Moodys Seasoned Aaa Corporate Bond Yield	lin	RW
GLB-USBAA	US Baa Corporate Bond Yield, credit proxy (global common)	lin	RW
FXTWM	Nominal Trade-Weighted Exch Value of US\$ vs AFE Currencies	log	RW
GLB-USEQ	US Equity Price Index (global common)	log	RW
PZTEXP	Spot Oil Price: West Texas Intermediate	log	RW
GSNE	Non-Fuel Commodity Price Index (IMF, non-energy proxy)	log	RW
SP500VOL	CBOE Volatility Index for S&P 500 Stock Price Index	lin	RW
HP	House Price	log	RW
LEV	Real nonfinancial business debt	log	RW
FREV_GDP	Federal Receipts (% of GDP)	lin	WN
FPEXP_GDP	Federal Primary Expenditure (% of GDP)	lin	WN

ID	Description	Trans.	Prior
EFFI	Federal Effective Interest Rate (% p.a.)	lin	RW
SFA	Stock-Flow Adjustment (% of GDP)	lin	WN
GLB_OIL	Spot Oil Price, WTI (global common)	log	RW
ROWDEM_US	US RoW demand (trade-weighted partner GDP growth)	lin	RW
ROWINF_US	US RoW inflation (trade-weighted partner inflation)	lin	RW
ROWFIN_US	US RoW financial conditions (trade-weighted partner rates)	lin	RW

The other coupled blocs follow the same estimation methodology with country-appropriate variable selections, replacing the US fiscal and detailed financial block with the three **ROW*** stars of Section 9.1. Across the deployed roster the partner blocs range from $n = 14$ (Costa Rica) to $n = 25$ (UK and Canada), with the euro-area aggregate at $n = 24$ and the core majors at $n \in \{18, \dots, 25\}$ (e.g. CH, NO = 18, CN = 20, JP = 23); the US is the largest at $n = 41$. The China bloc uses a tighter Minnesota tightness to discipline its shorter, noisier sample.

9.8 Variable Transformations, Priors, and Financial Tags by Economy

Section 9.7 tabulates the US bloc in full; this subsection extends that detail to *every* deployed economy. For all 50 blocs (the 33 GVAR members of Section 9.1 and the 17 euro-area standalones of Section 9.2) each variable’s *transformation* (**log** = log level, growth displayed; **lin** = level), its *financial* tag (which governs whether a series is held observed through the COVID window, Section 9.4), and its *prior* mean (RW = random walk in levels; WN = white noise / stationary) are read directly from the deployed `public/data/bvar/<CC>/spec.json` specifications, so the tables below mirror the current re-estimated state of the tool exactly.

Cross-economy coherence matrix. Table 4 collects the common concepts that recur across blocs. Because the transformation, financial tag, and prior for each concept are assigned *identically* in every economy that carries it, a single row characterises the concept tool-wide. Every variable is given a random-walk (RW) prior *except* the three fiscal GDP-ratios (**FREV_GDP**, **FPEXP_GDP**, **SFA**), which are bounded mean-reverting shares and therefore carry a white-noise / stationary (WN) prior; the effective-interest series **EFFI** remains RW.

The 33 GVAR blocs. Each carries a domestic block, the four global commons (**GLB_***), and the three trade-weighted **ROW*_<CC>** stars. Tables follow in alphabetical order.

Table 4: Cross-economy coherence matrix for the common concepts. “Trans.” is `log/lin`; “Fin.” marks financial series (✓); “Prior” is RW (random walk) or WN (white noise / stationary). Each concept takes the same triple in every bloc that carries it.

Concept	Trans.	Fin.	Prior
Real GDP (level)	<code>log</code>	–	RW
Price index (CPI/HICP/deflator)	<code>log</code>	–	RW
Policy rate	<code>lin</code>	✓	RW
Short rate (3M/money-mkt)	<code>lin</code>	✓	RW
10Y government yield	<code>lin</code>	✓	RW
Corporate / Baa spread	<code>lin</code>	✓	RW
House prices	<code>log</code>	✓	RW
Equity price index	<code>log</code>	✓	RW
Oil price (WTI, global)	<code>log</code>	✓	RW
FX / NEER	<code>log</code>	✓	RW
Fiscal revenue ratio (FREV_GDP)	<code>lin</code>	–	WN
Fiscal primary-exp. ratio (FPEXP_GDP)	<code>lin</code>	–	WN
Stock-flow adjustment (SFA)	<code>lin</code>	–	WN
Effective interest (EFFI)	<code>lin</code>	✓	RW
RoW demand star (ROWDEM)	<code>lin</code>	✓	RW
RoW inflation star (ROWINF)	<code>lin</code>	✓	RW
RoW financial star (ROWFIN)	<code>lin</code>	✓	RW

Table 5: Argentina (`AR`, $n = 18$): variable transformations, financial tags, and priors.

ID	Trans.	Fin.	Prior
GDP_AR	<code>log</code>	–	RW
CONS_AR	<code>log</code>	–	RW
INV_AR	<code>log</code>	–	RW
GOV_AR	<code>log</code>	–	RW
EXP_AR	<code>log</code>	–	RW
IMP_AR	<code>log</code>	–	RW
CPI_AR	<code>log</code>	–	RW
LR_AR	<code>lin</code>	–	RW
RATE_AR	<code>lin</code>	✓	RW
NEER_AR	<code>log</code>	✓	RW
POLRATE_AR	<code>lin</code>	✓	RW
GLB_OIL	<code>log</code>	✓	RW
GLB_USLR	<code>lin</code>	✓	RW
GLB_USBAA	<code>lin</code>	✓	RW
GLB_USEQ	<code>log</code>	✓	RW
ROWDEM_AR	<code>lin</code>	✓	RW
ROWINF_AR	<code>lin</code>	✓	RW
ROWFIN_AR	<code>lin</code>	✓	RW

Table 6: Australia (AU, $n = 19$): variable transformations, financial tags, and priors.

ID	Trans.	Fin.	Prior
GDP_AU	log	–	RW
CONS_AU	log	–	RW
GOV_AU	log	–	RW
INV_AU	log	–	RW
EXP_AU	log	–	RW
IMP_AU	log	–	RW
CPI_AU	log	–	RW
LR_AU	lin	–	RW
NEER_AU	log	✓	RW
RATE_AU	lin	✓	RW
YIELD10_AU	lin	✓	RW
HP_AU	log	✓	RW
GLB_OIL	log	✓	RW
GLB_USLR	lin	✓	RW
GLB_USBAA	lin	✓	RW
GLB_USEQ	log	✓	RW
ROWDEM_AU	lin	✓	RW
ROWINF_AU	lin	✓	RW
ROWFIN_AU	lin	✓	RW

Table 7: Brazil (BR, $n = 18$): variable transformations, financial tags, and priors.

ID	Trans.	Fin.	Prior
GDP_BR	log	–	RW
CONS_BR	log	–	RW
INV_BR	log	–	RW
GOV_BR	log	–	RW
EXP_BR	log	–	RW
IMP_BR	log	–	RW
CPI_BR	log	–	RW
LR_BR	lin	–	RW
NEER_BR	log	✓	RW
POLRATE_BR	lin	✓	RW
HP_BR	log	✓	RW
GLB_OIL	log	✓	RW
GLB_USLR	lin	✓	RW
GLB_USBAA	lin	✓	RW
GLB_USEQ	log	✓	RW
ROWDEM_BR	lin	✓	RW
ROWINF_BR	lin	✓	RW
ROWFIN_BR	lin	✓	RW

Table 8: Canada (CA, $n = 25$): variable transformations, financial tags, and priors.

ID	Trans.	Fin.	Prior
GDP_CA	log	–	RW
CONS_CA	log	–	RW
INV_CA	log	–	RW
GOV_CA	log	–	RW
EXP_CA	log	–	RW
IMP_CA	log	–	RW
IP_CA	log	–	RW
LR_CA	lin	–	RW
CPI_CA	log	–	RW
RATE_CA	lin	✓	RW
POLRATE_CA	lin	✓	RW
YIELD10_CA	lin	✓	RW
HP_CA	log	✓	RW
NEER_CA	log	✓	RW
GLB_OIL	log	✓	RW
GLB_USLR	lin	✓	RW
GLB_USBAA	lin	✓	RW
GLB_USEQ	log	✓	RW
FREV_GDP	lin	–	WN
FPEXP_GDP	lin	–	WN
EFFI	lin	✓	RW
SFA	lin	–	WN
ROWDEM_CA	lin	✓	RW
ROWINF_CA	lin	✓	RW
ROWFIN_CA	lin	✓	RW

Table 9: Switzerland (CH, $n = 18$): variable transformations, financial tags, and priors.

ID	Trans.	Fin.	Prior
GDP_CH	log	–	RW
CONS_CH	log	–	RW
INV_CH	log	–	RW
GOV_CH	log	–	RW
EXP_CH	log	–	RW
IMP_CH	log	–	RW
HICP_CH	log	–	RW
LR_CH	lin	–	RW
NEER_CH	log	✓	RW
POLRATE_CH	lin	✓	RW
HP_CH	log	✓	RW
GLB_OIL	log	✓	RW
GLB_USLR	lin	✓	RW
GLB_USBAA	lin	✓	RW

Switzerland (CH), continued

ID	Trans.	Fin.	Prior
GLB_USEQ	log	✓	RW
ROWDEM_CH	lin	✓	RW
ROWINF_CH	lin	✓	RW
ROWFIN_CH	lin	✓	RW

Table 10: Chile (CL, $n = 20$): variable transformations, financial tags, and priors.

ID	Trans.	Fin.	Prior
GDP_CL	log	–	RW
CONS_CL	log	–	RW
INV_CL	log	–	RW
GOV_CL	log	–	RW
EXP_CL	log	–	RW
IMP_CL	log	–	RW
CPI_CL	log	–	RW
LR_CL	lin	–	RW
RATE_CL	lin	✓	RW
POLRATE_CL	lin	✓	RW
YIELD10_CL	lin	✓	RW
HP_CL	log	✓	RW
NEER_CL	log	✓	RW
GLB_OIL	log	✓	RW
GLB_USLR	lin	✓	RW
GLB_USBAA	lin	✓	RW
GLB_USEQ	log	✓	RW
ROWDEM_CL	lin	✓	RW
ROWINF_CL	lin	✓	RW
ROWFIN_CL	lin	✓	RW

Table 11: China (CN, $n = 20$): variable transformations, financial tags, and priors.

ID	Trans.	Fin.	Prior
GDP_CN	log	–	RW
CPI_CN	log	–	RW
RATE_CN	lin	✓	RW
EXP_CN	log	–	RW
IMP_CN	log	–	RW
REER_CN	log	✓	RW
EQ_CN	log	✓	RW
POLRATE_CN	lin	✓	RW
HP_CN	log	✓	RW
GLB_OIL	log	✓	RW

China (CN), continued

ID	Trans.	Fin.	Prior
GLB_USLR	lin	✓	RW
GLB_USBAA	lin	✓	RW
GLB_USEQ	log	✓	RW
FREV_GDP	lin	–	WN
FPEXP_GDP	lin	–	WN
EFFI	lin	✓	RW
SFA	lin	–	WN
ROWDEM_CN	lin	✓	RW
ROWINF_CN	lin	✓	RW
ROWFIN_CN	lin	✓	RW

Table 12: Colombia (CO, $n = 20$): variable transformations, financial tags, and priors.

ID	Trans.	Fin.	Prior
GDP_CO	log	–	RW
CONS_CO	log	–	RW
INV_CO	log	–	RW
GOV_CO	log	–	RW
EXP_CO	log	–	RW
IMP_CO	log	–	RW
CPI_CO	log	–	RW
LR_CO	lin	–	RW
RATE_CO	lin	✓	RW
POLRATE_CO	lin	✓	RW
YIELD10_CO	lin	✓	RW
HP_CO	log	✓	RW
NEER_CO	log	✓	RW
GLB_OIL	log	✓	RW
GLB_USLR	lin	✓	RW
GLB_USBAA	lin	✓	RW
GLB_USEQ	log	✓	RW
ROWDEM_CO	lin	✓	RW
ROWINF_CO	lin	✓	RW
ROWFIN_CO	lin	✓	RW

Table 13: Costa Rica (CR, $n = 14$): variable transformations, financial tags, and priors.

ID	Trans.	Fin.	Prior
GDP_CR	log	–	RW
EXP_CR	log	–	RW
IMP_CR	log	–	RW
CPI_CR	log	–	RW

Costa Rica (CR), continued

ID	Trans.	Fin.	Prior
LR_CR	lin	–	RW
RATE_CR	lin	✓	RW
NEER_CR	log	✓	RW
GLB_OIL	log	✓	RW
GLB_USLR	lin	✓	RW
GLB_USBAA	lin	✓	RW
GLB_USEQ	log	✓	RW
ROWDEM_CR	lin	✓	RW
ROWINF_CR	lin	✓	RW
ROWFIN_CR	lin	✓	RW

Table 14: Czechia (CZ, $n = 19$): variable transformations, financial tags, and priors.

ID	Trans.	Fin.	Prior
GDP_CZ	log	–	RW
CONS_CZ	log	–	RW
INV_CZ	log	–	RW
GOV_CZ	log	–	RW
EXP_CZ	log	–	RW
IMP_CZ	log	–	RW
HICP_CZ	log	–	RW
LR_CZ	lin	–	RW
YIELD10_CZ	lin	✓	RW
NEER_CZ	log	✓	RW
POLRATE_CZ	lin	✓	RW
HP_CZ	log	✓	RW
GLB_OIL	log	✓	RW
GLB_USLR	lin	✓	RW
GLB_USBAA	lin	✓	RW
GLB_USEQ	log	✓	RW
ROWDEM_CZ	lin	✓	RW
ROWINF_CZ	lin	✓	RW
ROWFIN_CZ	lin	✓	RW

Table 15: Denmark (DK, $n = 20$): variable transformations, financial tags, and priors.

ID	Trans.	Fin.	Prior
GDP_DK	log	–	RW
CONS_DK	log	–	RW
INV_DK	log	–	RW
GOV_DK	log	–	RW
EXP_DK	log	–	RW

Denmark (DK), continued

ID	Trans.	Fin.	Prior
IMP_DK	log	–	RW
HICP_DK	log	–	RW
LR_DK	lin	–	RW
RATE_DK	lin	✓	RW
YIELD10_DK	lin	✓	RW
NEER_DK	log	✓	RW
POLRATE_DK	lin	✓	RW
HP_DK	log	✓	RW
GLB_OIL	log	✓	RW
GLB_USLR	lin	✓	RW
GLB_USBAA	lin	✓	RW
GLB_USEQ	log	✓	RW
ROWDEM_DK	lin	✓	RW
ROWINF_DK	lin	✓	RW
ROWFIN_DK	lin	✓	RW

Table 16: Euro Area (EA, $n = 24$): variable transformations, financial tags, and priors.

ID	Trans.	Fin.	Prior
GDP_EA	log	–	RW
CONS_EA	log	–	RW
INV_EA	log	–	RW
GOV_EA	log	–	RW
EXP_EA	log	–	RW
IMP_EA	log	–	RW
HICP_EA	log	–	RW
RATE_EA	lin	✓	RW
YIELD10_EA	lin	✓	RW
NEER_EA	log	✓	RW
POLRATE_EA	lin	✓	RW
HPI_EA	log	✓	RW
EACORP	lin	✓	RW
GLB_OIL	log	✓	RW
GLB_USLR	lin	✓	RW
GLB_USBAA	lin	✓	RW
GLB_USEQ	log	✓	RW
FREV_GDP	lin	–	WN
FPEXP_GDP	lin	–	WN
EFFI	lin	✓	RW
SFA	lin	–	WN
ROWDEM_EA	lin	✓	RW
ROWINF_EA	lin	✓	RW
ROWFIN_EA	lin	✓	RW

Table 17: Hungary (HU, $n = 19$): variable transformations, financial tags, and priors.

ID	Trans.	Fin.	Prior
GDP_HU	log	–	RW
CONS_HU	log	–	RW
INV_HU	log	–	RW
GOV_HU	log	–	RW
EXP_HU	log	–	RW
IMP_HU	log	–	RW
HICP_HU	log	–	RW
LR_HU	lin	–	RW
YIELD10_HU	lin	✓	RW
NEER_HU	log	✓	RW
POLRATE_HU	lin	✓	RW
HP_HU	log	✓	RW
GLB_OIL	log	✓	RW
GLB_USLR	lin	✓	RW
GLB_USBAA	lin	✓	RW
GLB_USEQ	log	✓	RW
ROWDEM_HU	lin	✓	RW
ROWINF_HU	lin	✓	RW
ROWFIN_HU	lin	✓	RW

Table 18: Indonesia (ID, $n = 18$): variable transformations, financial tags, and priors.

ID	Trans.	Fin.	Prior
GDP_ID	log	–	RW
CONS_ID	log	–	RW
INV_ID	log	–	RW
GOV_ID	log	–	RW
EXP_ID	log	–	RW
IMP_ID	log	–	RW
CPI_ID	log	–	RW
RATE_ID	lin	✓	RW
NEER_ID	log	✓	RW
POLRATE_ID	lin	✓	RW
HP_ID	log	✓	RW
GLB_OIL	log	✓	RW
GLB_USLR	lin	✓	RW
GLB_USBAA	lin	✓	RW
GLB_USEQ	log	✓	RW
ROWDEM_ID	lin	✓	RW
ROWINF_ID	lin	✓	RW
ROWFIN_ID	lin	✓	RW

Table 19: Israel (IL, $n = 19$): variable transformations, financial tags, and priors.

ID	Trans.	Fin.	Prior
GDP_IL	log	–	RW
CONS_IL	log	–	RW
INV_IL	log	–	RW
GOV_IL	log	–	RW
EXP_IL	log	–	RW
IMP_IL	log	–	RW
CPI_IL	log	–	RW
LR_IL	lin	–	RW
RATE_IL	lin	✓	RW
POLRATE_IL	lin	✓	RW
NEER_IL	log	✓	RW
HP_IL	log	✓	RW
GLB_OIL	log	✓	RW
GLB_USLR	lin	✓	RW
GLB_USBAA	lin	✓	RW
GLB_USEQ	log	✓	RW
ROWDEM_IL	lin	✓	RW
ROWINF_IL	lin	✓	RW
ROWFIN_IL	lin	✓	RW

Table 20: India (IN, $n = 20$): variable transformations, financial tags, and priors.

ID	Trans.	Fin.	Prior
GDP_IN	log	–	RW
CONS_IN	log	–	RW
INV_IN	log	–	RW
GOV_IN	log	–	RW
EXP_IN	log	–	RW
IMP_IN	log	–	RW
CPI_IN	log	–	RW
RATE_IN	lin	✓	RW
RATE3M_IN	lin	✓	RW
YIELD10_IN	lin	✓	RW
NEER_IN	log	✓	RW
POLRATE_IN	lin	✓	RW
HP_IN	log	✓	RW
GLB_OIL	log	✓	RW
GLB_USLR	lin	✓	RW
GLB_USBAA	lin	✓	RW
GLB_USEQ	log	✓	RW
ROWDEM_IN	lin	✓	RW
ROWINF_IN	lin	✓	RW
ROWFIN_IN	lin	✓	RW

Table 21: Iceland (IS, $n = 19$): variable transformations, financial tags, and priors.

ID	Trans.	Fin.	Prior
GDP_IS	log	–	RW
CONS_IS	log	–	RW
INV_IS	log	–	RW
GOV_IS	log	–	RW
EXP_IS	log	–	RW
IMP_IS	log	–	RW
CPI_IS	log	–	RW
LR_IS	lin	–	RW
RATE_IS	lin	✓	RW
NEER_IS	log	✓	RW
POLRATE_IS	lin	✓	RW
HP_IS	log	✓	RW
GLB_OIL	log	✓	RW
GLB_USLR	lin	✓	RW
GLB_USBAA	lin	✓	RW
GLB_USEQ	log	✓	RW
ROWDEM_IS	lin	✓	RW
ROWINF_IS	lin	✓	RW
ROWFIN_IS	lin	✓	RW

Table 22: Japan (JP, $n = 23$): variable transformations, financial tags, and priors.

ID	Trans.	Fin.	Prior
GDP_JP	log	–	RW
CONS_JP	log	–	RW
INV_JP	log	–	RW
GOV_JP	log	–	RW
EXP_JP	log	–	RW
IMP_JP	log	–	RW
LR_JP	lin	–	RW
CPI_JP	log	–	RW
RATE_JP	lin	✓	RW
POLRATE_JP	lin	✓	RW
YIELD10_JP	lin	✓	RW
HP_JP	log	✓	RW
GLB_OIL	log	✓	RW
GLB_USLR	lin	✓	RW
GLB_USBAA	lin	✓	RW
GLB_USEQ	log	✓	RW
FREV_GDP	lin	–	WN
FPEXP_GDP	lin	–	WN
EFFI	lin	✓	RW
SFA	lin	–	WN

Japan (JP), continued

ID	Trans.	Fin.	Prior
ROWDEM_JP	lin	✓	RW
ROWINF_JP	lin	✓	RW
ROWFIN_JP	lin	✓	RW

Table 23: Korea (KR, $n = 19$): variable transformations, financial tags, and priors.

ID	Trans.	Fin.	Prior
GDP_KR	log	–	RW
CONS_KR	log	–	RW
INV_KR	log	–	RW
GOV_KR	log	–	RW
EXP_KR	log	–	RW
IMP_KR	log	–	RW
CPI_KR	log	–	RW
LR_KR	lin	–	RW
RATE_KR	lin	✓	RW
POLRATE_KR	lin	✓	RW
YIELD10_KR	lin	✓	RW
HP_KR	log	✓	RW
GLB_OIL	log	✓	RW
GLB_USLR	lin	✓	RW
GLB_USBAA	lin	✓	RW
GLB_USEQ	log	✓	RW
ROWDEM_KR	lin	✓	RW
ROWINF_KR	lin	✓	RW
ROWFIN_KR	lin	✓	RW

Table 24: Mexico (MX, $n = 20$): variable transformations, financial tags, and priors.

ID	Trans.	Fin.	Prior
GDP_MX	log	–	RW
CONS_MX	log	–	RW
INV_MX	log	–	RW
GOV_MX	log	–	RW
EXP_MX	log	–	RW
IMP_MX	log	–	RW
CPI_MX	log	–	RW
LR_MX	lin	–	RW
RATE_MX	lin	✓	RW
POLRATE_MX	lin	✓	RW
YIELD10_MX	lin	✓	RW
HP_MX	log	✓	RW

Mexico (MX), continued

ID	Trans.	Fin.	Prior
NEER_MX	log	✓	RW
GLB_OIL	log	✓	RW
GLB_USLR	lin	✓	RW
GLB_USBAA	lin	✓	RW
GLB_USEQ	log	✓	RW
ROWDEM_MX	lin	✓	RW
ROWINF_MX	lin	✓	RW
ROWFIN_MX	lin	✓	RW

Table 25: Malaysia (MY, $n = 19$): variable transformations, financial tags, and priors.

ID	Trans.	Fin.	Prior
GDP_MY	log	–	RW
CONS_MY	log	–	RW
INV_MY	log	–	RW
GOV_MY	log	–	RW
EXP_MY	log	–	RW
IMP_MY	log	–	RW
CPI_MY	log	–	RW
LR_MY	lin	–	RW
RATE_MY	lin	✓	RW
NEER_MY	log	✓	RW
POLRATE_MY	lin	✓	RW
HP_MY	log	✓	RW
GLB_OIL	log	✓	RW
GLB_USLR	lin	✓	RW
GLB_USBAA	lin	✓	RW
GLB_USEQ	log	✓	RW
ROWDEM_MY	lin	✓	RW
ROWINF_MY	lin	✓	RW
ROWFIN_MY	lin	✓	RW

Table 26: Norway (NO, $n = 18$): variable transformations, financial tags, and priors.

ID	Trans.	Fin.	Prior
GDP_NO	log	–	RW
CONS_NO	log	–	RW
INV_NO	log	–	RW
GOV_NO	log	–	RW
EXP_NO	log	–	RW
IMP_NO	log	–	RW
HICP_NO	log	–	RW

Norway (NO), continued

ID	Trans.	Fin.	Prior
LR_NO	lin	–	RW
NEER_NO	log	✓	RW
POLRATE_NO	lin	✓	RW
HP_NO	log	✓	RW
GLB_OIL	log	✓	RW
GLB_USLR	lin	✓	RW
GLB_USBAA	lin	✓	RW
GLB_USEQ	log	✓	RW
ROWDEM_NO	lin	✓	RW
ROWINF_NO	lin	✓	RW
ROWFIN_NO	lin	✓	RW

Table 27: New Zealand (NZ, $n = 19$): variable transformations, financial tags, and priors.

ID	Trans.	Fin.	Prior
GDP_NZ	log	–	RW
CONS_NZ	log	–	RW
INV_NZ	log	–	RW
GOV_NZ	log	–	RW
EXP_NZ	log	–	RW
IMP_NZ	log	–	RW
CPI_NZ	log	–	RW
LR_NZ	lin	–	RW
RATE_NZ	lin	✓	RW
YIELD10_NZ	lin	✓	RW
HP_NZ	log	✓	RW
NEER_NZ	log	✓	RW
GLB_OIL	log	✓	RW
GLB_USLR	lin	✓	RW
GLB_USBAA	lin	✓	RW
GLB_USEQ	log	✓	RW
ROWDEM_NZ	lin	✓	RW
ROWINF_NZ	lin	✓	RW
ROWFIN_NZ	lin	✓	RW

Table 28: Poland (PL, $n = 19$): variable transformations, financial tags, and priors.

ID	Trans.	Fin.	Prior
GDP_PL	log	–	RW
CONS_PL	log	–	RW
INV_PL	log	–	RW
GOV_PL	log	–	RW

Poland (PL), continued

ID	Trans.	Fin.	Prior
EXP_PL	log	–	RW
IMP_PL	log	–	RW
HICP_PL	log	–	RW
LR_PL	lin	–	RW
YIELD10_PL	lin	✓	RW
NEER_PL	log	✓	RW
POLRATE_PL	lin	✓	RW
HP_PL	log	✓	RW
GLB_OIL	log	✓	RW
GLB_USLR	lin	✓	RW
GLB_USBAA	lin	✓	RW
GLB_USEQ	log	✓	RW
ROWDEM_PL	lin	✓	RW
ROWINF_PL	lin	✓	RW
ROWFIN_PL	lin	✓	RW

Table 29: Russia (RU, $n = 15$): variable transformations, financial tags, and priors.

ID	Trans.	Fin.	Prior
GDP_RU	log	–	RW
CPI_RU	log	–	RW
EXP_RU	log	–	RW
LR_RU	lin	–	RW
RATE_RU	lin	✓	RW
NEER_RU	log	✓	RW
POLRATE_RU	lin	✓	RW
HP_RU	log	✓	RW
GLB_OIL	log	✓	RW
GLB_USLR	lin	✓	RW
GLB_USBAA	lin	✓	RW
GLB_USEQ	log	✓	RW
ROWDEM_RU	lin	✓	RW
ROWINF_RU	lin	✓	RW
ROWFIN_RU	lin	✓	RW

Table 30: Saudi Arabia (SA, $n = 18$): variable transformations, financial tags, and priors.

ID	Trans.	Fin.	Prior
GDP_SA	log	–	RW
CONS_SA	log	–	RW
INV_SA	log	–	RW
GOV_SA	log	–	RW

Saudi Arabia (SA), continued

ID	Trans.	Fin.	Prior
EXP_SA	log	–	RW
IMP_SA	log	–	RW
CPI_SA	log	–	RW
LR_SA	lin	–	RW
RATE_SA	lin	✓	RW
NEER_SA	log	✓	RW
POLRATE_SA	lin	✓	RW
GLB_OIL	log	✓	RW
GLB_USLR	lin	✓	RW
GLB_USBAA	lin	✓	RW
GLB_USEQ	log	✓	RW
ROWDEM_SA	lin	✓	RW
ROWINF_SA	lin	✓	RW
ROWFIN_SA	lin	✓	RW

Table 31: Sweden (SE, $n = 20$): variable transformations, financial tags, and priors.

ID	Trans.	Fin.	Prior
GDP_SE	log	–	RW
CONS_SE	log	–	RW
INV_SE	log	–	RW
GOV_SE	log	–	RW
EXP_SE	log	–	RW
IMP_SE	log	–	RW
HICP_SE	log	–	RW
LR_SE	lin	–	RW
RATE_SE	lin	✓	RW
YIELD10_SE	lin	✓	RW
NEER_SE	log	✓	RW
POLRATE_SE	lin	✓	RW
HP_SE	log	✓	RW
GLB_OIL	log	✓	RW
GLB_USLR	lin	✓	RW
GLB_USBAA	lin	✓	RW
GLB_USEQ	log	✓	RW
ROWDEM_SE	lin	✓	RW
ROWINF_SE	lin	✓	RW
ROWFIN_SE	lin	✓	RW

Table 32: Thailand (TH, $n = 20$): variable transformations, financial tags, and priors.

ID	Trans.	Fin.	Prior
GDP_TH	log	–	RW
CONS_TH	log	–	RW
INV_TH	log	–	RW
GOV_TH	log	–	RW
EXP_TH	log	–	RW
IMP_TH	log	–	RW
CPI_TH	log	–	RW
LR_TH	lin	–	RW
RATE_TH	lin	✓	RW
YIELD10_TH	lin	✓	RW
NEER_TH	log	✓	RW
POLRATE_TH	lin	✓	RW
HP_TH	log	✓	RW
GLB_OIL	log	✓	RW
GLB_USLR	lin	✓	RW
GLB_USBAA	lin	✓	RW
GLB_USEQ	log	✓	RW
ROWDEM_TH	lin	✓	RW
ROWINF_TH	lin	✓	RW
ROWFIN_TH	lin	✓	RW

Table 33: Türkiye (TR, $n = 18$): variable transformations, financial tags, and priors.

ID	Trans.	Fin.	Prior
GDP_TR	log	–	RW
CONS_TR	log	–	RW
INV_TR	log	–	RW
GOV_TR	log	–	RW
EXP_TR	log	–	RW
IMP_TR	log	–	RW
CPI_TR	log	–	RW
LR_TR	lin	–	RW
RATE_TR	lin	✓	RW
POLRATE_TR	lin	✓	RW
HP_TR	log	✓	RW
GLB_OIL	log	✓	RW
GLB_USLR	lin	✓	RW
GLB_USBAA	lin	✓	RW
GLB_USEQ	log	✓	RW
ROWDEM_TR	lin	✓	RW
ROWINF_TR	lin	✓	RW
ROWFIN_TR	lin	✓	RW

Table 34: Taiwan (TW, $n = 18$): variable transformations, financial tags, and priors.

ID	Trans.	Fin.	Prior
GDP_TW	log	–	RW
CONS_TW	log	–	RW
GOV_TW	log	–	RW
INV_TW	log	–	RW
EXP_TW	log	–	RW
IMP_TW	log	–	RW
CPI_TW	log	–	RW
LR_TW	lin	–	RW
RATE_TW	lin	✓	RW
MMRATE_TW	lin	✓	RW
YIELD10_TW	lin	✓	RW
GLB_OIL	log	✓	RW
GLB_USLR	lin	✓	RW
GLB_USBAA	lin	✓	RW
GLB_USEQ	log	✓	RW
ROWDEM_TW	lin	✓	RW
ROWINF_TW	lin	✓	RW
ROWFIN_TW	lin	✓	RW

Table 35: United Kingdom (UK, $n = 25$): variable transformations, financial tags, and priors.

ID	Trans.	Fin.	Prior
GDP_UK	log	–	RW
CONS_UK	log	–	RW
INV_UK	log	–	RW
GOV_UK	log	–	RW
EXP_UK	log	–	RW
IMP_UK	log	–	RW
IP_UK	log	–	RW
LR_UK	lin	–	RW
CPI_UK	log	–	RW
RATE_UK	lin	✓	RW
POLRATE_UK	lin	✓	RW
YIELD10_UK	lin	✓	RW
HP_UK	log	✓	RW
NEER_UK	log	✓	RW
GLB_OIL	log	✓	RW
GLB_USLR	lin	✓	RW
GLB_USBAA	lin	✓	RW
GLB_USEQ	log	✓	RW
FREV_GDP	lin	–	WN
FPEXP_GDP	lin	–	WN
EFFI	lin	✓	RW

United Kingdom (UK), continued

ID	Trans.	Fin.	Prior
SFA	lin	–	WN
ROWDEM_UK	lin	✓	RW
ROWINF_UK	lin	✓	RW
ROWFIN_UK	lin	✓	RW

Table 36: United States (US, $n = 41$): variable transformations, financial tags, and priors.

ID	Trans.	Fin.	Prior
GDPH	log	–	RW
CH	log	–	RW
FRH	log	–	RW
FNH	log	–	RW
XH	log	–	RW
MH	log	–	RW
GH	log	–	RW
JGDP	log	–	RW
JC	log	–	RW
JCXFE	log	–	RW
UI	log	–	RW
UIXFDG	log	–	RW
LXBC	log	–	RW
LANAGRA	log	–	RW
LR	lin	–	RW
IP	log	–	RW
CUMFG	lin	–	RW
HST	log	–	RW
YPDH	log	–	RW
CSENT	lin	–	RW
POLRATE	lin	✓	RW
FCM2	lin	✓	RW
FCM5	lin	✓	RW
GLB_USLR	lin	✓	RW
FAAA	lin	✓	RW
GLB_USBAA	lin	✓	RW
FXTWM	log	✓	RW
GLB_USEQ	log	✓	RW
PZTEXP	log	✓	RW
GSNE	log	✓	RW
SP500VOL	lin	✓	RW
HP	log	✓	RW
LEV	log	✓	RW
FREV_GDP	lin	–	WN
FPEXP_GDP	lin	–	WN
EFFI	lin	✓	RW

United States (US), continued

ID	Trans.	Fin.	Prior
SFA	lin	–	WN
GLB_OIL	log	✓	RW
ROWDEM_US	lin	✓	RW
ROWINF_US	lin	✓	RW
ROWFIN_US	lin	✓	RW

Table 37: South Africa (ZA, $n = 20$): variable transformations, financial tags, and priors.

ID	Trans.	Fin.	Prior
GDP_ZA	log	–	RW
CONS_ZA	log	–	RW
INV_ZA	log	–	RW
GOV_ZA	log	–	RW
EXP_ZA	log	–	RW
IMP_ZA	log	–	RW
CPI_ZA	log	–	RW
LR_ZA	lin	–	RW
RATE_ZA	lin	✓	RW
YIELD10_ZA	lin	✓	RW
NEER_ZA	log	✓	RW
POLRATE_ZA	lin	✓	RW
HP_ZA	log	✓	RW
GLB_OIL	log	✓	RW
GLB_USLR	lin	✓	RW
GLB_USBAA	lin	✓	RW
GLB_USEQ	log	✓	RW
ROWDEM_ZA	lin	✓	RW
ROWINF_ZA	lin	✓	RW
ROWFIN_ZA	lin	✓	RW

The 17 euro-area standalones. These omit the ROW* stars and instead carry US-block foreign variables (*_US) plus the shared euro-area corporate spread (EACORP); they are estimated in separation (Section 9.2).

Table 38: Austria (AT, $n = 23$): variable transformations, financial tags, and priors.

ID	Trans.	Fin.	Prior
GDP_AT	log	–	RW
CONS_AT	log	–	RW
INV_AT	log	–	RW
GOV_AT	log	–	RW
EXP_AT	log	–	RW

Austria (AT), continued

ID	Trans.	Fin.	Prior
IMP_AT	log	–	RW
HICP_AT	log	–	RW
LR_AT	lin	–	RW
POLRATE_AT	lin	✓	RW
HPI_AT	log	✓	RW
EACORP	lin	✓	RW
RATE_AT	lin	✓	RW
YIELD10_AT	lin	✓	RW
NEER_AT	log	✓	RW
GDPH_US	log	–	RW
GH_US	log	–	RW
LR_US	lin	–	RW
JGDP_US	log	–	RW
FCM2_US	lin	✓	RW
FCM10_US	lin	✓	RW
EQ_US	log	✓	RW
FBAA_US	lin	✓	RW
WTI_US	log	✓	RW

Table 39: Belgium (BE, $n = 23$): variable transformations, financial tags, and priors.

ID	Trans.	Fin.	Prior
GDP_BE	log	–	RW
CONS_BE	log	–	RW
INV_BE	log	–	RW
GOV_BE	log	–	RW
EXP_BE	log	–	RW
IMP_BE	log	–	RW
HICP_BE	log	–	RW
LR_BE	lin	–	RW
POLRATE_BE	lin	✓	RW
HPI_BE	log	✓	RW
EACORP	lin	✓	RW
RATE_BE	lin	✓	RW
YIELD10_BE	lin	✓	RW
NEER_BE	log	✓	RW
GDPH_US	log	–	RW
GH_US	log	–	RW
LR_US	lin	–	RW
JGDP_US	log	–	RW
FCM2_US	lin	✓	RW
FCM10_US	lin	✓	RW
EQ_US	log	✓	RW
FBAA_US	lin	✓	RW

Belgium (BE), continued

ID	Trans.	Fin.	Prior
WTI_US	log	✓	RW

Table 40: Germany (DE, $n = 23$): variable transformations, financial tags, and priors.

ID	Trans.	Fin.	Prior
GDP_DE	log	–	RW
CONS_DE	log	–	RW
INV_DE	log	–	RW
GOV_DE	log	–	RW
EXP_DE	log	–	RW
IMP_DE	log	–	RW
HICP_DE	log	–	RW
LR_DE	lin	–	RW
POLRATE_DE	lin	✓	RW
HPI_DE	log	✓	RW
EACORP	lin	✓	RW
RATE_DE	lin	✓	RW
YIELD10_DE	lin	✓	RW
NEER_DE	log	✓	RW
GDPH_US	log	–	RW
GH_US	log	–	RW
LR_US	lin	–	RW
JGDP_US	log	–	RW
FCM2_US	lin	✓	RW
FCM10_US	lin	✓	RW
EQ_US	log	✓	RW
FBAA_US	lin	✓	RW
WTI_US	log	✓	RW

Table 41: Estonia (EE, $n = 22$): variable transformations, financial tags, and priors.

ID	Trans.	Fin.	Prior
GDP_EE	log	–	RW
CONS_EE	log	–	RW
INV_EE	log	–	RW
GOV_EE	log	–	RW
EXP_EE	log	–	RW
IMP_EE	log	–	RW
HICP_EE	log	–	RW
LR_EE	lin	–	RW
POLRATE_EE	lin	✓	RW
HPI_EE	log	✓	RW

Estonia (EE), continued

ID	Trans.	Fin.	Prior
EACORP	lin	✓	RW
RATE_EE	lin	✓	RW
NEER_EE	log	✓	RW
GDPH_US	log	–	RW
GH_US	log	–	RW
LR_US	lin	–	RW
JGDP_US	log	–	RW
FCM2_US	lin	✓	RW
FCM10_US	lin	✓	RW
EQ_US	log	✓	RW
FBAA_US	lin	✓	RW
WTI_US	log	✓	RW

Table 42: Spain (ES, $n = 23$): variable transformations, financial tags, and priors.

ID	Trans.	Fin.	Prior
GDP_ES	log	–	RW
CONS_ES	log	–	RW
INV_ES	log	–	RW
GOV_ES	log	–	RW
EXP_ES	log	–	RW
IMP_ES	log	–	RW
HICP_ES	log	–	RW
LR_ES	lin	–	RW
POLRATE_ES	lin	✓	RW
HPI_ES	log	✓	RW
EACORP	lin	✓	RW
RATE_ES	lin	✓	RW
YIELD10_ES	lin	✓	RW
NEER_ES	log	✓	RW
GDPH_US	log	–	RW
GH_US	log	–	RW
LR_US	lin	–	RW
JGDP_US	log	–	RW
FCM2_US	lin	✓	RW
FCM10_US	lin	✓	RW
EQ_US	log	✓	RW
FBAA_US	lin	✓	RW
WTI_US	log	✓	RW

Table 43: Finland (FI, $n = 23$): variable transformations, financial tags, and priors.

ID	Trans.	Fin.	Prior
GDP_FI	log	—	RW
CONS_FI	log	—	RW
INV_FI	log	—	RW
GOV_FI	log	—	RW
EXP_FI	log	—	RW
IMP_FI	log	—	RW
HICP_FI	log	—	RW
LR_FI	lin	—	RW
POLRATE_FI	lin	✓	RW
HPI_FI	log	✓	RW
EACORP	lin	✓	RW
RATE_FI	lin	✓	RW
YIELD10_FI	lin	✓	RW
NEER_FI	log	✓	RW
GDPH_US	log	—	RW
GH_US	log	—	RW
LR_US	lin	—	RW
JGDP_US	log	—	RW
FCM2_US	lin	✓	RW
FCM10_US	lin	✓	RW
EQ_US	log	✓	RW
FBAA_US	lin	✓	RW
WTI_US	log	✓	RW

Table 44: France (FR, $n = 23$): variable transformations, financial tags, and priors.

ID	Trans.	Fin.	Prior
GDP_FR	log	—	RW
CONS_FR	log	—	RW
INV_FR	log	—	RW
GOV_FR	log	—	RW
EXP_FR	log	—	RW
IMP_FR	log	—	RW
HICP_FR	log	—	RW
LR_FR	lin	—	RW
POLRATE_FR	lin	✓	RW
HPI_FR	log	✓	RW
EACORP	lin	✓	RW
RATE_FR	lin	✓	RW
YIELD10_FR	lin	✓	RW
NEER_FR	log	✓	RW
GDPH_US	log	—	RW
GH_US	log	—	RW

France (FR), continued

ID	Trans.	Fin.	Prior
LR_US	lin	–	RW
JGDP_US	log	–	RW
FCM2_US	lin	✓	RW
FCM10_US	lin	✓	RW
EQ_US	log	✓	RW
FBAA_US	lin	✓	RW
WTI_US	log	✓	RW

Table 45: Greece (GR, $n = 22$): variable transformations, financial tags, and priors.

ID	Trans.	Fin.	Prior
GDP_GR	log	–	RW
CONS_GR	log	–	RW
INV_GR	log	–	RW
GOV_GR	log	–	RW
EXP_GR	log	–	RW
IMP_GR	log	–	RW
HICP_GR	log	–	RW
LR_GR	lin	–	RW
POLRATE_GR	lin	✓	RW
EACORP	lin	✓	RW
RATE_GR	lin	✓	RW
YIELD10_GR	lin	✓	RW
NEER_GR	log	✓	RW
GDPH_US	log	–	RW
GH_US	log	–	RW
LR_US	lin	–	RW
JGDP_US	log	–	RW
FCM2_US	lin	✓	RW
FCM10_US	lin	✓	RW
EQ_US	log	✓	RW
FBAA_US	lin	✓	RW
WTI_US	log	✓	RW

Table 46: Ireland (IE, $n = 23$): variable transformations, financial tags, and priors.

ID	Trans.	Fin.	Prior
GDP_IE	log	–	RW
CONS_IE	log	–	RW
INV_IE	log	–	RW
GOV_IE	log	–	RW
EXP_IE	log	–	RW

Ireland (IE), continued

ID	Trans.	Fin.	Prior
IMP_IE	log	–	RW
HICP_IE	log	–	RW
LR_IE	lin	–	RW
POLRATE_IE	lin	✓	RW
HPI_IE	log	✓	RW
EACORP	lin	✓	RW
RATE_IE	lin	✓	RW
YIELD10_IE	lin	✓	RW
NEER_IE	log	✓	RW
GDPH_US	log	–	RW
GH_US	log	–	RW
LR_US	lin	–	RW
JGDP_US	log	–	RW
FCM2_US	lin	✓	RW
FCM10_US	lin	✓	RW
EQ_US	log	✓	RW
FBAA_US	lin	✓	RW
WTI_US	log	✓	RW

Table 47: Italy (IT, $n = 23$): variable transformations, financial tags, and priors.

ID	Trans.	Fin.	Prior
GDP_IT	log	–	RW
CONS_IT	log	–	RW
INV_IT	log	–	RW
GOV_IT	log	–	RW
EXP_IT	log	–	RW
IMP_IT	log	–	RW
HICP_IT	log	–	RW
LR_IT	lin	–	RW
POLRATE_IT	lin	✓	RW
HPI_IT	log	✓	RW
EACORP	lin	✓	RW
RATE_IT	lin	✓	RW
YIELD10_IT	lin	✓	RW
NEER_IT	log	✓	RW
GDPH_US	log	–	RW
GH_US	log	–	RW
LR_US	lin	–	RW
JGDP_US	log	–	RW
FCM2_US	lin	✓	RW
FCM10_US	lin	✓	RW
EQ_US	log	✓	RW
FBAA_US	lin	✓	RW

Italy (IT), continued

ID	Trans.	Fin.	Prior
WTI_US	log	✓	RW

Table 48: Lithuania (LT, $n = 23$): variable transformations, financial tags, and priors.

ID	Trans.	Fin.	Prior
GDP_LT	log	–	RW
CONS_LT	log	–	RW
INV_LT	log	–	RW
GOV_LT	log	–	RW
EXP_LT	log	–	RW
IMP_LT	log	–	RW
HICP_LT	log	–	RW
LR_LT	lin	–	RW
POLRATE_LT	lin	✓	RW
HPI_LT	log	✓	RW
EACORP	lin	✓	RW
RATE_LT	lin	✓	RW
YIELD10_LT	lin	✓	RW
NEER_LT	log	✓	RW
GDPH_US	log	–	RW
GH_US	log	–	RW
LR_US	lin	–	RW
JGDP_US	log	–	RW
FCM2_US	lin	✓	RW
FCM10_US	lin	✓	RW
EQ_US	log	✓	RW
FBAA_US	lin	✓	RW
WTI_US	log	✓	RW

Table 49: Luxembourg (LU, $n = 23$): variable transformations, financial tags, and priors.

ID	Trans.	Fin.	Prior
GDP_LU	log	–	RW
CONS_LU	log	–	RW
INV_LU	log	–	RW
GOV_LU	log	–	RW
EXP_LU	log	–	RW
IMP_LU	log	–	RW
HICP_LU	log	–	RW
LR_LU	lin	–	RW
POLRATE_LU	lin	✓	RW
HPI_LU	log	✓	RW

Luxembourg (LU), continued

ID	Trans.	Fin.	Prior
EACORP	lin	✓	RW
RATE_LU	lin	✓	RW
YIELD10_LU	lin	✓	RW
NEER_LU	log	✓	RW
GDPH_US	log	–	RW
GH_US	log	–	RW
LR_US	lin	–	RW
JGDP_US	log	–	RW
FCM2_US	lin	✓	RW
FCM10_US	lin	✓	RW
EQ_US	log	✓	RW
FBAA_US	lin	✓	RW
WTI_US	log	✓	RW

Table 50: Latvia (LV, $n = 20$): variable transformations, financial tags, and priors.

ID	Trans.	Fin.	Prior
GDP_LV	log	–	RW
CONS_LV	log	–	RW
INV_LV	log	–	RW
GOV_LV	log	–	RW
EXP_LV	log	–	RW
IMP_LV	log	–	RW
HICP_LV	log	–	RW
LR_LV	lin	–	RW
RATE_LV	lin	✓	RW
YIELD10_LV	lin	✓	RW
NEER_LV	log	✓	RW
GDPH_US	log	–	RW
GH_US	log	–	RW
LR_US	lin	–	RW
JGDP_US	log	–	RW
FCM2_US	lin	✓	RW
FCM10_US	lin	✓	RW
EQ_US	log	✓	RW
FBAA_US	lin	✓	RW
WTI_US	log	✓	RW

Table 51: Netherlands (NL, $n = 23$): variable transformations, financial tags, and priors.

ID	Trans.	Fin.	Prior
GDP_NL	log	–	RW

Netherlands (NL), continued

ID	Trans.	Fin.	Prior
CONS_NL	log	–	RW
INV_NL	log	–	RW
GOV_NL	log	–	RW
EXP_NL	log	–	RW
IMP_NL	log	–	RW
HICP_NL	log	–	RW
LR_NL	lin	–	RW
POLRATE_NL	lin	✓	RW
HPI_NL	log	✓	RW
EACORP	lin	✓	RW
RATE_NL	lin	✓	RW
YIELD10_NL	lin	✓	RW
NEER_NL	log	✓	RW
GDPH_US	log	–	RW
GH_US	log	–	RW
LR_US	lin	–	RW
JGDP_US	log	–	RW
FCM2_US	lin	✓	RW
FCM10_US	lin	✓	RW
EQ_US	log	✓	RW
FBAA_US	lin	✓	RW
WTI_US	log	✓	RW

Table 52: Portugal (PT, $n = 23$): variable transformations, financial tags, and priors.

ID	Trans.	Fin.	Prior
GDP_PT	log	–	RW
CONS_PT	log	–	RW
INV_PT	log	–	RW
GOV_PT	log	–	RW
EXP_PT	log	–	RW
IMP_PT	log	–	RW
HICP_PT	log	–	RW
LR_PT	lin	–	RW
POLRATE_PT	lin	✓	RW
HPI_PT	log	✓	RW
EACORP	lin	✓	RW
RATE_PT	lin	✓	RW
YIELD10_PT	lin	✓	RW
NEER_PT	log	✓	RW
GDPH_US	log	–	RW
GH_US	log	–	RW
LR_US	lin	–	RW
JGDP_US	log	–	RW

Portugal (PT), continued

ID	Trans.	Fin.	Prior
FCM2_US	lin	✓	RW
FCM10_US	lin	✓	RW
EQ_US	log	✓	RW
FBAA_US	lin	✓	RW
WTI_US	log	✓	RW

Table 53: Slovenia (SI, $n = 23$): variable transformations, financial tags, and priors.

ID	Trans.	Fin.	Prior
GDP_SI	log	–	RW
CONS_SI	log	–	RW
INV_SI	log	–	RW
GOV_SI	log	–	RW
EXP_SI	log	–	RW
IMP_SI	log	–	RW
HICP_SI	log	–	RW
LR_SI	lin	–	RW
POLRATE_SI	lin	✓	RW
HPI_SI	log	✓	RW
EACORP	lin	✓	RW
RATE_SI	lin	✓	RW
YIELD10_SI	lin	✓	RW
NEER_SI	log	✓	RW
GDPH_US	log	–	RW
GH_US	log	–	RW
LR_US	lin	–	RW
JGDP_US	log	–	RW
FCM2_US	lin	✓	RW
FCM10_US	lin	✓	RW
EQ_US	log	✓	RW
FBAA_US	lin	✓	RW
WTI_US	log	✓	RW

Table 54: Slovakia (SK, $n = 23$): variable transformations, financial tags, and priors.

ID	Trans.	Fin.	Prior
GDP_SK	log	–	RW
CONS_SK	log	–	RW
INV_SK	log	–	RW
GOV_SK	log	–	RW
EXP_SK	log	–	RW
IMP_SK	log	–	RW

Slovakia (SK), continued

ID	Trans.	Fin.	Prior
HICP_SK	log	–	RW
LR_SK	lin	–	RW
POLRATE_SK	lin	✓	RW
HPI_SK	log	✓	RW
EACORP	lin	✓	RW
RATE_SK	lin	✓	RW
YIELD10_SK	lin	✓	RW
NEER_SK	log	✓	RW
GDPH_US	log	–	RW
GH_US	log	–	RW
LR_US	lin	–	RW
JGDP_US	log	–	RW
FCM2_US	lin	✓	RW
FCM10_US	lin	✓	RW
EQ_US	log	✓	RW
FBAA_US	lin	✓	RW
WTI_US	log	✓	RW

Coherence review. A cross-economy audit of all 1048 variable rows confirms that every shared concept is assigned the same transformation, financial tag, and prior in every bloc that carries it: real activity, price indices, house prices, equities, oil, and FX enter in **log**; interest rates, yields, spreads, unemployment, and the fiscal ratios enter **lin**; all financial series share the financial tag; and all priors are RW apart from the three stationary fiscal GDP-ratios (**FREV_GDP**, **FPEXP_GDP**, **SFA**, prior **WN**). Canada’s industrial-production series **IP_CA** has been rebuilt as a **log**-level index, consistent with the activity convention used for the United States and the United Kingdom. The FRED level index (**CANPROINDMISMEI**, 2015 = 100) was discontinued at 2024-02, so the series is the quarterly average of that index through 2024-Q1 and is then growth-chained over 2024-Q2–2025-Q4 (eight quarters) using the FRED q-on-q growth series **PRINT001CAQ657S** ($\text{level}_t = \text{level}_{t-1}(1 + g_t)$), the same chaining used to extend the other Canadian series). With this reconstruction the audit finds no remaining transformation-coherence exceptions.

10 Per-Country Data Sources and Transformations

This section documents, variable by variable, the data underlying every one of the 50 economies in the system — the six core GVAR blocs, the euro-area members, the EU non-euro and other advanced economies, and the emerging markets. For each economy a table lists *every* variable in its deployed specification together with its provider, the exact series identifier (FRED mnemonic, ECB/Eurostat SDMX key, BIS dataflow key, or national central-bank code), native frequency, transformation, COVID treatment, and any imputation or splicing performed at build time. The authoritative variable list (id, description, transformation, **isFinancial** tag, prior, block) is taken verbatim from

each economy’s staged `spec.json`; the source/series-id/frequency/COVID/imputation columns are traced from the data builders and credit fetchers.

Transformation and COVID conventions (system-wide). Two transformation tags appear throughout. `log` denotes a series entered as $100 \cdot \log(\text{level})$; because such a variable carries a random-walk (RW) prior unless flagged otherwise, it *enters the dynamics as* $4 \cdot \Delta \log$, i.e. a quarter-on-quarter *annualized* growth rate (Section 2.3). `lin` denotes a series entered in levels (rates in per cent, ratios in per cent of GDP, spreads in percentage points, diffusion indices, etc.). The *COVID-as-missing* rule (Section 9.4) masks the macro block (`isFinancial= 0`) over the pandemic window — the canonical window is 2020Q2–Q4, though a few builds use a self-calibrated four-quarter 2020Q1–Q4 window (flagged where it applies) — and lets the data-augmentation/Durbin–Koopman E-step impute it, while *financial* variables (`isFinancial= 1`) are held observed through the pandemic. Per-variable priors are RW except the stationary fiscal GDP-ratios (`FREV_GDP`, `FPEXP_GDP`, `SFA`), which carry the white-noise (WN) prior.

The new credit block (flagged throughout). Each bloc is augmented with a credit *quantity* variable `CREDIT_GDP_<CC> = BIS Total Credit to Non-Financial Corporations, % of GDP` (the BIS *broad* measure: bank loans + debt securities + cross-border lending), pulled from FRED as `Q<CC>NAM770A` where a BIS reporting series exists, transformed `lin`, prior RW, `isFinancial= 1`. Although tagged financial, the credit quantity is *masked* over the COVID window because it is a %-of-GDP ratio whose denominator (GDP) collapses in 2020, so it inherits the macro GDP mask. On the credit *price* side the blocs gain either a representative NFC bank *lending rate* (`LENDR_<CC>`) or, where no lending rate is reachable, a corporate or sovereign-proxy yield (`CORP_<CC>`); where no price could be sourced it is dropped (quantity only). Economies whose credit quantity is *constructed* (non-BIS) or whose price is a sovereign proxy, spliced, dropped, or a non-canonical variant are flagged in-line in the relevant table and prose. All credit *price* variables are held observed through COVID.

10.1 Per-Economy Data Documentation: the Six Core Blocs

This subsection documents, variable by variable, the data underlying the six core GVAR blocs — the United States anchor and its five major partners (euro area, Japan, United Kingdom, Canada, China). For each economy the table lists *every* variable in the deployed specification, together with its provider, the exact series identifier (FRED mnemonic, ECB/Eurostat SDMX key, BIS dataflow key, or national central-bank code), native frequency, transformation, COVID treatment, and any imputation or splicing performed at build time.

Transformation and COVID conventions. Two transformation tags appear throughout. `log` denotes a series entered as $100 \cdot \log(\text{level})$; because every variable carries a random-walk (RW) prior unless flagged otherwise, a `log` variable *enters the dynamics as* $4 \cdot \Delta \log$, i.e. as a quarter-on-quarter *annualized growth rate*. `lin` denotes a series entered in levels (rates in per cent, ratios in per cent

of GDP, diffusion indices, etc.). The *COVID-as-missing* rule (Section 9.4) masks the macro block (`isFinancial` = 0) over 2020Q2–2020Q4 and lets the data-augmentation Gibbs sampler impute it, while *financial* variables (`isFinancial` = 1) are held observed through the pandemic window. “Fin.” in the tables below reproduces the `isFinancial` tag.

The new credit block (flagged). Each of the six blocs gains a credit *quantity* variable `CREDIT_GDP_<CC>` = BIS *Total Credit to Non-Financial Corporations, % of GDP* — the BIS *broad* credit measure (bank loans + debt securities + cross-border), pulled from FRED as `Q<CC>NAM770A`, transformed `lin`, prior RW, `isFinancial` = 1. Although tagged financial, the credit *quantity* is *masked* over 2020Q2–Q4 because it is a %-of-GDP ratio whose denominator (GDP) collapses in 2020; it therefore inherits the macro GDP mask rather than the financial observe-rule. On the credit *price* side: the US and euro area already carry a corporate-yield channel and gain *no* new price variable; Japan, the United Kingdom, and Canada gain a representative NFC bank *lending rate* (`LEND_<CC>`); and China gains a *sovereign 10Y yield proxy* (`CORP_CN`, flagged — no corporate spread exists for China). All credit *price* variables are held observed through COVID.

10.2 United States (US)

The US anchor bloc carries $n = 39$ quarterly variables on a 1998Q3–2025Q4 grid (110 quarters) at $p = 3$ lags, comprising a rich domestic macro/financial block, four global commons, and a four-variable fiscal block; the COVID-as-missing window 2020Q2–Q4 masks the macro (`isFinancial` = 0) series. Macro and financial series are pulled from FRED at native quarterly frequency (`frequency=q`, `aggregation_method=avg`); the deployed bloc additionally appends the three trade-weighted `ROW*_US` stars (Section 6.1), which are not data series and are omitted here. The new credit quantity `CREDIT_GDP` is BIS broad NFC credit (% of GDP); **no new credit price is added — the existing Aaa (FAAA) and Baa (GLB_USBAA) corporate yields already span the corporate credit-price channel.**

Table 55: US bloc variables. Trans.: `log` = 100 log level (enters as $4\Delta \log$, q/q annualized); `lin` = level. Fin. = `isFinancial`. COVID: “mask” = 2020Q2–Q4 set missing (Gibbs-imputed); “obs” = held observed.

Variable (id)	Description	Source	Series ID/key	Freq	Fin.	Trans.	COVID / Notes
<i>Domestic macro block (FRED, quarterly avg; isFinancial=0 ⇒ COVID-masked)</i>							
GDPH	Real Gross Domestic Product	FRED	GDPC1	Q	0	log	mask
CH	Real Personal Consumption Exp.	FRED	PCECC96	Q	0	log	mask
FRH	Real Private Residential Fixed Inv.	FRED	PRFI	Q	0	log	mask; deflated by GDPDEF

Variable (id)	Description	Source	Series ID/key	Freq	Fin.	Trans.	COVID / Notes
FNH	Real Private Nonres. Fixed Inv.	FRED	PNFI	Q	0	log	mask; deflated by GDPDEF
XH	Real Exports of Goods & Services	FRED	EXPGSC1	Q	0	log	mask
MH	Real Imports of Goods & Services	FRED	IMPGSC1	Q	0	log	mask
GH	Real Gov. Consumption & Gross Inv.	FRED	GCEC1	Q	0	log	mask
JGDP	GDP: Chain Price Index	FRED	GDPDEF	Q	0	log	mask
JC	PCE: Chain Price Index	FRED	PCEPI	Q	0	log	mask
JCXFE	PCE less Food & Energy: Chain Price Idx	FRED	PCEPILFE	Q	0	log	mask
UI	CPI-U: All Items	FRED	CPIAUCSL	Q	0	log	mask
UIXFDG	CPI-U: All Items less Food & Energy	FRED	CPILFESL	Q	0	log	mask
LXBC	Business Sector: Compensa- tion/Hour	FRED	HCOMPBS	Q	0	log	mask
LANAGRA	All Employees: Total Nonfarm	FRED	PAYEMS	M→Q	0	log	mask
LR	Civilian Unemployment Rate 16+	FRED	UNRATE	M→Q	0	lin	mask
IP	Industrial Production Index	FRED	INDPRO	M→Q	0	log	mask
CUMFG	Capacity Utilization: Mfg [SIC]	FRED	MCUMFN	M→Q	0	lin	mask
HST	Housing Starts	FRED	HOUST	M→Q	0	log	mask
YPDH	Real Disposable Personal Income	FRED	DPIC96	Q	0	log	mask
CSENT	U. Michigan Consumer Sentiment	FRED	UMCSENT	M→Q	0	lin	mask
<i>Domestic financial block (FRED; isFinancial=1 ⇒ observed through COVID)</i>							
POLRATE	Effective Federal Funds Rate (policy)	FRED	FEDFUNDS	M→Q	1	lin	obs
FCM2	2Y Treasury Yield, const. maturity	FRED	DGS2	D→Q	1	lin	obs

Variable (id)	Description	Source	Series ID/key	Freq	Fin.	Trans.	COVID / Notes
FCM5	5Y Treasury Yield, const. maturity	FRED	DGS5	D→Q	1	lin	obs
FAAA	Moody's Seasoned Aaa Corp. Bond Yield	FRED	AAA	M→Q	1	lin	obs; high-grade corp. price
FXTWM	Nominal Trade-Wtd US\$ vs AFE	FRED	DTWEXAFEGS / DTWEXM	D→Q	1	log	obs; AFE spliced onto legacy Major-\$ at 2006 overlap
PZTEXP	Spot Oil Price: WTI	FRED	WTISPLC	M→Q	1	log	obs
GSNE	Non-Fuel Commodity Price Idx (IMF proxy)	FRED	PNFUELINDEXM	M→Q	1	log	obs
SP500VOL	CBOE Volatility Index (VIX)	FRED	VIXCLS	D→Q	1	lin	obs
HP	House Price	FRED	CSUSHPISA	M→Q	1	log	obs
LEV	Real nonfinancial business debt	FRED	BCNSDODNS	Q	1	log	obs; deflated by GDPDEF/100
<i>Global commons (US-determined common shocks; isFinancial=1, observed)</i>							
GLB-USLR	10Y Treasury Yield, const. maturity	FRED	DGS10	D→Q	1	lin	obs; global common
GLB-USBAAM	Moody's Baa Corp. Bond Yield (credit)	FRED	BAA	M→Q	1	lin	obs; risk-spread corp. price
GLB-USEQ	US Equity Price Index (S&P / OECD US)	FRED	SPASTT01USM661N	M→Q	1	log	obs; global common
GLB-OIL	WTI crude oil, q-avg (\$/bbl)	FRED	WTISPLC	M→Q	1	log	obs; global common
<i>Fiscal block (FRED-derived, 4q-annualized; GDP-ratio drivers carry a WN/stationary prior)</i>							
FREV_GDP	Federal Receipts (% of GDP)	FRED	FGRECPT/GDP	Q	0	lin	mask; WN prior
FPEXP_GDP	Federal Primary Expenditure (% of GDP)	FRED	(FGEXPND—A091RC1Q027SBEA)/GDP	Q	0	lin	WN prior; 2020Q2–2021Q4 spike linearly interpolated at build time
EFFI	Federal Effective Interest Rate (% p.a.)	FRED	A091RC1Q027SBEA /lag (GFDEGDQ188S)	Q	1	lin	obs; RW prior

Variable (id)	Description	Source	Series ID/key	Freq	Fin.	Trans.	COVID / Notes
SFA	Stock-Flow Adjustment (% of GDP)	derived	recursion residual (GFDEGDQ188S)	Q	0	lin	WN prior; closes the debt recursion by construction
<i>NEW credit block</i>							
CREDIT_GDP	BIS Credit to NFCs (% of GDP)	FRED / BIS	QUSNAM770A	Q	1	lin	mask (inherits GDP-denominator mask); BIS broad credit, 1998Q3=61.3, 2025Q4=72.2; <i>no new price (FAAA/GLB_USBAA cover it)</i>

10.3 Euro Area (EA)

The euro-area aggregate carries $n = 22$ quarterly variables on a 1999Q1–2025Q4 grid ($T = 108$). Real-activity and price series come from FRED (which mirrors Eurostat/OECD/ECB feeds for the euro aggregate), short and long rates from FRED OECD/ECB series, and the financial overlay — policy rate (POLRATE_EA), real house prices (HPI_EA), and the all-bonds corporate-risk yield (EACORP) — from ECB SDMX and Eurostat. The new credit quantity is BIS broad NFC credit (% of GDP); **no new credit price is added** — EACORP already provides the euro-area corporate-yield **channel** (and GLB_USBAA a second global credit-risk channel).

Table 56: Euro-area bloc variables. Columns as in Table 55.

Variable (id)	Description	Source	Series ID/key	Freq	Fin.	Trans.	COVID / Notes
GDP_EA	EA Real GDP	FRED (Euro- stat)	CLVMNACSCAB1GQEA19	Q	0	log	mask
CONS_EA	EA Consumption	FRED (OECD)	NAEXKP02EZQ661S	Q	0	log	mask
INV_EA	EA Investment	FRED (OECD)	NAEXKP04EZQ661S	Q	0	log	mask
GOV_EA	EA Gov. Spending	FRED (OECD)	NAEXKP03EZQ661S	Q	0	log	mask
EXP_EA	EA Exports	FRED (OECD)	NAEXKP06EZQ661S	Q	0	log	mask
IMP_EA	EA Imports	FRED (OECD)	NAEXKP07EZQ661S	Q	0	log	mask

Variable (id)	Description	Source	Series ID/key	Freq	Fin.	Trans.	COVID / Notes
HICP_EA	EA HICP Index (2015=100)	FRED (Eurostat)	CP0000EZ19M086NEST	M→Q	0	log	mask
RATE_EA	EA 3M Interbank Rate	FRED (OECD)	IR3TIB01EZQ156N	Q	1	lin	obs
YIELD10_EA	EA 10Y Gov. Bond Yield	FRED (OECD)	IRLTLT01EZM156N	M→Q	1	lin	obs
NEER_EA	EA Nominal Effective Exch. Rate	FRED (OECD)	CCUSMA02EZQ618N	Q	1	log	obs
POLRATE_EA	EA ECB Deposit Facility Rate (policy)	ECB SDW	FM/D.U2.EUR.4F. KR.DFR.LEV	D→Q	1	lin	obs
HPI_EA	EA Real House Price Index (HPI/HICP, 2015=100)	Eurostat	prc_hpi_q. TOTAL.I15.Q. <geo> (nominal HPI) ÷ HICP_EA	Q	1	log	obs; deflated by HICP
EACORP	EA Corporate Bond Yield (Bundesbank non-MFI)	Bundesbank	BBSIS/M.I.UMR.RD.EUR.Q2000.B.knA.R.Als; GenZine	Q	1	lin	non-MFI corporate bond yield
GLB_OIL	WTI crude oil, q-avg (\$/bbl)	FRED	WTISPLC	M→Q	1	log	obs; global common
GLB_USLR	US 10Y Treasury Yield (%)	FRED	DGS10	D→Q	1	lin	obs; global common
GLB_USBAA	AUS Baa Corporate Yield (%)	FRED	BAA	M→Q	1	lin	obs; global common
GLB_USEQ	US Equity (S&P 500)	FRED	SPASTT01USM661N	M→Q	1	log	obs; global common
FREV_GDP	EA Revenue (% of GDP)	<i>see flag</i>	4q-annualized rev./GDP	Q	0	lin	mask; WN prior
FPEXP_GDP	EA Primary Expenditure (% of GDP)	<i>see flag</i>	4q-annualized (exp.−int.)/GDP	Q	0	lin	mask; WN prior
EFFI	EA Effective Interest Rate (% p.a.)	<i>see flag</i>	4×int./lag(debt)	Q	1	lin	obs; RW prior
SFA	EA Stock-Flow Adjustment (% of GDP)	derived	recursion residual	Q	0	lin	WN prior; closes the debt recursion

[FLAG: the EA fiscal flows (revenue, total expenditure, interest, gross debt, nominal GDP) are read from `pipeline/fiscal_raw/fiscal_EA.json`, which records no provenance; the consuming builders (`build_fiscal_drivers.py`, `build_fiscal_ea_4var.py`) do not fetch them. Upstream provider (Eurostat `gov_10q_ggnfa/gov_10dd_edpt1` is the most likely candidate) NOT verified from the code; series IDs left blank rather than fabricated.]

Variable (id)	Description	Source	Series ID/key	Freq	Fin.	Trans.	COVID / Notes
CREDIT_GDP_JP	Real Credit to NFCs (% of GDP)	FRED / BIS	QXMNAM770A	Q	1	lin	mask (GDP-denominator mask); BIS broad credit, 1999Q1=66.7, 2025Q4=103.1; <i>no new price (EACORP covers it)</i>

10.4 Japan (JP)

The Japan bloc carries $n = 22$ quarterly variables on a 2002Q2–2025Q4 grid ($T = 95$). Real-activity, price, rate, and exchange-rate series come from FRED; the policy rate (POLRATE_JP) and real house prices (HP_JP) are appended from the BIS (central-bank policy-rate dataflow and the real residential property-price dataflow). The new credit block adds BIS broad NFC credit (% of GDP) on the quantity side and a representative *bank lending rate* on the price side: LENDR_JP is the Bank of Japan average contract interest rate on new loans, fetched from the BoJ Time-Series Data Search REST API (not on FRED).

Table 57: Japan bloc variables. Columns as in Table 55.

Variable (id)	Description	Source	Series ID/key	Freq	Fin.	Trans.	COVID / Notes
GDP_JP	JP Real GDP	FRED	JPNRGDPEXP	Q	0	log	mask
CONS_JP	JP Consumption	FRED	NAEXKP02JPQ661S	Q	0	log	mask
		(OECD)					
INV_JP	JP Investment	FRED	NAEXKP04JPQ661S	Q	0	log	mask
		(OECD)					
GOV_JP	JP Gov. Spending	FRED	NAEXKP03JPQ661S	Q	0	log	mask
		(OECD)					
EXP_JP	JP Exports	FRED	NAEXKP06JPQ661S	Q	0	log	mask
		(OECD)					
IMP_JP	JP Imports	FRED	NAEXKP07JPQ661S	Q	0	log	mask
		(OECD)					
LR_JP	JP Unemployment Rate	FRED	LRHUTTTTJPM156S	M→Q	0	lin	mask
		(OECD)					
CPI_JP	JP CPI Index	FRED	JPNCPIALLMINMEI	M→Q	0	log	mask
		(OECD)					
RATE_JP	JP 3M Interbank Rate	FRED	IR3TIB01JPQ156N	Q	1	lin	obs
		(OECD)					

Variable (id)	Description	Source	Series ID/key	Freq	Fin.	Trans.	COVID / Notes
POLRATE_JBP	Central Bank Policy Rate (BIS CBPOL)	BIS	WS_CBPOL M.JP	M→Q	1	lin	obs
YIELD10_JP	JP 10Y JGB Yield	FRED (OECD)	IRLTLT01JPM156N	M→Q	1	lin	obs
HP_JP	JP Real Resid. House-Price Idx (BIS RPP, 2010=100)	BIS	WS_SPP Q.JP. R.628	Q	1	log	obs
GLB_OIL	WTI crude oil, q-avg (\$/bbl)	FRED	WTISPLC	M→Q	1	log	obs; global common
GLB_USLR	US 10Y Treasury Yield (%)	FRED	DGS10	D→Q	1	lin	obs; global common
GLB_USBAAUS	Baa Corporate Yield (%)	FRED	BAA	M→Q	1	lin	obs; global common
GLB_USEQ	US Equity (S&P 500)	FRED	SPASTT01USM661N	M→Q	1	log	obs; global common
FREV_GDP	JP Revenue (% of GDP)	OECD SDMX + FRED	4q-annualized rev./GDP; nominal GDP JPNNGDP	Q	0	lin	mask; WN prior
FPEXP_GDPJP	Primary Expenditure (% of GDP)	OECD SDMX + FRED	4q-annualized (exp.-int.)/GDP	Q	0	lin	mask; WN prior
EFFI	JP Effective Interest Rate (% p.a.)	OECD SDMX + FRED	4×int./lag(gross debt, OECD F2/F3/F4)	Q	1	lin	obs; RW prior
SFA	JP Stock-Flow Adjustment (% of GDP)	derived	recursion residual	Q	0	lin	WN prior; COVID 2020Q1–Q4 set NaN (Gibbs-imputed)
CREDIT_GDPJP	JP Credit to NFCs (% of GDP)	FRED / BIS	QJPNAM770A	Q	1	lin	mask (GDP-denominator mask); BIS broad credit, 2002Q2=111.0, 2025Q4=113.9
LENDR_JP	JP Avg Contract Rate on New Loans (% p.a.)	BoJ Time- Series API	IR04 / DLLR2CIDBNL1	M→Q	1	lin	obs ; domestically- licensed banks, new loans total; NOT on FRED (dedicated BoJ fetcher)

10.5 United Kingdom (UK)

The UK bloc carries $n = 24$ quarterly variables on a 1997Q1–2025Q4 grid ($T = 116$). Real-activity, price, rate, industrial-production, NEER, policy-rate, and house-price series come from FRED (plus the BIS for the BIS-RPP house-price and CBPOL policy-rate concepts where used). The new credit block adds BIS broad NFC credit (% of GDP) on the quantity side and the Bank of England effective rate on new sterling loans to PNFs (LENDR_UK) on the price side, fetched from the BoE Interactive Database (IADB) CSV API (not on FRED).

Table 58: UK bloc variables. Columns as in Table 55.

Variable (id)	Description	Source	Series ID/key	Freq	Fin.	Trans.	COVID / Notes
GDP_UK	UK Real GDP	FRED (ONS)	NGDPRSAXDCGBQ	Q	0	log	mask
CONS_UK	UK Consumption	FRED (OECD)	NAEXKP02GBQ661S	Q	0	log	mask
INV_UK	UK Investment	FRED (OECD)	NAEXKP04GBQ661S	Q	0	log	mask
GOV_UK	UK Gov. Spending	FRED (OECD)	NAEXKP03GBQ661S	Q	0	log	mask
EXP_UK	UK Exports	FRED (OECD)	NAEXKP06GBQ661S	Q	0	log	mask
IMP_UK	UK Imports	FRED (OECD)	NAEXKP07GBQ661S	Q	0	log	mask
IP_UK	UK Industrial Production Index	FRED (OECD)	GBRPROINDMISMEI	M→Q	0	log	mask
LR_UK	UK Unemployment Rate	FRED (OECD)	LRHUTTTTGBM156S	M→Q	0	lin	mask
CPI_UK	UK CPI Index	FRED (OECD)	GBRCPIALLMINMEI	M→Q	0	log	mask
RATE_UK	UK 3M Interbank Rate	FRED (OECD)	IR3TIB01GBQ156N	Q	1	lin	obs
POLRATE_UK	UK Central Bank Policy Rate (BIS CBPOL)	BIS	WS_CBPOL M.GB	M→Q	1	lin	obs
YIELD10_UK	UK 10Y Gilt Yield	FRED (OECD)	IRLTLT01GBM156N	M→Q	1	lin	obs
HP_UK	UK Real Resid. House-Price Idx (BIS RPP, 2010=100)	BIS	WS_SPP Q.GB. R.628	Q	1	log	obs
NEER_UK	UK Nominal Effective Exch. Rate	FRED (OECD)	CCUSMA02GBQ618N	Q	1	log	obs

Variable (id)	Description	Source	Series ID/key	Freq	Fin.	Trans.	COVID / Notes
GLB_OIL	WTI crude oil, q-avg (\$/bbl)	FRED	WTISPLC	M→Q	1	log	obs; global common
GLB_USLR	US 10Y Treasury Yield (%)	FRED	DGS10	D→Q	1	lin	obs; global common
GLB_USBAAUS	Baa Corporate Yield (%)	FRED	BAA	M→Q	1	lin	obs; global common
GLB_USEQ	US Equity (S&P 500)	FRED	SPASTT01USM661N	M→Q	1	log	obs; global common
FREV_GDP	UK Revenue (% of GDP)	<i>see flag</i>	4q-annualized rev./GDP	Q	0	lin	mask; WN prior
FPEXP_GDPUK	Primary Expenditure (% of GDP)	<i>see flag</i>	4q-annualized (exp.–int.)/GDP	Q	0	lin	mask; WN prior
EFFI	UK Effective Rate (% p.a.)	<i>see flag</i>	4×int./lag(debt)	Q	1	lin	obs; RW prior
SFA	UK Stock-Flow Adj. (% of GDP)	derived	recursion residual	Q	0	lin	WN prior; closes the debt recursion
<i>[FLAG: the UK fiscal flows are read from <code>pipeline/fiscal_raw/fiscal.UK.json</code>, which records no provenance; the consuming builders do not fetch them. Upstream provider (ONS public-sector finances / OECD QNA government finance) NOT verified from the code; series IDs left blank rather than fabricated. Note: an index-linked-gilt interest spike is documented separately in Section 7.6.]</i>							
CREDIT_GDPUK	UK Credit to NFCs (% of GDP)	FRED / BIS	QGBNAM770A	Q	1	lin	mask (GDP-denominator mask); BIS broad credit, NSA, 1997Q1=51.5, 2025Q4=59.0
LENDR_UK	UK Effective Rate on New Loans to PNFCs (% p.a.)	Bank of England IADB	CFMBJ82	M→Q	1	lin	obs ; new-business sterling lending rate; starts 2004Q1 (ragged lead, Gibbs-completed); NOT on FRED (dedicated BoE fetcher)

10.6 Canada (CA)

The Canada bloc carries $n = 24$ quarterly variables on a 1998Q1–2025Q4 grid ($T = 112$). Real-activity, price, rate, NEER, policy-rate, and house-price series come from FRED (with BIS for the CBPOL policy-rate and BIS-RPP house-price concepts). The new credit block adds BIS broad NFC credit (% of GDP) on the quantity side and the Bank of Canada weekly effective business interest rate (LENDR_CA) on the price side, fetched from the BoC Valet API (not on FRED).

Table 59: Canada bloc variables. Columns as in Table 55.

Variable (id)	Description	Source	Series ID/key	Freq	Fin.	Trans.	COVID / Notes
GDP_CA	CA Real GDP	FRED (OECD)	NAEXKP01CAQ661S	Q	0	log	mask
CONS_CA	CA Consumption	FRED (OECD)	NAEXKP02CAQ661S	Q	0	log	mask
INV_CA	CA Investment	FRED (OECD)	NAEXKP04CAQ661S	Q	0	log	mask
GOV_CA	CA Gov. Spending	FRED (OECD)	NAEXKP03CAQ661S	Q	0	log	mask
EXP_CA	CA Exports	FRED (OECD)	NAEXKP06CAQ661S	Q	0	log	mask
IMP_CA	CA Imports	FRED (OECD)	NAEXKP07CAQ661S	Q	0	log	mask
IP_CA	CA Industrial Production (Idx 2015=100)	FRED / StatCan	log-level IP (rebuilt)	Q	0	log	mask; rebuilt as a log-level index
LR_CA	CA Unemployment Rate	FRED (OECD)	LRHUTTTTCAM156S	M→Q	0	lin	mask
CPI_CA	CA CPI Index	FRED (OECD)	CANCPIALLMINMEI	M→Q	0	log	mask
RATE_CA	CA 3M Interbank Rate	FRED (OECD)	IR3TIB01CAQ156N	Q	1	lin	obs
POLRATE_CA	CA Central Bank Policy Rate (BIS CBPOL)	BIS	WS_CBPOL M.CA	M→Q	1	lin	obs
YIELD10_CA	CA 10Y Gov. Bond Yield	FRED (OECD)	IRLTTLT01CAM156N	M→Q	1	lin	obs
HP_CA	CA Real Resid. House-Price Idx (BIS RPP, 2010=100)	BIS	WS_SPP Q.CA. R.628	Q	1	log	obs
NEER_CA	CA Nominal Effective Exch. Rate	FRED (OECD)	CCUSMA02CAQ618N	Q	1	log	obs
GLB_OIL	WTI crude oil, q-avg (\$/bbl)	FRED	WTISPLC	M→Q	1	log	obs; global common
GLB_USLR	US 10Y Treasury Yield (%)	FRED	DGS10	D→Q	1	lin	obs; global common
GLB_USBAA	AUS Baa Corporate Yield (%)	FRED	BAA	M→Q	1	lin	obs; global common
GLB_USEQ	US Equity (S&P 500)	FRED	SPASTT01USM661N	M→Q	1	log	obs; global common
FREV_GDP	CA Revenue (% of GDP)	<i>see flag</i>	4q-annualized rev./GDP	Q	0	lin	mask; WN prior

Variable (id)	Description	Source	Series ID/key	Freq	Fin.	Trans.	COVID / Notes
FPEXP_GDPCA	Primary Expenditure (% of GDP)	<i>see flag</i>	4q-annualized (exp.–int.)/GDP	Q	0	lin	mask; WN prior
EFFI	CA Effective Interest Rate (% p.a.)	<i>see flag</i>	4×int./lag(debt)	Q	1	lin	obs; RW prior
SFA	CA Stock-Flow Adjustment (% of GDP)	derived	recursion residual	Q	0	lin	WN prior; closes the debt recursion
[FLAG: the CA fiscal flows are read from <code>pipeline/fiscal_raw/fiscal_CA.json</code> , which records no provenance; the consuming builders do not fetch them. Upstream provider (Statistics Canada government-finance / OECD QNA) NOT verified from the code; series IDs left blank rather than fabricated.]							
CREDIT_GDP_CA	CA Credit to NFCs (% of GDP)	FRED / BIS	QCANAM770A	Q	1	lin	mask (GDP-denominator mask); BIS broad credit, NSA, 1998Q1=84.2, 2025Q4=118.3
LENDR_CA	CA Bank Lending Rate (effective business rate, % p.a.)	Bank of Canada Valet	BUSEFFRATE	W→Q	1	lin	obs ; weekly effective business interest rate, q-avg; starts 1999Q1; NOT on FRED (dedicated BoC fetcher)

10.7 China (CN)

The China bloc carries $n = 19$ quarterly variables on a 1999Q1–2025Q4 grid ($T = 108$). Because clean quarterly *real* GDP is not available for China, real activity is anchored by official quarterly *nominal* GDP plus trade volumes and the BVAR/CPI separation; all domestic series are genuine FRED series (no placeholders). The policy rate (POLRATE.CN) and real property prices (HP.CN) are appended from the BIS. The new credit block adds BIS broad NFC credit (% of GDP) on the quantity side; on the price side CORP.CN is a *sovereign 10Y yield used as a corporate-price proxy* — **[FLAG: CN price is a SOVEREIGN 10Y benchmark government yield, not a corporate yield or NFC lending rate; no corporate spread exists for China]**.

Table 60: China bloc variables. Columns as in Table 55.

Variable (id)	Description	Source	Series ID/key	Freq	Fin.	Trans.	COVID / Notes
GDP_CN	CN Nominal GDP (quarterly)	FRED (OECD MEI)	CHNGDPNQDSMEI	Q	0	log	mask; nominal (no clean real q GDP)
CPI_CN	CN CPI	FRED (OECD)	CHNCPIALLMINMEI	M→Q	0	log	mask
RATE_CN	CN 3M Interbank Rate	FRED (OECD)	IR3TIB01CNM156N	M→Q	1	lin	obs
EXP_CN	CN Exports (USD)	FRED (OECD)	XTEXVA01CNM667S	M→Q	0	log	mask
IMP_CN	CN Imports (USD)	FRED (OECD)	XTIMVA01CNM667S	M→Q	0	log	mask
REER_CN	CN Real Effective FX	FRED (BIS)	RBCNBIS	M→Q	1	log	obs
EQ_CN	Shanghai Share Prices	FRED (OECD)	SPASTT01CNM661N	M→Q	1	log	obs
POLRATE_CN	CN Policy Rate (BIS CBPOL)	BIS	WS_CBPOL M.CN	M→Q	1	lin	obs
HP_CN	CN Real Resid. Property Price (BIS)	FRED (BIS)	QCNR628BIS	Q	1	log	obs
GLB_OIL	WTI crude oil, q-avg (\$/bbl)	FRED	WTISPLC	M→Q	1	log	obs; global common
GLB_USLR	US 10Y Treasury Yield (%)	FRED	DGS10	D→Q	1	lin	obs; global common
GLB_USBAA	US Baa Corporate Yield (%)	FRED	BAA	M→Q	1	lin	obs; global common
GLB_USEQ	US Equity (S&P 500)	FRED	SPASTT01USM661N	M→Q	1	log	obs; global common
FREV_GDP	CN Revenue (% of GDP)	IMF WEO (2025-04)	CHN.GGR. national_currency (annual)÷GDP	A→Q	0	lin	mask; WN prior; annual, temporally disaggregated
FPEXP_GDP	CN Primary Expenditure (% of GDP)	IMF WEO (2025-04)	CHN.GGX less derived interest, ÷GDP	A→Q	0	lin	mask; WN prior
EFFI	CN Effective Interest Rate (% p.a.)	IMF WEO + BIS debt	4×int./lag(QCNGANXDCA)		1	lin	obs; RW prior; interest derived as overall—primary balance
SFA	CN Stock-Flow Adjustment (% of GDP)	derived	recursion residual	Q	0	lin	WN prior; gross debt = BIS GG credit QCNGANXDCA

Variable (id)	Description	Source	Series ID/key	Freq	Fin.	Trans.	COVID / Notes
CREDIT_GDP_ENN	Credit to NFCs (% of GDP)	FRED / BIS	QCNNAM770A	Q	1	lin	mask (GDP-denominator mask); BIS broad credit, NSA, starts 2006Q1=99.4 (leading NaNs pre-2006), 2025Q4=142.8
CORP_CN	CN 10Y Government Bond Yield (sovereign proxy)	OECD KEI (SDMX)	DSD_KEI@DF_KEI / CHN.M.IRLT	M→Q	1	lin	obs ; <i>[FLAG: SOVEREIGN 10Y proxy, no corporate spread]</i> ; starts 2014Q1=4.48 (leading NaNs pre-2014), 2025Q4=1.83

10.8 Euro-core single-country BVARs (AT, BE, DE, FR, NL)

The five euro-core economies are estimated as *standalone* single-country Bayesian VARs (not GVAR blocs): the rest-of-euro-area and the area-wide rates enter as a weakly-exogenous foreign block rather than through trade-weighted stars. All real macro aggregates are Eurostat-sourced (the redesigned `namq_10_gdp` / `prc_hicp_midx` / `une_rt_q` / `irt_*` / `ert_eff_ic_q` pulls, current to 2025Q4); the financial block (ECB Deposit Facility Rate, the EA all-bonds 10Y yield, the 3M EURIBOR, the Bund and the sovereign spread) is fetched from the ECB Data Portal; the fiscal block is built from Eurostat `gov_10q` on the 4-variable {revenue, primary expenditure, effective rate, stock-flow adjustment} recipe; and the credit block (`CREDIT_GDP_<CC>`, `LENDR_<CC>`) is appended last from BIS-via-FRED and the ECB MIR statistics.

Transformation conventions. Throughout, `log` denotes the model space $100 \cdot \log(\text{level})$; such a variable enters growth-rate reporting as $4 \cdot \Delta(100 \log) = \text{quarter-on-quarter annualised growth}$ (§2.3). `lin` denotes the raw level (a rate in % p.a., a ratio in % of GDP, or an index read in levels). The own-lag prior is random-walk (RW) for every macro and financial variable; the fiscal flow ratios (`FREV_GDP`, `FPEXP_GDP`, `SFA`) carry the stationary white-noise prior (WN).

COVID treatment. Under the COVID-as-missing convention only the non-financial macro columns (`isFinancial`= 0: the six national-accounts aggregates, HICP, the unemployment rate, and the two rest-of-EA macro series) are masked (NaN) over the pandemic window and imputed by the COVID-as-missing Durbin–Koopman smoother; financial columns are kept observed. *[FLAG: window mismatch.] Two distinct COVID windows coexist in these builds. The macro/ROW block*

(and the fiscal *SFA*) is masked over the four quarters **2020Q1–Q4** — the self-calibrated window *build_euro_member_redesign.py* selects as the largest $100 \log(\text{GDP})$ drop in 2019Q4–2021Q1, i.e. *trough* – 1 ... *trough* + 2 — whereas the newly-appended credit quantity *CREDIT_GDP_<CC>* is masked over the three quarters **2020Q2–Q4** (the hard-coded *COVID_MASK* in *build_credit_eurocore.py*). The two masks therefore differ by one quarter (2020Q1). All financial columns (rates, spreads, *EA-CORP*, *EFFI*, house prices, and the credit price *LENDR_<CC>*) remain observed throughout. The fiscal flow ratios *FREV_GDP*, *FPEXP_GDP* and the effective rate *EFFI* are *not* masked: they are interpolated/forward-filled to a NaN-free column before estimation; only the derived *SFA* inherits the four-quarter macro mask.

Imputation of ragged edges. Financial and exogenous columns are leading-back-filled (first finite value carried back) and forward-filled so every column is NaN-free over its estimation window; the national-accounts/HICP/unemployment columns drive the common contiguous window. The credit quantity has no leading/trailing gaps (BIS history predates every base start); its only missing cells are the three COVID-masked quarters.

The credit block (NEW). Each economy gains exactly two domestic financial variables appended after all existing columns (*isFinancial*= 1, *lin*, *RW*, *block*=domestic):

- **Quantity** *CREDIT_GDP_<CC>* = BIS “Total credit to non-financial corporations, % of GDP” (break-adjusted *broad* credit: bank loans *plus* debt securities *plus* cross-border lending), pulled via FRED under the uniform mnemonic *Q<CC>NAM770A*. Native quarterly (FRED dates the observations at the quarter start). COVID-masked 2020Q2–Q4 (it inherits the collapse of the GDP denominator).
- **Price** *LENDR_<CC>* = the ECB MIR (MFI Interest Rate statistics) NFC *new-business* bank lending rate, SDMX key *M.<CC>.B.A2A.A.R.A.2240.EUR.N* (title “Bank interest rates — loans to corporations (new business)”; S.11 NFC counterparty, AAR/NDER, all-maturity, EUR, new-business coverage). Monthly → quarter average; observed through COVID.

Because the shipped quantity is the BIS *broad* total-credit series, the euro-core members carry the canonical BIS measure (not a bank-loans-only proxy), and the credit price is a true national NFC new-business lending rate (not a sovereign-yield or corporate-spread proxy). The only price-coverage caveat is Belgium (see below).

10.9 Austria (AT)

A 23-variable single-country BVAR estimated over **2009Q1–2025Q4** ($T = 68$). COVID-as-missing: macro block masked 2020Q1–Q4, credit quantity masked 2020Q2–Q4, financials observed (§10.8).

Table 61: Austria (AT) — variable documentation.

Variable (id)	Description	Source	Series ID/key	Freq	Transf.	COVID	Notes
GDP_AT	Real GDP (chain-linked vol., SCA)	Eurostat	namq_10_gdp Q.CLV15_MEUR.SCA.B1GQ	Q	log	masked 20Q1– Q4	domestic
CONS_AT	Real household final consumption	Eurostat	namq_10_gdp Q.CLV15_MEUR.SCA.P31.S14	Q	log	masked 20Q1– Q4	domestic
INV_AT	Real gross fixed capital formation	Eurostat	namq_10_gdp Q.CLV15_MEUR.SCA.P51G	Q	log	masked 20Q1– Q4	domestic
GOV_AT	Real government consumption	Eurostat	namq_10_gdp Q.CLV15_MEUR.SCA.P3.S13	Q	log	masked 20Q1– Q4	domestic
EXP_AT	Real exports	Eurostat	namq_10_gdp Q.CLV15_MEUR.SCA.P6	Q	log	masked 20Q1– Q4	domestic
IMP_AT	Real imports	Eurostat	namq_10_gdp Q.CLV15_MEUR.SCA.P7	Q	log	masked 20Q1– Q4	domestic
HICP_AT	HICP index (all-items, 2015=100)	Eurostat	prc_hicp_midx M.I15.CP00.A3avg	M→Q	log	masked 20Q1– Q4	domestic
LR_AT	Unemployment rate (SA, % active pop.)	Eurostat	une_rt_q Q.SA.TOTAL.PC_ACT.T	Q	lin	masked 20Q1– Q4	domestic
HPI_AT	Real house-price index (nominal HPI/HICP, 2015=100)	Eurostat	prc_hpi_q Q.TOTAL.I15_Q.AT ÷ HICP	Q	log	observed (financial) 20Q1– Q4	domestic; nominal HPI deflated by HICP

continued on next page

Table 61 (Austria), continued

Variable (id)	Description	Source	Series ID/key	Freq	Transf.	COVID	Notes
NEER_AT	Nominal effective ex- change rate, IC42 partners	Eurostat	ert_eff_ic_q Q.NEER.IC42.I15.AT	Q	log	observed	domestic; broad 42- partner group
SPREAD_AT	10Y sovereign spread vs Bund	Eurostat (com- puted)	irt_lt_mcbym M.MCBY.AT – M.MCBY.DE	M→Q	avg	observed	domestic; own 10Y minus Bund
POLRATE_AT	ECB Deposit Facil- ity Rate (policy)	ECB Data Portal	FM/D.U2.EUR.DF-X avg	Q	lin	observed	external (shared)
EACORP	EA corporate bond yield (non-MFI)	Bundesbank	BBSIS/M.I.UMB.MQ.EUR.X2000 avg	Q	lin	observed	external; IG-corp risk proxy
RATE_AT	3-month EURIBOR	Eurostat	irt_st_q Q.IRT.M3.EA	Q	lin	observed	external (shared)
YIELD10_DE	DE 10Y Bund yield (spread base)	Eurostat	irt_lt_mcbym M.MCBY.DE	M→Q	avg	observed	external (in- jected)
ROWEA_GDP_AT	Rest-of-EA GDP (GDP- weighted de-agg.)	real (com- puted)	Eurostat EA20 less AT, namq_10_gdp	Q	log	masked	row 20Q1– Q4
ROWEA_HICP_AT	Rest-of-EA HICP (GDP-weighted de-agg.)	HICP (com- puted)	Eurostat EA20 less AT, prc_hicp_midx	Q	log	masked	row 20Q1– Q4
FREV_GDP	General-gov. enue (% GDP, 4q- trailing)	rev- Eurostat	gov_10q_ggnfa Q.MIO.EUR.NSA.S13.TR.AT	Q	lin	observed	fiscal; (forward- filled) WN
FPEXP_GDP	Primary diture (% GDP, 4q-trailing)	expen- Eurostat	gov_10q_ggnfa ...TE – ...D41PAY .AT	Q	lin	observed	fiscal; (forward- filled) WN prior

continued on next page

Table 61 (Austria), continued

Variable (id)	Description	Source	Series ID/key	Freq	Transf.	COVID	Notes
EFFI	Effective interest rate on gov. debt	Eurostat (computed)	...D41PAY / Q gov_10q_ggdebt ...GD.AT	Q	lin	observed	fiscal; financial
SFA	Stock-flow adjustment (budget-identity residual)	Eurostat (computed)	debt-recursion residual	Q	lin	masked 20Q1– Q4	fiscal; WN prior
CREDIT_GDP_AT	[NEW] BIS total credit to NFCs, % of GDP (broad: loans+bonds+cross-border)	BIS via FRED	QATNAM770A	Q	lin	masked 20Q2– Q4	domestic; src 1995Q4:69.7..2025Q4:83.9
LENDR_AT	[NEW] NFC new-business lending rate	ECB MIR	M.AT.B.A2A.AM.AQ2240.FUB	Q	avg	observed	domestic; src 2000Q1:4.95..2026Q2:3.57

10.10 Belgium (BE)

A 23-variable single-country BVAR estimated over **2009Q1–2025Q4** ($T = 68$). COVID-as-missing: macro block masked 2020Q1–Q4, credit quantity masked 2020Q2–Q4, financials observed.

Table 62: Belgium (BE) — variable documentation.

Variable (id)	Description	Source	Series ID/key	Freq	Transf.	COVID	Notes
GDP_BE	Real GDP (chain-linked vol., SCA)	Eurostat	namq_10_gdp Q.CLV15.MEUR.SCA.B1GQ	Q	log	masked 20Q1– Q4	domestic
CONS_BE	Real household final consumption	Eurostat	namq_10_gdp ...P31_S14_S15.BE	Q	log	masked 20Q1– Q4	domestic
INV_BE	Real gross fixed capital formation	Eurostat	namq_10_gdp ...P51G.BE	Q	log	masked 20Q1– Q4	domestic

continued on next page

Table 62 (Belgium), continued

Variable (id)	Description	Source	Series ID/key	Freq	Transf.	COVID	Notes
GOV_BE	Real government consumption	Eurostat	namq_10_gdp ...P3.S13.BE	Q	log	masked 20Q1– Q4	domestic
EXP_BE	Real exports	Eurostat	namq_10_gdp ...P6.BE	Q	log	masked 20Q1– Q4	domestic
IMP_BE	Real imports	Eurostat	namq_10_gdp ...P7.BE	Q	log	masked 20Q1– Q4	domestic
HICP_BE	HICP index (all-items, 2015=100)	Eurostat	prc_hicp_midxM→Q M.I15.CP00.BE	Q	log	masked 20Q1– Q4	domestic
LR_BE	Unemployment rate (SA, % active pop.)	Eurostat	une_rt_q Q.SA.TOTAL.PC.ACT.T.BE	Q	lin	masked 20Q1– Q4	domestic
HPI_BE	Real house-price index (nominal HPI/HICP, 2015=100)	Eurostat	prc_hpi_q Q.TOTAL.I15.Q.BE ÷ HICP	Q	log	observed (financial)	domestic
NEER_BE	Nominal effective exchange rate, IC42 partners	Eurostat	ert_eff_ic_q Q.NEER.IC42.I15.BE	Q	log	observed	domestic
SPREAD_BE	10Y sovereign spread vs Bund	Eurostat (computed)	irt_lt_mcbymM→Q M.MCBY.BE – avg M.MCBY.DE	Q	lin	observed	domestic
POLRATE_BE	ECB Deposit Facility Rate (policy)	ECB Data Portal	FM/D.U2.EUR.DFR.LFN DF-X01.DFR.LFN	Q	lin	observed	external (shared)
EACORP	EA corporate bond yield (non-MFI)	Bundesbank	BBSIS/M.I.UMB.MQ.EUR.X2009.BdA.BrAaA.Z.Z.A	Q	lin	observed	external (shared)
RATE_BE	3-month EURIBOR	Eurostat	irt_st_q Q.IRT.M3.EA	Q	lin	observed	external (shared)
YIELD10_DE	DE 10Y Bund yield (spread base)	Eurostat	irt_lt_mcbymM→Q M.MCBY.DE	Q	avg	observed	external (injected)

continued on next page

Table 62 (Belgium), continued

Variable (id)	Description	Source	Series ID/key	Freq	Transf.	COVID	Notes
ROWEA_GDP_BE	Rest-of-EA GDP (GDP-weighted de-aggr.)	real (GDP- de-aggr.)	Eurostat (com- puted)	EA20 less namq_10_gdp	Q	log	masked 20Q1– Q4
ROWEA_HICP_BE	Rest-of-EA (GDP-weighted de-aggr.)	HICP (com- puted)	Eurostat (com- puted)	EA20 less prc_hicp_midx	Q	log	masked 20Q1– Q4
FREV_GDP	General-gov. enue (% GDP, 4q- trailing)	rev- GDP, 4q-	Eurostat	gov_10q_ggnfa Q.MIO.EUR.NSA.S13.TR.BE	Q	lin	observed fiscal; (forward- WN filled)
FPEXP_GDP	Primary diture (% GDP, 4q-trailing)	expen- GDP,	Eurostat	gov_10q_ggnfa ...TE – ...D41PAY .BE	Q	lin	observed fiscal; (forward- WN filled)
EFFI	Effective rate on gov. debt	interest debt	Eurostat (com- puted)	...D41PAY / gov_10q_ggdebt ...GD.BE	Q	lin	observed fiscal; fi- nancial
SFA	Stock-flow ment identity residual	adjust- (budget- residual)	Eurostat (com- puted)	debt- recursion residual	Q	lin	masked 20Q1– Q4
CREDIT_GDP_BE	[NEW] BIS credit to NFCs, % of GDP (broad)	total FRED	BIS via Q	BENAM770A	Q	lin	masked 20Q2– Q4

continued on next page

Table 62 (Belgium), continued

Variable (id)	Description	Source	Series ID/key	Freq	Transf.	COVID	Notes
LENDR_BE	[NEW] NFC new- business bank lend- ing rate	ECB MIR	M.BE.B.A2A.AM-Q	Q	log	2020Q1–2020Q4	Observed domestic; [FLAG: price coverage starts 2003Q1, not 2000Q1 as for the oth- ers; src 2003Q1:3.66..2026Q2:3.48 — ear- liest pre- 2009 values are back- filled.]

10.11 Germany (DE)

A 22-variable single-country BVAR estimated over **2009Q1–2025Q4** ($T = 68$). Germany is the spread *base*: it carries the Bund 10Y yield YIELD10_DE as an *endogenous domestic* variable and therefore has *no* SPREAD_DE variable (hence $n = 22$, one fewer than the other euro-core members). COVID-as-missing: macro block masked 2020Q1–Q4, credit quantity masked 2020Q2–Q4, financials observed.

Table 63: Germany (DE) — variable documentation.

Variable (id)	Description	Source	Series ID/key	Freq	Transf.	COVID	Notes
GDP_DE	Real GDP (chain- linked vol., SCA)	Eurostat	namq_10_gdp Q.CLV15_MEUR.SCA.B1GQ.DE	Q	log	masked 2020Q1– Q4	domestic

continued on next page

Table 63 (Germany), continued

Variable (id)	Description	Source	Series ID/key	Freq	Transf.	COVID	Notes
CONS_DE	Real household final consumption	Eurostat	namq_10_gdp ...P31_S14_S15.DE	Q	log	masked 20Q1– Q4	domestic
INV_DE	Real gross fixed capital formation	Eurostat	namq_10_gdp ...P51G.DE	Q	log	masked 20Q1– Q4	domestic
GOV_DE	Real government consumption	Eurostat	namq_10_gdp ...P3_S13.DE	Q	log	masked 20Q1– Q4	domestic
EXP_DE	Real exports	Eurostat	namq_10_gdp ...P6.DE	Q	log	masked 20Q1– Q4	domestic
IMP_DE	Real imports	Eurostat	namq_10_gdp ...P7.DE	Q	log	masked 20Q1– Q4	domestic
HICP_DE	HICP index (all-items, 2015=100)	Eurostat	prc_hicp_midxM→Q M.I15.CP00.DEavg	Q	log	masked 20Q1– Q4	domestic
LR_DE	Unemployment rate (SA, % active pop.)	Eurostat	une_rt_q Q.SA.TOTAL.PC_ACT.T.DE	Q	lin	masked 20Q1– Q4	domestic
HPI_DE	Real house-price index (nominal HPI/HICP, 2015=100)	Eurostat	prc_hpi_q Q.TOTAL.I15_Q.DE ÷ HICP	Q	log	observed (financial)	domestic
NEER_DE	Nominal effective exchange rate, IC42 partners	Eurostat	ert_eff_ic_q Q.NEER_IC42.I15.DE	Q	log	observed	domestic
YIELD10_DE	10Y Bund yield (the spread base)	Eurostat	irt_lt_mcbymM→Q M.MCBY.DEavg	Q	lin	observed	domestic (DE special case; no SPREAD.DE)

continued on next page

Table 63 (Germany), continued

Variable (id)	Description	Source	Series ID/key	Freq	Transf.	COVID	Notes
POLRATE_DE	ECB Deposit Facility Rate (policy)	ECB Data Portal	FM/D.U2.EUR.DF.AQ	Q	lin	observed	external (shared)
EACORP	EA corporate bond yield (non-MFI)	Bundesbank	BBSIS/M.I.UMD.MQ.EUR.X2000	Q	lin	observed	external
RATE_DE	3-month EURIBOR	Eurostat	irt_st_q Q.IRT.M3.EA	Q	lin	observed	external (shared)
ROWEA_GDP_DE	Rest-of-EA GDP (GDP-weighted de-agg.)	real Eurostat	EA20 less DE, namq_10_gdp	Q	log	masked	row 20Q1–Q4
ROWEA_HICP_DE	Rest-of-EA HICP (GDP-weighted de-agg.)	Eurostat	EA20 less DE, prc_hicp_midx	Q	log	masked	row 20Q1–Q4
FREV_GDP	General-gov. revenue (% GDP, 4q-trailing)	rev- Eurostat	gov_10q_ggnfa Q.MIO.EUR.NSA.S13.TR.DE	Q	lin	observed	fiscal; (forward- WN filled)
FPEXP_GDP	Primary expenditure (% GDP, 4q-trailing)	expen- Eurostat	gov_10q_ggnfa ...TE – ...D41PAY .DE	Q	lin	observed	fiscal; (forward- WN filled)
EFFI	Effective interest rate on gov. debt	interest Eurostat (com-puted)	gov_10q_ggdebt ...GD.DE	Q	lin	observed	fiscal; financial
SFA	Stock-flow adjustment (budget-identity residual)	adjust- Eurostat (com-puted)	debt-recursion residual	Q	lin	masked	fiscal; WN 20Q1–Q4
CREDIT_GDP_DE	[NEW] BIS credit to NFCs, % of GDP (broad)	BIS total BIS via FRED	QDENAM770A	Q	lin	masked	domestic; src 20Q2–Q4 1990Q1:59.5..2025Q4:87.8
LENDR_DE	[NEW] NFC business lending rate	new- ECB MIR	M.DE.B.A2A.AM.AQ	Q	lin	observed	domestic; src 2000Q1:5.85..2026Q2:3.55

10.12 France (FR)

A 23-variable single-country BVAR estimated over **2003Q1–2025Q4** ($T = 92$) — the longest euro-core sample (the others start 2009Q1). COVID-as-missing: macro block masked 2020Q1–Q4, credit quantity masked 2020Q2–Q4, financials observed.

Table 64: France (FR) — variable documentation.

Variable (id)	Description	Source	Series ID/key	Freq	Transf.	COVID	Notes
GDP_FR	Real GDP (chain-linked vol., SCA)	Eurostat	namq_10_gdp Q.CLV15_MEUR.SCA.B1GQ.FR	Q	log	masked 20Q1–Q4	domestic
CONS_FR	Real household final consumption	Eurostat	namq_10_gdp ...P31_S14_S15.FR	Q	log	masked 20Q1–Q4	domestic
INV_FR	Real gross fixed capital formation	Eurostat	namq_10_gdp ...P51G.FR	Q	log	masked 20Q1–Q4	domestic
GOV_FR	Real government consumption	Eurostat	namq_10_gdp ...P3_S13.FR	Q	log	masked 20Q1–Q4	domestic
EXP_FR	Real exports	Eurostat	namq_10_gdp ...P6.FR	Q	log	masked 20Q1–Q4	domestic
IMP_FR	Real imports	Eurostat	namq_10_gdp ...P7.FR	Q	log	masked 20Q1–Q4	domestic
HICP_FR	HICP index (all-items, 2015=100)	Eurostat	prc_hicp_midxM→Q M.I15.CP00.FRavg	Q	log	masked 20Q1–Q4	domestic
LR_FR	Unemployment rate (SA, % active pop.)	Eurostat	une_rt_q Q.SA.TOTAL.PC_ACT.T.FR	Q	lin	masked 20Q1–Q4	domestic
HPI_FR	Real house-price index (nominal HPI/HICP, 2015=100)	Eurostat	prc_hpi_q Q.TOTAL.I15_Q.FR ÷ HICP	Q	log	observed (financial)	domestic

continued on next page

Table 64 (France), continued

Variable (id)	Description	Source	Series ID/key	Freq	Transf.	COVID	Notes
NEER_FR	Nominal effective ex- change rate, IC42 partners	Eurostat	ert_eff_ic_q Q.NEER.IC42.I15.FR	Q	log		observed domestic
SPREAD_FR	10Y sovereign spread vs Bund	Eurostat (com- puted)	irt_lt_mcbymM→Qn M.MCBY.FR – avg M.MCBY.DE	Q	lin		observed domestic
POLRATE_FR	ECB Deposit Facil- ity Rate (policy)	ECB Data Portal	FM/D.U2.EUR.DF→Qn DF→Qn	Q	lin		observed external (shared)
EACORP	EA corporate bond yield (non-MFI)	Bundesbank	BBSIS/M.I.UMD.MQEUR.X2009BdAActBrAaA.Z..Z.A	Q	log		observed external
RATE_FR	3-month EURIBOR	Eurostat	irt_st_q Q.IRT.M3.EA	Q	lin		observed external (shared)
YIELD10_DE	DE 10Y Bund yield (spread base)	Eurostat	irt_lt_mcbymM→Qn M.MCBY.DE	Q	lin		observed external (in- jected)
ROWEA_GDP_FR	Rest-of-EA GDP (GDP- weighted de-agg.)	real Eurostat (com- puted)	EA20 less FR, namq_10_gdp	Q	log		masked row 20Q1– Q4
ROWEA_HICP_FR	Rest-of-EA (GDP-weighted de-agg.)	HICP Eurostat (com- puted)	EA20 less FR, prc_hicp_midx	Q	log		masked row 20Q1– Q4
FREV_GDP	General-gov. enue (% GDP, 4q- trailing)	rev- Eurostat	gov_10q_ggnfaQ Q.MIO.EUR.NSA.S13.TR.FR	Q	lin		observed fiscal; (forward- WN filled)
FPEXP_GDP	Primary diture (% GDP, 4q-trailing)	expen- Eurostat	gov_10q_ggnfaQ ...TE – ...D41PAY .FR	Q	lin		observed fiscal; (forward- WN filled)
EFFI	Effective rate on gov. debt	interest Eurostat (com- puted)	...D41PAY / Q gov_10q_ggdebt ...GD.FR	Q	lin		observed fiscal; fi- nancial
SFA	Stock-flow ment identity residual)	adjust- Eurostat (budget- (com- puted)	debt- recursion residual	Q	lin		masked fiscal; 20Q1– WN Q4

continued on next page

Table 64 (France), continued

Variable (id)	Description	Source	Series ID/key	Freq	Transf.	COVID	Notes
CREDIT_GDP_FR	[NEW] BIS total BIS via QFRNAM770A credit to NFCs, % FRED of GDP (broad: loans+bonds+cross-border)			Q	lin	masked 20Q2–Q4	domestic; src 1990Q1:90.8..2025Q4:155.8. Broad BIS measure includes France's large bond-financed NFC debt (the bank-loans-only alternative was <i>not</i> used)
LENDR_FR	[NEW] NFC new- ECB business bank lend- MIR ing rate		M.FR.B.A2A.AMR-4Q2240.FB	Q	avg	masked 20Q2–Q4	domestic; src 2000Q1:4.01..2026Q2:3.56

10.13 Netherlands (NL)

A 23-variable single-country BVAR estimated over **2009Q1–2025Q4** ($T = 68$). COVID-as-missing: macro block masked 2020Q1–Q4, credit quantity masked 2020Q2–Q4, financials observed.

Table 65: Netherlands (NL) — variable documentation.

Variable (id)	Description	Source	Series ID/key	Freq	Transf.	COVID	Notes
GDP_NL	Real GDP (chain-linked vol., SCA)	Eurostat	namq_10_gdp Q.CLV15.MEUR.SCA.B1GQ.N20Q1–	Q	log	masked 20Q1– Q4	domestic
CONS_NL	Real household final consumption	Eurostat	namq_10_gdp ...P31.S14.S15.NL	Q	log	masked 20Q1– Q4	domestic
INV_NL	Real gross fixed capital formation	Eurostat	namq_10_gdp ...P51G.NL	Q	log	masked 20Q1– Q4	domestic
GOV_NL	Real government consumption	Eurostat	namq_10_gdp ...P3.S13.NL	Q	log	masked 20Q1– Q4	domestic
EXP_NL	Real exports	Eurostat	namq_10_gdp ...P6.NL	Q	log	masked 20Q1– Q4	domestic
IMP_NL	Real imports	Eurostat	namq_10_gdp ...P7.NL	Q	log	masked 20Q1– Q4	domestic
HICP_NL	HICP index (all items, 2015=100)	Eurostat	prc_hicp_midxM→Q M.I15.CP00.NLavg	Q	log	masked 20Q1– Q4	domestic
LR_NL	Unemployment rate (SA, % active pop.)	Eurostat	une_rt_q Q.SA.TOTAL.PC.ACT.T.NL	Q	lin	masked 20Q1– Q4	domestic
HPI_NL	Real house-price index (nominal HPI/HICP, 2015=100)	Eurostat	prc_hpi_q Q.TOTAL.I15-Q.NL ÷ HICP	Q	log	observed (financial)	domestic
NEER_NL	Nominal effective exchange rate, IC42 partners	Eurostat	ert_eff_ic_q Q.NEER_IC42.I15.NL	Q	log	observed	domestic
SPREAD_NL	10Y sovereign spread vs Bund	Eurostat (computed)	irt_lt_mcbymM→Q M.MCBY.NL – avg M.MCBY.DE	Q	lin	observed	domestic

continued on next page

Table 65 (Netherlands), continued

Variable (id)	Description	Source	Series ID/key	Freq	Transf.	COVID	Notes
POLRATE_NL	ECB Deposit Facility Rate (policy)	ECB Data Portal	FM/D.U2.EUR.DF.AQ	Q	lin	observed	external (shared)
EACORP	EA corporate bond yield (non-MFI)	Bundesbank	BBSIS/M.I.UMD.MQ.EUR.X2000	Q	lin	observed	external
RATE_NL	3-month EURIBOR	Eurostat	irt_st_q	Q	lin	observed	external (shared)
YIELD10_DE	DE 10Y Bund yield (spread base)	Eurostat	irt_lt_mcbymM→Q M.MCBY.DE	Q	lin	observed	external (injected)
ROWEA_GDP_NL	Rest-of-EA GDP (GDP-weighted de-agg.)	real Eurostat	EA20 less NL, namq_10_gdp	Q	log	masked	row 20Q1–Q4
ROWEA_HICP_NL	Rest-of-EA HICP (GDP-weighted de-agg.)	HICP Eurostat	EA20 less NL, prc_hicp_midx	Q	log	masked	row 20Q1–Q4
FREV_GDP	General-gov. revenue (% GDP, 4q-trailing)	rev- Eurostat	gov_10q_ggnfaQ Q.MIO.EUR.NSA.S13.TR.NL	Q	lin	observed	fiscal; (forward-WN filled)
FPEXP_GDP	Primary expenditure (% GDP, 4q-trailing)	expen- Eurostat	gov_10q_ggnfaQ ...TE – ...D41PAY .NL	Q	lin	observed	fiscal; (forward-WN filled)
EFFI	Effective interest rate on gov. debt	interest Eurostat (com-puted)	gov_10q_ggdebt ...GD.NL	Q	lin	observed	fiscal; financial
SFA	Stock-flow adjustment (budget-identity residual)	adjust- Eurostat (com-puted)	debt-recursion residual	Q	lin	masked	fiscal; WN 20Q1–Q4
CREDIT_GDP_NL	[NEW] BIS credit to NFCs, % of GDP (broad)	total BIS via FRED	QNLNAM770A	Q	lin	masked	domestic; src 20Q2–Q4 1990Q4:152.6..2025Q4:166.3
LENDR_NL	[NEW] NFC business lending rate	new- ECB MIR	M.NL.B.A2A.AM→Q2240.EUR	Q	lin	observed	domestic; src 2000Q1:6.17..2026Q2:3.37

10.14 Euro-area periphery: data conventions (ES, IT, IE, PT, GR)

Transformation and COVID conventions (all five blocs). Two transformation codes appear in the staged specs. `log` denotes $100 \cdot \log(\text{level})$; because the VAR enters in (annualised) growth space, a `log` variable contributes $4 \cdot \Delta[100 \log]$, i.e. the quarter-on-quarter *annualised* growth rate. `lin` denotes the raw level (rates in % p.a., ratios in % of GDP, spreads in percentage points, indices in points). COVID-as-missing masks every *macro* column (`isFinancial` = 0) over the fixed window 2020Q2–Q4 (set to JSON `null`/NaN and imputed inside the Durbin–Koopman E-step); financial columns (`isFinancial` = 1) are kept observed throughout. All five blocs are estimated on a common 2009Q1–2025Q4 quarterly grid ($T = 68$), $p = 3$ lags, with the hierarchical (GLP) Minnesota prior; per-variable own-lag prior means are random walk (RW) except the three fiscal white-noise variables (WN: `FREV_GDP`, `FPEXP_GDP`, `SFA`).

New credit block (flagged). Each periphery bloc gains two new domestic financial variables, appended after the redesign vars by `build_credit_euro_periph.py`:

- **Quantity** `CREDIT_GDP_<CC>` — BIS “Total Credit to Non-Financial Corporations, % of GDP” (the *broad* credit measure: bank loans *plus* debt securities *plus* cross-border lending), pulled from FRED under the mnemonic `Q<CC>NAM770A`. Native quarterly, `lin`, RW, `isFinancial`. Despite being a financial series it is COVID-masked over 2020Q2–Q4 because it *inherits the GDP-denominator collapse* (a ratio whose denominator is a masked macro flow).
- **Price** `LENDR_<CC>` — ECB MIR (MFI Interest Rate statistics) NFC bank lending rate, monthly → quarter by simple month-average, `lin`, RW, `isFinancial`, kept observed through COVID (financial-observed rule).

[FLAG: the research scoping report (*wrqpqko5z*) recommended an ECB BSI MFI “loans-to-NFC” stock (`BSI/M.<CC>.N.A.A20.A.1....`) as the quantity series. The shipped builder instead uses the BIS broad Total-Credit-to-NFCs %-of-GDP series via FRED (`Q<CC>NAM770A`), which is the convention the task brief and the build audit confirm. The staged data are authoritative; the BSI recommendation was superseded.]

[FLAG: GR `LENDR_GR` is a non-canonical MIR variant. The standard composite cost-of-borrowing key `M.GR.B.A2A.A.R.A.2240.EUR.N` is sparse for Greece (only ≈ 54 obs, with ≈ 29 internal gaps over the 2010–2016 crisis years), so the builder substitutes the fully dense new-business floating-rate / IRF-up-to-1yr variant `M.GR.B.A2A.F.R.A.2240.EUR.N`. Its definition differs from the A2A composite used by the other four periphery members — reduced cross-economy comparability of the price-of-credit variable.]

[FLAG: all five periphery `DataRaw_<CC>_glob.json` bases store the COVID-masked macro cells as bare NaN literals (a MATLAB `jsondecode` convention inherited from *euro_redesign_staging*), which fail a strict `JSON.parse`; the appended credit cells correctly use `null`. This is a pre-existing format quirk, not a data error.]

10.15 Spain (ES)

A 23-variable single-country bloc (10 domestic macro/financial + 1 sovereign spread, 4 external exogenous, 2 rest-of-EA, 4 fiscal, 2 credit), estimated on 2009Q1–2025Q4 ($T = 68$, $p = 3$) with COVID-as-missing over 2020Q2–Q4 (macro masked, financials observed). The credit quantity (QESNAM770A) shows a large post-2008 boom-bust cycle (peak $\approx 148\%$ of GDP, falling to $\approx 77\%$ by 2025Q4); the log-level RW absorbs the trend.

Table 66: Spain (ES) — variable documentation. $n = 23$, 2009Q1–2025Q4.

Variable (id)	Description	Source	Series ID / key	Freq	Transf.	COVID	Notes
GDP_ES	Real GDP (chain-vol, SCA)	Eurostat	namq_10_gdp Q.CLV15_MEUR.SCA.B1GQ.ES	Q	log	masked	domestic
CONS_ES	Household NPISH final cons.	+ Eurostat	namq_10_gdp ...P31_S14_S15.ES	Q	log	masked	domestic
INV_ES	Gross fixed capital formation	Eurostat	namq_10_gdp ...P51G.ES	Q	log	masked	domestic
GOV_ES	Government final consumption	Eurostat	namq_10_gdp ...P3_S13.ES	Q	log	masked	domestic
EXP_ES	Exports of goods & services	Eurostat	namq_10_gdp ...P6.ES	Q	log	masked	domestic
IMP_ES	Imports of goods & services	Eurostat	namq_10_gdp ...P7.ES	Q	log	masked	domestic
HICP_ES	HICP, all-items index (2015=100)	Eurostat	prc_hicp_midx M.I15.CP00.ES	M→Q	log	masked	domestic; m→q avg
LR_ES	Unemployment rate (SA, total)	Eurostat	une_rt_q Q.SA.TOTAL.PC.ACT.T.ES	Q	lin	masked	domestic; % of labour force
HPI_ES	Real house-price index (HPI/HICP, 2015=100)	Eurostat	prc_hpi_q Q.TOTAL.I15.Q.ES, deflated by HICP_ES	Q	log	observed	domestic; isFin=1
NEER_ES	Nominal effective FX (IC42 partners, 2015=100)	Eurostat	ert_eff_ic_q Q.NEER.IC42.I15.ES	Q	log	observed	domestic; isFin=1

continued on next page

Table 66 – continued

Variable (id)	Description	Source	Series ID / key	Freq	Transf.	COVID	Notes
SPREAD_ES	10Y sovereign spread vs Bund (pp)	Eurostat	irt_lt_mcbym M.MCBY.ES — M.MCBY.DE	M	Qn	observed	domestic; Maas-tricht 10Y
POLRATE_ES	ECB Deposit Facility Rate (policy)	ECB Data Portal	FM D.U2.EUR.4F.KR.DFR.LEV	D	Qn	observed	external; % p.a.
EACORP	EA corporate bond yield (%)	Bundesbank	BSIS/M.I.UMR.BD→EUR.X2000sBrAdeRtAnl.Z..Z.A	D	Qn	observed	external
RATE_ES	3-month EURIBOR (%)	Eurostat	irt_st_q Q.IRT_M3.EA	Q	lin	observed	external; EA-wide 3M
YIELD10_DE	German 10Y Bund yield (spread base)	Eurostat	irt_lt_mcbym M.MCBY.DE	M	Qn	observed	external
ROWEA_GDP_ES	Rest-of-EA GDP (GDP-wtd ex-ES)	real Eurostat	EA20 namq_10_gdp de-aggregated by ES weight	Q	log	masked	row; index 2015=100
ROWEA_HICP_ES	Rest-of-EA HICP (GDP-wtd ex-ES)	HICP Eurostat	EA20 prc_hicp_midx de-aggregated by ES weight	M	Qg	masked	row; index 2015=100
FREV_GDP	General-gov't revenue (% of GDP, 4q-trailing)	rev- Eurostat	gov_10q_ggnfa Q.MIO_EUR.NSA.S13.TR.ES / CP GDP	Q	lin	observed	fiscal; WN
FPEXP_GDP	Primary expenditure (% of GDP, 4q-trailing)	Eurostat	gov_10q_ggnfa TE-D41PAY, S13.ES / CP GDP	Q	lin	observed	fiscal; WN
EFFI	Effective interest rate on debt (%)	Eurostat	4-D41PAY _t /debt _{t-1} gov_10q_ggnfa/ggdebt.ES	Q	lin	observed	fiscal; isFin=1
SFA	Stock-flow adjustment (% of GDP)	adjust- Eurostat (de-rived)	debt-identity residual from ggdebt.ES	Q	lin	masked	fiscal; WN; COVID-NaN'd

continued on next page

Table 66 – continued

Variable (id)	Description	Source	Series ID / key	Freq	Transf.	COVID	Notes
CREDIT_GDP_ES	BIS Total Credit to BIS NFCs (% of GDP)	via QESNAM770A FRED		Q	lin	masked	domestic; isFin=1; broad credit; GDP- denom mask
LENDR_ES	NFC bank lending rate (ECB composite, %)	ECB MIR Data Portal	MIR M.ES.B.A2A.A.R.A.2240	M→Q	lin	observed	domestic; isFin=1; new business

[FLAG: *EACORP* is shared across euro members and labelled “EA IG Corporate Bond Yield” in the spec, but the underlying ECB series (*YC/B.U2.EUR.4F.G.N.C.SV.C.YM.SR.10Y*) is the euro-area all-bonds/AAA 10Y government yield-curve point, not an investment-grade corporate index. The id name overstates the credit content; the series itself is correctly traced.] Credit-block coverage (ES): CREDIT_GDP_ES 65/68 obs (2009Q1=141.3 %, 2025Q4=76.5 %, range 76.5–148.2), 2020Q2–Q4 null; LENDR_ES 68/68 obs (2009Q1=3.63 %, 2025Q4=3.27 %, COVID observed 1.57–1.70 %).

10.16 Italy (IT)

A 23-variable bloc with the identical structure to Spain (10 domestic + spread, 4 external, 2 rest-of-EA, 4 fiscal, 2 credit), estimated on 2009Q1–2025Q4 ($T = 68$, $p = 3$), COVID-masked 2020Q2–Q4. The Italian sovereign spread and the own MIR lending rate both carry the periphery fragmentation premium that the shared EACORP benchmark misses.

Table 67: Italy (IT) — variable documentation. $n = 23$, 2009Q1–2025Q4.

Variable (id)	Description	Source	Series ID / key	Freq	Transf.	COVID	Notes
GDP_IT	Real GDP (chain- vol, SCA)	Eurostat	namq_10_gdp Q.CLV15_MEUR.SCA.B1GQ.IT	Q	log	masked	domestic
CONS_IT	Household NPISH final cons.	+ Eurostat	namq_10_gdp ... P31.S14.S15.IT	Q	log	masked	domestic
INV_IT	Gross fixed capital formation	Eurostat	namq_10_gdp ... P51G.IT	Q	log	masked	domestic

continued on next page

Table 67 – continued

Variable (id)	Description	Source	Series ID / key	Freq	Transf.	COVID	Notes
GOV_IT	Government consumption	final Eurostat	namq-10_gdp...P3_S13.IT	Q	log	masked	domestic
EXP_IT	Exports of goods & services	Eurostat	namq-10_gdp...P6.IT	Q	log	masked	domestic
IMP_IT	Imports of goods & services	Eurostat	namq-10_gdp...P7.IT	Q	log	masked	domestic
HICP_IT	HICP, all-items index (2015=100)	in- Eurostat	prc_hicp_midx M.I15.CP00.IT	M→Q	log	masked	domestic; m→q avg
LR_IT	Unemployment rate (SA, total)	Eurostat	une_rt_q Q.SA.TOTAL.PC_ACT.T.IT	Q	lin	masked	domestic; % of labour force
HPI_IT	Real house-price index (HPI/HICP, 2015=100)	Eurostat	prc_hpi_q Q.TOTAL.I15.Q.IT, deflated by HICP_IT	Q	log	observed	domestic; isFin=1
NEER_IT	Nominal effective FX (IC42 partners, 2015=100)	Eurostat	ert_eff_ic_q Q.NEER.IC42.I15.IT	Q	log	observed	domestic; isFin=1
SPREAD_IT	10Y sovereign spread vs Bund (pp)	Eurostat	irt_lt_mcbym M.MCBY.IT – M.MCBY.DE	M→Q	lin	observed	domestic; Maas-tricht 10Y
POLRATE_IT	ECB Deposit Facility Rate (policy)	Facil- ECB Data Portal	FM D.U2.EUR.4F.KR.DFR.LEV	D→Q	lin	observed	external; % p.a.
EACORP	EA corporate bond yield (%)	Bundesbank	BBSIS/M.I.UMR.BD→EUR.X2000sBrAderArAl Z..Z.A	Q	lin	observed	external
RATE_IT	3-month EURIBOR (%)	Eurostat	irt_st_q Q.IRT_M3.EA	Q	lin	observed	external; EA-wide 3M
YIELD10_DE	German 10Y Bund yield (spread base)	Eurostat	irt_lt_mcbym M.MCBY.DE	M→Q	lin	observed	external

continued on next page

Table 67 – continued

Variable (id)	Description	Source	Series ID / key	Freq	Transf.	COVID	Notes
ROWEA_GDP_IT	Rest-of-EA GDP (GDP-wtd ex-IT)	real Eurostat	EA20 namq_10_gdp de-aggregated by IT weight	Q	log	masked	row; index 2015=100
ROWEA_HICP_IT	Rest-of-EA (GDP-wtd ex-IT)	HICP Eurostat	EA20 prc_hicp_midx de-aggregated by IT weight	M→Q	log	masked	row; index 2015=100
FREV_GDP	General-gov't enue (% of GDP, 4q-trailing)	rev- Eurostat	gov_10q_ggnfa Q.MIO_EUR.NSA.S13.TR.IT / CP GDP	Q	lin	observed	fiscal; WN
FPEXP_GDP	Primary ture (% of GDP, 4q-trailing)	expendi- Eurostat	gov_10q_ggnfa TE–D41PAY, S13.IT / CP GDP	Q	lin	observed	fiscal; WN
EFFI	Effective rate on debt (%)	interest Eurostat	4·D41PAY _t /debt _{t-1} , gov_10q_ggnfa/ggdebt.IT	Q	lin	observed	fiscal; isFin=1
SFA	Stock-flow ment (% of GDP)	adjust- Eurostat	debt-identity residual from ggdebt.IT	Q	lin	masked	fiscal; WN; COVID- NaN'd
CREDIT_GDP_IT	BIS Total Credit to NFCs (% of GDP)	BIS via FRED	QITNAM770A	Q	lin	masked	domestic; isFin=1; broad credit; GDP- denom mask
LENDR_IT	NFC bank lending rate (ECB MIR Data composite, %)	ECB MIR Data Portal	MIR M.IT.B.A2A.A.R.A.2240.EUR.N	M→Q	lin	observed	domestic; isFin=1; new business

Credit-block coverage (IT): CREDIT_GDP_IT 65/68 obs (2009Q1=78.7 %, 2025Q4=58.5 %, range 58.4–81.6), 2020Q2–Q4 null; LENDR_IT 68/68 obs (2009Q1=3.47 %, 2025Q4=3.55 %, COVID observed 1.18–1.33 %). The EACORP naming caveat (ES table) applies identically.

10.17 Ireland (IE)

A 23-variable bloc, same template, 2009Q1–2025Q4 ($T = 68$, $p = 3$), COVID-masked 2020Q2–Q4. Two Irish-specific data caveats are worth noting: the chain-volume GDP aggregate is distorted by multinational balance-sheet relocation (the “modified” domestic-demand concept is the cleaner activity gauge but is not the Eurostat **B1GQ** used here), and the BIS credit-to-GDP series exhibits a very large boom-bust path.

Table 68: Ireland (IE) — variable documentation. $n = 23$, 2009Q1–2025Q4.

Variable (id)	Description	Source	Series ID / key	Freq	Transf.	COVID	Notes
GDP_IE	Real GDP (chain-vol, SCA)	Eurostat	namq_10_gdp Q.CLV15_MEUR.SCA.B1GQ.IE	Q	log	masked	domestic; MNC- distorted
CONS_IE	Household NPISH final cons.	+ Eurostat	namq_10_gdp ...P31_S14_S15.IE	Q	log	masked	domestic
INV_IE	Gross fixed capital formation	Eurostat	namq_10_gdp ...P51G.IE	Q	log	masked	domestic; MNC- volatile
GOV_IE	Government final consumption	Eurostat	namq_10_gdp ...P3_S13.IE	Q	log	masked	domestic
EXP_IE	Exports of goods & services	Eurostat	namq_10_gdp ...P6.IE	Q	log	masked	domestic; MNC- volatile
IMP_IE	Imports of goods & services	Eurostat	namq_10_gdp ...P7.IE	Q	log	masked	domestic
HICP_IE	HICP, all-items in-dex (2015=100)	Eurostat	prc_hicp_midx M.I15.CP00.IE	M→Q	log	masked	domestic; m→q avg
LR_IE	Unemployment rate (SA, total)	Eurostat	une_rt_q Q.SA.TOTAL.PC_ACT.T.IE	Q	lin	masked	domestic; % of labour force
HPI_IE	Real house-price index (HPI/HICP, 2015=100)	Eurostat	prc_hpi_q Q.TOTAL.I15_Q.IE, deflated by HICP_IE	Q	log	observed	domestic; isFin=1

continued on next page

Variable (id)	Description	Source	Series ID / key	Freq	Transf.	COVID	Notes
NEER_IE	Nominal effective FX (IC42 partners, 2015=100)	Eurostat	ert_eff_ic_q Q.NEER.IC42.I15.IE	Q	log		observed domestic; isFin=1
SPREAD_IE	10Y sovereign spread vs Bund (pp)	Eurostat	irt_lt_mcbym M.MCBY.IE – M.MCBY.DE	M→Q	lin		observed domestic; Maas-tricht 10Y
POLRATE_IE	ECB Deposit Facility Rate (policy)	ECB Data Portal	FM D.U2.EUR.4F.KR.DFR.LEV	D→Q	lin		observed external; % p.a.
EACORP	EA corporate bond yield (%)	Bundesbank	BBSIS/M.I.UMR.BD→EUR.X2000s.BrAcA.RtAnl.Z	Q	lin		observed external
RATE_IE	3-month EURIBOR (%)	Eurostat	irt_st_q Q.IRT_M3.EA	Q	lin		observed external; EA-wide 3M
YIELD10_DE	German 10Y yield (spread base)	Bund Eurostat	irt_lt_mcbym M.MCBY.DE	M→Q	lin		observed external
ROWEA_GDP_IE	Rest-of-EA GDP (GDP-wtd ex-IE)	real Eurostat	EA20 namq_10_gdp de-aggregated by IE weight	Q	log		masked row; index 2015=100
ROWEA_HICP_IE	Rest-of-EA (GDP-wtd ex-IE)	HICP Eurostat	EA20 prc_hicp_midx de-aggregated by IE weight	M→Q	log		masked row; index 2015=100
FREV_GDP	General-gov't enue (% of GDP, 4q-trailing)	rev- Eurostat	gov_10q_ggnfa Q.MIO_EUR.NSA.S13.TR.IE / CP GDP	Q	lin		observed fiscal; WN
FPEXP_GDP	Primary expendi- ture (% of GDP, 4q-trailing)	Eurostat	gov_10q_ggnfa TE–D41PAY, S13.IE / CP GDP	Q	lin		observed fiscal; WN
EFFI	Effective interest rate on debt (%)	Eurostat	4-D41PAY _t /debt _{t-1} gov_10q_ggnfa/ggdebt.IE	Q	lin		observed fiscal; isFin=1

113

Table 68 – continued

Variable (id)	Description	Source	Series ID / key	Freq	Transf.	COVID	Notes
SFA	Stock-flow adjust- ment (% of GDP)	Eurostat (de- rived)	debt-identity residual from ggdebt.IE	Q	lin	masked	fiscal; WN; COVID- NaN'd
CREDIT_GDP_IE	BIS Total Credit to NFCs (% of GDP)	BIS via FRED	QIENAM770A	Q	lin	masked	domestic; isFin=1; broad credit; severe boom- bust
LENDR_IE	NFC bank lending rate (ECB MIR composite, %)	ECB MIR Data Portal	MIR M.IE.B.A2A.A.R.A.2240	M→Q	lin	observed	domestic; isFin=1; new business

[FLAG: CREDIT_GDP_IE has an extreme boom-bust path ($\approx 305\%$ of GDP near 2009/2010 collapsing to $\approx 92\%$ by 2025Q4 — property-bubble deleveraging, NAMA asset transfers, bank wind-downs and multinational reclassification). It is the most volatile credit-quantity series of the periphery group; the BIS broad measure and the log-level RW absorb the trend, but the path is non-stationary in level. The research report recommended considering a post-2014 start or growth-rate space; the shipped build uses the full sample.] Credit-block coverage (IE): CREDIT_GDP_IE 65/68 obs (2009Q1=177.3 %, 2025Q4=92.4 %, range 92.4–305.8), 2020Q2–Q4 null; LENDR_IE 68/68 obs (2009Q1=4.10 %, 2025Q4=4.67 %, COVID observed 2.32–2.81 %). The EACORP naming caveat (ES table) applies.

10.18 Portugal (PT)

A 23-variable bloc, same template, 2009Q1–2025Q4 ($T = 68$, $p = 3$), COVID-masked 2020Q2–Q4. Portuguese bank lending rates carry a clear sovereign-stress / fragmentation premium over the 2009–2012 crisis years (own LENDR_PT peaks near 6.7%), which the shared EACORP benchmark does not capture.

Table 69: Portugal (PT) — variable documentation. $n = 23$, 2009Q1–2025Q4.

Variable (id)	Description	Source	Series ID / key	Freq	Transf.	COVID	Notes
GDP_PT	Real GDP (chain- vol, SCA)	Eurostat	namq_10_gdp Q.CLV15_MEUR.SCA.B1GQ.PT	Q	log	masked	domestic
CONS_PT	Household + Eurostat NPISH final cons.		namq_10_gdp ...P31_S14_S15.PT	Q	log	masked	domestic
INV_PT	Gross fixed capital formation	Eurostat	namq_10_gdp ...P51G.PT	Q	log	masked	domestic
GOV_PT	Government final consumption	Eurostat	namq_10_gdp ...P3_S13.PT	Q	log	masked	domestic
EXP_PT	Exports of goods & services	Eurostat	namq_10_gdp ...P6.PT	Q	log	masked	domestic
IMP_PT	Imports of goods & services	Eurostat	namq_10_gdp ...P7.PT	Q	log	masked	domestic
HICP_PT	HICP, all-items in- dex (2015=100)	Eurostat	prc_hicp_midx M.II15.CP00.PT	M→Q	log	masked	domestic; m→q avg
LR_PT	Unemployment rate (SA, total)	Eurostat	une_rt_q Q.SA.TOTAL.PC.ACT.T.PT	Q	lin	masked	domestic; % of labour force
HPI_PT	Real house-price index (HPI/HICP, 2015=100)	Eurostat	prc_hpi_q Q.TOTAL.II15.Q.PT, deflated by HICP_PT	Q	log	observed	domestic; isFin=1
NEER_PT	Nominal effective FX (IC42 partners, 2015=100)	Eurostat	ert_eff_ic_q Q.NEER.IC42.II15.PT	Q	log	observed	domestic; isFin=1
SPREAD_PT	10Y sovereign spread vs Bund (pp)	Eurostat	irt_lt_mcbym M.MCBY.PT — M.MCBY.DE	M→Q	lin	observed	domestic; Maas- tricht 10Y
POLRATE_PT	ECB Deposit Facil- ity Rate (policy)	ECB Data Portal	FM D.U2.EUR.4F.KR.DFR.LEV	D→Q	lin	observed	external; % p.a.
EACORP	EA corporate bond yield (%)	Bundesbank	BBSIS/M.I.UMR.BD→EUR.X2000s.Br.Adr.Rtr.AA	D→Q	lin	observed	external; Z.Z.A

continued on next page

Table 69 – continued

Variable (id)	Description	Source	Series ID / key	Freq	Transf.	COVID	Notes
RATE_PT	3-month EURIBOR (%)	Eurostat	irt_st_q Q.IRT_M3.EA	Q	lin	observed	external; EA-wide 3M
YIELD10_DE	German 10Y Bund yield (spread base)	Eurostat	irt_lt_mcbym M.MCBY.DE	M→Q	lin	observed	external
ROWEA_GDP_PT	Rest-of-EA GDP (GDP-wtd ex-PT)	real Eurostat	EA20 namq_10_gdp de-aggregated by PT weight	Q	log	masked	row; index 2015=100
ROWEA_HICP_PT	Rest-of-EA HICP (GDP-wtd ex-PT)	HICP Eurostat	EA20 prc_hicp_midx de-aggregated by PT weight	M→Q	log	masked	row; index 2015=100
FREV_GDP	General-gov't revenue (% of GDP, 4q-trailing)	rev- Eurostat	gov_10q_ggnfa Q.MIO_EUR.NSA.S13.TR.PT / CP GDP	Q	lin	observed	fiscal; WN
FPEXP_GDP	Primary expenditure (% of GDP, 4q-trailing)	expendi- Eurostat	gov_10q_ggnfa TE–D41PAY, S13.PT / CP GDP	Q	lin	observed	fiscal; WN
EFFI	Effective interest rate on debt (%)	interest Eurostat	4·D41PAY _t /debt _{t-1} gov_10q_ggnfa/ggdebt.PT	Q	lin	observed	fiscal; isFin=1
SFA	Stock-flow adjustment (% of GDP)	adjust- Eurostat	debt-identity residual from ggdebt.PT	Q	lin	masked	fiscal; WN; COVID- NaN'd
CREDIT_GDP_PT	BIS Total Credit to NFCs (% of GDP)	BIS via FRED	QPTNAM770A	Q	lin	masked	domestic; isFin=1; broad credit; GDP- denom mask

continued on next page

Table 69 – continued

Variable (id)	Description	Source	Series ID / key	Freq	Transf.	COVID	Notes
LENDR_PT	NFC bank lending rate (ECB composite, %)	ECB MIR Data Portal	MIR M.PT.B.A2A.A.R.A.2240	Q	M→Q	observed	domestic; isFin=1; new business

Credit-block coverage (PT): CREDIT.GDP_PT 65/68 obs (2009Q1=121.7 %, 2025Q4=74.2 %, range 74.2–140.9), 2020Q2–Q4 null; LENDR_PT 68/68 obs (2009Q1=5.62 %, 2025Q4=3.67 %, COVID observed 1.82–2.05 %). The EACORP naming caveat (ES table) applies.

10.19 Greece (GR)

A **22-variable** bloc — one fewer than the other four because HPI_GR is dropped (no Eurostat prc_hpi_q house-price index is published for Greece). The structure is otherwise the periphery template (9 domestic + spread, 4 external, 2 rest-of-EA, 4 fiscal, 2 credit), estimated on 2009Q1–2025Q4 ($T = 68$, $p = 3$), COVID-masked 2020Q2–Q4. All Eurostat keys use the Greece geo code EL.

Table 70: Greece (GR) — variable documentation. $n = 22$ (no HPI), 2009Q1–2025Q4.

Variable (id)	Description	Source	Series ID / key	Freq	Transf.	COVID	Notes
GDP_GR	Real GDP (chain-vol, SCA)	Eurostat	namq_10_gdp Q.CLV15_MEUR.SCA.B1G	Q	log	masked	domestic; geo=EL
CONS_GR	Household NPISH final cons.	+ Eurostat	namq_10_gdp ... P31_S14.S15	Q	log	masked	domestic
INV_GR	Gross fixed capital formation	Eurostat	namq_10_gdp ... P51G	Q	log	masked	domestic
GOV_GR	Government final consumption	Eurostat	namq_10_gdp ... P3_S13	Q	log	masked	domestic
EXP_GR	Exports of goods & services	Eurostat	namq_10_gdp ... P6	Q	log	masked	domestic
IMP_GR	Imports of goods & services	Eurostat	namq_10_gdp ... P7	Q	log	masked	domestic
HICP_GR	HICP, all-items dex (2015=100)	in- Eurostat	prc_hicp_midx M.I15.CP00	M→Q	log	masked	domestic; m→q avg

continued on next page

Table 70 – continued

Variable (id)	Description	Source	Series ID / key	Freq	Transf.	COVID	Notes
LR_GR	Unemployment rate (SA, total)	Eurostat	une_rt_q Q.SA.TOTAL.PC_ACT.T.EL	Q	lin	masked	domestic; % of labour force
NEER_GR	Nominal effective FX (IC42 partners, 2015=100)	Eurostat	ert_eff_ic_q Q.NEER.IC42.I15.EL	Q	log	observed	domestic; isFin=1
SPREAD_GR	10Y sovereign spread vs Bund (pp)	Eurostat	irt_lt_mcbym M.MCBY.EL — M.MCBY.DE	M→Q	lin	observed	domestic; Maas- tricht 10Y
POLRATE_GR	ECB Deposit Facil- ity Rate (policy)	ECB Data Portal	FM D.U2.EUR.4F.KR.DFR.LEV	D→Q	lin	observed	external; % p.a.
EACORP	EA corporate bond yield (%)	Bundesbank	BSIS/M.I.UMR.BD→EUR.X2000sBrVedLrtArAl.Z.Z.A	Q	lin	observed	external
RATE_GR	3-month EURIBOR (%)	Eurostat	irt_st_q Q.IRT_M3.EA	Q	lin	observed	external; EA-wide 3M
YIELD10_DE	German 10Y Bund yield (spread base)	Eurostat	irt_lt_mcbym M.MCBY.DE	M→Q	lin	observed	external
ROWEA_GDP_GR	Rest-of-EA real GDP (GDP-wtd (de- ex-GR) rived)	Eurostat	EA20 namq_10_gdp de-aggregated by GR weight	Q	log	masked	row; index 2015=100
ROWEA_HICP_GR	Rest-of-EA HICP (GDP-wtd ex-GR) (de- rived)	Eurostat	EA20 prc_hicp_midx de-aggregated by GR weight	M→Q	log	masked	row; index 2015=100
FREV_GDP	General-gov't rev- enue (% of GDP, 4q-trailing)	Eurostat	gov_10q_ggnfa Q.MIO_EUR.NSA.S13.TR.EL / CP GDP	Q	lin	observed	fiscal; WN

continued on next page

Table 70 – continued

Variable (id)	Description	Source	Series ID / key	Freq	Transf.	COVID	Notes
FPEXP_GDP	Primary expenditure (% of GDP, 4q-trailing)	Eurostat	gov_10q_ggnfa TE–D41PAY, S13.EL / CP GDP	Q	lin	observed	fiscal; WN
EFFI	Effective interest rate on debt (%)	Eurostat	4·D41PAY _t /debt _{t-1} , lin gov_10q_ggnfa/ggdebt.EL	Q	lin	observed	fiscal; isFin=1
SFA	Stock-flow adjustment (% of GDP)	Eurostat (derived)	debt-identity residual from ggdebt.EL	Q	lin	masked	fiscal; WN; COVID-NaN'd
CREDIT_GDP_GR	BIS Total Credit to NFCs (% of GDP)	BIS via FRED	QGRNAM770A	Q	lin	masked	domestic; isFin=1; broad credit; GDP-denom mask
LENDR_GR	NFC bank lending rate (ECB MIR new-business floating, %)	ECB Data Portal	MIR M.GR.B.A2A.F.R.A.2240.EUR.N	Q	lin	observed	domestic; isFin=1; non-canonical MIR variant

[FLAG: LENDR_GR uses the floating-rate / IRF-up-to-1yr new-business MIR key M.GR.B.A2A.F.R.A.2240.EUR.N rather than the A2A.R.A composite cost-of-borrowing used by the other four periphery members, because the composite is sparse for Greece (≈ 54 obs with crisis-era gaps). The floating variant is gap-free on the full grid but is a different instrument — reduced cross-economy comparability.] [FLAG: HPI_GR is absent ($n = 22$, not 23) — Eurostat publishes no house-price index (*prc_hpi_q*) for Greece, so the variable is dropped rather than imputed.] Credit-block coverage (GR): CREDIT_GDP_GR 65/68 obs (2009Q1=63.6 %, 2025Q4=59.3 %, range 52.5–71.6), 2020Q2–Q4 null; LENDR_GR 68/68 obs (2009Q1=4.50 %, 2025Q4=3.66 %, COVID observed 2.86–3.07 %). The EACORP naming caveat (ES table) applies.

10.20 Euro-area small members: data conventions

This section documents the seven smaller euro-area member blocs (Estonia, Finland, Luxembourg, Latvia, Lithuania, Slovakia, Slovenia). Each is a standalone single-country Bayesian VAR ($p =$

3 lags, hierarchical Minnesota/GLP prior) estimated on a common quarterly sample **2009Q1–2025Q4** ($T = 68$), re-estimated by `export_bvar_euro_redesign.m`. The domestic real national accounts, prices, labour, and house prices come from **Eurostat** SDMX-CSV; the financial block (policy rate, the shared euro corporate yield anchor, the 3M rate, and the German 10Y spread base) comes from the **ECB Data Portal** and Eurostat; the four-variable fiscal block is built from **Eurostat** government finance statistics (`gov_10q`); and the rest-of-EA macro pair is a GDP-weighted de-aggregation of the EA20 aggregate.

Two transformation conventions are used throughout. `log` means the variable is stored as $100 \cdot \log(\text{level})$ and therefore enters the model in growth space (displayed as $4\Delta \log =$ quarter-on-quarter annualized growth); `lin` means the level (percent, percentage points, or %-of-GDP). Under the COVID-as-missing treatment, all macro variables (`isFinancial=0`) are masked over **2020Q2–2020Q4** and re-imputed by the Durbin–Koopman simulation smoother, while financial variables (`isFinancial=1`) are kept observed through COVID — with one deliberate exception: the credit *quantity* `CREDIT_GDP_<CC>` is masked 2020Q2–Q4 because it inherits the collapse of its GDP denominator.

The new credit block (flagged). Each bloc gains a two-variable credit block appended after the fiscal block: a *quantity*, `CREDIT_GDP_<CC>` (credit to non-financial corporations, % of GDP; `lin`, prior RW, masked 2020Q2–Q4), and — where data permit — a *price*, `LEND_<CC>` (bank lending rate to NFCs; `lin`, prior RW, observed through COVID). For **FI** and **LU** the quantity is the BIS “Total credit to NFCs, % of GDP” (broad credit: loans, debt securities, and cross-border lending) drawn from FRED. For **EE**, **LV**, **LT**, **SK**, **SI** the BIS %-of-GDP table has no entry, so the quantity is **constructed** (non-BIS) as $100 \times (\text{ECB BSI MFI loans to NFCs}) / (4\text{-quarter rolling sum of Eurostat nominal GDP})$; this is a *narrow* (domestic-MFI-loans-only) measure whose levels are far below the BIS broad measure, reducing cross-country comparability — see the per-economy *[FLAG]* notes. Two further idiosyncrasies are flagged: **LU**’s price uses the ECB MIR *outstanding-amounts* rate (the new-business cost-of-borrowing series is too sparse pre-2015), and **LV**’s price is **dropped** entirely (the ECB MIR NFC rate is too sparse over the 2009Q1 grid); for LV the shared **EACORP** remains the external corporate-yield anchor.

10.21 Estonia (EE)

Estonia is a standalone euro-member BVAR with $n = 23$ variables, $p = 3$ lags, estimated on 2009Q1–2025Q4 ($T = 68$). Macro variables (`isFinancial=0`) are COVID-masked 2020Q2–Q4 and re-imputed by the Durbin–Koopman smoother; financials are kept observed.

Variable	Description	Source	Series ID/key	Freq	Trans.	COVID	Notes
RATE_EE	3M rate (EURIBOR)	Eurostat	irt_st_q Q.IRT.M3.EA	Q	lin	observed (financial)	External; shared 3M EURIBOR
YIELD10_DE	DE 10Y (spread base)	Bund Eurostat	irt_lt_mcbym M.MCBY.DE	M→Q	lin	observed (financial)	External
ROWEA_GDP_EE	Rest-of-EA GDP	real Eurostat	EA20 namq.10_gdp de-aggregated (GDP-weighted)	Q	log	masked 2020Q2–Q4	EA20 ex-EE, GDP-weighted
ROWEA_HICP_EE	Rest-of-EA HICP	Eurostat	EA20 prc_hicp_midx de-aggregated	Q	log	masked 2020Q2–Q4	EA20 ex-EE, GDP-weighted
FREV_GDP	Revenue GDP (%)	of Eurostat	gov_10q_ggnfa Q...TR.EE (4q sum/GDP)	Q	lin	masked 2020Q2–Q4	Prior WN; 4q-trailing/nom. GDP
FPEXP_GDP	Primary expenditure (% of GDP)	Eurostat	gov_10q_ggnfa TE–D41PAY (4q sum/GDP)	Q	lin	masked 2020Q2–Q4	Prior WN
EFFI	Effective interest rate (% p.a.)	Eurostat	gov_10q inter-est/lagged debt	Q	lin	observed (financial)	Prior RW
SFA	Stock-flow adjustment (% of GDP)	Eurostat	gov_10q_ggdebt debt recursion residual	Q	lin	masked 2020Q2–Q4	Prior WN; budget-identity residual
CREDIT_GDP_EE	Credit to NFCs (% of GDP)	ECB BSI + Eurostat (constructed)	100×BSI + M.EE.N.A.A20.A.1.U2.2240.Z01.E2020Q2– ÷ 4q Σnamq.10_gdp CP	Q	lin	masked 2020Q2–Q4	NEW quantity. [FLAG: CONSTRUCTED (non-BIS) NARROW MFI-loan measure; not directly comparable to BIS broad credit (FI/LU); not break-adjusted. obs 2009Q1=45.2..2025Q4=29.2.]
LENDR_EE	ECB MIR composite cost of borrowing, NFCs (new business, % p.a.)	ECB MIR	MIR M.EE.B.A2A.A.R.A.2240.EUR.N	M→Q	lin	observed (financial)	NEW price. 63/68 grid quarters; 5 scattered single-quarter interior gaps filled by the Gibbs. obs 5.90..4.54

10.22 Finland (FI)

Finland is a standalone euro-member BVAR with $n = 23$ variables, $p = 3$ lags, estimated on 2009Q1–2025Q4 ($T = 68$). Macro variables are COVID-masked 2020Q2–Q4 and re-imputed by the Durbin–Koopman smoother; financials are kept observed.

Table 72: Finland (FI) — variable coverage.

Variable	Description	Source	Series ID/key	Freq	Trans.	COVID	Notes
GDP_FI	FI real GDP	Eurostat	namq_10_gdp Q.CLV15.MEUR.SCA.B1GQ.FI	Q	log	masked 2020Q2– Q4	Chain-linked volume, SCA
CONS_FI	Real household consumption	Eurostat	namq_10_gdp ...P31_S14_S15.FI	Q	log	masked 2020Q2– Q4	
INV_FI	Real gross fixed capital form.	Eurostat	namq_10_gdp ...P51G.FI	Q	log	masked 2020Q2– Q4	
GOV_FI	Real government consumption	Eurostat	namq_10_gdp ...P3_S13.FI	Q	log	masked 2020Q2– Q4	
EXP_FI	Real exports	Eurostat	namq_10_gdp ...P6.FI	Q	log	masked 2020Q2– Q4	
IMP_FI	Real imports	Eurostat	namq_10_gdp ...P7.FI	Q	log	masked 2020Q2– Q4	
HICP_FI	HICP index (2015=100)	Eurostat	prc_hicp_midx M.I15.CP00.FI	M→Q	log	masked 2020Q2– Q4	Monthly avg. to quarterly
LR_FI	Unemployment rate (%)	Eurostat	une_rt_q Q.SA.TOTAL.PC.ACT.T.FI	Q	lin	masked 2020Q2– Q4	
HPI_FI	Real house price index (HPI/HICP, 2015=100)	Eurostat	prc_hpi_q Q.TOTAL.I15_Q.FI ÷ HICP	Q	log	observed (financial)	Nominal HPI deflated by HICP
NEER_FI	NEER, IC42 trade-weighted	Eurostat	ert_eff_ic_q Q.NEER.IC42.I15.FI	Q	log	observed (financial)	42-partner nominal eff. FX
SPREAD_FI	10Y sovereign spread vs Bund	Eurostat	irt_lt_mcby_m M.MCBY.FI M.MCBY.DE	M→Q	lin	observed (financial)	Own 10Y minus Bund
POLRATE_FI	ECB Deposit Facility Rate (policy)	ECB	FM D.U2.EUR.4F.KR.DFR.LEV	D→Q	lin	observed (financial)	External; shared by all members
EACORP	EA corporate bond yield (non-MFI)	Bundesbank	BBSIS/M.I.UMR.RD.EUR.Q2000.B.AktivaAdA.Z	D→Q	lin	observed (financial)	External; shared anchor. (Corrected 2026-06-18: genuine corporate yield — Bundesbank non-MFI enterprises, issuer class X2000.)
RATE_FI	3M rate (EURIBOR)	Eurostat	irt_st_q Q.IRT_M3.EA	Q	lin	observed (financial)	External; shared 3M EURIBOR

Variable	Description	Source	Series ID/key	Freq	Trans.	COVID	Notes
YIELD10_DE	DE 10Y Bund (spread base)	Eurostat	irt.lt.mcby.m M.MCBY.DE	M→Q	lin	observed (financial)	External
ROWEA_GDP_FI	Rest-of-EA GDP	real Eurostat	EA20 namq_10_gdp de-aggregated (GDP-weighted)	Q	log	masked 2020Q2– Q4	EA20 ex-FI, GDP-weighted
ROWEA_HICP_FI	Rest-of-EA HICP	Eurostat	EA20 prc_hicp_midx de-aggregated	Q	log	masked 2020Q2– Q4	EA20 ex-FI, GDP-weighted
FREV_GDP	Revenue GDP (%)	of Eurostat	gov_10q_ggnfa Q...TR.FI (4q sum/GDP)	Q	lin	masked 2020Q2– Q4	Prior WN
FPEXP_GDP	Primary expenditure (% of GDP)	Eurostat	gov_10q_ggnfa TE–D41PAY (4q sum/GDP)	Q	lin	masked 2020Q2– Q4	Prior WN
EFFI	Effective interest rate (% p.a.)	Eurostat	gov_10q inter- est/lagged debt	Q	lin	observed (financial)	Prior RW
SFA	Stock-flow adjustment (% of GDP)	Eurostat	gov_10q_ggdebt debt recursion residual	Q	lin	masked 2020Q2– Q4	Prior WN
CREDIT_GDP_FI	Credit to NFCs (% of GDP)	BIS via FRED	QFINAM770A (Total credit to NFCs, % GDP)	Q	lin	masked 2020Q2– Q4	NEW quantity. BIS broad credit (loans+bonds+cross- border). obs 2009Q1=107.6..2025Q4=114.5
LENDR_FI	ECB MIR composite cost of borrowing, NFCs (new business, % p.a.)	ECB MIR	MIR M.FI.B.A2A.A.R.A.2240.EUR.N	M→Q	lin	observed (financial)	NEW price. 67/68 grid (one gap 2019Q3 filled by Gibbs). obs 2.88..3.54

10.23 Luxembourg (LU)

Luxembourg is a standalone euro-member BVAR with $n = 23$ variables, $p = 3$ lags, estimated on 2009Q1–2025Q4 ($T = 68$). Macro variables are COVID-masked 2020Q2–Q4 and re-imputed by the Durbin–Koopman smoother; financials are kept observed.

Table 73: Luxembourg (LU) — variable coverage.

Variable	Description	Source	Series ID/key	Freq	Trans.	COVID	Notes
GDP_LU	LU real GDP	Eurostat	namq_10_gdp Q.CLV15.MEUR.SCA.B1GQ.LU	Q	log	masked 2020Q2– Q4	Chain-linked volume, SCA
CONS_LU	Real household consumption	Eurostat	namq_10_gdp ...P31_S14_S15.LU	Q	log	masked 2020Q2– Q4	

Variable	Description	Source	Series ID/key	Freq	Trans.	COVID	Notes
INV_LU	Real gross fixed capital form.	Eurostat	namq_10_gdp...P51G.LU	Q	log	masked 2020Q2–Q4	
GOV_LU	Real government consumption	Eurostat	namq_10_gdp...P3.S13.LU	Q	log	masked 2020Q2–Q4	
EXP_LU	Real exports	Eurostat	namq_10_gdp...P6.LU	Q	log	masked 2020Q2–Q4	
IMP_LU	Real imports	Eurostat	namq_10_gdp...P7.LU	Q	log	masked 2020Q2–Q4	
HICP_LU	HICP index (2015=100)	Eurostat	prc_hicp_midx M.I15.CP00.LU	M→Q	log	masked 2020Q2–Q4	Monthly avg. to quarterly
LR_LU	Unemployment rate (%)	Eurostat	une_rt_q Q.SA.TOTAL.PC.ACT.T.LU	Q	lin	masked 2020Q2–Q4	
HPI_LU	Real house price index (HPI/HICP, 2015=100)	Eurostat	prc_hpi_q Q.TOTAL.I15-Q.LU ÷ HICP	Q	log	observed (financial)	Nominal HPI deflated by HICP
NEER_LU	NEER, IC42 trade-weighted	Eurostat	ert_eff_ic_q Q.NEER.IC42.I15.LU	Q	log	observed (financial)	42-partner nominal eff. FX
SPREAD_LU	10Y sovereign spread vs Bund	Eurostat	irt_lt_mcby_m M.MCBY.LU – M.MCBY.DE	M→Q	lin	observed (financial)	Own 10Y minus Bund
POLRATE_LU	ECB Deposit Facility Rate (policy)	ECB	FM D.U2.EUR.4F.KR.DFR.LEV	D→Q	lin	observed (financial)	External; shared by all members
EACORP	EA corporate bond yield (non-MFI)	Bundesbank	BBSIS/M.I.UMR.RD.EUR.Q2000.B.AktivaAdA.Z	D→Q	lin	observed (financial)	External; shared anchor. (<i>Corrected 2026-06-18: genuine corporate yield — Bundesbank non-MFI enterprises, issuer class X2000.</i>)
RATE_LU	3M rate (EURIBOR)	Eurostat	irt_st_q Q.IRT.M3.EA	Q	lin	observed (financial)	External; shared 3M EURIBOR
YIELD10_DE	DE 10Y (spread base)	Bund Eurostat	irt_lt_mcby_m M.MCBY.DE	M→Q	lin	observed (financial)	External
ROWEA_GDP_LU	Rest-of-EA real GDP	Eurostat	EA20 namq_10_gdp de-aggregated (GDP-weighted)	Q	log	masked 2020Q2–Q4	EA20 ex-LU, GDP-weighted
ROWEA_HICP_LU	Rest-of-EA HICP	Eurostat	EA20 prc_hicp_midx de-aggregated	Q	log	masked 2020Q2–Q4	EA20 ex-LU, GDP-weighted

Variable	Description	Source	Series ID/key	Freq	Trans.	COVID	Notes
FREV_GDP	Revenue (% of GDP)	Eurostat	gov_10q_ggnfa Q...TR.LU sum/GDP)	Q (4q)	lin	masked 2020Q2– Q4	Prior WN
FPEXP_GDP	Primary expenditure (% of GDP)	Eurostat	gov_10q_ggnfa TE–D41PAY sum/GDP)	Q (4q)	lin	masked 2020Q2– Q4	Prior WN
EFFI	Effective interest rate (% p.a.)	Eurostat	gov_10q inter- est/lagged debt	Q	lin	observed (financial)	Prior RW
SFA	Stock-flow adjustment (% of GDP)	Eurostat	gov_10q_ggdebt debt recursion residual	Q	lin	masked 2020Q2– Q4	Prior WN
CREDIT_GDP_LU	Credit to NFCs (% of GDP)	BIS via FRED	QLUNAM770A (Total credit to NFCs, % GDP)	Q	lin	masked 2020Q2– Q4	NEW quantity. BIS broad credit; very high (~270–360% GDP, captive-finance hub). obs 2009Q1=268.5..2025Q4=358.8
LENDR_LU	ECB MIR bank lending rate to NFCs (outstanding amounts, % p.a.)	ECB MIR	MIR M.LU.B.A20.A.R.A.2240.EUR.O	M→Q lin	observed (financial)	observed (financial)	NEW price. [FLAG: NON-CANONICAL MIR variant — OUTSTANDING-amounts rate (A20...O), not the new-business cost-of-borrowing (A2A...N), which is sparse pre-2015. Full 68/68 coverage.] obs 3.65..3.40

10.24 Latvia (LV)

Latvia is a standalone euro-member BVAR with $n = 22$ variables (one fewer than the group — its credit *price* is dropped, see below), $p = 3$ lags, estimated on 2009Q1–2025Q4 ($T = 68$). Macro variables are COVID-masked 2020Q2–Q4 and re-imputed by the Durbin–Koopman smoother; financials are kept observed.

Table 74: Latvia (LV) — variable coverage.

Variable	Description	Source	Series ID/key	Freq	Trans.	COVID	Notes
GDP_LV	LV real GDP	Eurostat	namq_10.gdp Q.CLV15.MEUR.SCA.B1GQ.LV	Q	log	masked 2020Q2– Q4	Chain-linked volume, SCA
CONS_LV	Real household consumption	Eurostat	namq_10.gdp ...P31_S14_S15.LV	Q	log	masked 2020Q2– Q4	
INV_LV	Real gross fixed capital form.	Eurostat	namq_10.gdp ...P51G.LV	Q	log	masked 2020Q2– Q4	
GOV_LV	Real government consumption	Eurostat	namq_10.gdp ...P3_S13.LV	Q	log	masked 2020Q2– Q4	
EXP_LV	Real exports	Eurostat	namq_10.gdp ...P6.LV	Q	log	masked 2020Q2– Q4	
IMP_LV	Real imports	Eurostat	namq_10.gdp ...P7.LV	Q	log	masked 2020Q2– Q4	
HICP_LV	HICP index (2015=100)	Eurostat	prc_hicp_midx M.I15.CP00.LV	M→Q	log	masked 2020Q2– Q4	Monthly avg. to quarterly
LR_LV	Unemployment rate (%)	Eurostat	une_rt_q Q.SA.TOTAL.PC_ACT.T.LV	Q	lin	masked 2020Q2– Q4	
HPI_LV	Real house price index (HPI/HICP, 2015=100)	Eurostat	prc_hpi_q Q.TOTAL.I15_Q.LV ÷ HICP	Q	log	observed (financial)	Nominal HPI deflated by HICP
NEER_LV	NEER, IC42 trade-weighted	Eurostat	ert_eff_ic_q Q.NEER_IC42.I15.LV	Q	log	observed (financial)	42-partner nominal eff. FX
SPREAD_LV	10Y sovereign spread vs Bund	Eurostat	irt_lt_mcbym M.MCBY.LV M.MCBY.DE	M→Q	lin	observed (financial)	Own 10Y minus Bund
POLRATE_LV	ECB Deposit Facility Rate (policy)	ECB	FM D.U2.EUR.4F.KR.DFR.LEV	D→Q	lin	observed (financial)	External; shared by all members

Variable	Description	Source	Series ID/key	Freq	Trans.	COVID	Notes
EACORP	EA corporate bond yield (non-MFI)	Bundesbank	BBSIS/M.I.UMR.RD.EUR.X2000.B.AlsatvAdA.ZExZ	Q	lin	observed (financial)	External; shared anchor (also the LV credit-price proxy, since LENDR.LV is dropped). <i>(Corrected 2026-06-18: genuine corporate yield — Bundesbank non-MFI enterprises, issuer class X2000.)</i>
RATE_LV	3M rate (EURI-BOR)	Eurostat	irt.st-q Q.IRT.M3.EA	Q	lin	observed (financial)	External; shared 3M EURIBOR
YIELD10_DE	DE 10Y Bund (spread base)	Eurostat	irt.lt_mcby_m M.MCBY.DE	M→Q	lin	observed (financial)	External
ROWEA_GDP_LV	Rest-of-EA real GDP	Eurostat	EA20 namq_10_gdp de-aggregated (GDP-weighted)	Q	log	masked 2020Q2–Q4	EA20 ex-LV, GDP-weighted
ROWEA_HICP_LV	Rest-of-EA HICP	Eurostat	EA20 prc.hicp_midx de-aggregated	Q	log	masked 2020Q2–Q4	EA20 ex-LV, GDP-weighted
FREV_GDP	Revenue (% of GDP)	Eurostat	gov_10q_ggnfa Q...TR.LV (4q sum/GDP)	Q	lin	masked 2020Q2–Q4	Prior WN
FPEXP_GDP	Primary expenditure (% of GDP)	Eurostat	gov_10q_ggnfa TE–D41PAY (4q sum/GDP)	Q	lin	masked 2020Q2–Q4	Prior WN
EFFI	Effective interest rate (% p.a.)	Eurostat	gov_10q inter-est/lagged debt	Q	lin	observed (financial)	Prior RW
SFA	Stock-flow adjustment (% of GDP)	Eurostat	gov_10q_ggdebt debt recursion residual	Q	lin	masked 2020Q2–Q4	Prior WN

Variable	Description	Source	Series ID/key	Freq	Trans.	COVID	Notes
CREDIT.GDP.LV	Credit to NFCs (% of GDP)	ECB BSI Eurostat (constructed)	100×BSI + M.LV.N.A.A20.A.1.U2.2240.Z01.E ÷ 4q Σnamq_10_gdp CP	Q	lin	masked 2020Q2–Q4	NEW quantity. [FLAG: CON- STRUCTED (non-BIS) NAR- ROW MFI-loan measure; re- duced compa- rability; large secular decline (56→16% GDP) is real post-2009 deleveraging, not a denomina- tor artifact. obs 2010Q3=56.0..2025Q4=16.6.]
[FLAG: LV credit PRICE DROPPED. The ECB MIR NFC lending rate has only 41/67 grid quarters (2009–2017 riddled with gaps), so no LENDR.LV variable is included (n = 22, one fewer than the group). EACORP is the external corporate-yield anchor.]							

10.25 Lithuania (LT)

Lithuania is a standalone euro-member BVAR with $n = 23$ variables, $p = 3$ lags, estimated on 2009Q1–2025Q4 ($T = 68$). Macro variables are COVID-masked 2020Q2–Q4 and re-imputed by the Durbin–Koopman smoother; financials are kept observed.

Table 75: Lithuania (LT) — variable coverage.

Variable	Description	Source	Series ID/key	Freq	Trans.	COVID	Notes
GDP.LT	LT real GDP	Eurostat	namq_10_gdp Q.CLV15_MEUR.SCA.B1GQ.LT	Q	log	masked 2020Q2–Q4	Chain-linked volume, SCA
CONS.LT	Real household consumption	Eurostat	namq_10_gdp ...P31_S14_S15.LT	Q	log	masked 2020Q2–Q4	
INV.LT	Real gross fixed capital form.	Eurostat	namq_10_gdp ...P51G.LT	Q	log	masked 2020Q2–Q4	
GOV.LT	Real government consumption	Eurostat	namq_10_gdp ...P3_S13.LT	Q	log	masked 2020Q2–Q4	
EXP.LT	Real exports	Eurostat	namq_10_gdp ...P6.LT	Q	log	masked 2020Q2–Q4	

Variable	Description	Source	Series ID/key	Freq	Trans.	COVID	Notes
IMP_LT	Real imports	Eurostat	namq_10_gdp ...P7.LT	Q	log	masked 2020Q2– Q4	
HICP_LT	HICP index (2015=100)	Eurostat	prc_hicp_midx M.I15.CP00.LT	M→Q	log	masked 2020Q2– Q4	Monthly avg. to quarterly
LR_LT	Unemployment rate (%)	Eurostat	une_rt_q Q.SA.TOTAL.PC.ACT.T.LT	Q	lin	masked 2020Q2– Q4	
HPI_LT	Real house price index (HPI/HICP, 2015=100)	Eurostat	prc_hpi_q Q.TOTAL.I15_Q.LT ÷ HICP	Q	log	observed (financial)	Nominal HPI de- flated by HICP
NEER_LT	NEER, IC42 trade-weighted	Eurostat	ert_eff_ic_q Q.NEER.IC42.I15.LT	Q	log	observed (financial)	42-partner nomi- nal eff. FX
SPREAD_LT	10Y sovereign spread vs Bund	Eurostat	irt_lt_mcby_m M.MCBY.LT M.MCBY.DE	M→Q	lin	observed (financial)	Own 10Y minus Bund
POLRATE_LT	ECB Deposit Facility Rate (policy)	ECB	FM D.U2.EUR.4F.KR.DFR.LEV	D→Q	lin	observed (financial)	External; shared by all members
EACORP	EA corporate bond yield (non-MFI)	Bundesbank	BBSIS/M.I.UMR.RD.EUR.Q2000.B.AktivaAdA.Z	D→Q	lin	observed (financial)	External; shared anchor. (<i>Cor- rected 2026-06- 18: genuine corporate yield — Bundesbank non-MFI enter- prises, issuer class X2000.</i>)
RATE_LT	3M rate (EURIBOR)	Eurostat	irt_st_q Q.IRT.M3.EA	Q	lin	observed (financial)	External; shared 3M EURIBOR
YIELD10_DE	DE 10Y (spread base)	Bund Eurostat	irt_lt_mcby_m M.MCBY.DE	M→Q	lin	observed (financial)	External
ROWEA_GDP_LT	Rest-of-EA real GDP	Eurostat	EA20 namq_10_gdp de-aggregated (GDP-weighted)	Q	log	masked 2020Q2– Q4	EA20 ex-LT, GDP-weighted
ROWEA_HICP_LT	Rest-of-EA HICP	Eurostat	EA20 prc_hicp_midx de-aggregated	Q	log	masked 2020Q2– Q4	EA20 ex-LT, GDP-weighted
FREV_GDP	Revenue (% of GDP)	Eurostat	gov_10q_ggnfa Q...TR.LT (4q sum/GDP)	Q	lin	masked 2020Q2– Q4	Prior WN
FPEXP_GDP	Primary expenditure (% of GDP)	Eurostat	gov_10q_ggnfa TE–D41PAY (4q sum/GDP)	Q	lin	masked 2020Q2– Q4	Prior WN
EFFI	Effective interest rate (% p.a.)	Eurostat	gov_10q inter- est/lagged debt	Q	lin	observed (financial)	Prior RW

Variable	Description	Source	Series ID/key	Freq	Trans.	COVID	Notes
SFA	Stock-flow adjust- ment (% of GDP)	Eurostat	gov_10q_ggdebt debt recursion residual	Q	lin	masked 2020Q2– Q4	Prior WN
CREDIT_GDP_LT	Credit to NFCs (% of GDP)	ECB BSI + Eurostat (con- structed)	100×BSI M.LT.N.A.A20.A.1.U2.2240.Z01.E ÷ 4q Σnamq_10_gdp CP	Q	lin	masked 2020Q2– Q4	NEW quantity. [FLAG: CON- STRUCTED (non-BIS) NAR- ROW MFI-loan measure; reduced comparability; secular decline (32→17% GDP) is real post-2009 deleveraging. obs 2009Q1=32.3..2025Q4=17.1.]
LENDR_LT	ECB MIR compos- ite cost of borrowing, NFCs (new business, % p.a.)	ECB MIR	MIR M.LT.B.A2A.A.R.A.2240.EUR.N	M→Q	lin	observed (financial)	NEW price. Full 68/68 grid. obs 5.36..4.61

10.26 Slovakia (SK)

Slovakia is a standalone euro-member BVAR with $n = 23$ variables, $p = 3$ lags, estimated on 2009Q1–2025Q4 ($T = 68$). Macro variables are COVID-masked 2020Q2–Q4 and re-imputed by the Durbin–Koopman smoother; financials are kept observed.

Table 76: Slovakia (SK) — variable coverage.

Variable	Description	Source	Series ID/key	Freq	Trans.	COVID	Notes
GDP_SK	SK real GDP	Eurostat	namq_10_gdp Q.CLV15_MEUR.SCA.B1GQ.SK	Q	log	masked 2020Q2– Q4	Chain-linked vol- ume, SCA
CONS_SK	Real household con- sumption	Eurostat	namq_10_gdp ...P31_S14_S15.SK	Q	log	masked 2020Q2– Q4	
INV_SK	Real gross fixed cap- ital form.	Eurostat	namq_10_gdp ...P51G.SK	Q	log	masked 2020Q2– Q4	
GOV_SK	Real government consumption	Eurostat	namq_10_gdp ...P3_S13.SK	Q	log	masked 2020Q2– Q4	
EXP_SK	Real exports	Eurostat	namq_10_gdp ...P6.SK	Q	log	masked 2020Q2– Q4	

Variable	Description	Source	Series ID/key	Freq	Trans.	COVID	Notes
IMP_SK	Real imports	Eurostat	namq_10_gdp ...P7.SK	Q	log	masked 2020Q2– Q4	
HICP_SK	HICP index (2015=100)	Eurostat	prc_hicp_midx M.I15.CP00.SK	M→Q	log	masked 2020Q2– Q4	Monthly avg. to quarterly
LR_SK	Unemployment rate (%)	Eurostat	une_rt_q Q.SA.TOTAL.PC.ACT.T.SK	Q	lin	masked 2020Q2– Q4	
HPI_SK	Real house price index (HPI/HICP, 2015=100)	Eurostat	prc_hpi_q Q.TOTAL.I15_Q.SK ÷ HICP	Q	log	observed (financial)	Nominal HPI de- flated by HICP
NEER_SK	NEER, IC42 trade-weighted	Eurostat	ert_eff_ic_q Q.NEER.IC42.I15.SK	Q	log	observed (financial)	42-partner nomi- nal eff. FX
SPREAD_SK	10Y sovereign spread vs Bund	Eurostat	irt_lt_mcpy_m M.MCBY.SK M.MCBY.DE	M→Q	lin	observed (financial)	Own 10Y minus Bund
POLRATE_SK	ECB Deposit Facility Rate (policy)	ECB	FM D.U2.EUR.4F.KR.DFR.LEV	D→Q	lin	observed (financial)	External; shared by all members
EACORP	EA corporate bond yield (non-MFI)	Bundesbank	BBSIS/M.I.UMR.RD.EUR.Q2000.B.AktivaAdA.Z	D→Q	lin	observed (financial)	External; shared anchor. (<i>Cor- rected 2026-06- 18: genuine corporate yield — Bundesbank non-MFI enter- prises, issuer class X2000.</i>)
RATE_SK	3M rate (EURIBOR)	Eurostat	irt_st_q Q.IRT.M3.EA	Q	lin	observed (financial)	External; shared 3M EURIBOR
YIELD10_DE	DE 10Y (spread base)	Bund Eurostat	irt_lt_mcpy_m M.MCBY.DE	M→Q	lin	observed (financial)	External
ROWEA_GDP_SK	Rest-of-EA real GDP	Eurostat	EA20 namq_10_gdp de-aggregated (GDP-weighted)	Q	log	masked 2020Q2– Q4	EA20 ex-SK, GDP-weighted
ROWEA_HICP_SK	Rest-of-EA HICP	Eurostat	EA20 prc_hicp_midx de-aggregated	Q	log	masked 2020Q2– Q4	EA20 ex-SK, GDP-weighted
FREV_GDP	Revenue (% of GDP)	Eurostat	gov_10q_ggnfa Q...TR.SK (4q sum/GDP)	Q	lin	masked 2020Q2– Q4	Prior WN
FPEXP_GDP	Primary expenditure (% of GDP)	Eurostat	gov_10q_ggnfa TE–D41PAY (4q sum/GDP)	Q	lin	masked 2020Q2– Q4	Prior WN
EFFI	Effective interest rate (% p.a.)	Eurostat	gov_10q inter- est/lagged debt	Q	lin	observed (financial)	Prior RW

Variable	Description	Source	Series ID/key	Freq	Trans.	COVID	Notes
SFA	Stock-flow adjust- ment (% of GDP)	Eurostat	gov_10q_ggdebt debt recursion residual	Q	lin	masked 2020Q2– Q4	Prior WN
CREDIT_GDP_SK	Credit to NFCs (% of GDP)	ECB BSI + Eurostat (con- structed)	100×BSI M.SK.N.A.A20.A.1.U2.2240.Z01.E2020Q2– ÷ 4q Σnamq_10_gdp CP	Q	lin	masked 2020Q2– Q4	NEW quantity. [FLAG: CON- STRUCTED (non-BIS) NAR- ROW MFI- loan measure; reduced com- parability. obs 2009Q1=23.6..2025Q4=18.6.]
LENDR_SK	ECB MIR compos- ite cost of borrowing, NFCs (new business, % p.a.)	ECB MIR	MIR M.SK.B.A2A.A.R.A.2240.EUR.N	M→Q	lin	observed (financial)	NEW price. Full 68/68 grid. obs 3.43..4.15

10.27 Slovenia (SI)

Slovenia is a standalone euro-member BVAR with $n = 23$ variables, $p = 3$ lags, estimated on 2009Q1–2025Q4 ($T = 68$). Macro variables are COVID-masked 2020Q2–Q4 and re-imputed by the Durbin–Koopman smoother; financials are kept observed.

Table 77: Slovenia (SI) — variable coverage.

Variable	Description	Source	Series ID/key	Freq	Trans.	COVID	Notes
GDP_SI	SI real GDP	Eurostat	namq_10_gdp Q.CLV15.MEUR.SCA.B1GQ.SI	Q	log	masked 2020Q2– Q4	Chain-linked vol- ume, SCA
CONS_SI	Real household con- sumption	Eurostat	namq_10_gdp ...P31_S14_S15.SI	Q	log	masked 2020Q2– Q4	
INV_SI	Real gross fixed cap- ital form.	Eurostat	namq_10_gdp ...P51G.SI	Q	log	masked 2020Q2– Q4	
GOV_SI	Real government consumption	Eurostat	namq_10_gdp ...P3.S13.SI	Q	log	masked 2020Q2– Q4	
EXP_SI	Real exports	Eurostat	namq_10_gdp ...P6.SI	Q	log	masked 2020Q2– Q4	
IMP_SI	Real imports	Eurostat	namq_10_gdp ...P7.SI	Q	log	masked 2020Q2– Q4	

Variable	Description	Source	Series ID/key	Freq	Trans.	COVID	Notes
HICP_SI	HICP (2015=100)	index Eurostat	prc_hicp_midx M.I15.CP00.SI	M→Q	log	masked 2020Q2– Q4	Monthly avg. to quarterly
LR_SI	Unemployment rate (%)	Eurostat	une_rt_q Q.SA.TOTAL.PC_ACT.T.SI	Q	lin	masked 2020Q2– Q4	
HPI_SI	Real house price index (HPI/HICP, 2015=100)	Eurostat	prc_hpi_q Q.TOTAL.I15_Q.SI ÷ HICP	Q	log	observed (financial)	Nominal HPI de- flated by HICP
NEER_SI	NEER, IC42 trade-weighted	Eurostat	ert_eff_ic_q Q.NEER.IC42.I15.SI	Q	log	observed (financial)	42-partner nomi- nal eff. FX
SPREAD_SI	10Y sovereign spread vs Bund	Eurostat	irt_lt_mcbym M.MCBY.SI M.MCBY.DE	M→Q	lin	observed (financial)	Own 10Y minus Bund
POLRATE_SI	ECB Deposit Facility Rate (policy)	ECB	FM D.U2.EUR.4F.KR.DFR.LEV	D→Q	lin	observed (financial)	External; shared by all members
EACORP	EA corporate bond yield (non-MFI)	Bundesbank	BBSIS/M.I.UMR.RD.EUR.Q2000.B.AnsrAdA.Z	M→Q	lin	observed (financial)	External; shared anchor. <i>(Cor- rected 2026-06- 18: genuine corporate yield — Bundesbank non-MFI enter- prises, issuer class X2000.)</i>
RATE_SI	3M rate (EURIBOR)	Eurostat	irt_st_q Q.IRT.M3.EA	Q	lin	observed (financial)	External; shared 3M EURIBOR
YIELD10_DE	DE 10Y (spread base)	Bund Eurostat	irt_lt_mcbym M.MCBY.DE	M→Q	lin	observed (financial)	External
ROWEA_GDP_SI	Rest-of-EA real GDP	Eurostat	EA20 namq_10_gdp de-aggregated (GDP-weighted)	Q	log	masked 2020Q2– Q4	EA20 ex-SI, GDP-weighted
ROWEA_HICP_SI	Rest-of-EA HICP	Eurostat	EA20 prc_hicp_midx de-aggregated	Q	log	masked 2020Q2– Q4	EA20 ex-SI, GDP-weighted
FREV_GDP	Revenue (% of GDP)	Eurostat	gov_10q_ggnfa Q...TR.SI (4q sum/GDP)	Q	lin	masked 2020Q2– Q4	Prior WN
FPEXP_GDP	Primary expenditure (% of GDP)	Eurostat	gov_10q_ggnfa TE–D41PAY (4q sum/GDP)	Q	lin	masked 2020Q2– Q4	Prior WN
EFFI	Effective interest rate (% p.a.)	Eurostat	gov_10q est/lagged debt	inter- Q	lin	observed (financial)	Prior RW
SFA	Stock-flow adjustment (% of GDP)	Eurostat	gov_10q_ggdebt debt recursion residual	Q	lin	masked 2020Q2– Q4	Prior WN

Variable	Description	Source	Series ID/key	Freq	Trans.	COVID	Notes
CREDIT_GDP_SIC	Credit to NFCs (% of GDP)	ECB BSI Eurostat (con- structed)	100×BSI + M.SI.N.A.A.20.A.1.U2.2240.Z01.E ÷ 4q Σnamq-10_gdp CP	Q	lin	masked 2020Q2– Q4	NEW quantity. [FLAG: CON- STRUCTED (non-BIS) NAR- ROW MFI-loan measure; reduced comparability. The 2013–14 cliff (56→16% GDP) is the BAMC “bad bank” NPL transfer off MFI books, not a denomina- tor artifact. obs 2009Q1=56.2..2025Q4=16.5.]
LENDR_SI	ECB MIR compos- ite cost of borrowing, NFCs (new business, % p.a.)	ECB MIR	MIR M.SI.B.A2A.A.R.A.2240.EUR.N	M→Q	lin	observed (financial)	NEW price. Full 68/68 grid. obs 5.57..3.54

10.28 EU non-euro and hard-gap small blocs: data conventions

This section documents five coupled GVAR member blocs: the three EU *non-euro* economies (Czechia, Hungary, Poland) and two non-EU blocs (Costa Rica, Iceland) whose credit quantity required a *constructed* (non-BIS) measure. Each is a Bayesian VAR with $p = 3$ lags and a hierarchical Minnesota/GLP prior, coupled into the GVAR through trade-weighted foreign *star* variables that are added at solve time. The per-bloc tables below list the domestic block plus the four global commons; the three ROW*-<CC> stars are not part of the staged `spec.json` and are appended by the estimator.

Two transformation conventions are used throughout. `log` means the variable is stored as $100 \cdot \log(\text{level})$ and therefore enters the model in growth space (displayed as $4\Delta \log =$ quarter-on-quarter annualized growth); `lin` means the level (percent, percentage points, or %-of-GDP). Under the COVID-as-missing treatment, all macro variables (`isFinancial=0`) are masked over **2020Q2–2020Q4** and re-imputed by the Durbin–Koopman simulation smoother, while financial variables (`isFinancial=1`) are kept observed through COVID — with one deliberate exception: the credit *quantity* CREDIT_GDP-<CC> is masked 2020Q2–Q4 because it inherits the collapse of its GDP denominator.

The new credit block (flagged). Each bloc gains a two-variable credit block appended after all existing variables: a *quantity*, CREDIT_GDP-<CC> (credit to non-financial corporations as

a % of GDP; `lin`, prior RW, masked 2020Q2–Q4), and a *price*, `LENDR.<CC>` (bank lending rate to NFCs; `lin`, prior RW, observed through COVID). For **CZ**, **HU**, **PL** the quantity is the BIS “Total credit to NFCs, % of GDP” (broad credit: bank loans, debt securities, and cross-border lending) pulled from FRED, mnemonic `Q<IS02>NAM770A`; the price is the ECB MIR new-business NFC bank lending rate in local currency. For **CR**, **IS** neither a BIS %-of-GDP table nor a clean national bank lending rate is published, so the quantity is **constructed** (non-BIS) as $100 \times (\text{IMF IFS ODC claims on the private sector}) / (4\text{-quarter trailing nominal GDP})$ and the price is the IMF IFS `FILR.PA` lending rate. The salient caveats — the broad/narrow-credit comparability gap, the non-canonical MIR variant used for HU and PL, and the constructed-quantity flag for CR and IS — are recorded in the per-economy *[FLAG]* notes below.

10.29 Czechia (CZ)

Czechia is a coupled GVAR bloc with $n = 18$ domestic-plus-global variables (the three foreign `ROW*.CZ` stars are added at GVAR-estimation time), $p = 3$ lags, estimated on the common quarterly sample **2009Q1–2025Q4** ($T = 68$). Macro variables (`isFinancial=0`) are COVID-masked 2020Q2–Q4 and re-imputed by the Durbin–Koopman smoother; financials are kept observed. The domestic real national accounts, prices, and labour come from Eurostat; the policy rate and house price from the BIS; and the new credit block from BIS (quantity, via FRED) and the ECB MIR (price).

Table 78: Czechia (CZ) — variable coverage. `log=` 100 log level (displayed as $4\Delta \log$, q/q ann. growth); `lin=` level.

Variable	Description	Source	Series ID/key	Freq	Trans.	COVID	Notes
<code>GDP.CZ</code>	Real GDP	Eurostat	<code>namq_10_gdp</code> <code>Q.CLV15.MEUR.SCA.B1GQ.CZ</code>	Q	log	masked 2020Q2– Q4	Chain-linked volume, SCA
<code>CONS.CZ</code>	Real household consumption	Eurostat	<code>namq_10_gdp</code> <code>...P31.S14.S15.CZ</code>	Q	log	masked 2020Q2– Q4	
<code>INV.CZ</code>	Real gross fixed cap. form.	Eurostat	<code>namq_10_gdp</code> <code>...P51G.CZ</code>	Q	log	masked 2020Q2– Q4	
<code>GOV.CZ</code>	Real government consumption	Eurostat	<code>namq_10_gdp</code> <code>...P3.S13.CZ</code>	Q	log	masked 2020Q2– Q4	
<code>EXP.CZ</code>	Real exports	Eurostat	<code>namq_10_gdp</code> <code>...P6.CZ</code>	Q	log	masked 2020Q2– Q4	
<code>IMP.CZ</code>	Real imports	Eurostat	<code>namq_10_gdp</code> <code>...P7.CZ</code>	Q	log	masked 2020Q2– Q4	

Variable	Description	Source	Series ID/key	Freq	Trans.	COVID	Notes
HICP_CZ	HICP index (2015=100)	Eurostat	prc_hicp_midx M.I15.CP00.CZ	M→Q	log	masked 2020Q2– Q4	Monthly avg. to quarterly
LR_CZ	Unemployment rate (%)	Eurostat	une_rt_q Q.SA.TOTAL.PC.ACT.T.CZ	Q	lin	masked 2020Q2– Q4	
YIELD10_CZ	10Y government bond yield	Eurostat	irt_lt_mcby_m M.MCBY.CZ	M→Q	lin	observed (financial)	Maastricht 10Y
NEER_CZ	Nominal effective FX (IC42)	Eurostat	ert_eff_ic_q Q.NEER.IC42.I15.CZ	Q	log	observed (financial)	42-partner nominal eff. FX
POLRATE_CZ	Central-bank policy rate	pol- BIS	WS_CBPOL M.CZ	M→Q	lin	observed (financial)	3-month q-avg of monthly EoP
HP_CZ	Real residential property price	BIS (via FRED)	QCZR628BIS	Q	log	observed (financial)	Real RPP, 2010=100
GLB_OIL	WTI crude oil price	FRED	WTISPLC (q-avg)	M→Q	log	observed (financial)	Global common (spot WTI, \$/bbl)
GLB_USLR	US 10Y Treasury yield	FRED	DGS10 [FLAG]	D→Q	lin	observed (financial)	Global common
GLB_USBAA	US Baa corporate yield	FRED	BAA [FLAG]	M→Q	lin	observed (financial)	Global common (credit proxy)
GLB_USEQ	US equity (S&P 500)	FRED	SP500-class [FLAG]	D→Q	log	observed (financial)	Global common
CREDIT_GDP_CZ	Credit to NFCs (% of GDP)	BIS (via FRED)	QCZNAM770A	Q	lin	masked 2020Q2– Q4	NEW. BIS broad NFC credit %GDP; 65 present, 3 masked, range 50.1–61.6%; inherits GDP-denominator mask
LENDR_CZ	NFC bank lending rate (% p.a.)	ECB MIR	M.CZ.B.A2A.F.R.O.2006-01-01	M→Q	lin	observed (financial)	NEW. New-business, ≤EUR1M, floating-rate/IRF≤1y, S.11 NFC, CZK; 68 present, range 2.14–9.24%

[FLAG: CREDIT_GDP_CZ/LENDR_CZ variant.] The price uses the local-currency *floating-rate* ≤EUR1M new-business MIR series (...A2A.F.R.O..., available from 2006-01, the LOCKED-spec pinned key) rather than the cleaner total-maturity composite (...A2A.A.R.A...), which only begins 2010-06 and would leave a leading gap on the 2009Q1 grid. No price was dropped.

10.30 Hungary (HU)

Hungary is a coupled GVAR bloc with $n = 18$ domestic-plus-global variables (three ROW*_HU stars added at estimation), $p = 3$ lags, estimated on **2009Q1–2025Q4** ($T = 68$). Macro variables are COVID-masked 2020Q2–Q4 and re-imputed; financials are kept observed. Sourcing mirrors CZ.

Table 79: Hungary (HU) — variable coverage. log= 100 log level (displayed as 4Δ log, q/q ann. growth); lin= level.

Variable	Description	Source	Series ID/key	Freq	Trans.	COVID	Notes
GDP_HU	Real GDP	Eurostat	namq_10_gdp Q.CLV15.MEUR.SCA.B1GQ.HU	Q	log	masked 2020Q2– Q4	Chain-linked volume, SCA
CONS_HU	Real household consumption	Eurostat	namq_10_gdp ...P31.S14.S15.HU	Q	log	masked 2020Q2– Q4	
INV_HU	Real gross fixed cap. form.	Eurostat	namq_10_gdp ...P51G.HU	Q	log	masked 2020Q2– Q4	
GOV_HU	Real government consumption	Eurostat	namq_10_gdp ...P3.S13.HU	Q	log	masked 2020Q2– Q4	
EXP_HU	Real exports	Eurostat	namq_10_gdp ...P6.HU	Q	log	masked 2020Q2– Q4	
IMP_HU	Real imports	Eurostat	namq_10_gdp ...P7.HU	Q	log	masked 2020Q2– Q4	
HICP_HU	HICP index (2015=100)	Eurostat	prc_hicp_midx M.I15.CP00.HU	M→Q	log	masked 2020Q2– Q4	Monthly avg. to quarterly
LR_HU	Unemployment rate (%)	Eurostat	une_rt_q Q.SA.TOTAL.PC.ACT.T.HU	Q	lin	masked 2020Q2– Q4	
YIELD10_HU	10Y government bond yield	Eurostat	irt_lt_mcby_m M.MCBY.HU	M→Q	lin	observed (financial)	Maastricht 10Y
NEER_HU	Nominal effective FX (IC42)	Eurostat	ert_eff_ic_q Q.NEER.IC42.I15.HU	Q	log	observed (financial)	42-partner nominal eff. FX
POLRATE_HU	Central-bank policy rate	BIS	WS_CBPOL M.HU	M→Q	lin	observed (financial)	3-month q-avg of monthly EoP
HP_HU	Real residential property price	BIS (via FRED)	QHUR628BIS	Q	log	observed (financial)	Real RPP, 2010=100
GLB_OIL	WTI crude oil price	FRED	WTISPLC (q-avg)	M→Q	log	observed (financial)	Global common (spot WTI, \$/bbl)
GLB_USLR	US 10Y Treasury yield	FRED	DGS10 [FLAG]	D→Q	lin	observed (financial)	Global common

Variable	Description	Source	Series ID/key	Freq	Trans.	COVID	Notes
GLB_USBAA	US Baa corporate yield	FRED	BAA <i>[FLAG]</i>	M→Q	lin	observed (financial)	Global common (credit proxy)
GLB_USEQ	US equity (S&P 500)	FRED	SP500-class <i>[FLAG]</i>	D→Q	log	observed (financial)	Global common
CREDIT_GDP_HU	Credit to NFCs (% of GDP)	BIS (via FRED)	QHUNAM770A	Q	lin	masked 2020Q2–Q4	NEW. BIS broad NFC credit %GDP; 65 present, 3 masked, range 64.2–94.9%
LENDR_HU	NFC bank lending rate (% p.a.)	ECB MIR	M.HU.B.A2A.F.R.O.2240.Q	Q	lin	observed (financial)	NEW. New-business floating-rate ≤EUR1M, S.11 NFC, HUF; starts 2010Q1 (4 leading nulls handled by Gibbs); range 2.64–16.37%

[FLAG: LENDR_HU non-canonical MIR variant.] The LOCKED spec gave only “from ~2010” without pinning an exact MIR key. The standard total-maturity variant (...A2A.A.R.A...) is EUR-denominated and starts only 2017-09 (too short); the local-currency floating-rate variant used here starts 2010-01, consistent with the hint and with the CZ-pinned key form, and is a genuine NFC new-business lending rate. Four leading quarters (2009Q1–Q4) are null and are handled by the COVID-as-missing Gibbs ragged-edge machinery. *[FLAG: CREDIT_GDP_HU composition.]* The forint NFC credit level is partly inflated by FX revaluation of FX-denominated corporate loans and by high HU inflation; the %-of-GDP ratio dampens but does not fully remove this nominal component.

10.31 Poland (PL)

Poland is a coupled GVAR bloc with $n = 18$ domestic-plus-global variables (three ROW*_PL stars added at estimation), $p = 3$ lags, estimated on **2009Q1–2025Q4** ($T = 68$). Macro variables are COVID-masked 2020Q2–Q4 and re-imputed; financials are kept observed. Sourcing mirrors CZ.

Table 80: Poland (PL) — variable coverage. log= 100 log level (displayed as 4Δ log, q/q ann. growth); lin= level.

Variable	Description	Source	Series ID/key	Freq	Trans.	COVID	Notes
GDP_PL	Real GDP	Eurostat	namq.10_gdp Q.CLV15.MEUR.SCA.B1GQ	Q	log	masked 2020Q2–Q4	Chain-linked volume, SCA

Variable	Description	Source	Series ID/key	Freq	Trans.	COVID	Notes
CONS_PL	Real household consumption	Eurostat	namq_10_gdp ...P31_S14_S15.PL	Q	log	masked 2020Q2– Q4	
INV_PL	Real gross fixed cap. form.	Eurostat	namq_10_gdp ...P51G.PL	Q	log	masked 2020Q2– Q4	
GOV_PL	Real government consumption	Eurostat	namq_10_gdp ...P3_S13.PL	Q	log	masked 2020Q2– Q4	
EXP_PL	Real exports	Eurostat	namq_10_gdp ...P6.PL	Q	log	masked 2020Q2– Q4	
IMP_PL	Real imports	Eurostat	namq_10_gdp ...P7.PL	Q	log	masked 2020Q2– Q4	
HICP_PL	HICP index (2015=100)	Eurostat	prc_hicp_midx M.I15.CP00.PL	M→Q	log	masked 2020Q2– Q4	Monthly avg. to quarterly
LR_PL	Unemployment rate (%)	Eurostat	une_rt_q Q.SA.TOTAL.PC.ACT.T.PL	Q	lin	masked 2020Q2– Q4	
YIELD10_PL	10Y government bond yield	Eurostat	irt_lt_mcbym M.MCBY.PL	M→Q	lin	observed (financial)	Maastricht 10Y
NEER_PL	Nominal effective FX (IC42)	Eurostat	ert_eff_ic_q Q.NEER.IC42.I15.PL	Q	log	observed (financial)	42-partner nominal eff. FX
POLRATE_PL	Central-bank policy rate	BIS	WS-CBPOL M.PL	M→Q	lin	observed (financial)	3-month q-avg of monthly EoP
HP_PL	Real residential property price	BIS (via FRED)	QPLR628BIS	Q	log	observed (financial)	Real RPP, 2010=100
GLB_OIL	WTI crude oil price	FRED	WTISPLC (q-avg)	M→Q	log	observed (financial)	Global common (spot WTI, \$/bbl)
GLB_USLR	US 10Y Treasury yield	FRED	DGS10 [FLAG]	D→Q	lin	observed (financial)	Global common
GLB_USBAA	US Baa corporate yield	FRED	BAA [FLAG]	M→Q	lin	observed (financial)	Global common (credit proxy)
GLB_USEQ	US equity (S&P 500)	FRED	SP500-class [FLAG]	D→Q	log	observed (financial)	Global common
CREDIT_GDP_PL	Credit to NFCs (% of GDP)	BIS (via FRED)	QPLNAM770A	Q	lin	masked 2020Q2–Q4	NEW. BIS broad NFC credit %GDP; 65 present, 3 masked, range 34.7–49.5%

Variable	Description	Source	Series ID/key	Freq	Trans.	COVID	Notes
LENDR_PL	NFC bank lending rate (% p.a.)	ECB MIR	M.PL.B.A2A.F.R.O.2240.QLN	Q	log	observed (financial)	NEW. New-business floating-rate $\leq$ EUR1M, S.11 NFC, PLN; 68 present, range 1.82–8.62%

[*FLAG: LENDR_PL non-canonical MIR variant.*] The clean total-maturity composite (`...A2A.A.R.A...`) begins only 2017-08 — far too short for the 2009Q1 grid — so the local-currency floating-rate $\leq$ EUR1M new-business series (full coverage from 2005, gap-free, and the dominant WIBOR-linked PL corporate-loan type) is used instead. Overlap checks confirm it tracks the composite within ~ 30 bp; no splice and no back-fill needed.

10.32 Costa Rica (CR)

Costa Rica is a coupled GVAR bloc with $n = 13$ domestic-plus-global variables (three `ROW*_CR` stars added at estimation), $p = 3$ lags, estimated on **2010Q3–2025Q4** ($T = 62$). It is a compact emerging-market specification: real activity is proxied by industrial production, external trade is in USD, and there is no separate 10Y yield or house-price variable. Macro variables are COVID-masked 2020Q2–Q4 and re-imputed; financials (including the policy rate and the new credit price) are kept observed.

Table 81: Costa Rica (CR) — variable coverage. `log`= 100 log level (displayed as 4Δ log, q/q ann. growth); `lin`= level.

Variable	Description	Source	Series ID/key	Freq	Trans.	COVID	Notes
GDP_CR	Real GDP (IP proxy)	[FLAG]	[FLAG: not determinable from current builders]	Q	log	masked 2020Q2–Q4	Activity proxied by industrial production
EXP_CR	Goods exports (USD)	[FLAG]	[FLAG: not determinable from current builders]	Q	log	masked 2020Q2–Q4	USD goods exports
IMP_CR	Goods imports (USD)	[FLAG]	[FLAG: not determinable from current builders]	Q	log	masked 2020Q2–Q4	USD goods imports
CPI_CR	CPI index	[FLAG]	[FLAG: not determinable from current builders]	Q	log	masked 2020Q2–Q4	
LR_CR	Unemployment rate (%)	[FLAG]	[FLAG: not determinable from current builders]	Q	lin	masked 2020Q2–Q4	

Variable	Description	Source	Series ID/key	Freq	Trans.	COVID	Notes
RATE_CR	Monetary policy rate	[FLAG]	[FLAG: not determinable from current builders]	Q	lin	observed (financial)	National monetary policy rate (BCCR); POLRATE/HP not added (CR not in BIS CBPOL/RPP)
NEER_CR	Nominal effective FX	[FLAG]	[FLAG: not determinable from current builders]	Q	log	observed (financial)	
GLB_OIL	WTI crude oil price	FRED	WTISPLC (q-avg)	M→Q	log	observed (financial)	Global common (spot WTI, \$/bbl)
GLB_USLR	US 10Y Treasury yield	FRED	DGS10 [FLAG]	D→Q	lin	observed (financial)	Global common
GLB_USBAA	US Baa corporate yield	FRED	BAA [FLAG]	M→Q	lin	observed (financial)	Global common (credit proxy)
GLB_USEQ	US equity (S&P 500)	FRED	SP500-class [FLAG]	D→Q	log	observed (financial)	Global common
CREDIT_GDP_CR	Credit to NFCs (% of GDP)	IMF IFS (CON-STRUCTED)	100×FOSAOP_XDC/4Q-Q trailing NGDP_SA_XDC	Q	lin	masked 2020Q2–Q4	NEW. 56 present, range 42.8–59.4%; last 2025Q1; broad private-sector credit used as NFC proxy
LENDR_CR	Bank lending rate (% p.a.)	IMF IFS	FILR_PA (DBnomics mirror)	Q	lin	observed (financial)	NEW. IMF-compiled national lending rate (source BCCR); 60 present, range 5.0–18.8%; last 2025Q2

[FLAG: CREDIT_GDP_CR constructed / non-BIS.] Costa Rica is not a BIS total-credit reporter (FRED QCRNAM770A and BIS WS.TC both return “no series”), so the quantity is constructed as $100 \times$ (IMF IFS ODC claims on the private sector, FOSAOP_XDC, CRC) over a 4-quarter trailing sum of nominal GDP (NGDP_SA_XDC). This is *broad private-sector* credit (households + NFCs), used as the NFC proxy — the same validated construction as Colombia — so its *level* is not directly comparable to the BIS broad-NFC %GDP series used for CZ/HU/PL. The price LENDR_CR is the IMF IFS FILR_PA (IMF-compiled national lending rate sourced from the BCCR), used in lieu of a direct BCCR scrape; both new series have ragged trailing edges (credit to 2025Q1, price to 2025Q2) handled by the COVID-as-missing Gibbs. [FLAG: CR base macro provenance.] The exact source and series

identifiers for the *base* CR macro/financial variables (GDP_CR IP proxy, EXP_CR/IMP_CR USD trade, CPI_CR, LR_CR, RATE_CR, NEER_CR) could not be determined from the builders present in the current pipeline tree (these blocs were assembled by an earlier exporter/IMF flow not retained here); they are most plausibly IMF (IFS/IMTS) and BCCR series but the precise keys are not verifiable from the available scripts and are *not* fabricated here.

10.33 Iceland (IS)

Iceland is a coupled GVAR bloc with $n = 18$ domestic-plus-global variables (three ROW*_IS stars added at estimation), $p = 3$ lags, estimated on the long sample **2000Q2–2025Q4** ($T = 103$). It carries a full real national-accounts block, a 3-month interbank rate, the BIS policy rate and real house price, and the new credit block. Macro variables are COVID-masked 2020Q2–Q4 and re-imputed; financials are kept observed.

Table 82: Iceland (IS) — variable coverage. **log**= 100log level (displayed as 4Δ log, q/q ann. growth); **lin**= level.

Variable	Description	Source	Series ID/key	Freq	Trans.COVID	Notes
GDP_IS	Real GDP	[FLAG]	[FLAG: not deter- minable from cur- rent builders]	Q	log	masked 2020Q2– Q4
CONS_IS	Real private con- sumption	[FLAG]	[FLAG: not deter- minable from cur- rent builders]	Q	log	masked 2020Q2– Q4
INV_IS	Real investment	[FLAG]	[FLAG: not deter- minable from cur- rent builders]	Q	log	masked 2020Q2– Q4
GOV_IS	Real government consumption	[FLAG]	[FLAG: not deter- minable from cur- rent builders]	Q	log	masked 2020Q2– Q4
EXP_IS	Real exports	[FLAG]	[FLAG: not deter- minable from cur- rent builders]	Q	log	masked 2020Q2– Q4
IMP_IS	Real imports	[FLAG]	[FLAG: not deter- minable from cur- rent builders]	Q	log	masked 2020Q2– Q4
CPI_IS	CPI index	[FLAG]	[FLAG: not deter- minable from cur- rent builders]	Q	log	masked 2020Q2– Q4
LR_IS	Unemployment rate (%)	[FLAG]	[FLAG: not deter- minable from cur- rent builders]	Q	lin	masked 2020Q2– Q4
RATE_IS	3-month interbank rate	[FLAG]	[FLAG: not deter- minable from cur- rent builders]	Q	lin	observed (financial) 3M interbank (likely FRED IR3TIB01ISQ156N; unverified here)

Variable	Description	Source	Series ID/key	Freq	Trans.	COVID	Notes
NEER_IS	Nominal effective FX	[FLAG]	[FLAG: not determinable from current builders]	Q	log	observed (financial)	
POLRATE_IS	Central-bank policy rate	pol- BIS	WS-CBPOL M.IS	M→Q	lin	observed (financial)	3-month q-avg of monthly EoP
HP_IS	Real residential property price	BIS (via FRED)	QISR628BIS	Q	log	observed (financial)	Real RPP, 2010=100
GLB_OIL	WTI crude oil price	FRED	WTISPLC (q-avg)	M→Q	log	observed (financial)	Global common (spot WTI, \$/bbl)
GLB_USLR	US 10Y Treasury yield	FRED	DGS10 [FLAG]	D→Q	lin	observed (financial)	Global common
GLB_USBAA	US Baa corporate yield	FRED	BAA [FLAG]	M→Q	lin	observed (financial)	Global common (credit proxy)
GLB_USEQ	US equity (S&P 500)	FRED	SP500-class [FLAG]	D→Q	log	observed (financial)	Global common
CREDIT_GDP_IS	Credit to NFCs (% of GDP)	IMF IFS (CON-STRUCTED)	100×FOSAOP_XDC/4Q-Q trailing NGDP_SA_XDC	Q	lin	masked 2020Q2–Q4	NEW. 91 present, range 83.0–317.7% (318% = 2008 banking-crisis peak); last 2025Q1
LENDR_IS	Bank lending rate (% p.a.)	IMF IFS	FILR_PA (DBnomics mirror)	Q	lin	observed (financial)	NEW. IMF-compiled national lending rate (source CBI); 100 present, range 5.2–23.0%; last 2025Q2

[FLAG: *CREDIT_GDP_IS* constructed / non-BIS.] Iceland is not a BIS total-credit reporter, so the quantity is constructed identically to Costa Rica: $100 \times$ (IMF IFS ODC claims on the private sector, FOSAOP_XDC, ISK) over a 4-quarter trailing sum of nominal GDP (NGDP_SA_XDC). This is *broad private-sector* credit (HH + NFC) used as the NFC proxy — its level (and the historically accurate ~318% 2008-crisis peak) is not directly comparable to the BIS broad-NFC %GDP series for CZ/HU/PL. The price LENDR_IS is the IMF IFS FILR_PA (IMF-compiled national lending rate sourced from the Central Bank of Iceland); both new series have ragged trailing edges handled by the Gibbs. [FLAG: *IS base macro provenance.*] The exact source and series identifiers for the *base* IS macro/financial variables (real national accounts, CPI_IS, LR_IS, RATE_IS 3M interbank, NEER_IS) could not be determined from the builders present in the current pipeline tree; RATE_IS is plausibly the FRED OECD 3M interbank series IR3TIB01ISQ156N and the national accounts are plausibly OECD QNA / Statistics Iceland, but the precise keys are not verifiable from the available scripts

and are *not* fabricated here. Only POLRATE_IS and HP_IS (added by `build_polrate_hp_group.py`) and the two new credit variables have fully traceable sources.

10.34 Advanced-Economy Blocs: Per-Variable Data Provenance

This subsection documents, variable by variable, the eight advanced-economy GVAR blocs that carry the new credit block: Australia, Switzerland, Denmark, Norway, Sweden, New Zealand, Korea, and Israel. Each is a coupled GVAR bloc: the staged `spec.json` lists the domestic and global-commons variables; the estimator additionally appends the three trade-weighted ROW*-<CC> star variables (Section 9.1) at re-estimation, so the deployed n is the staged n plus three.

Conventions used in every table. The *Transformation* column follows the manual-wide convention: `log` denotes $100 \cdot \log(\text{level})$, which enters the VAR as $4 \Delta \log$ (quarter-on-quarter annualized growth), and `lin` denotes the raw level (percent per annum for rates and ratios; an index for NEER and house prices). The *COVID* column records the COVID-as-missing treatment of Section 9.4: macro variables (those with `isFinancial` = 0) are masked over 2020Q2–2020Q4 and imputed by the data-augmentation Gibbs sampler, whereas financial variables are held observed through the pandemic. The four global commons (GLB_OIL, GLB_USLR, GLB_USBAA, GLB_USEQ) are US-sourced series shared across all blocs (Section 6.6): WTI spot oil (FRED WTISPLC, quarter-average), the US 10Y Treasury yield (FRED DGS10), the US Baa corporate yield (FRED BAA), and the US equity price index (FRED SPASTT01USM661N, OECD US share-price/S&P proxy). They are kept observed through COVID. Where the exact upstream series identifier for a domestic *base* macro/financial variable (real-activity aggregates, CPI/HICP, unemployment, NEER, short rate, long yield, policy rate, house price) cannot be recovered from a machine-readable builder in the pipeline, the source is attributed at the provider level documented in Section 9.3 (OECD Quarterly National Accounts, the IMF SDMX flows, Eurostat, BIS, and national central banks) and the per-series id is marked [*FLAG: series id not recoverable from a readable builder*] rather than fabricated. The new **credit-block** variables (CREDIT_GDP-<CC> and LENDR-<CC>/CORP-<CC>) carry fully traced, live-probed identifiers from the dedicated fetchers (`fetch_rba_f7.py`, `fetch_snb_lendr.py`, `fetch_dk_lendr.py`, `fetch_ssb_jsonstat.py`, `fetch_scb_pxweb.py`, `fetch_rbnz_b3.py`, `fetch_credit_kr.py`, `fetch_credit_il.py`, `fetch_bis_credit_gdp.py`) and are flagged prominently below.

The new credit block (all eight economies). Each bloc gains two appended variables, both `lin` with a random-walk (RW) prior and `isFinancial` = 1 in the `domestic` block:

- **Quantity**, CREDIT_GDP-<CC> – BIS “Total credit to non-financial corporations, % of GDP” (broad credit: bank loans + debt securities + cross-border lending, adjusted for breaks), pulled via FRED under the uniform mnemonic Q<IS02>NAM770A (`fetch_bis_credit_gdp.py`). It is *COVID-masked* 2020Q2–Q4 (it inherits the GDP-denominator mask: treated as a macro-like quantity even though tagged financial), the three quarters set to JSON `null` and imputed by the Gibbs.

- **Price**, LENDR_<CC> (a bank/corporate lending rate) or CORP_<CC> (a corporate-bond yield) – *observed through COVID* (not masked), % p.a.

None of these eight economies has a dropped price or a constructed (non-BIS) quantity; the quantity is the canonical BIS broad-credit series for all eight. Economy-specific caveats (short price history, leading ragged edges, a price-source substitution, a transport workaround, and one reconstructed corporate yield) are flagged in the relevant tables.

10.35 Australia (AU)

The Australian bloc has $n = 18$ staged variables (deployed $n = 21$ with the three ROW*_AU stars), estimated on the quarterly window 1999Q1–2025Q4. The eight macro variables (real GDP, consumption, government consumption, investment, exports, imports, CPI, unemployment) are masked over 2020Q2–Q4 and imputed; all financial variables are kept observed.

Table 83: Australia (AU): per-variable data provenance. Trans. key: **log** = 100 log level (enters as $4\Delta \log$, q/q-annualized growth); **lin** = level.

Variable	Description	Source	Series id/key	Freq	Trans.	COVID	Notes
GDP_AU	Real GDP	OECD QNA	[FLAG]	Q	log	masked	macro
CONS_AU	Real household consumption	OECD QNA	[FLAG]	Q	log	masked	macro
GOV_AU	Real govt consumption	OECD QNA	[FLAG]	Q	log	masked	macro
INV_AU	Real investment (GFCF)	OECD QNA	[FLAG]	Q	log	masked	macro
EXP_AU	Real exports	OECD QNA	[FLAG]	Q	log	masked	macro
IMP_AU	Real imports	OECD QNA	[FLAG]	Q	log	masked	macro
CPI_AU	CPI, all items (index)	IMF SDMX	[FLAG]	Q	log	masked	macro
LR_AU	Unemployment rate	IMF SDMX	[FLAG]	Q	lin	masked	macro
NEER_AU	NEER (index)	IMF SDMX	[FLAG]	Q	log	observed	financial
RATE_AU	Cash rate (interbank o/n)	RBA / IMF	[FLAG]	Q	lin	observed	policy short rate
YIELD10_AU	10Y govt bond yield	RBA / IMF	[FLAG]	Q	lin	observed	financial

Australia (AU), continued

Variable	Description	Source	Series id/key	Freq	Trans.	COVID	Notes
HP_AU	Real residential house-price index (2010=100)	BIS RPP	FRED QAUR628BIS (BIS real RPP)	Q	log	observed	financial; [FLAG: <i>id</i> by convention]
GLB_OIL	WTI crude, q-avg (\$/bbl)	FRED	WTISPLC	Q	log	observed	global common
GLB_USLR	US 10Y Treasury yield	FRED	DGS10	Q	lin	observed	global common
GLB_USBAA	US Baa corporate yield	FRED	BAA	Q	lin	observed	global common
GLB_USEQ	US equity (S&P 500)	FRED	SPASTT01USM661N	Q	log	observed	global common
CREDIT_GDP_AU	Credit to NFCs (% of GDP)	BIS (via FRED)	QAUNAM770A	Q	lin	masked	NEW broad credit; 105 obs, 1999Q1=63.5→2025Q4=62 (58.8–80.6)
LENDR_AU	Business lending rate (RBA F7, F7 outstanding large-business total, % p.a.)	RBA Table	FLRBFOLBT (fetch_rba_f7.py)	M→Q-avg	lin	observed	NEW price; [FLAG: <i>short history — starts 2019Q3, 26 obs; 1999Q1–2019Q2 leading null, Gibbs-imputed</i>]

10.36 Switzerland (CH)

The Swiss bloc has $n = 17$ staged variables (deployed $n = 20$), estimated on 2009Q2–2025Q4. Switzerland is one of the rate-optional blocs: it carries a policy rate (POLRATE_CH) but no separate domestic short rate. Seven macro variables are COVID-masked.

Table 84: Switzerland (CH): per-variable data provenance.

Variable	Description	Source	Series id/key	Freq	Trans.	COVID	Notes
GDP_CH	Real GDP	OECD QNA	[FLAG]	Q	log	masked	macro
CONS_CH	Real consumption	OECD QNA	[FLAG]	Q	log	masked	macro
INV_CH	Real investment	OECD QNA	[FLAG]	Q	log	masked	macro

Switzerland (CH), continued

Variable	Description	Source	Series id/key	Freq	Trans.	COVID	Notes
GOV_CH	Real govt con- sumption	OECD QNA	[FLAG]	Q	log	masked	macro
EXP_CH	Real exports	OECD QNA	[FLAG]	Q	log	masked	macro
IMP_CH	Real imports	OECD QNA	[FLAG]	Q	log	masked	macro
HICP_CH	HICP (index)	IMF SDMX	[FLAG]	Q	log	masked	macro
LR_CH	Unemployment rate	IMF SDMX	[FLAG]	Q	lin	masked	macro
NEER_CH	NEER (index)	IMF SDMX	[FLAG]	Q	log	observed	financial
POLRATE_CH	Policy rate (BIS CBPOL)	BIS	BIS WS.CBPOL M.CH	M→Q-avg	lin	observed	policy rate; [FLAG: id by convention]
HP_CH	Real residential property price	BIS RPP	FRED QCHR628BIS	Q	log	observed	financial; [FLAG: id by convention]
GLB_OIL	WTI crude, q-avg (\$/bbl)	FRED	WTISPLC	Q	log	observed	global common
GLB_USLR	US 10Y Treasury yield	FRED	DGS10	Q	lin	observed	global common
GLB_USBAA	US Baa corporate yield	FRED	BAA	Q	lin	observed	global common
GLB_USEQ	US equity (S&P 500)	FRED	SPASTT01USM661N	Q	log	observed	global common
CREDIT_GDP_CH	Credit to NFCs (% of GDP)	BIS (via FRED)	QCHNAM770A	Q	lin	masked	NEW broad credit; 64 obs, 2009Q2=99.6→2025Q4=12 (99.6–149.4)
LENDR_CH	Corporate lending rate (SNB new portal investment loans, fixed rate, % p.a.)	SNB data cube	zikreddet item FI (fetch_snb_lendr.py)	M→Q-avg	lin	observed	NEW price; 67 obs, 2009Q2=1.80→2025Q4=1. (1.46–3.20); clean

10.37 Denmark (DK)

The Danish bloc has $n = 19$ staged variables (deployed $n = 22$), estimated on 2009Q1–2025Q4. Seven macro variables are COVID-masked. Denmark carries both a 3-month rate and a policy rate.

Table 85: Denmark (DK): per-variable data provenance.

Variable	Description	Source	Series id/key	Freq	Trans.	COVID	Notes
GDP_DK	Real GDP	Eurostat OECD	/ [FLAG]	Q	log	masked	macro
CONS_DK	Real consumption	Eurostat OECD	/ [FLAG]	Q	log	masked	macro
INV_DK	Real investment	Eurostat OECD	/ [FLAG]	Q	log	masked	macro
GOV_DK	Real govt con- sumption	Eurostat OECD	/ [FLAG]	Q	log	masked	macro
EXP_DK	Real exports	Eurostat OECD	/ [FLAG]	Q	log	masked	macro
IMP_DK	Real imports	Eurostat OECD	/ [FLAG]	Q	log	masked	macro
HICP_DK	HICP (index)	Eurostat	[FLAG]	Q	log	masked	macro
LR_DK	Unemployment rate	Eurostat	[FLAG]	Q	lin	masked	macro
RATE_DK	3-month rate	IMF / Euro- stat	[FLAG]	Q	lin	observed	short rate
YIELD10_DK	10Y govt bond yield	IMF / Euro- stat	[FLAG]	Q	lin	observed	financial
NEER_DK	NEER (index)	IMF SDMX	[FLAG]	Q	log	observed	financial
POLRATE_DK	Policy rate (BIS CBPOL)	BIS	BIS WS_CBPOL M.DK	M→Q-avg	lin	observed	policy rate; [FLAG: id by convention]
HP_DK	Real residential property price	BIS RPP	FRED QDKR628BIS	Q	log	observed	financial; [FLAG: id by convention]
GLB_OIL	WTI crude, q-avg (\$/bbl)	FRED	WTISPLC	Q	log	observed	global common
GLB_USLR	US 10Y Treasury yield	FRED	DGS10	Q	lin	observed	global common
GLB_USBAA	US Baa corporate yield	FRED	BAA	Q	lin	observed	global common
GLB_USEQ	US equity (S&P 500)	FRED	SPASTT01USM661N	Q	log	observed	global common
CREDIT_GDP_DK	Credit to NFCs (% of GDP)	BIS (via FRED)	QDKNAM770A	Q	lin	masked	NEW broad credit; 65 obs, 2009Q1=134.5→2025Q4=1 (127.4–151.6)

Denmark (DK), continued

Variable	Description	Source	Series id/key	Freq	Trans.	COVID	Notes
LENDR_DK	Corporate lending rate (ECB MIR Portal loans to NFCs, (MIR) new business, % p.a.)	ECB Data M.DK.B.A2A.AMR-Q2249.DKK.N	(<code>fetch.dk_lendr.py</code>)	Q	log	observed	NEW price; 34 obs, 2017Q3=1.01→2025Q4=3. (0.73–5.00); [FLAG: source substitution + leading gap — see below]

[FLAG: DK price source substitution.] Danmarks Nationalbank’s StatBank exposes no machine-readable JSON/PX-Web API for the MFI interest-rate return. The ECB Data Portal (SDW MIR) compiles the same Nationalbanken return (source agency DK2, harmonized under ECB Reg. 290/2009) and is used as the authoritative substitute — the data are genuine, only the transport differs. The ECB MIR Danish NFC rate starts 2017Q3, so *LENDR_DK* is null for 2009Q1–2017Q2 (leading ragged edge, Gibbs-imputed).

10.38 Norway (NO)

The Norwegian bloc has $n = 17$ staged variables (deployed $n = 20$), estimated on 2006Q1–2025Q4. Norway is rate-optional: it carries a policy rate (*POLRATE_NO*) but no separate domestic short rate. Seven macro variables are COVID-masked.

Table 86: Norway (NO): per-variable data provenance.

Variable	Description	Source	Series id/key	Freq	Trans.	COVID	Notes
GDP_NO	Real GDP	OECD QNA	[FLAG]	Q	log	masked	macro
CONS_NO	Real consumption	OECD QNA	[FLAG]	Q	log	masked	macro
INV_NO	Real investment	OECD QNA	[FLAG]	Q	log	masked	macro
GOV_NO	Real govt consumption	OECD QNA	[FLAG]	Q	log	masked	macro
EXP_NO	Real exports	OECD QNA	[FLAG]	Q	log	masked	macro
IMP_NO	Real imports	OECD QNA	[FLAG]	Q	log	masked	macro
HICP_NO	HICP (index)	IMF SDMX	[FLAG]	Q	log	masked	macro

Norway (NO), continued

Variable	Description	Source	Series id/key	Freq	Trans.	COVID	Notes
LR_NO	Unemployment rate	IMF SDMX	[FLAG]	Q	lin	masked	macro
NEER_NO	NEER (index)	IMF SDMX	[FLAG]	Q	log	observed	financial
POLRATE_NO	Policy rate (BIS CBPOL)	BIS	BIS WS_CBPOL M.NO	M→Q-avg	lin	observed	policy rate; [FLAG: id by convention]
HP_NO	Real residential property price	BIS RPP	FRED QNOR628BIS	Q	log	observed	financial; [FLAG: id by convention]
GLB_OIL	WTI crude, q-avg (\$/bbl)	FRED	WTISPLC	Q	log	observed	global common
GLB_USLR	US 10Y Treasury yield	FRED	DGS10	Q	lin	observed	global common
GLB_USBAA	US Baa corporate yield	FRED	BAA	Q	lin	observed	global common
GLB_USEQ	US equity (S&P 500)	FRED	SPASTT01USM661N	Q	log	observed	global common
CREDIT_GDP_NO	Credit to NFCs (% of GDP)	BIS (via FRED)	QNONAM770A	Q	lin	masked	NEW broad credit; 77 obs, 2006Q1=111.5→2025Q4=1 (111.5–157.9)
LENDR_NO	NFC bank lending rate (% p.a.)	Statistics Norway (SSB)	table 10746 (Ut-lanstype=02, Sektor=03 NFC, Rente-binding=99, RenterUtNedbLan; fetch_ssb_jsonstat.py)	M→Q-avg	lin	observed	NEW price; 49 obs, 2013Q4=4.13→2025Q4=6. (2.42–6.81); [FLAG: leading gap 2006Q1–2013Q3, Gibbs-imputed]

10.39 Sweden (SE)

The Swedish bloc has $n = 19$ staged variables (deployed $n = 22$), estimated on 2005Q1–2025Q4. Seven macro variables are COVID-masked. Sweden carries a 3-month rate, a 10Y yield, and a policy rate.

Table 87: Sweden (SE): per-variable data provenance.

Variable	Description	Source	Series id/key	Freq	Trans.	COVID	Notes
GDP_SE	Real GDP	Eurostat OECD	/ [FLAG]	Q	log	masked	macro
CONS_SE	Real consumption	Eurostat OECD	/ [FLAG]	Q	log	masked	macro
INV_SE	Real investment	Eurostat OECD	/ [FLAG]	Q	log	masked	macro
GOV_SE	Real govt con- sumption	Eurostat OECD	/ [FLAG]	Q	log	masked	macro
EXP_SE	Real exports	Eurostat OECD	/ [FLAG]	Q	log	masked	macro
IMP_SE	Real imports	Eurostat OECD	/ [FLAG]	Q	log	masked	macro
HICP_SE	HICP (index)	Eurostat	[FLAG]	Q	log	masked	macro
LR_SE	Unemployment rate	Eurostat	[FLAG]	Q	lin	masked	macro
RATE_SE	3-month rate	IMF / Euro- stat	[FLAG]	Q	lin	observed	short rate
YIELD10_SE	10Y govt bond yield	IMF / Euro- stat	[FLAG]	Q	lin	observed	financial
NEER_SE	NEER (index)	IMF SDMX	[FLAG]	Q	log	observed	financial
POLRATE_SE	Policy rate (BIS CBPOL)	BIS	BIS WS_CBPOL M.SE	M→Q-avg	lin	observed	policy rate; [FLAG: id by convention]
HP_SE	Real residential property price	BIS RPP	FRED QSER628BIS	Q	log	observed	financial; [FLAG: id by convention]
GLB_OIL	WTI crude, q-avg (\$/bbl)	FRED	WTISPLC	Q	log	observed	global common
GLB_USLR	US 10Y Treasury yield	FRED	DGS10	Q	lin	observed	global common
GLB_USBAA	US Baa corporate yield	FRED	BAA	Q	lin	observed	global common
GLB_USEQ	US equity (S&P 500)	FRED	SPASTT01USM661N	Q	log	observed	global common
CREDIT_GDP_SE	Credit to NFCs (% of GDP)	BIS (via FRED)	QSENAM770A	Q	lin	masked	NEW broad credit; 81 obs, 2005Q1=102.1→2025Q4=1 (102.1–185.5)

Sweden (SE), continued

Variable	Description	Source	Series id/key	Freq	Trans.	COVID	Notes
LENDR_SE	NFC bank lending rate (% p.a.)	Statistics Sweden (SCB)	PxWeb FM/FM5001/FM5001C/RantaT01N (Referenssek- tor=1.1.1, Motpartssek- tor=1 NFC, Avtal=0200, Con- tentsCode=000004ZT; fetch_scb_pxweb.py)	M→Q-avg	lin	observed	NEW price; 84 obs (full), 2005Q1=3.57→2025Q4=3. (1.58–5.72); clean

10.40 New Zealand (NZ)

The New Zealand bloc has $n = 18$ staged variables (deployed $n = 21$), estimated on 2000Q1–2025Q4. Eight macro variables are COVID-masked. NZ carries a policy rate (OCR) and a 10Y yield.

Table 88: New Zealand (NZ): per-variable data provenance.

Variable	Description	Source	Series id/key	Freq	Trans.	COVID	Notes
GDP_NZ	Real GDP	OECD QNA	[FLAG]	Q	log	masked	macro
CONS_NZ	Real consumption	OECD QNA	[FLAG]	Q	log	masked	macro
INV_NZ	Real investment	OECD QNA	[FLAG]	Q	log	masked	macro
GOV_NZ	Real govt consumption	OECD QNA	[FLAG]	Q	log	masked	macro
EXP_NZ	Real exports	OECD QNA	[FLAG]	Q	log	masked	macro
IMP_NZ	Real imports	OECD QNA	[FLAG]	Q	log	masked	macro
CPI_NZ	CPI (index)	IMF SDMX	[FLAG]	Q	log	masked	macro
LR_NZ	Unemployment rate	IMF SDMX	[FLAG]	Q	lin	masked	macro
RATE_NZ	Policy rate (OCR)	RBNZ IMF	/ [FLAG]	Q	lin	observed	policy short rate
YIELD10_NZ	10Y govt bond yield	RBNZ IMF	/ [FLAG]	Q	lin	observed	financial

New Zealand (NZ), continued

Variable	Description	Source	Series id/key	Freq	Trans.	COVID	Notes
HP_NZ	Real residential house-price index (2010=100)	BIS RPP	FRED QNZR628BIS	Q	log	observed	financial; <i>[FLAG: id by convention]</i>
NEER_NZ	NEER (index)	IMF SDMX	<i>[FLAG]</i>	Q	log	observed	financial
GLB_OIL	WTI crude, q-avg (\$/bbl)	FRED	WTISPLC	Q	log	observed	global common
GLB_USLR	US 10Y Treasury yield	FRED	DGS10	Q	lin	observed	global common
GLB_USBAA	US Baa corporate yield	FRED	BAA	Q	lin	observed	global common
GLB_USEQ	US equity (S&P 500)	FRED	SPASTT01USM661N	Q	log	observed	global common
CREDIT_GDP_NZ	Credit to NFCs (% of GDP)	BIS (via FRED)	QNZNAM770A	Q	lin	masked	NEW broad credit; 101 obs, 2000Q1=83.0→2025Q4=70 (69.5–108.8)
LENDR_NZ	Business lending rate (% p.a.)	RBNZ Table B3	series A1 M→Q-avg “SME new over-draft rate” INR.MBI06.D3 (fetch_rbnz_b3.py)	lin	observed	NEW	price; 104 obs (full), 2000Q1=8.53→2025Q4=9. (8.16–12.59); <i>[FLAG: transport workaround — see below]</i>

[FLAG: NZ price transport workaround.] The live RBNZ host is bot-walled (HTTP 403, Aka-mai/Imperva) to non-interactive clients. The genuine RBNZ B3 hb3.xlsx was retrieved via a Way-back Machine snapshot (2026-01-28) and cached as pipeline/credit_staging/NZ/_rbnz_b3A1_quarterly.json; the data are the real RBNZ B3 A1 series (only the transport differs). The older E1 business lending rate was discontinued Dec-2016; A1 covers the full sample with no gaps.

10.41 Korea (KR)

The Korean bloc has $n = 18$ staged variables (deployed $n = 21$), estimated on 2000Q1–2025Q4. Eight macro variables are COVID-masked. Korea carries a 3-month money-market rate, a policy rate, and a 10Y yield, plus a *true* corporate bond yield as its credit-block price.

Table 89: Korea (KR): per-variable data provenance.

Variable	Description	Source	Series id/key	Freq	Trans.	COVID	Notes
GDP_KR	Real GDP	OECD QNA	[FLAG]	Q	log	masked	macro
CONS_KR	Real household consumption	OECD QNA	[FLAG]	Q	log	masked	macro
INV_KR	Real GFCF	OECD QNA	[FLAG]	Q	log	masked	macro
GOV_KR	Real govt consumption	OECD QNA	[FLAG]	Q	log	masked	macro
EXP_KR	Real exports	OECD QNA	[FLAG]	Q	log	masked	macro
IMP_KR	Real imports	OECD QNA	[FLAG]	Q	log	masked	macro
CPI_KR	CPI, all items (index)	IMF SDMX	[FLAG]	Q	log	masked	macro
LR_KR	Unemployment rate	IMF SDMX	[FLAG]	Q	lin	masked	macro
RATE_KR	Money-market rate (3M, short-term)	IMF SDMX	[FLAG]	Q	lin	observed	short rate
POLRATE_KR	Policy rate (BIS CBPOL)	BIS	BIS WS_CBPOL M.KR	M→Q-avg	lin	observed	policy rate; [FLAG: id by convention]
YIELD10_KR	Govt bond yield (long-term)	IMF OECD	/ [FLAG]	Q	lin	observed	financial
HP_KR	Real residential house-price index (2010=100)	BIS RPP	FRED QKRR628BIS	Q	log	observed	financial; [FLAG: id by convention]
GLB_OIL	WTI crude, q-avg (\$/bbl)	FRED	WTISPLC	Q	log	observed	global common
GLB_USLR	US 10Y Treasury yield	FRED	DGS10	Q	lin	observed	global common
GLB_USBAA	US Baa corporate yield	FRED	BAA	Q	lin	observed	global common
GLB_USEQ	US equity (S&P 500)	FRED	SPASTT01USM661N	Q	log	observed	global common
CREDIT_GDP_KR	Credit to NFCs (% of GDP)	BIS (via FRED)	QKRNAM770A	Q	lin	masked	NEW broad credit; 101 obs, 2000Q1=89.1→2025Q4=114.6 (70.5–114.6)

Korea (KR), continued

Variable	Description	Source	Series id/key	Freq	Trans.	COVID	Notes
CORP_KR	Corporate bond yield (3Y, AA-)	Bank of Korea ECOS	table 721Y001 item 7020000 (fetch.credit_kr.py)	M→Q-avg	lin	observed	NEW price = true AA- corporate yield ; 104 obs, 2000Q1=10.09→2025Q4=3 (1.67–10.09)

10.42 Israel (IL)

The Israeli bloc has $n = 18$ staged variables (deployed $n = 21$), estimated on 2000Q1–2024Q4. Eight macro variables are COVID-masked. Israel carries a 3-month T-bill rate and a policy rate; its credit-block price is a *reconstructed* Tel-Bond corporate yield.

Table 90: Israel (IL): per-variable data provenance.

Variable	Description	Source	Series id/key	Freq	Trans.	COVID	Notes
GDP_IL	Real GDP	OECD QNA	[FLAG]	Q	log	masked	macro
CONS_IL	Real household consumption	OECD QNA	[FLAG]	Q	log	masked	macro
INV_IL	Real GFCF	OECD QNA	[FLAG]	Q	log	masked	macro
GOV_IL	Real govt consumption	OECD QNA	[FLAG]	Q	log	masked	macro
EXP_IL	Real exports	OECD QNA	[FLAG]	Q	log	masked	macro
IMP_IL	Real imports	OECD QNA	[FLAG]	Q	log	masked	macro
CPI_IL	CPI, all items (index)	IMF SDMX	[FLAG]	Q	log	masked	macro
LR_IL	Unemployment rate	IMF SDMX	[FLAG]	Q	lin	masked	macro
RATE_IL	3M Treasury-bill yield	IMF / BoI	[FLAG]	Q	lin	observed	short rate
POLRATE_IL	Policy rate (Bank of Israel)	BoI / BIS	[FLAG]	Q	lin	observed	policy rate
NEER_IL	NEER (index)	IMF SDMX	[FLAG]	Q	log	observed	financial

Israel (IL), continued

Variable	Description	Source	Series id/key	Freq	Trans.	COVID	Notes
HP_IL	Real residential property price	BIS RPP	FRED QILR628BIS	Q	log	observed	financial; <i>[FLAG: id by convention]</i>
GLB_OIL	WTI crude, q-avg (\$/bbl)	FRED	WTISPLC	Q	log	observed	global common
GLB_USLR	US 10Y Treasury yield	FRED	DGS10	Q	lin	observed	global common
GLB_USBAA	US Baa corporate yield	FRED	BAA	Q	lin	observed	global common
GLB_USEQ	US equity (S&P 500)	FRED	SPASTT01USM661N	Q	log	observed	global common
CREDIT_GDP_IL	Credit to NFCs (% of GDP)	BIS (via FRED)	QILNAM770A	Q	lin	masked	NEW broad credit; 97 obs, 2000Q1=69.4→2024Q4=69.4 (66.8–93.1)
CORP_IL	Tel-Bond rate (5Y gov't Bond spread)	Bank of Israel (SDMX)	DWH_SRC_0356_MA + DWH_SRC_0299_MA (fetch_credit_il.py)	Q-avg	lin	observed	NEW price; 56 obs, 2009Q4=7.03→2024Q4=5.03 (1.89–7.03); <i>[FLAG: RECONSTRUCTED + leading gap — see below]</i>

[FLAG: IL price is RECONSTRUCTED.] CORP_IL is built as the sum of the Bank of Israel 5-year non-indexed fixed-rate government bond yield (DWH_SRC_0356_MA) and the BoI quoted Tel-Bond-vs-govt yield spread (DWH_SRC_0299_MA); the Tel-Bond Index is Israel's main corporate-bond index (~5Y duration), so the sum is a genuine corporate yield, not a single published outright series. It was preferred over the BoI BIR bank business-loan rate, which carries a systematic source anomaly 2017Q2–2018Q1 (values spike to 30–50%). The Tel-Bond spread starts 2009-12, so CORP_IL is null for 2000Q1–2009Q3 (44 leading nulls, Gibbs-imputed); it is observed continuously through COVID and to end-of-sample.

Summary of data flags (advanced-economy blocs). The genuinely indeterminate sources — where the exact upstream series id could not be recovered from a machine-readable builder and is therefore *not* fabricated — are the domestic *base* macro and financial variables (real-activity aggregates, CPI/HICP, unemployment, NEER, short rate, 10Y yield, and for IL the policy rate), attributed at the provider level (OECD QNA, IMF SDMX, Eurostat for DK/SE, BIS, national cen-

tral banks) per Section 9.3. The BIS real house-price ($HP_{<CC>}$) and BIS policy-rate ($POLRATE_{<CC>}$) ids are stated by the established FRED-mnemonic / BIS-dataflow convention (marked “id by convention”). All credit-block ids (quantity and price) are live-probed and fully traced.

10.43 Emerging-market blocs I: BR, MX, CL, CO, IN, ID

This subsection documents six emerging-market GVAR blocs in full, including the new credit block (a quantity $CREDIT_GDP_{<CC>}$ and a price, $LENDR_{<CC>}$ or $CORP_{<CC>}$) appended to each bloc. Throughout, “log” denotes a series entered as $100 \ln(\text{level})$ and displayed as $4 \Delta(100 \ln)$, i.e. q/q-annualized growth, while “lin” denotes a level (percent, ratio, or index). Under the *COVID-as-missing* treatment (Section 9.4) every *macro* variable (those tagged $isFinancial=0$) is masked over the fixed window 2020Q2–2020Q4 and imputed by the augmentation Gibbs sampler; *financial* variables ($isFinancial=1$) are held observed.

The new credit block (all six blocs). Each bloc gains two variables, appended after the existing domestic and global-commons columns:

- **Quantity** $CREDIT_GDP_{<CC>}$ — BIS Total Credit to Non-Financial Corporations, percentage of GDP (the broad credit measure: loans + debt securities + cross-border), pulled from FRED with the BIS mnemonic $Q_{<CC>NAM770A$ (“Total Credit to Non-Financial Corporations, Adjusted for Breaks”, units: % of GDP, native quarterly). Transformation **lin**, prior **RW**, $isFinancial=1$. Because the %-of-GDP ratio inherits the collapsing-GDP denominator artefact in 2020, the three cells 2020Q2–Q4 are set to **null** (masked), consistent with the macro COVID mask, and imputed by the Gibbs sampler. **Colombia is the exception:** BIS publishes no Colombia NFC total credit, so $CREDIT_GDP_CO$ is *constructed* (non-BIS); see below and the flag list.
- **Price** — a domestic credit-cost series, observed through COVID (**lin**, prior **RW**, $isFinancial=1$). The price source differs by bloc: a bank *lending rate* ($LENDR_{<CC>}$) where one is available, or a 10-year *government* bond yield used as a *sovereign proxy* ($CORP_{<CC>}$) where no corporate/lending series exists. Per-bloc provenance and caveats are given in each table and flagged where they apply (CL and IN use the sovereign 10Y proxy).

10.44 Brazil (BR)

The Brazil bloc is an $n = 17$ quarterly BVAR estimated over 2012Q2–2025Q4 ($T = 55$); macro variables are treated as missing over 2020Q2–Q4 (COVID-as-missing), while the financial block (including the new credit price) is held observed. The credit block is *clean*: BIS NFC credit-to-GDP for the quantity and the Banco Central do Brasil corporate lending rate for the price.

Table 91: Brazil (BR) bloc: variable coverage and provenance. $\log = 100 \ln$ level (q/q-annualized growth displayed); $\text{lin} = \text{level}$. COVID mask = 2020Q2–Q4.

Variable	Description	Source	Series ID/key	Freq	Trans.	COVID	Notes
GDP_BR	Real GDP	National counts (GVAR panel)	ac- $[FLAG]$	Q	log	masked	macro
CONS_BR	Real household consumption	National counts (GVAR panel)	ac- $[FLAG]$	Q	log	masked	macro
INV_BR	Real gross fixed capital formation	National counts (GVAR panel)	ac- $[FLAG]$	Q	log	masked	macro
GOV_BR	Real government consumption	National counts (GVAR panel)	ac- $[FLAG]$	Q	log	masked	macro
EXP_BR	Real exports	National counts (GVAR panel)	ac- $[FLAG]$	Q	log	masked	macro
IMP_BR	Real imports	National counts (GVAR panel)	ac- $[FLAG]$	Q	log	masked	macro
CPI_BR	CPI index	National / IMF (GVAR panel)	source $[FLAG]$	Q	log	masked	macro
LR_BR	Unemployment rate	National / IMF (GVAR panel)	source $[FLAG]$	Q	lin	masked	macro
NEER_BR	Nominal effective exch. rate	BIS / IMF (GVAR panel)	$[FLAG]$	Q	log	observed	financial
POLRATE_BR	Policy rate	BIS CBPOL	WS_CBPOL M.BR	M→Q avg	lin	observed	3-month
HP_BR	Real residential property price (2010=100)	FRED / BIS real RPP	QBRR628BIS	Q	log	observed	financial
GLB_OIL	WTI crude oil, q-avg (\$/bbl)	FRED	WTISPLC	M→Q avg	log	observed	global c
GLB_USLR	US 10Y Treasury yield (%)	FRED	DGS10	Q	lin	observed	global c
GLB_USBAA	US Baa corporate yield (%)	FRED	BAA	Q	lin	observed	global c
GLB_USEQ	US equity (S&P 500)	FRED	SPASTT01USM661N	M→Q avg	log	observed	global c
CREDIT_GDP_BR	Credit to NFCs (% of GDP)	BIS via FRED	QBRNAM770A	Q	lin	masked	credit tity ; non-nul 57.0% c the 3 C null (cle

Variable	Description	Source	Series ID/key	Freq	Trans.	COVID	Notes
LENDR_BR	Corporate bank lending rate (% p.a.)	Banco do Brasil (BCB SGS)	SGS 20718	M→Q avg	lin	observed	credit Pessoas avg rate; non-nul 31.24%; fetch.b

10.45 Mexico (MX)

The Mexico bloc is an $n = 19$ quarterly BVAR estimated over 2001Q4–2025Q4 ($T = 97$); macro variables are masked over 2020Q2–Q4. The credit block uses BIS NFC credit-to-GDP and the IMF IFS lending rate; the price has a ragged trailing edge (the IFS series ends 2025Q2, so 2025Q3–Q4 are null and handled by the Gibbs ragged-edge treatment).

Table 92: Mexico (MX) bloc: variable coverage and provenance.

Variable	Description	Source	Series ID/key	Freq	Trans.	COVID	Notes
GDP_MX	Real GDP (chain volume, SA)	National counts / QNA panel)	ac- [FLAG] (GVAR	Q	log	masked	mac
CONS_MX	Real household consumption (chain vol, SA)	National counts panel)	ac- [FLAG] (GVAR	Q	log	masked	mac
INV_MX	Real gross fixed capital formation (SA)	National counts panel)	ac- [FLAG] (GVAR	Q	log	masked	mac
GOV_MX	Real government consumption (chain vol, SA)	National counts panel)	ac- [FLAG] (GVAR	Q	log	masked	mac
EXP_MX	Real exports (chain volume, SA)	National counts panel)	ac- [FLAG] (GVAR	Q	log	masked	mac
IMP_MX	Real imports (chain volume, SA)	National counts panel)	ac- [FLAG] (GVAR	Q	log	masked	mac
CPI_MX	CPI (all items, index)	National / IMF panel)	source [FLAG] (GVAR	Q	log	masked	mac
LR_MX	Unemployment rate	National / IMF panel)	source [FLAG] (GVAR	Q	lin	masked	mac

Variable	Description	Source	Series ID/key	Freq	Trans.	COVID	Notes
RATE_MX	Money-market rate (short-term)	National source [FLAG] / IMF (GVAR panel)		Q	lin	observed	financial
POLRATE_MX	Central-bank policy rate	BIS CBPOL	WS_CBPOL M.MX	M→Q avg	lin	observed	3-mo
YIELD10_MX	Government bond yield (long-term)	National source [FLAG] / IMF (GVAR panel)		Q	lin	observed	financial
HP_MX	Real residential house-price index (2010=100)	FRED / BIS real RPP	QMXR628BIS	Q	log	observed	financial
NEER_MX	Nominal effective exch. rate (index)	BIS / IMF [FLAG] (GVAR panel)		Q	log	observed	financial
GLB_OIL	WTI crude oil, q-avg (\$/bbl)	FRED	WTISPLC	M→Q avg	log	observed	global
GLB_USLR	US 10Y Treasury yield (%)	FRED	DGS10	Q	lin	observed	global
GLB_USBAA	US Baa corporate yield (%)	FRED	BAA	Q	lin	observed	global
GLB_USEQ	US equity (S&P 500)	FRED	SPASTT01USM661N	M→Q avg	log	observed	global
CREDIT_GDP_MX	Credit to NFCs (% of GDP)	BIS via FRED	QMXNAM770A	Q	lin	masked	credit quantity non-28.3 the null
LENDR_MX	Bank lending rate (% p.a.)	IMF IFS (via DB-nomics)	FILR_PA	Q	lin	observed	credit quantity 95/9 3.34 rags ing ends 2025

10.46 Chile (CL)

The Chile bloc is an $n = 19$ quarterly BVAR estimated over 2004Q3–2025Q4 ($T = 86$); macro variables are masked over 2020Q2–Q4. The credit *quantity* is the BIS NFC credit-to-GDP series, but the credit *price* is a **sovereign 10Y proxy** (CORP_CL): the IMF IFS Chilean lending rate was discontinued in 2018, so the OECD KEI 10-year *government* bond yield stands in — there is no corporate spread.

Table 93: Chile (CL) bloc: variable coverage and provenance.

Variable	Description	Source	Series ID/key	Freq	Trans.	COVID	Notes
GDP_CL	Real GDP	National counts / QNA (GVAR panel)	ac- [FLAG] OECD	Q	log	masked	macro

Variable	Description	Source	Series ID/key	Freq	Trans.	COVID	Notes
CONS_CL	Real household consumption	National counts panel)	ac- <i>[FLAG]</i> (GVAR	Q	log	masked	macro
INV_CL	Real gross fixed capital formation	National counts panel)	ac- <i>[FLAG]</i> (GVAR	Q	log	masked	macro
GOV_CL	Real government consumption	National counts panel)	ac- <i>[FLAG]</i> (GVAR	Q	log	masked	macro
EXP_CL	Real exports	National counts panel)	ac- <i>[FLAG]</i> (GVAR	Q	log	masked	macro
IMP_CL	Real imports	National counts panel)	ac- <i>[FLAG]</i> (GVAR	Q	log	masked	macro
CPI_CL	CPI index	National / IMF panel)	source <i>[FLAG]</i> (GVAR	Q	log	masked	macro
LR_CL	Unemployment rate	National / IMF panel)	source <i>[FLAG]</i> (GVAR	Q	lin	masked	macro
RATE_CL	3M interbank rate	National / IMF panel)	source <i>[FLAG]</i> (GVAR	Q	lin	observed	financial
POLRATE_CL	Central-bank policy rate	BIS CBPOL	WS_CBPOL M.CL	M→Q avg	lin	observed	3-month
YIELD10_CL	10Y government bond yield	National / IMF panel)	source <i>[FLAG]</i> (GVAR	Q	lin	observed	financial
HP_CL	Real residential house-price index (2010=100)	FRED / BIS RPP	real QCLR628BIS	Q	log	observed	financial
NEER_CL	Nominal effective exch. rate	BIS / IMF (GVAR panel)	<i>[FLAG]</i>	Q	log	observed	financial
GLB_OIL	WTI crude oil, q-avg (\$/bbl)	FRED	WTISPLC	M→Q avg	log	observed	global
GLB_USLR	US 10Y Treasury yield (%)	FRED	DGS10	Q	lin	observed	global
GLB_USBAA	US Baa corporate yield (%)	FRED	BAA	Q	lin	observed	global
GLB_USEQ	US equity (S&P 500)	FRED	SPASTT01USM661N	M→Q avg	log	observed	global
CREDIT_GDP_CL	Credit to NFCs (% of GDP)	BIS via FRED	QCLNAM770A	Q	lin	masked	credit non- 111.9 COV

Variable	Description	Source	Series ID/key	Freq	Trans.	COVID	Not
CORP_CL	10Y government bond yield (sovereign proxy)	OECD KEI long-term interest rates (IRLT)	IRLT, country M→Q avg CHL	lin	observed	crec	SOV 10Y — rate 86/8 2.51

10.47 Colombia (CO)

The Colombia bloc is an $n = 19$ quarterly BVAR estimated over 2005Q1–2025Q4 ($T = 84$); macro variables are masked over 2020Q2–Q4. Colombia is *not* a BIS total-credit reporter, so the credit *quantity* is a **constructed (non-BIS) broad-credit proxy** (reducing cross-country comparability); the credit *price* is the IMF IFS lending rate. Both credit columns carry trailing nulls from the input cut-off (2025Q1 for the constructed quantity inputs; the IFS price ends mid-2025).

Table 94: Colombia (CO) bloc: variable coverage and provenance.

Variable	Description	Source	Series ID/key	Freq	Trans.	COVID	Not
GDP_CO	Real GDP	National counts / OECD QNA (GVAR panel)	ac- [FLAG]	Q	log	masked	macr
CONS_CO	Real household consumption	National counts (GVAR panel)	ac- [FLAG]	Q	log	masked	macr
INV_CO	Real gross fixed capital formation	National counts (GVAR panel)	ac- [FLAG]	Q	log	masked	macr
GOV_CO	Real government consumption	National counts (GVAR panel)	ac- [FLAG]	Q	log	masked	macr
EXP_CO	Real exports	National counts (GVAR panel)	ac- [FLAG]	Q	log	masked	macr
IMP_CO	Real imports	National counts (GVAR panel)	ac- [FLAG]	Q	log	masked	macr
CPI_CO	CPI index	National / IMF (GVAR panel)	source [FLAG]	Q	log	masked	macr
LR_CO	Unemployment rate	National / IMF (GVAR panel)	source [FLAG]	Q	lin	masked	macr

Variable	Description	Source	Series ID/key	Freq	Trans.	COVID	Notes
RATE_CO	3M interbank rate	National source [FLAG] / IMF (GVAR panel)		Q	lin	observed	financial
POLRATE_CO	Central-bank policy rate	BIS CBPOL	WS_CBPOL M.CO	M→Q avg	lin	observed	3-mo
YIELD10_CO	10Y government bond yield	National source [FLAG] / IMF (GVAR panel)		Q	lin	observed	financial
HP_CO	Real residential house-price index (2010=100)	FRED / BIS real RPP	QCOR628BIS	Q	log	observed	financial
NEER_CO	Nominal effective exch. rate	BIS / IMF [FLAG] (GVAR panel)		Q	log	observed	financial
GLB_OIL	WTI crude oil, q-avg (\$/bbl)	FRED	WTISPLC	M→Q avg	log	observed	global
GLB_USLR	US 10Y Treasury yield (%)	FRED	DGS10	Q	lin	observed	global
GLB_USBAA	US Baa corporate yield (%)	FRED	BAA	Q	lin	observed	global
GLB_USEQ	US equity (S&P 500)	FRED	SPASTT01USM661N	M→Q avg	log	observed	global
CREDIT_GDP_CO	Credit to NFCs (% of GDP), constructed	IMF IFS (constructed: ÷ 4Q-trailing NGDP)	FOSAOP_XDC / NGDP_SA_XDC	Q	lin	masked	credit quantity (non-financial) = claims on private NFC companies 75/82.7 = 3 C-trail end
LENDR_CO	Bank lending rate (% p.a.)	IMF IFS (via DB-nomics)	FILR_PA	Q	lin	observed	credit 82/88.59 ragg edge null)

10.48 India (IN)

The India bloc is an $n = 19$ quarterly BVAR estimated over 2011Q4–2025Q4 ($T = 57$); macro variables are masked over 2020Q2–Q4. The credit *quantity* is the BIS NFC credit-to-GDP series; the credit *price* is a **sovereign 10Y proxy** (CORP_IN): no Indian corporate or bank lending rate is

currently maintained on FRED/IMF/World Bank (the IFS lending rate was discontinued in 2022), so the OECD KEI 10-year *government* bond yield stands in — there is no corporate spread.

Table 95: India (IN) bloc: variable coverage and provenance.

Variable	Description	Source	Series ID/key	Freq	Trans.	COVID	
GDP_IN	Real GDP (chain volume, SA)	National counts / OECD QNA (GVAR panel)	ac- <i>[FLAG]</i>	Q	log	masked	n
CONS_IN	Real household consumption (chain vol, SA)	National counts (GVAR panel)	ac- <i>[FLAG]</i>	Q	log	masked	n
INV_IN	Real gross fixed capital formation (chain vol, SA)	National counts (GVAR panel)	ac- <i>[FLAG]</i>	Q	log	masked	n
GOV_IN	Real government consumption (chain vol, SA)	National counts (GVAR panel)	ac- <i>[FLAG]</i>	Q	log	masked	n
EXP_IN	Real exports of goods & services (chain vol, SA)	National counts (GVAR panel)	ac- <i>[FLAG]</i>	Q	log	masked	n
IMP_IN	Real imports of goods & services (chain vol, SA)	National counts (GVAR panel)	ac- <i>[FLAG]</i>	Q	log	masked	n
CPI_IN	CPI (all items, index)	National / IMF source (GVAR panel)	<i>[FLAG]</i>	Q	log	masked	n
RATE_IN	Call money / immediate interbank rate	FRED (OECD MEI)	IRSTCI01INM	M→Q avg	lin	observed	f
RATE3M_IN	Short-term 3M interbank rate	FRED (OECD MEI)	INDIR3TIB01STM	M→Q avg	lin	observed	f
YIELD10_IN	Long-term 10Y government bond yield	FRED (OECD MEI)	INDIRLT01STM	M→Q avg	lin	observed	f
NEER_IN	Nominal effective exch. rate (BIS broad, index)	BIS / IMF (GVAR panel)	<i>[FLAG]</i>	Q	log	observed	f
POLRATE_IN	Policy rate	BIS CBPOL	WS.CBPOL M.IN	M→Q avg	lin	observed	3
HP_IN	Real residential property price (2010=100)	FRED / BIS real RPP	QINR628BIS	Q	log	observed	f
GLB_OIL	WTI crude oil, q-avg (\$/bbl)	FRED	WTISPLC	M→Q avg	log	observed	g
GLB_USLR	US 10Y Treasury yield (%)	FRED	DGS10	Q	lin	observed	g
GLB_USBAA	US Baa corporate yield (%)	FRED	BAA	Q	lin	observed	g
GLB_USEQ	US equity (S&P 500)	FRED	SPASTT01USM661N	M→Q avg	log	observed	g
CREDIT_GDP_IN	Credit to NFCs (% of GDP)	BIS via FRED	QINNAM770A	Q	lin	masked	c

Variable	Description	Source	Series ID/key	Freq	Trans.	COVID	Notes
CORP_IN	10Y government bond yield (sovereign proxy)	OECD KEI long-term interest rates (IRLT)	IRLT, country IND	M→Q avg	lin	observed	macro

10.49 Indonesia (ID)

The Indonesia bloc is an $n = 17$ quarterly BVAR estimated over 2005Q3–2025Q4 ($T = 82$); macro variables are masked over 2020Q2–Q4. The credit block is *clean*: BIS NFC credit-to-GDP for the quantity and the IMF IFS lending rate (fully observed, current to 2025Q4) for the price.

Table 96: Indonesia (ID) bloc: variable coverage and provenance.

Variable	Description	Source	Series ID/key	Freq	Trans.	COVID	Notes
GDP_ID	Real GDP (SA, constant prices)	National counts / OECD QNA (GVAR panel)	ac- [FLAG]	Q	log	masked	macro
CONS_ID	Real private consumption (SA)	National counts (GVAR panel)	ac- [FLAG]	Q	log	masked	macro
INV_ID	Real gross fixed capital formation (SA)	National counts (GVAR panel)	ac- [FLAG]	Q	log	masked	macro
GOV_ID	Real government consumption (SA)	National counts (GVAR panel)	ac- [FLAG]	Q	log	masked	macro
EXP_ID	Real exports of goods & services (SA)	National counts (GVAR panel)	ac- [FLAG]	Q	log	masked	macro
IMP_ID	Real imports of goods & services (SA)	National counts (GVAR panel)	ac- [FLAG]	Q	log	masked	macro
CPI_ID	CPI (all items, index)	National / IMF source (GVAR panel)	[FLAG]	Q	log	masked	macro
RATE_ID	Monetary policy-related rate (BI rate)	National / IMF source (GVAR panel)	[FLAG]	Q	lin	observed	financial
NEER_ID	Nominal effective exch. rate (BIS broad)	BIS / IMF (GVAR panel)	[FLAG]	Q	log	observed	financial
POLRATE_ID	Policy rate	BIS CBPOL	WS_CBPOL M.ID	M→Q avg	lin	observed	3-month

Variable	Description	Source	Series ID/key	Freq	Trans.	COVID	Notes
HP_ID	Real residential property price (2010=100)	FRED / BIS real RPP	QIDR628BIS	Q	log	observed	financial
GLB_OIL	WTI crude oil, q-avg (\$/bbl)	FRED	WTISPLC	M→Q avg	log	observed	global oil
GLB_USLR	US 10Y Treasury yield (%)	FRED	DGS10	Q	lin	observed	global oil
GLB_USBAA	US Baa corporate yield (%)	FRED	BAA	Q	lin	observed	global oil
GLB_USEQ	US equity (S&P 500)	FRED	SPASTT01USM661N	M→Q avg	log	observed	global oil
CREDIT_GDP_ID	Credit to NFCs (% of GDP)	BIS via FRED	QIDNAM770A	Q	lin	masked	credit tity ; non-nul 27.7% COVID
LENDR_ID	IFS lending rate (% p.a.)	IMF IFS (MFS_IR, MFS162_RT_PT_A_PT new SDMX 2.1 (= IFS FILR_PA) API)		Q	lin	observed	credit 82/82 8.21–16 observe to 2025 fetch.i

Caveats and provenance flags for this group. The *credit-block* provenance above is traced to specific FRED/IMF/OECD/BCB series and verified against the staged data (first/last/coverage all reconcile). The flagged items are:

- CORP_CL, CORP_IN: credit *price* is a **sovereign 10Y proxy** (OECD KEI government-bond yield), not a corporate/lending rate — no corporate spread is captured (CL: IFS lending rate discontinued 2018; IN: IFS lending rate discontinued 2022).
- CREDIT_GDP_CO: **constructed, non-BIS** broad-credit proxy (100× IMF IFS ODC claims on the private sector ÷ 4Q-trailing nominal GDP); it mixes households + NFCs (not NFC-specific) and so has reduced cross-country comparability. (BIS publishes no Colombia NFC total-credit series; the mnemonic QCONAM770A on FRED is *Argentina*, not Colombia — a verified mnemonic trap.)
- LENDR_MX, LENDR_CO: credit-price series have a **ragged trailing edge** (IFS FILR_PA ends mid-2025), so 2025Q3–Q4 are null and imputed by the ragged-edge Gibbs treatment.

[FLAG: base macro / domestic-financial series IDs not recorded in the staged artefacts.] For the pre-existing domestic block (real GDP, the national-accounts expenditure components, CPI, unemployment, NEER, and the domestic short/long rates), the staged `DataRaw.<CC>.glob.json` files carry only the variable list, dates, and values — no per-series source mnemonic. These columns originate from the upstream GVAR national-source quarterly panel (national statistical offices / OECD QNA / IMF flows, balanced to a common 2025Q4 endpoint via WEO-benchmarked temporal disaggregation in `build_gvar_2025q4.py`); the exact provider series IDs are therefore *not* determinable from the artefacts inspected and are left flagged rather than fabricated. The cells documented with a concrete ID — POLRATE (BIS CBPOL WS.CBPOL M.<CC>), HP (FRED BIS real RPP

Q<CC>R628BIS), the four global commons (FRED), the India domestic rates (FRED/OECD MEI IRSTCI01INM, INDIR3TIB01STM, INDIRLTLT01STM, traced in the research report), and the whole credit block — are confirmed against their builders.

[FLAG: India domestic-rate IDs are inferred from the research report’s FRED enumeration (IRSTCI01INM, INDIR3TIB01STM, INDIRLTLT01STM); they match the OECD-MEI rate definitions used for IN but were not independently re-fetched in this documentation pass.]

10.50 Emerging markets II: data conventions (TR, ZA, SA, AR, RU, TH, MY, TW)

Transformation and COVID conventions. Throughout these tables the *transformation* column follows the tool-wide convention of Section 9.8: `log` denotes a series stored as $100 \cdot \log(\text{level})$, which enters the VAR and is reported as $4 \cdot \Delta \log =$ the quarter-on-quarter *annualized* growth rate; and `lin` denotes a level (rates in percent, ratios in percent of GDP, index points). The *prior* on every variable’s own first lag is the random walk (RW) unless noted. *COVID-as-missing* (Section 9.4) masks the three quarters 2020Q2–2020Q4 of every *macro* variable (those tagged `isFinancial=0`); these holes are completed inside the Durbin–Koopman E-step rather than fitted. *Financial* variables (`isFinancial=1`) are held *observed* through the COVID window. The four *global commons* are identical US-determined columns shared by every bloc and pinned 1:1 onto each RoW bloc at solve time (Section 6.6): `GLB_OIL` = FRED WTISPLC (WTI spot, \$/bbl, quarterly average); `GLB_USLR` = FRED DGS10 (US 10Y Treasury constant-maturity yield); `GLB_USBAA` = FRED BAA (Moody’s seasoned Baa corporate-bond yield); `GLB_USEQ` = US equity price index (FRED SPASTT01USM661N, OECD US share-price/S&P proxy). All four are financial and observed through COVID.

The new credit block. Each of these eight blocs is augmented with a *credit block* of one or two variables. The *quantity* leg `CREDIT_GDP_<CC>` is, wherever a BIS reporting series exists, the BIS *Total Credit to Non-Financial Corporations*, percent of GDP (the *broad* credit concept: bank loans plus debt securities plus cross-border lending), pulled from FRED under the uniform mnemonic Q<IS02>NAM770A (end-of-quarter, % of GDP). It is stored `lin/RW` and is tagged *financial*, but because its denominator is nominal GDP it inherits the macro COVID mask: its 2020Q2–Q4 cells are set `null`. The *price* leg is either a bank lending rate (`LEND_<CC>`) or, where no lending rate is reachable, a corporate/sovereign-proxy yield (`CORP_<CC>`); it is `lin/RW`, financial, and *observed* through COVID. Where the price could not be sourced it was *dropped* (quantity only). Bloc-specific caveats are flagged in-line below.

10.51 Türkiye (TR)

Türkiye is a coupled GVAR bloc estimated with $n = 16$ quarterly domestic + global variables (the staged credit spec; the deployed GVAR bloc adds the three `ROW*.TR` trade-weighted stars) at $p = 3$ lags over a quarterly grid running 2000Q1–2025Q4 ($T = 104$). The eight real-activity, price and labour series are masked over COVID 2020Q2–Q4 and completed by the Durbin–Koopman

E-step; financials are kept observed. The credit block is *quantity-only*: the bank-lending-rate / corporate-yield price leg was dropped (flag below).

Table 97: Türkiye (TR) variable documentation.

Variable (id)	Description	Source	Series ID/key	Freq	Trans.	COVID	Notes
GDP_TR	Real GDP	IMF/OECD QNA (GVAR base)	[FLAG]	Q	log	masked	WEO-extended to 2025Q4
CONS_TR	Real household consumption	IMF/OECD QNA	[FLAG]	Q	log	masked	
INV_TR	Real gross fixed capital formation	IMF/OECD QNA	[FLAG]	Q	log	masked	
GOV_TR	Real government consumption	IMF/OECD QNA	[FLAG]	Q	log	masked	
EXP_TR	Real exports	IMF/OECD QNA	[FLAG]	Q	log	masked	
IMP_TR	Real imports	IMF/OECD QNA	[FLAG]	Q	log	masked	
CPI_TR	CPI, all items (index)	IMF IFS / national	[FLAG]	Q	log	masked	
LR_TR	Unemployment rate	IMF IFS / national	[FLAG]	Q	lin	masked	
RATE_TR	Monetary-policy-related (short) rate	IMF IFS / national	[FLAG]	Q	lin	observed	financial
POLRATE_TR	Policy rate (BIS CBPOL)	BIS SDMX M.TR (3M avg) WS.CBPOL		Q	lin	observed	financial
HP_TR	Real residential property price	FRED (BIS real RPP)	QTRR628BIS	Q	log	observed	financial
GLB_OIL	WTI crude (q-avg, \$/bbl)	FRED	WTISPLC	Q	log	observed	global common
GLB_USLR	US 10Y Treasury yield (%)	FRED	DGS10	Q	lin	observed	global common
GLB_USBAA	US Baa corporate yield (%)	FRED	BAA	Q	lin	observed	global common
GLB_USEQ	US equity (S&P 500 proxy)	FRED	SPASTT01USM661M	Q	log	observed	global common
CREDIT_GDP_TR	Credit to NFCs (% of GDP) [NEW: credit quantity]	BIS via FRED	QTRNAM770A	Q	lin	masked	RW; nonnull 101/104, range 17.5–78.1%; 2000Q1=23.2, 2025Q4=38.9; COVID cells null

[FLAG: TR credit PRICE DROPPED] No reachable free NFC lending rate or corporate/sovereign

yield: OECD IRLT/FINMARK is empty for TUR; the IMF MFS_IR flow carries only discount/deposit/policy-related rates for TUR (which would duplicate RATE_TR/POLRATE_TR); the CBRT EVDS commercial-loan rate (TP.KTF12) needs an EVDS API key not available; FRED Turkey rate series are stale (≤ 2022). Per the locked spec the TR credit block is *quantity only*.

[FLAG: TR base-macro series IDs] The provider-class for the eight base real-activity/price/labour series and the short rate is documented at the GVAR roster level (Section 9.3: OECD QNA / IMF flows, extended to 2025Q4 by `build_gvar_2025q4.py` via WEO temporal disaggregation), but the upstream raw file `country_rawdata/DataRaw.TR_glob.json` carries no per-variable source/series-id stamp, so the exact provider series codes for GDP/CONS/INV/GOV/EXP/IMP/CPI/LR/RATE cannot be asserted from the artifacts and are not fabricated here.

10.52 South Africa (ZA)

South Africa is a coupled GVAR bloc, $n = 19$ (staged credit spec; deployed bloc adds the three ROW*_ZA stars), $p = 3$, over 2002Q1–2025Q4 ($T = 96$). Macro series are COVID-masked 2020Q2–Q4 and DK-completed; financials observed. The credit block carries both legs: a BIS quantity and an IMF/IFS lending-rate price.

Table 98: South Africa (ZA) variable documentation.

Variable (id)	Description	Source	Series ID/key	Freq	Trans.	COVID	Notes
GDP_ZA	Real GDP	IMF/OECD QNA (GVAR base)	[FLAG]	Q	log	masked	WEO-extended
CONS_ZA	Real household consumption	IMF/OECD QNA	[FLAG]	Q	log	masked	
INV_ZA	Real gross fixed capital formation	IMF/OECD QNA	[FLAG]	Q	log	masked	
GOV_ZA	Real government consumption	IMF/OECD QNA	[FLAG]	Q	log	masked	
EXP_ZA	Real exports	IMF/OECD QNA	[FLAG]	Q	log	masked	
IMP_ZA	Real imports	IMF/OECD QNA	[FLAG]	Q	log	masked	
CPI_ZA	CPI index	IMF IFS / national	[FLAG]	Q	log	masked	
LR_ZA	Unemployment rate	IMF IFS / national	[FLAG]	Q	lin	masked	
RATE_ZA	SARB policy rate (repo)	IMF IFS / SARB	[FLAG]	Q	lin	observed	financial
YIELD10_ZA	10Y government bond yield	IMF IFS / national	[FLAG]	Q	lin	observed	financial
NEER_ZA	Nominal effective exchange rate	IMF IFS / BIS	[FLAG]	Q	log	observed	financial

Variable (id)	Description	Source	Series ID/key	Freq	Trans.	COVID	Notes
POLRATE_ZA	Policy rate (BIS CBPOL)	BIS SDMX WS.CBPOL	M.ZA (3M avg)	Q	lin	observed	financial
HP_ZA	Real residential property price	FRED (BIS real RPP)	QZAR628BIS	Q	log	observed	financial
GLB_OIL	WTI crude oil (\$/bbl)	(q-avg, FRED)	WTISPLC	Q	log	observed	global common
GLB_USLR	US 10Y Treasury yield (%)	FRED	DGS10	Q	lin	observed	global common
GLB_USBAA	US Baa corporate yield (%)	FRED	BAA	Q	lin	observed	global common
GLB_USEQ	US equity (S&P 500 proxy)	FRED	SPASTT01USM661M	Q	log	observed	global common
CREDIT_GDP_ZA	Credit to NFCs (% of GDP) [NEW: credit quantity]	BIS via FRED	QZANAM770A	Q	lin	masked	RW; nonnull 69/96 (BIS series starts 2008Q1 $\Rightarrow$ 24 leading nulls), range 32.9–41.4%; last 33.2; COVID cells null
LENDR_ZA	IFS lending rate (% p.a.) [NEW: credit price]	IMF IFS via MFS_IR	ZAF.MFS162_RT_PQA.P11nQ	Q	lin	observed	RW; SARB-prime-linked representative bank lending rate; nonnull 96/96, range 7.0–17.0%; last 10.33

[FLAG: ZA credit quantity leading gap] The BIS NFC series begins 2008Q1, leaving 24 leading nulls (2002Q1–2007Q4); these are a normal ragged leading edge handled by the COVID-as-missing Gibbs.

[FLAG: ZA base-macro series IDs] As for TR, the per-variable provider series codes for the eight base macro series, the SARB repo rate, the 10Y yield and the NEER are not stamped in `country_rawdata/DataRaw_ZA_glob.json`; provider class is per Section 9.3 (IMF flows / OECD QNA / national, WEO-extended).

10.53 Saudi Arabia (SA)

Saudi Arabia is a coupled GVAR bloc, $n = 16$ (staged credit spec; deployed bloc adds the three ROW*.SA stars), $p = 3$, over 2010Q1–2025Q4 ($T = 64$). Macro series are COVID-masked 2020Q2–Q4 and DK-completed; financials observed. The credit block is *quantity-only* (price dropped; flag

below).

Table 99: Saudi Arabia (SA) variable documentation.

Variable (id)	Description	Source	Series ID/key	Freq	Trans.	COVID	Notes
GDP_SA	Real GDP	IMF/OECD QNA (GVAR base)	[FLAG]	Q	log	masked	WEO-extended
CONS_SA	Real household consumption	IMF/OECD QNA	[FLAG]	Q	log	masked	
INV_SA	Real gross fixed capital formation	IMF/OECD QNA	[FLAG]	Q	log	masked	
GOV_SA	Real government consumption	IMF/OECD QNA	[FLAG]	Q	log	masked	
EXP_SA	Real exports	IMF/OECD QNA	[FLAG]	Q	log	masked	
IMP_SA	Real imports	IMF/OECD QNA	[FLAG]	Q	log	masked	
CPI_SA	CPI index	IMF IFS / national	[FLAG]	Q	log	masked	
LR_SA	Unemployment rate	IMF IFS / national	[FLAG]	Q	lin	masked	
RATE_SA	Policy rate (SAMA repo)	IMF IFS / SAMA	[FLAG]	Q	lin	observed	financial
NEER_SA	Nominal effective exchange rate	IMF IFS / BIS	[FLAG]	Q	log	observed	financial
POLRATE_SA	Policy rate (BIS CBPOL)	BIS SDMX M.SA (3M avg) WS.CBPOL		Q	lin	observed	financial
GLB_OIL	WTI crude (q-avg, \$/bbl)	FRED	WTISPLC	Q	log	observed	global common
GLB_USLR	US 10Y Treasury yield (%)	FRED	DGS10	Q	lin	observed	global common
GLB_USBAA	US Baa corporate yield (%)	FRED	BAA	Q	lin	observed	global common
GLB_USEQ	US equity (S&P 500 proxy)	FRED	SPASTT01USM6611Q	Q	log	observed	global common
CREDIT_GDP_SA	Credit to NFCs (% of GDP) [NEW: credit quantity]	BIS via FRED	QSANAM770A	Q	lin	masked	RW; nonnull 61/64, range 19.3–45.9%; 2010Q1=27.7, 2025Q4=45.9; COVID cells null

[FLAG: SA credit PRICE DROPPED] SAMA publishes no NFC lending rate (SAIBOR is an interbank money-market rate, conceptually covered by RATE_SA); SA is absent from IMF MFS_IR;

OECD/FRED/WB carry no Saudi lending rate. Saudi banking is largely Islamic-finance, so no conventional IFS-style “lending rate” is published. Per the locked spec the SA credit block is *quantity only*.

[*FLAG: SA base-macro series IDs*] Per-variable provider series codes for the base macro series, the SAMA repo rate and the NEER are not stamped in `country_rawdata/DataRaw_SA_glob.json`; provider class per Section 9.3.

10.54 Argentina (AR)

Argentina is a coupled GVAR bloc, $n = 17$ (staged credit spec; deployed bloc adds the three ROW*`_AR` stars), $p = 3$, over 2004Q1–2025Q4 ($T = 88$). Macro series are COVID-masked 2020Q2–Q4 and DK-completed; financials observed. Note that the entire AR bloc uses an RW own-lag prior on *every* variable (including the macro logs), reflecting the highly non-stationary, high-inflation Argentine sample. The credit block carries both legs.

Table 100: Argentina (AR) variable documentation.

Variable (id)	Description	Source	Series ID/key	Freq	Trans.	COVID	Notes
GDP_AR	Real GDP	IMF/OECD QNA (GVAR base)	[FLAG]	Q	log	masked	RW; WEO-extended
CONS_AR	Real household consumption	IMF/OECD QNA	[FLAG]	Q	log	masked	RW
INV_AR	Real gross fixed capital formation	IMF/OECD QNA	[FLAG]	Q	log	masked	RW
GOV_AR	Real government consumption	IMF/OECD QNA	[FLAG]	Q	log	masked	RW
EXP_AR	Real exports	IMF/OECD QNA	[FLAG]	Q	log	masked	RW
IMP_AR	Real imports	IMF/OECD QNA	[FLAG]	Q	log	masked	RW
CPI_AR	CPI index	IMF IFS / national	[FLAG]	Q	log	masked	RW
LR_AR	Unemployment rate	IMF IFS / national	[FLAG]	Q	lin	masked	RW
RATE_AR	BADLAR short-term rate	IMF IFS / BCRA	[FLAG]	Q	lin	observed	financial, RW
NEER_AR	Nominal effective exchange rate	IMF IFS / BIS	[FLAG]	Q	log	observed	financial, RW
POLRATE_AR	Policy rate (BIS CBPOL)	BIS SDMX M.AR (3M avg) WS.CBPOL	[FLAG]	Q	lin	observed	financial, RW
GLB_OIL	WTI crude (\$/bbl)	(q-avg, FRED)	WTISPLC	Q	log	observed	global common
GLB_USLR	US 10Y Treasury yield (%)	FRED	DGS10	Q	lin	observed	global common

Variable (id)	Description	Source	Series ID/key	Freq	Trans.	COVID	Notes
GLB_USBAA	US Baa corporate yield (%)	FRED	BAA	Q	lin	observed	global common
GLB_USEQ	US equity (S&P 500 proxy)	FRED	SPASTT01USM6611Q	Q	log	observed	global common
CREDIT_GDP_AR	Credit to NFCs (% of GDP) [NEW: credit quantity]	BIS via FRED	QARNAM770A	Q	lin	masked	RW; nonnull 85/88, range 13.5–38.0%; 2004Q1=38.0, 2025Q4=23.2; COVID cells null
LENDR_AR	Lending rate (% p.a.; IMF extreme volatility) [NEW: credit price]	IFS via MFS_IR	ARG.MFS162_RT_PT_A_PT	Q	lin	observed	RW; peso lending rate (not the FC variant); nonnull 64/88 (24 leading nulls, IFS starts 2010Q1); range 9.85–120.54% (120.54% @ 2023Q4 = genuine hyperinflation peak); observed through COVID

[*FLAG: AR credit price extreme volatility*] The peso lending rate swings enormously with the inflation regime (peak 120.5% in 2023Q4); under an RW lin prior it produces very large innovations. The series has 24 leading nulls (2004Q1–2009Q4; IFS begins 2010Q1), handled by the ragged-edge Gibbs. The peso variant MFS162_RT_PT_A_PT is used, not the much-lower foreign-currency variant. [FLAG: AR base-macro series IDs] Per-variable provider codes for the base macro series, the BAD-LAR rate and the NEER are not stamped in `country_rawdata/DataRaw_AR_glob.json`; provider class per Section 9.3.

10.55 Russia (RU)

Russia is a coupled GVAR bloc, $n = 14$ (staged credit spec; deployed bloc adds the three ROW*RU stars), $p = 3$, over 2005Q1–2025Q4 ($T = 84$). It is a compact bloc (no consumption/investment/government/import expenditure split): its real-side activity is summarised by real GDP and goods exports. Macro series are COVID-masked 2020Q2–Q4 and DK-completed; financials observed. The credit block carries both legs; the price leg is *spliced* (flag below).

Table 101: Russia (RU) variable documentation.

Variable (id)	Description	Source	Series ID/key	Freq	Trans.	COVID	Notes
GDP_RU	Real GDP (const. 2010 LCU, SA)	IMF/OECD QNA (GVAR base)	[FLAG]	Q	log	masked	RW; WEO-extended
CPI_RU	CPI index (all-items)	IMF IFS / national	[FLAG]	Q	log	masked	RW
EXP_RU	Goods exports (FOB, USD)	IMF IFS / national	[FLAG]	Q	log	masked	RW
LR_RU	Unemployment rate	IMF IFS / national	[FLAG]	Q	lin	masked	RW
RATE_RU	Interbank call-money rate	IMF IFS / CBR	[FLAG]	Q	lin	observed	financial, RW
NEER_RU	Nominal effective exchange rate	IMF IFS / BIS	[FLAG]	Q	log	observed	financial, RW
POLRATE_RU	Policy rate (BIS CBPOL)	BIS SDMX M.RU (3M avg) WS.CBPOL		Q	lin	observed	financial, RW
HP_RU	Real residential property price	FRED (BIS QRUR628BIS real RPP)		Q	log	observed	financial, RW
GLB_OIL	WTI crude (q-avg, \$/bbl)	FRED	WTISPLC	Q	log	observed	global common
GLB_USLR	US 10Y Treasury yield (%)	FRED	DGS10	Q	lin	observed	global common
GLB_USBAA	US Baa corporate yield (%)	FRED	BAA	Q	lin	observed	global common
GLB_USEQ	US equity (S&P 500 proxy)	FRED	SPASTT01USM66100	Q	log	observed	global common
CREDIT_GDP_RU	Credit to NFCs (% of GDP) [NEW: credit quantity]	BIS via FRED	QRUNAM770A	Q	lin	masked	RW; nonnull 81/84, range 31.1–90.3%; 2005Q1=31.7, 2025Q4=84.7; COVID cells null
LENDR_RU	Lending rate (% p.a.; IMF spliced w/ CBR) [NEW: credit price]	IMF IFS RUS.MFS162_RT_PQA_P11nQ MFS_IR spliced + CBR with CBR loans_nonfin.e.xlsx		Q	lin	observed	RW; full coverage 84/84, range 6.05–22.57%; 2005Q1=10.63, 2025Q4=18.23

[FLAG: RU credit price SPLICED] IMF MFS162 ends 2024Q2 (= 17.21). For 2024Q3–2025Q4 the series is spliced with the CBR weighted-average rate on ruble loans to non-financial organisations, “up to 1 year (incl. call loans)” (loans_nonfin.e.xlsx, sheet 1 col. 5; values 18.97, 22.57, 22.19, 22.28, 19.64, 18.23). This is the same underlying CBR survey the IMF reports as MFS162: validated MAD = 0.003 over 42 overlapping quarters (2014–2024Q2), with a continuous seam at 2024Q2 (IMF

17.21 vs CBR 17.15).

[*FLAG: RU base-macro series IDs*] Per-variable provider codes for GDP, CPI, goods exports, unemployment, the interbank rate and the NEER are not stamped in `country_rawdata/DataRaw_RU_glob.json`; provider class per Section 9.3.

10.56 Thailand (TH)

Thailand is a coupled GVAR bloc, $n = 19$ (staged credit spec; deployed bloc adds the three `ROW*_TH` stars), $p = 3$, over 2003Q1–2025Q4 ($T = 92$). All expenditure components and prices are seasonally adjusted. Macro series are COVID-masked 2020Q2–Q4 and DK-completed; financials observed. The credit block carries both legs.

Table 102: Thailand (TH) variable documentation.

Variable (id)	Description	Source	Series ID/key	Freq	Trans.	COVID	Notes
GDP_TH	Real GDP (SA, const. prices)	IMF/OECD QNA (GVAR base)	[<i>FLAG</i>]	Q	log	masked	WEO-extended
CONS_TH	Real private consumption (SA)	IMF/OECD QNA	[<i>FLAG</i>]	Q	log	masked	
INV_TH	Real gross fixed capital formation (SA)	IMF/OECD QNA	[<i>FLAG</i>]	Q	log	masked	
GOV_TH	Real government consumption (SA)	IMF/OECD QNA	[<i>FLAG</i>]	Q	log	masked	
EXP_TH	Real exports of g&s (SA)	IMF/OECD QNA	[<i>FLAG</i>]	Q	log	masked	
IMP_TH	Real imports of g&s (SA)	IMF/OECD QNA	[<i>FLAG</i>]	Q	log	masked	
CPI_TH	CPI, all items (index)	IMF IFS / national	[<i>FLAG</i>]	Q	log	masked	
LR_TH	Unemployment rate	IMF IFS / national	[<i>FLAG</i>]	Q	lin	masked	
RATE_TH	Policy rate (BOT)	IMF IFS / BOT	[<i>FLAG</i>]	Q	lin	observed	financial
YIELD10_TH	10Y government bond yield	IMF IFS / national	[<i>FLAG</i>]	Q	lin	observed	financial
NEER_TH	Nominal effective exchange rate (BIS broad)	BIS / IMF	[<i>FLAG</i>]	Q	log	observed	financial
POLRATE_TH	Policy rate (BIS CBPOL)	BIS SDMX M.TH (3M avg) WS.CBPOL		Q	lin	observed	financial
HP_TH	Real residential property price	FRED (BIS real RPP)	QTHR628BIS	Q	log	observed	financial
GLB_OIL	WTI crude (q-avg, \$/bbl)	FRED	WTISPLC	Q	log	observed	global common

Variable (id)	Description	Source	Series ID/key	Freq	Trans.	COVID	Notes
GLB_USLR	US 10Y Treasury yield (%)	FRED	DGS10	Q	lin	observed	global common
GLB_USBAA	US Baa corporate yield (%)	FRED	BAA	Q	lin	observed	global common
GLB_USEQ	US equity (S&P 500 proxy)	FRED	SPASTT01USM6610Q	Q	log	observed	global common
CREDIT_GDP_TH	Credit to NFCs (% of GDP) [NEW: credit quantity]	BIS via FRED	QTHNAM770A	Q	lin	masked	RW; nonnull 89/92, range 68.4–95.9%; last 68.5; COVID cells null
LENDR_TH	IFS lending rate (% p.a.) [NEW: credit price]	IMF IFS via MFS_IR	THA.MFS162_RT_PQA.P1nQ	Q	lin	observed	RW; BOT announced/minimum loan rate; non-null 88/92 (4 leading nulls in 2003, IFS starts 2004Q1); range 3.03–6.58%; last 3.44

[FLAG: TH credit price leading gap] The IFS lending rate has 4 leading nulls (2003; the series begins 2004Q1), a normal ragged leading edge handled by the Gibbs.

[FLAG: TH base-macro series IDs] Per-variable provider codes for the base macro series, the BOT policy rate, the 10Y yield and the NEER are not stamped in `country_rawdata/DataRaw_TH_glob.json`; provider class per Section 9.3.

10.57 Malaysia (MY)

Malaysia is a coupled GVAR bloc, $n = 18$ (staged credit spec; deployed bloc adds the three ROW* MY stars), $p = 3$, over a short 2015Q1–2025Q4 grid ($T = 44$). Macro series are COVID-masked 2020Q2–Q4 and DK-completed; financials observed. The credit block carries both legs; note the short sample (flag).

Table 103: Malaysia (MY) variable documentation.

Variable (id)	Description	Source	Series ID/key	Freq	Trans.	COVID	Notes
GDP_MY	Real GDP (chain volume, SA)	IMF/OECD QNA (GVAR base)	[FLAG]	Q	log	masked	WEO-extended
CONS_MY	Real private consumption (SA)	IMF/OECD QNA	[FLAG]	Q	log	masked	

Variable (id)	Description	Source	Series ID/key	Freq	Trans.	COVID	Notes
INV_MY	Real gross fixed capital formation (SA)	IMF/OECD QNA	[FLAG]	Q	log	masked	
GOV_MY	Real government consumption (SA)	IMF/OECD QNA	[FLAG]	Q	log	masked	
EXP_MY	Real exports of g&s (SA)	IMF/OECD QNA	[FLAG]	Q	log	masked	
IMP_MY	Real imports of g&s (SA)	IMF/OECD QNA	[FLAG]	Q	log	masked	
CPI_MY	CPI index (all items)	IMF IFS / national	[FLAG]	Q	log	masked	
LR_MY	Unemployment rate	IMF IFS / national	[FLAG]	Q	lin	masked	
RATE_MY	Overnight policy rate (BNM)	IMF IFS / BNM	[FLAG]	Q	lin	observed	financial
NEER_MY	Nominal effective exchange rate	IMF IFS / BIS	[FLAG]	Q	log	observed	financial
POLRATE_MY	Policy rate (BIS CBPOL)	BIS SDMX M.MY (3M avg) WS_CBPOL		Q	lin	observed	financial
HP_MY	Real residential property price	FRED (BIS real RPP)	QMYR628BIS	Q	log	observed	financial
GLB_OIL	WTI crude (q-avg, \$/bbl)	FRED	WTISPLC	Q	log	observed	global common
GLB_USLR	US 10Y Treasury yield (%)	FRED	DGS10	Q	lin	observed	global common
GLB_USBAA	US Baa corporate yield (%)	FRED	BAA	Q	lin	observed	global common
GLB_USEQ	US equity (S&P 500 proxy)	FRED	SPASTT01USM6611Q	Q	log	observed	global common
CREDIT_GDP_MY	Credit to NFCs (% of GDP) [NEW: credit quantity]	BIS via FRED	QMYNAM770A	Q	lin	masked	RW; nonnull 41/44, range 86.1–106.3%; last 88.6; COVID cells null
LENDR_MY	IFS lending rate (% p.a.) [NEW: credit price]	IMF IFS via MFS_IR	MYS.MFS162_RT_PQA.P11nQ	Q	lin	observed	RW; BNM average lending/base rate; nonnull 44/44, range 3.31–5.47%; last 4.62

[FLAG: MY short sample] The MY grid is only $T = 44$ (2015Q1–2025Q4) with 41 credit observations – thin for a bloc that adds two credit variables; predictive bands are correspondingly wide.

[FLAG: MY base-macro series IDs] Per-variable provider codes for the base macro series, the BNM overnight policy rate and the NEER are not stamped in `country_rawdata/DataRaw_MY_glob.json`;

provider class per Section 9.3.

10.58 Taiwan, Province of China (TW)

Taiwan is a coupled GVAR bloc, $n = 17$ (staged credit spec; deployed bloc adds the three ROW*.TW stars), $p = 3$, over a long 1997Q1–2025Q4 grid ($T = 116$). It carries an unusually rich short-end financial block (CBC discount rate, interbank call-loan rate and 10Y yield) but *no* BIS house-price series. Macro series are COVID-masked 2020Q2–Q4 and DK-completed; financials observed. Both credit legs are present but *constructed/national*, not BIS (flags below).

Table 104: Taiwan (TW) variable documentation.

Variable (id)	Description	Source	Series ID/key	Freq	Trans.	COVID	Notes
GDP_TW	Real GDP	DGBAS OECD-style QNA	/ [FLAG]	Q	log	masked	RW; WEO-extended
CONS_TW	Real household consumption	DGBAS	[FLAG]	Q	log	masked	RW
GOV_TW	Real government consumption	DGBAS	[FLAG]	Q	log	masked	RW
INV_TW	Real gross fixed capital formation	DGBAS	[FLAG]	Q	log	masked	RW
EXP_TW	Real exports of g&s	DGBAS	[FLAG]	Q	log	masked	RW
IMP_TW	Real imports of g&s	DGBAS	[FLAG]	Q	log	masked	RW
CPI_TW	CPI, all items (index, 2021=100)	DGBAS	[FLAG]	Q	log	masked	RW
LR_TW	Unemployment rate	DGBAS	[FLAG]	Q	lin	masked	RW
RATE_TW	CBC discount rate (policy)	CBC	[FLAG]	Q	lin	observed	financial, RW
MMRATE_TW	Interbank call-loan rate (money mkt)	CBC	[FLAG]	Q	lin	observed	financial, RW
YIELD10_TW	10Y government bond yield	CBC / national	na- [FLAG]	Q	lin	observed	financial, RW
GLB_OIL	WTI crude (q-avg, \$/bbl)	FRED	WTISPLC	Q	log	observed	global common
GLB_USLR	US 10Y Treasury yield (%)	FRED	DGS10	Q	lin	observed	global common
GLB_USBAA	US Baa corporate yield (%)	FRED	BAA	Q	lin	observed	global common
GLB_USEQ	US equity (S&P 500 proxy)	FRED	SPASTT01USM6610Q	Q	log	observed	global common

Variable (id)	Description	Source	Series ID/key	Freq	Trans.	COVID	Notes
CREDIT_GDP_TW	Credit to NFCs (% of GDP) [NEW: DGBAS credit quantity, CONSTRUCTED]	CBC EI86 / 100× DGBAS GDP	CBC EI86 private-enterprise loans (NT\$mn, monthly stock → q-avg) ÷ 4Q-trailing DGBAS nominal GDP	Q	lin	masked	RW; nonnull 113/116, range 46.3–65.4%; 2025Q4=60.6; COVID cells null
LENDR_TW	5-major-banks loan rate (% p.a.) [NEW: credit price]	new CBC (total, interest rate)	EH45 CBC EH45 total (monthly → q-avg)	Q	lin	observed	RW; CBC realised five-major-banks new-loan rate; nonnull 116/116, range 1.16–8.62%; 2020Q2=1.26; 2025Q4=2.16

[*FLAG: TW credit quantity CONSTRUCTED, not BIS*] BIS publishes no NFC credit-to-GDP for Taiwan (FRED QTWNAM770A returns HTTP 400; BIS WS_TC SDMX returns “no structures match”). CREDIT_GDP_TW is therefore constructed as 100× CBC EI86 private-enterprise loans (private enterprises; excludes household and public-enterprise loans) over 4-quarter-trailing DGBAS nominal GDP. This is a private-enterprise-loan proxy, broader than the BIS market-value NFC concept used elsewhere, so cross-economy comparability of the credit variable is reduced.

[*FLAG: TW credit price provenance*] LENDR_TW is the CBC’s published headline five-major-banks total new-loan rate (EH45, “total”), used as the locked-spec “5-bank base lending rate” – a national CBC series, not an IMF or BIS feed. The Taiwan source files are fetched via `fetch_tw_cbc_dgbas.py`; the DGBAS GDP host ships an incomplete TLS chain and is fetched with verification relaxed for that host only (documented in the fetcher).

[*FLAG: TW house price*] Unlike the other blocs here, TW carries no HP_TW (no BIS real residential property price was added).

[*FLAG: TW base-macro series IDs*] The base real-activity, price, labour and short-rate series originate with Taiwan’s DGBAS (national accounts, CPI, unemployment) and the CBC (discount rate, call-loan rate, 10Y yield), but the upstream raw file `country_rawdata/DataRaw_TW_glob.json` carries no per-variable series-id stamp, so exact DGBAS/CBC table codes for GDP/CONS/GOV/INV/EXP/IMP/CPI/LR/RA are not asserted here.

11 User Interface Guide

11.1 Top Bar

The tool is deployed as a static site served at the `/bvar-scenario` path of the academic website, which is hosted on **GitHub Pages** (the `matyasfarkas/starter-hugo-academic` Hugo Academic site). The top bar provides global controls:

- **Country selector:** Switch between any of the roughly fifty estimated economies. The drop-down is grouped into two sections by actual GVAR status: a *GVAR* group listing the 33 coupled blocs, and an *Other countries* group listing the standalone single-country models (Section 9). Membership is conveyed purely by the section a country sits in—the names are shown plainly, with no per-option suffix or bold—and the split is driven by the live trade-weights roster rather than a static geographic grouping. Switching reloads all BVAR data and resets conditioning.
- **User Manual:** Download this document as a PDF.
- **Uncertainty mode:** “None” (deterministic), “Quick” ($D = 50$ draws), or “Full” ($D = 200$ draws).
- **Compute button:** Triggers the simulation smoother (Section 4.6). For the long-run anchor (the entropic tilt of Section 8.2), the Compute button is enabled whenever an anchor is active and auto-selects “Quick” mode if no uncertainty mode is chosen.

11.2 Chart Panel

The main panel displays a grid of time-series charts. The first row shows the primary macroeconomic triad—inflation, policy rate, GDP growth—plus government spending. Subsequent rows show core inflation, unemployment, consumption, investment, and any country-specific variables.

Each chart includes:

- A **dotted baseline** forecast with 68% credible bands (shaded).
- A **solid scenario line** showing the conditional forecast.
- A **dashed counterfactual line** (when active) showing the policy counterfactual.
- A **“Condition” button** to toggle conditioning on the variable.
- **Drag interaction:** Click and drag data points on conditioned charts to set target paths. Right-click to toggle individual quarters on or off.

GVAR solve badge and the “Solve full GVAR” button. On a GVAR-linked bloc a small badge above the charts reports the cross-country solve that produced the displayed scenario, together with its sweep count, spectral radius ρ , and time. Because the US is now endogenous (Section 6.5), dragging a pin shows a fast first-order *direct preview* (an amber “GVAR DIRECT preview” badge); when a preview is on screen a blue **Solve full GVAR** button is offered to compute the complete cross-country fixed point—all feedback loops, including the Rest-of-World feedback onto the US (a blue “GVAR full solve” badge). While a solve is running a “computing...” pill is shown so a structure-change re-probe does not look like a freeze. On standalone (non-GVAR) blocs no badge or button appears; the scenario is the ordinary single-country conditional forecast.

11.3 Control Panel (Sidebar)

11.3.1 Scenario Section

- **Scenario name:** Editable text field.
- **Reset:** Clear all conditioning to return to baseline.
- **Save/Load:** Export and import scenarios as Excel files. The Excel format includes sheets for settings, conditioning paths, active quarters, rate adjustments, and a variable map for reproducibility.
- **Example:** Download a pre-built inflation-surge scenario.
- **Conditioning sliders:** Per-variable, per-quarter deviation sliders (± 3 pp) for each conditioned variable.

11.3.2 Policy Counterfactual Section

- **Counterfactual toggle:** Enter/exit counterfactual mode.
- **DSGE model browser:** Select from 169 models, searchable by name or reference.
- **Policy rule selector:** Choose from 9 standardized monetary policy rules (Table in Section 5.1).
- **Rule-based / Optimal toggle:** Switch between specifying the rate path via a Taylor-type rule or via the [Barnichon and Mesters \(2023\)](#) optimal policy (Section 5.4).
- **Dual-mandate slider:** In optimal mode, adjust the relative weight on price stability versus full employment.

11.4 Export

The scenario bar at the bottom provides four export formats:

- **CSV:** Tabular data with columns for each variable’s Q16 (16th percentile), baseline (median), Q84 (84th percentile), active scenario, deviation from baseline, and counterfactual (if active). Rows span historical quarters through the full forecast horizon.
- **Excel:** A workbook with per-variable detail sheets (Q16, baseline, Q84, scenario, deviation, counterfactual) and a summary sheet showing all variables side by side.
- **PNG:** A ZIP archive of high-resolution chart images (1200×500 px) capturing the full visual state including all rendered scenarios, uncertainty bands, and counterfactuals.
- **PDF:** A multi-page PDF with one chart per page (landscape orientation), including metadata footers with country, timestamp, and chart title.

All tabular exports (CSV, Excel) respect the active growth-rate toggle: if year-over-year is selected, the exported values reflect the y/y transformation. Visual exports (PNG, PDF) capture exactly what is displayed on screen, including all visible scenarios and their uncertainty bands.

11.4.1 Multi-Scenario Support

The tool supports up to four simultaneous scenarios. Each scenario’s conditional forecast and uncertainty bands are cached when computed, so that switching between scenarios preserves their results. Tabular exports include data for the active scenario; visual exports capture all visible scenario lines and bands.

12 Technical Notes

12.1 Client-Side Computation

All matrix algebra, Kalman filtering, and simulation smoothing are performed in the browser using TypeScript with `Float64Array` storage. The linear algebra kernel implements:

- LU decomposition with partial pivoting for F_t^{-1} (singularity threshold: 10^{-15}).
- Cholesky decomposition for shock draws ($\Sigma = LL'$; diagonal check threshold: 10^{-14}).
- Row-major storage for cache-friendly matrix operations.

12.2 Performance Optimizations

- The unconditional smoother result is cached by a fingerprint of the model dimensions and a sample of coefficient values, avoiding recomputation when only the conditioning paths change.
- Conditioning input is debounced at 30ms to avoid recomputation during interactive drag operations.

- The 500 thinned posterior draws (≈ 34 MB for the US model) are loaded lazily—only when the user clicks “Compute.”
- Plotly.js is loaded via dynamic `import()` to avoid blocking initial page render.

12.3 Interactive GVAR Architecture: keeping the heavy solve off the drag path

The deployed 50-bloc, US-endogenous GVAR poses a hard interactivity problem: a single coupled mean solve costs ≈ 4.6 s warm and ≈ 13 s cold (every US-pinned structure misses the shipped $(I - M)^{-1}$ coupling cache and must re-probe the Durbin–Koopman smoother per coupled bloc). Firing that solve on every pointer-move during a chart drag would make the interface unusable. The deployed design therefore removes the solve from the live drag path entirely, and reconciles the exact answer afterwards.

No solve during drag; exactly one cheap solve on release. A module-level signal bus (`gvarDragState.ts`) carries the reactive drag flag between the chart drag handler and the GVAR solve hook (`useGvarForecast.ts`), kept *out* of the React/Zustand store so that a pointer-down does not re-render the grid. While a drag is in progress *no* GVAR solve fires—neither a value-only nor a structure-change solve, since both are multi-second on the live panel. On **pointer-up** a single solve is dispatched after a short (100 ms) release macrotask that also coalesces back-to-back click-drag-click gestures into one dispatch. That release solve uses the cheap ‘**direct**’ channel (one-hop local-elasticity preview, no 95 MB coupling-artifact structured-clone into the worker); the full cross-country fixed point with uncertainty bands (the ‘**full**’ channel) runs *only* on the explicit “Compute” / “Solve full GVAR” action. Non-drag edits (numeric entry, Reset, scenario switch) still dispatch live: full solves run immediately and direct previews debounce (≈ 120 ms).

Optimistic conditioned-point dot. So that the user sees *immediate* feedback while the line and bands reconcile, the pinned marker (the conditioned-point “dot”) is sourced from the *conditioning path* the click writes—not from the lagging solve output. The mapping lives in `conditionedDotY.ts`: `conditioningDeviationFromDisplayY` converts a clicked display *y*-value into the stored-space deviation, and its exact inverse `conditionedDotDisplayY` reconstructs the displayed dot, so the dot round-trips to the cursor to machine precision (`conditionedDotDisplayYoconditioningDeviationFromDisplayY` id). The same inverse pair backs the right-click “plant the bullet at the cursor” gesture. The smoothed forecast line and the uncertainty bands then catch up when the solve returns.

Per-chart render memoization. The render cost of a conditioning edit was historically dominated by the *render*, not the solve: a monolithic chart component repainted all ≈ 35 panels on any edit. The deployed grid (`ChartGrid.tsx`) wraps each panel in `React.memo` and reconciles the freshly-built chart objects against the prior render with a tight structural equality (`reconcileCharts/chartDeepEqual`): a panel whose trace data did not move keeps its *prior object reference*, so `React.memo` skips it. A single conditioning edit therefore rebuilds and repaints *exactly one* of the ≈ 35 panels rather than all of

them (asserted in `chartgrid-render-memo.test.ts`), with a stale-guard that still repaints every affected panel when the solve lands. The whole interactive contract—no solve on drag, one cheap solve on release, an optimistic dot, and single-panel repaints—is covered by the `gvar-drag-scheduling`, `conditioned-dot-optimistic`, `right-click-plant-at-cursor`, and `chartgrid-render-memo` test suites.

12.4 Numerical Conventions

- Measurement noise: $H = 10^{-12} I_n$, ensuring the Kalman filter treats conditioned values as near-exact observations while maintaining numerical stability.
- Growth-to-level conversion: $\Delta\ell = \Delta g/4$ for log-transformed variables, reflecting the annualization factor in (2).
- Tikhonov regularization for Barnichon–Mesters: $\alpha = 10^{-6}$.
- All matrix operations use IEEE 754 double precision (64-bit floating point).

12.5 Identifiability and Weak Exogeneity

The platform’s separable estimation–solution architecture rests on a chain of identifying assumptions; we state them explicitly and are candid about where each binds.

Weak exogeneity of the star variables. Each bloc is estimated *conditionally* on its three trade-weighted star variables (Section 6.2), treating the foreign environment as predetermined. This is the weak-exogeneity device of the GVAR literature (Pesaran et al., 2004; Dees et al., 2007): provided a domestic economy is small relative to its trading partners, its idiosyncratic shocks do not feed back contemporaneously into the aggregated foreign block, so the conditional likelihood factorizes and the bloc-specific parameters ($\hat{\beta}_c, \hat{\Sigma}_c$) are identified without a joint, system-wide estimator. Chudik and Pesaran (2016) formalize the conditions—a granular weighting scheme in which no single partner dominates the trade-weighted average and a bounded number of strongly correlated units—under which the cross-section-averaged stars remain valid weakly-exogenous regressors as the panel grows. Our renormalized, per-component weights (Section 6.3) are designed to keep this granularity: absent or excluded partners have their mass reabsorbed across the present partners rather than concentrated on any one.

The classical treatment fixes a *frozen dominant unit* (here the United States) outside the system. We relax this in two ways. First, the EM-style re-estimation loop (Section 6.6) rebuilds every bloc’s stars from its partners’ Durbin–Koopman-completed paths and re-estimates each bloc in parallel until the maximum star change falls below the tolerance $\text{TOL} = 0.03$, so the conditioning information is mutually consistent at the fixed point rather than taken as given once. This tolerance is set *tighter* than the Monte Carlo noise floor of the imputation (≈ 0.13 on the star scale); the COVID-window cells of the two macro stars, where cross-iteration churn is dominated by that imputation noise, are

excluded from the convergence metric. Second, in the deployed roster the US is made *endogenous*: it carries its own ROW*_US stars and is solved inside the cross-country fixed point like any other bloc. What keeps the relaxed system well posed is the contraction property of the solve-time coupling operator M of (37): its spectral radius is $\rho(M) \approx 0.59$ for the original six-bloc pilot, $\rho(M) \approx 0.726$ under the frozen-anchor configuration with the euro area aggregated into a single bloc, and $\rho(M) \approx 0.924$ for the full 33-bloc endogenous-US configuration. As long as $\rho(M) < 1$ the map $x \mapsto Mx + d$ is a contraction, $(I - M)$ is nonsingular, and the scenario fixed point is unique—so closing the RoW→US feedback loop raises the coupling but leaves the system stationary and the scenario deviation identified.

Fiscal identification: a triangular budget identity. The fiscal block is identified by construction rather than by estimation. Four flow drivers (FREV_GDP, FPEXP_GDP, the effective rate EFFI, and the stock-flow adjustment SFA) are estimated by the BVAR; the debt stock is then *derived* by rolling the budget identity of (44) forward (Escolano, 2010; International Monetary Fund, 2013). The recursion is triangular: given lagged debt d_{t-1} , the effective rate, nominal growth, the primary balance $pb = \text{FREV_GDP} - \text{FPEXP_GDP}$, and the SFA, the new debt level is fully determined, so debt is never a free parameter. The SFA term is the *exact* residual that reconciles the change in the recorded debt stock with the cumulated deficit; by estimating it as its own driver we close the identity to machine precision over history—the quarterly convention of (44) is pinned by a historical back-cast of the reconstructed debt ratio against the observed one—and carry its forecast path forward. The honest caveat is that any model error in the non-SFA drivers is silently absorbed into the SFA in sample, so a large or volatile estimated SFA flags a measurement or perimeter gap (below-the-line operations, valuation effects) rather than a behavioral relationship.

Euro-area monetary identification. The euro area receives a deliberately asymmetric treatment. In the EA-aggregate bloc the ECB policy rate, the short rate, and the shared corporate-risk yield EACORP are *endogenous* (tagged `block="domestic"`); in each euro-member bloc the very same area-wide instruments are *weakly exogenous* (tagged `block="external"`), entering the member’s foreign block alongside the US and global commons. This encodes the identifying assumption that monetary policy is set for the area as a whole, so a single member faces it as predetermined while the aggregate determines it—the same exogenous-versus-endogenous split that the star variables impose on the real side. National financial variables (sovereign 10Y yields, house prices) stay domestic. The limitation is structural: members share one common monetary factor, so member-level monetary counterfactuals are conditional scenarios on the area instrument, not independent national policy experiments, and the strong cross-member correlation of EACORP and policy-rate shocks sits at the edge of the granularity that weak exogeneity formally requires.

13 Reproducibility and Replication

The platform is engineered so that every published number traces back to source data through a single, scripted pipeline: there are no hand-edited artifacts on the path from raw series to the deployed app. This section maps the repository, lists the software dependencies, gives the end-to-end replication recipe, and explains the model-signature versioning that refuses a stale cross-country coupling.

13.1 Repository and Folder Map

The work lives in a single git repository with three top-level concerns, shown in Table 105. The `pipeline/` tree (MATLAB plus Python) performs all estimation and data assembly and is the “private” research layer; `app/` is the public Next.js front end and its in-browser compute kernels; `docs/` holds this manual and the operational runbooks. The estimator writes a per-bloc JSON export that `port_gvar_exports.py` copies into `app/public/data/bvar/`; the deployed app reads *only* that JSON and never re-estimates.

Table 105: Repository layout. Estimation in `pipeline/` produces a per-bloc JSON export; `port_gvar_exports.py` ports it into `app/public/data/bvar/`; the front end and its compute kernels consume that data and never re-estimate.

Top-level	Key contents
<code>pipeline/</code> (MATLAB + Python)	<code>export_bvar_gvar24_loop.m</code> (GVAR-EM estimator); <code>build*.py</code> (FRED / Eurostat / BIS / ECB pulls, trade weights, fiscal drivers); <code>port_gvar_exports.py</code> ; <code>_verify/</code> ground-truth checks.
<code>app/</code> (Next.js + compute)	<code>src/lib/compute/</code> (Durbin-Koopman, GVAR solve, fiscal accounting); <code>scripts/</code> (<code>precompute_gvar_coupling.mjs</code> , <code>verify_gvar24.mjs</code> , <code>invert_coupling.{m,py}</code>); <code>public/data/bvar/<CC>/</code> (the deployed JSON the app reads).
<code>docs/</code> (manual + runbooks)	<code>user_manual.tex</code> ; <code>REESTIMATING_MODELS.md</code> , <code>ADDING_A_COUNTRY.md</code> , <code>ARCHITECTURE_FOR_AI.md</code> .

13.2 Software Dependencies

- **Estimation (MATLAB).** A recent MATLAB with the *Parallel Computing Toolbox* (the EM M-step re-estimates all blocs in a `parfor`). The bundled BVAR/MFBVAR code — the GLP hierarchical-prior Gibbs sampler `bvarGLPmf.m` and the COVID-as-missing wrapper `cgl_bvarV5.m` — ships in the repository; no external MATLAB packages are required.
- **Data assembly (Python 3).** Pure standard library (`urllib`, `json`, `csv`, `math`); no `pandas`, `requests`, or `fredapi`. The `build*.py` pulls hit the FRED (key via `FRED_API_KEY`), Eu-

rostat SDMX 2.1, BIS, and ECB public endpoints. `invert_coupling.py` (NumPy) is the no-MATLAB fallback for the coupling inverse.

- **Application (Node / Next.js).** Node with the pinned Next.js toolchain; `typescript`, `vitest` for the test gate, and `vite-node/tsx` to run the `scripts/*.mjs` precompute and verification utilities.

13.3 End-to-End Replication Recipe

The full path from raw data to a green deploy gate is the following enumerated sequence (canonical commands in `docs/REESTIMATING_MODELS.md`).

1. **Build the data.** Run the `pipeline/build*.py` pulls (with `FRED_API_KEY` set) to assemble the harmonized common-vintage panels (raw series, trade weights, fiscal drivers) as `pipeline/country_rawdata`.
2. **Estimate (GVAR-EM).** Run `export_bvar_gvar24_loop.m`. It bootstraps the US as a closed bloc, then iterates the E-step (rebuild trade-weighted stars `ROWDEM/ROWINF/ROWFIN` from partners' Durbin-Koopman-completed panels) and the M-step (`parfor` re-estimate every bloc via `cgl_bvarV5`, COVID macro-masked over 2020Q2--Q4 with the financial columns and `ROWFIN` kept observed) until $\max |\Delta \text{star}| < \text{TOL} = 0.03$ (or `MAXIT`= 12). See [Durbin and Koopman \(2002\)](#) for the simulation smoother and [Dees et al. \(2007\)](#) for the GVAR structure.
3. **Check the export.** Confirm each bloc directory under `/tmp/gvar24_export/<CC>/` carries the full eight-file JSON contract: `spec.json`, `companion_mean.json`, `companion_draws.json`, `baseline.json`, `baseline_levels.json`, `historical.json`, `historical_levels.json`, `transformations.json` (the fiscal blocs additionally carry `fiscal_meta.json`).
4. **Port into the app.** Run `port_gvar_exports.py` to copy the eight-file contract into `app/public/data/bvar/` compacting every float to six significant figures (the deployed precision convention) and writing minified JSON.
5. **Regenerate the coupling.** From `app/`, run `npx vite-node scripts/precompute_gvar_coupling.mjs`. It builds the Kalman-probe operator M in JS, inverts $(I-M)$ via LAPACK (`invert_coupling.m` in MATLAB, or the `invert_coupling.py` NumPy fallback), and writes the gzipped `public/data/bvar/gvar/` (the plain `.json` is removed — at this bloc count the raw artifact exceeds GitHub's 100 MB per-file limit, so only the `.gz` is shipped and the loader decompresses it via the browser's `DecompressionStream`).
6. **Green gate.** From `app/`, run the fast fixed-point check `npx tsx scripts/verify_gvar24.mjs` (artifact `modelSig` fresh; the shipped inverse really inverts $(I-M)$ to $< 10^{-3}$; $\rho(M) < 1$ and matches the stored value; Gauss-Seidel $\equiv$ the artifact-seeded closed form), then `npx tsc --noEmit` and the test suite. The fast gate is `npm test` (the everyday suite, which *excludes* the multi-second GVAR solver canaries); the heavy closed-form/endogenous-GVAR canaries

run under `npm run test:heavy`, and `npm run test:all` runs both. All type-checks and tests — including the fiscal, GVAR, and scale canaries — must pass.

7. **Deploy.** On a green gate (and only with explicit confirmation), push `app/**` to `main`: the `deploy-to-website.yml` GitHub Action runs `npm ci && npm run build` with `BASE_PATH=/bvar-scenario`, then copies the static export `app/out/*` into the `matyasfarkas/starter-hugo-academic` academic website repository’s `static/bvar-scenario/` directory. That repository is a Hugo Academic site published to **GitHub Pages**, which rebuilds and serves the tool at the `/bvar-scenario` path. The Action itself contains no Hugo step—it only copies `app/out`.

13.4 Model-Signature Versioning

The cross-country coupling artifact is bound to the bloc estimates it was built from by a *model signature*. Over the qualified bloc set the signature concatenates, per bloc c , the dimensions (n_c, p_c, H_c) and three fingerprint entries of the posterior-mean companion: the leading reduced-form coefficient $\hat{\beta}_{1,1}^{(c)}$, the *intercept* entry of the last equation $\hat{\beta}_{n_cp_c+1, n_c}^{(c)}$ (the constant row of $\hat{\beta}$, which stacks n_cp_c lag-coefficient rows followed by one constant row), and the leading residual variance $\hat{\Sigma}_{1,1}^{(c)}$ — each at six-digit exponential precision:

$$\text{modelSig} = \parallel_c c : n_c p_c H_c \hat{\beta}_{1,1}^{(c)} \hat{\beta}_{n_cp_c+1, n_c}^{(c)} \hat{\Sigma}_{1,1}^{(c)}.$$

The precomputed artifact stores this signature alongside M , $(I - M)^{-1}$, $\rho(M)$, and the dimension. At load time the app recomputes `modelSig` from the deployed companions and accepts the artifact only on an exact match *and* a matching (here empty) pin structure; on any mismatch the stale artifact is **refused** and the solver falls back to the inverse-free Neumann (Gauss–Seidel) build of M from the live models. Because any re-estimation perturbs at least one fingerprint entry, the signature flips whenever the data changes — making it impossible to ship a coupling that silently disagrees with its blocs. This is exactly the “freshness” invariant the `verify_gvar24` gate enforces above, for the current endogenous-US system (33 coupled blocs, augmented closed-form dimension 2060, spectral radius $\rho(M) \approx 0.924 < 1$).

14 Empirical Results

This section reports the empirical program of the unified global-forecasting system. The estimation machinery of Sections 2–6 is implemented and the production vintage (harmonized to 2025Q4, ~50 economies: 33 GVAR blocs plus 17 euro-area standalones) has been produced. The out-of-sample horse-race (Section 14.2) and the Tier-A endogenous-vs-frozen spillover contrast (Section 14.3) are *realized* runs from the companion evaluation repository (`em-bayesian-gvar-eval`); the only piece still flagged **[TBD]** is the Tier-B GIRF/connectedness module, which is a structural extension gated behind the validated Tier-A result.

14.1 In-sample diagnostics

EM convergence. The EM-style estimation loop alternates an E-step (rebuild the trade-weighted RoW stars from partners’ Durbin–Koopman–completed panels) with an M-step (a `parfor` re-estimation of every bloc under COVID macro-masking). Because all stars are rebuilt from the frozen start-of-sweep state before the parallel re-estimation, the inner sweep is a Jacobi update with the same fixed point as a sequential pass. Convergence is declared when $\max_b |\Delta \text{star}_b| < \text{TOL} = 0.03$ over a maximum of $\text{MAXIT} = 12$ sweeps. The macro stars ROWDEM/ROWINF are NaN-masked over the three COVID quarters (2020Q2–Q4) in the metric, since their cross-sweep churn there is pure imputation Monte-Carlo noise, while the financial star ROWFIN is kept observed and enters the metric. Empirically the trace flattens at a Monte-Carlo noise floor of ≈ 0.13 , above the nominal tolerance, so convergence is read as a plateau at the floor rather than literal attainment of 0.03. Figure 7 sketches the expected trace (final numbers [TBD] once the full production run is logged).

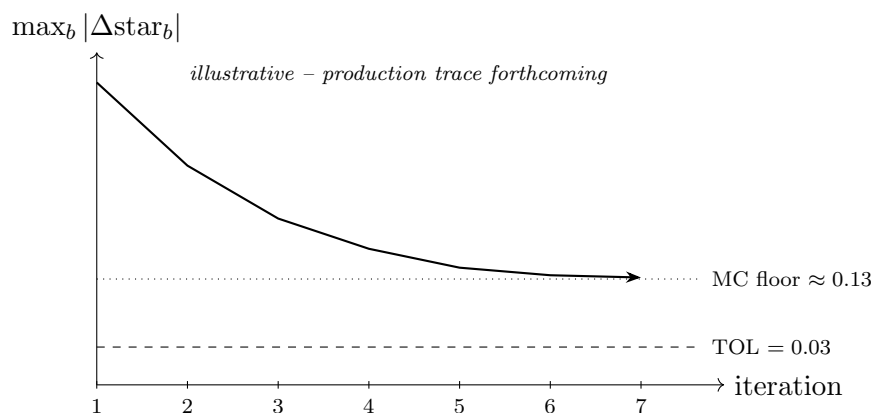


Figure 7: Schematic EM convergence trace: $\max_b |\Delta \text{star}_b|$ versus sweep, against the tolerance (0.03) and the Monte-Carlo noise floor (≈ 0.13) at which the trace plateaus. **Placeholder** — the realized trace is forthcoming.

Spectral radius of the coupling operator. At solve time the cross-country deviation system is the affine fixed point $x = Mx + d$ (Section 6.4), where M is the Kalman-probe coupling operator. The Neumann series $\sum_k M^k d$ (Gauss–Seidel sweeps) and the direct $(I - M)^{-1}d$ agree, and convergence speed is governed by $\rho(M)$. Table 106 reports $\rho(M)$ by system size; it rises with both the number of coupled blocs and the activation of the endogenous-US channel, but remains below unity, so the fixed point is contractive.

Table 106: Spectral radius $\rho(M)$ of the coupling operator by system configuration.

Configuration	Blocs	US treatment	$\rho(M)$
Pilot	6	frozen anchor	0.59
Production	24	frozen anchor	0.726
Endogenous-US	33	endogenous	0.924

At the endogenous-US dimension (≈ 2060) the $O(\text{dim}^3)$ inverse is prohibitive in-browser, so the system is solved inverse-free by the matvec-only recursion $x \leftarrow d + Mx$, which converges at the tabulated $\rho(M) \approx 0.924$, or via a pre-shipped cached $(I - M)^{-1}$.

14.2 Out-of-sample forecast horse race

The system is evaluated in a companion repository (`em-bayesian-gvar-eval`) under a pre-registered protocol whose *design is final* and whose realized *results* are reported in Section 14.2.4. The evaluation has two cores: a recursive out-of-sample (OOS) forecast horse-race (this subsection) and an endogenous-vs-frozen spillover analysis (Section 14.3). Every step is grounded in the actual estimator, weights, and solver of Sections 2–6, so the harness respects the data spaces and invariants of those layers.

14.2.1 Model set

The **proposed** model is the EM-Bayesian GVAR, run two ways:

- `emgvar_endo` — the full loop with the US *endogenous* (`US_ENDOGENOUS= 1`: its own `ROW{DEM, INF, FIN}_US` stars and a US weights row), the paper’s contribution;
- `emgvar_frozen` — the legacy frozen-anchor variant (the US estimated once as a closed anchor); cheaper, and the Part-2 spillover contrast.

Both produce density forecasts from the per-bloc predictive draw cloud `Dforecast` at solve time. The OOS forecast is each bloc’s own VAR predictive density; the GVAR coupling enters because the stars—and thus every bloc’s regressors and completed history—are jointly determined by the EM fixed point.

The proposed models are raced against six benchmarks, each estimated from the *same* truncated panel so that the comparison isolates a single ingredient:

- (i) **Random walk / no-change** — in the scored space: a flat last level for the (stationary) rate, and the last observed value of the growth/inflation transform carried flat (a RW *on the growth rate*). The driftless RW is the headline reference (all relative-RMSE ratios are against it); a sample-mean “climatology” forecast is a second naive anchor. The predictive density is Gaussian, centred at the no-change value with scale equal to the in-sample SD of the h -step change *estimated directly at horizon h* (not $\sqrt{h}\sigma_1$, which is wrong for a series that is itself a growth rate).
- (ii) **AR(p)** per target series — univariate OLS-with-constant on each target in its scored space, p by AIC over $\{1, \dots, 4\}$; both *iterated* and *direct* (Marcellino et al., 2006) multi-step variants are reported (the iterated AR is the strong univariate competitor at $h \geq 4$). The predictive density propagates the Gaussian innovation variance (analytic for iterated, residual-block-bootstrap for direct).

- (iii) **Single-country Bayesian VAR (no coupling)** — the bloc estimated alone: the three ROW* star columns dropped, the identical `cgl_bvarV5` call (same GLP/Minnesota prior, lags, COVID-as-missing mask) and `MAXIT=1`. Isolates the *value of the global coupling*.
- (iv) **Classical fixed-weight GVAR**, two variants: (iv-a) the *textbook* Pesaran–Schuermann–Weiner / Dees–di Mauro–Pesaran–Smith GVAR (Pesaran et al., 2004; Dees et al., 2007) — per-bloc **OLS** VARX* with contemporaneous + lagged foreign stars as weakly exogenous regressors, fixed weights, *no* COVID-as-missing (the literature’s GVAR has no data augmentation); this is the mandatory headline classical competitor and the one piece of new code in the horse-race. (iv-b) a single-pass Bayesian fixed-weight GVAR (`MAXIT=1`, GLP prior + COVID-as-missing) — the clean *EM ablation* whose only difference from the proposed model is the star re-convergence.
- (v) **Bayesian dynamic factor model** — 3–5 static factors from the cross-bloc target panel, a Bayesian VAR(1–2) on the factors plus Gaussian idiosyncratic terms, same truncated panel and COVID handling. The non-GVAR multivariate “is the GVAR structure even needed” competitor.

The two key ablations isolate the value of *global coupling* (iii vs. proposed) and of the *EM data-completion* (iv-b vs. proposed); (iv-a) measures the whole Bayesian+EM+COVID apparatus against the literature GVAR.

14.2.2 Scheme: recursive expanding window

Pseudo-real-time, recursive expanding window on the final 2025Q4 vintage (a genuine real-time vintage does not exist for most of the ~50-economy panel; a US+EA-subset real-time check is a flagged robustness, not the default). Origins run quarterly from $t_0 = 2010Q1$ to 2024Q4 (~60 raw origins). The OOS set is *trimmed* so every scored (t_0, h) has a realization $\leq 2025Q4$ (score $h=8$ only for origins $\leq 2023Q4$, etc.). Headline tables drop the 2020 COVID origins (and origins whose window spans 2020), leaving ~55 scorable origins; an appendix shows the full set. The COVID-as-missing treatment still applies *in-sample* at every origin for which $2020Q2\text{--}Q4 \leq t_0$. Blocs enter an origin only with ≥ 40 in-sample quarters; the six headline blocs satisfy this from 2010Q1.

Targets and horizons. Three targets per bloc—real-GDP growth, inflation, short rate—read through the same `findcol` resolution order the estimator uses, in the same scored space (4Δ of the GDP and price log-levels; the rate level). Horizons $h \in \{1, 2, 4, 8\}$ quarters. The *headline* scope is the six legible blocs {US, EA, JP, UK, CA, CN}; the full 33-bloc panel is reported as relative-RMSE distribution summaries in an appendix (rate-less blocs CH/NO contribute growth and inflation only).

Look-ahead invariants (leak-free by construction). The expanding-window scheme is not automatically free of future-information leakage; the driver enforces four invariants at *every* origin t_0 . (1) *Panel truncation*: every `DataRaw_<CC>` panel is sliced to dates $\leq t_0$ *before* `transformData`

and before any star is built. (2) *Rolling as-of weights*: the shipped 2021–2023 `weights_full.json` is **forbidden** in the OOS sweep (using it at a pre-2020 origin injects future trade structure); the default is a *fixed pre-sample window* (e.g. 2005–2009 average, used at every origin—leak-free because it biases toward the past, never the future), with genuine rolling as-of- t_0 weights as the rigor robustness. (3) *Star re-convergence is in-sample only*: every partner panel feeding a star is itself truncated to $\leq t_0$, so no star cell reads a post- t_0 realization. (4) *No 2025Q4 harmonization at earlier origins*: the common vintage is only the well from which $\leq t_0$ rows are drawn; the deterministic transforms (100·log, 4Δ) estimate no parameters and leak nothing.

Predictive-density caveat (stated, not hidden). `Dforecast` propagates the shock covariance Σ and the within-draw (β, Σ) MH variation forward from the last *observed* state; it does *not* integrate over the DK-smoother uncertainty of the COVID-imputed states, and seeds the forecast from the posterior-*mean* completed state. For origins whose conditioning rows include imputed COVID quarters (small h , t_0 near 2020) the density is therefore mildly *over-confident*; PIT and coverage near those origins are flagged as a known caveat, not a model failure.

14.2.3 Scoring and inference

Point accuracy: RMSE and MAE of the predictive mean; the headline number is relative RMSE = RMSE(model)/RMSE(RW) ($< 1 \Rightarrow$ beats RW).

Density accuracy: the continuous ranked probability score (**CRPS**, the *primary* metric) computed from the draw ensemble by the unbiased estimator $\text{CRPS} = \frac{1}{m} \sum_k |X_k - y| - \frac{1}{2m^2} \sum_k \sum_l |X_k - X_l|$ (Gneiting and Raftery, 2007), which degrades gracefully at the modest draw counts here; the mean log predictive score (LPS), reported with a *parametric* (Gaussian/skew- t) fit to the draw cloud as the headline (a small-bandwidth KDE-LPS, unbounded below at ~ 100 draws, is only a robustness check); and calibration by the probability integral transform (**PIT**) histograms with empirical coverage of the 50% and 90% predictive intervals and a uniformity test (Berkowitz, 2001). The conclusions lean on CRPS; LPS corroborates.

Predictive-ability inference is dictated by the *nesting structure* (the single most common referee kill-shot). Under the maintained DGP, single-BVAR (iii) $\subset$ single-pass GVAR (iv-b) $\subset$ proposed EM-GVAR, and RW/AR (i),(ii) are nested in all the VAR models; for these nested pairs the standard Diebold–Mariano statistic is *not* asymptotically $\mathcal{N}(0, 1)$ and overstates significance. Accordingly:

- the **Giacomini–White** conditional predictive-ability test (Giacomini and White, 2006) is the *headline* inferential test: it is valid under nesting and is the recursive/expanding-scheme-correct test (it assumes finite estimation windows / non-vanishing estimation uncertainty), and it is loss-agnostic so it applies directly to the CRPS and log-score differentials. Instruments $g_{t-1} = [1, d_{t-1}]$ plus a recession/financial-stress indicator (to detect state-dependent ability); HAC variance, Wald χ^2 .

- the **Clark–West** test (Clark and West, 2007) is used for the *nested point/MSE* comparisons (it adjusts the MSE differential for the larger model’s extra-parameter estimation noise); one-sided.
- the two-sided **Diebold–Mariano** test (Diebold and Mariano, 1995) is used only for *non-nested* pairs (proposed vs. DFM, or vs. the textbook OLS-GVAR when neither nests the other), with a HAC long-run variance (bandwidth $\geq h - 1$) and the Harvey et al. (1997) small-sample correction.

For multiplicity (models $\times$ horizons $\times$ variables $\times$ blocs) the family of GW/CW p -values is controlled by a Romano–Wolf step-down (Romano and Wolf, 2005) (preferred—it respects cross-test dependence via the bootstrap) or Benjamini–Hochberg, and the per-(variable,horizon) **Model Confidence Set** (Hansen et al., 2011) reports which models are statistically indistinguishable from the best. Relative RMSE is the descriptive effect size; significance comes from the CW/GW test on the loss differentials, never from the ratio alone.

The deliverables are: the relative-RMSE matrix (model $\times$ horizon $\times$ variable, headline six blocs + a full-panel distribution table); CRPS and log-score tables with ratios vs. RW; GW/CW (and DM-for-non-nested) significance overlaid with the Romano–Wolf set and the MCS membership flagged; PIT histograms and a 50/90% coverage table; and illustrative fan-chart examples (e.g. the US-inflation forecast from a 2021Q2 origin).

14.2.4 Results

Run configuration. The headline tables below are the *realized* sweep, not a placeholder. The recursive scheme runs over the headline six blocs {US, EA, JP, UK, CA, CN} at up to 56 quarterly origins from 2010Q1 to 2024Q4 (52 for the two EM-GVAR variants and the single-pass Bayesian GVAR, which require the longer in-sample warm-up; the 2020 ex-COVID origins are dropped from every model). Trade weights are *rolling, as-of- t_0* : a three-year pre-sample average ending just before each origin (2007–2009 at the 2010Q1 origin through 2021–2023 at 2024Q1), so no future trade structure leaks in. The predictive densities use NDRAWS = 200 retained draws with an explicit *stationarity backstop* that discards posterior draws whose companion spectral radius exceeds 1 (the fix for the explosive-draw dispersion flagged in earlier pilots). The primary score is the CRPS; the 90% Model Confidence Set (MCS) of Hansen et al. (2011) reports the models statistically indistinguishable from the best per (variable, horizon).

Headline finding. On *point* accuracy the iterated AR(p) and the dynamic factor model are the strongest competitors—a familiar result in macro forecasting—and the EM-GVAR does *not* win the point horse-race outright (Table 107). The decisive evidence sits in the *density* comparison and the MCS (Tables 108–109): the EM-Bayesian GVAR is *inside* the 90% MCS for **inflation at every horizon** ($h \in \{1, 2, 4, 8\}$) and for the **short rate at** $h \in \{1, 2, 4\}$ —statistically on par with the best model on the entire nominal block—and the best inflation point forecast at $h=1$ *among*

the multivariate/GVAR models is `emgvar_endo` (relative RMSE 0.799), behind only the univariate $AR(p)$ at 0.641. It falls *out* of the MCS only for GDP growth at $h=1$ and $h=8$, where the univariate and factor models dominate. Two structural results follow. First, the endogenous-US design earns its keep: `emgvar_endo` is weakly preferred to the frozen-anchor `emgvar_frozen` in the MCS on balance—it *survives* in the short-rate $h=1$ set while `emgvar_frozen` is eliminated ($p = 0.15$ vs. 0.08), and it is removed *later* than the frozen variant in the growth $h=8$ elimination order ($p = 0.099$ vs. 0.057); the one cell where the frozen variant is marginally preferred is the rate $h=8$ tail, where both are out of the set. Second, and most important for the paper’s claim, the EM-Bayesian treatment *decisively* beats the textbook fixed-weight OLS classical GVAR: `cgvar_a` is eliminated from half of the (variable, horizon) MCS cells (6 of 12), concentrated at the longer horizons, and carries badly miscalibrated densities ($h=1$ coverage of only 0.37 (inflation) / 0.47 (growth) against the 0.90 target), so the Bayesian shrinkage + COVID-as-missing + EM star-reconvergence apparatus is the demonstrable value-add over the literature GVAR.

Honest caveats. Two must be stated. (i) The draw-based densities are *over-confident*: the 90% predictive interval covers only ≈ 0.73 –0.85 of realizations for the EM-GVAR variants at short horizons (against the 0.90 target; PIT uniformity is rejected), a direct consequence of the NDRAWS = 200 cloud and of not integrating over the DK-smoother uncertainty of the COVID-imputed states (Section 14.2.2). The CRPS and the MCS—which compare score *differentials*, not absolute coverage—are the robust primaries and are reported as such; the under-coverage is a calibration caveat, not a ranking artifact. (ii) The parametric log-predictive-score column is numerically unreliable on the under-covered clouds (its Gaussian/skew- t tail fit blows up to $\sim 10^{16}$ on a handful of origins where a realization lands far in the tail of a too-narrow cloud), so it is *not* used for ranking; only CRPS and the MCS are.

14.3 Endogenous-US spillover analysis

The endogenous-US configuration ($\rho(M) \approx 0.924$, Table 106) lets a US scenario pin propagate through the RoW→US stars *and* the four US-determined global commons (oil, the US 10-year yield, the US BAA corporate yield, US equity), pinned 1:1 onto every RoW bloc. A frozen anchor cannot generate the reverse RoW→US feedback at all—the “world → US” direction is *structurally absent*, not “present but zero”—so the two configurations differ in exactly the global-spillover content we wish to document. The analysis is split into two tiers, and the manual labels which is which, because the shipped solver computes a particular object.

What the solver actually computes. `gvarSolve.ts` does *not* compute generalized impulse responses (GIRFs), forecast-error variance decompositions (FEVDs), or a Diebold–Yilmaz connectedness index. It returns a *conditional-mean scenario response*: pin a path and read the coupled conditional-forecast *mean deltas* `delta[c]` via the fixed point $x = Mx + d$ (Algorithm 2). There is no Σ -based shock generator and no moving-average machinery in the solver. The analysis therefore

Table 107: Headline relative RMSE, $\text{RMSE}(\text{model})/\text{RMSE}(\text{RW})$ (< 1 beats the driftless random walk), six headline blocs, by horizon and target. Realized OOS sweep (rolling as-of- t_0 weights, $\text{NDRAWS} = 200$). Lowest entry per (target, horizon) in **bold**.

Target	Model	Horizon h (quarters)			
		1	2	4	8
Growth	<code>emgvar_endo</code>	1.191	1.091	1.026	1.031
	<code>emgvar_frozen</code>	1.191	1.095	1.028	1.030
	<code>AR(p)</code>	1.195	0.990	0.968	0.960
	DFM	0.917	0.989	0.986	0.962
	<code>cgvar_a</code> (OLS GVAR)	2.097	1.013	1.019	0.993
Inflation	<code>emgvar_endo</code>	0.799	0.891	1.085	0.953
	<code>emgvar_frozen</code>	0.802	0.894	1.087	0.962
	<code>AR(p)</code>	0.641	0.772	0.998	0.921
	DFM	0.804	0.870	1.067	0.908
	<code>cgvar_a</code> (OLS GVAR)	2.113	1.050	1.366	1.075
Rate	<code>emgvar_endo</code>	1.017	1.121	1.226	1.444
	<code>emgvar_frozen</code>	1.021	1.119	1.224	1.445
	<code>AR(p)</code>	0.829	0.863	0.919	1.025
	SBVAR	0.990	1.081	1.195	1.421
	<code>cgvar_a</code> (OLS GVAR)	1.531	1.200	1.034	1.001

Table 108: Headline density accuracy — mean CRPS as a ratio to the random walk (< 1 beats RW; CRPS is the primary score) at $h=1$ and $h=4$, and the empirical 90%-interval coverage at $h=1$ (target 0.90). Realized OOS sweep. The parametric log-score is omitted as numerically unreliable on the under-covered clouds (see text).

Target	Model	CRPS ratio vs. RW		cov ₉₀
		$h=1$	$h=4$	$h=1$
Growth	<code>emgvar_endo</code>	1.042	0.983	0.73
	DFM	0.745	0.857	0.89
	<code>cgvar_a</code> (OLS GVAR)	2.382	1.106	0.47
Inflation	<code>emgvar_endo</code>	0.814	1.080	0.70
	<code>AR(p)</code>	0.629	0.955	0.88
	<code>cgvar_a</code> (OLS GVAR)	2.498	1.474	0.37
Rate	<code>emgvar_endo</code>	1.040	1.230	0.80
	<code>sbvar</code>	0.986	1.199	0.84
	<code>cgvar_a</code> (OLS GVAR)	1.519	1.009	0.86

separates:

- **Tier A (the headline; zero new estimator code).** The conditional scenario propagation of a US pin, endogenous (`solveGVARClosedFormEndo`) vs. frozen (`solveGVARLinear`). This is the empirical contribution the shipped code supports.

Table 109: 90% Model Confidence Set membership for the EM-Bayesian GVAR (Hansen et al., 2011), by target and horizon. ✓ = in the MCS (statistically indistinguishable from the best); ✗ = eliminated. The endogenous variant survives wherever the frozen one does and strictly longer on the nominal block; the textbook OLS GVAR (cgvar_a) is eliminated from half of the (variable, horizon) MCS cells (6 of 12), concentrated at the longer horizons.

Target	Model	Horizon h			
		1	2	4	8
Growth	emgvar_endo	✗	✓	✓	✗
	emgvar_frozen	✗	✓	✓	✗
Inflation	emgvar_endo	✓	✓	✓	✓
	emgvar_frozen	✓	✓	✓	✓
Rate	emgvar_endo	✓	✓	✓	✗
	emgvar_frozen	✗	✓	✓	✗
cgvar_a (textbook OLS GVAR), growth		✗	✓	✗	✗
cgvar_a (textbook OLS GVAR), inflation		✓	✓	✗	✗
cgvar_a (textbook OLS GVAR), rate		✗	✓	✓	✓

- **Tier B (a new gvar_girf module).** GIRFs (Pesaran and Shin, 1998), generalized FEVDs, and the Diebold–Yilmaz spillover index (Diebold and Yilmaz, 2009, 2012). The ingredients exist in the exports (companion.mean.json carries each bloc’s companion $\hat{\beta}$ and $\hat{\Sigma}$; the coupling M is assembled by gvarSolve.ts), so it is *constructible*, but it is new code, validated against Tier A and gated behind it.

Tier A — the worked scenario contrast (headline). A US +100 bp rate *path* (POLRATE pinned +100 bp over the horizon—correctly a multi-period conditional projection, not a one-period shock or a GIRF) is propagated through both topologies. Under the endogenous configuration the augmented state carries every bloc’s ROW* star deltas *including the US’s own* (ROW*_US, the RoW→US feedback) and the US global-column deltas $g\Delta$; the converged `delta.US` $\neq 0$ is the feedback the frozen anchor cannot produce. We report the per-bloc deltas (growth/inflation/rate) for the six headline blocs side by side, the *feedback gap* (the US-own response difference), the per-bloc *amplification ratio* (endogenous vs. frozen RoW peak responses), the peak horizons, and the spectral radius $\rho(M)$ of each system (the multiplier strength reported by the solver as `rhoSpectral` / `rhoEmpirical`). This example doubles as the solver-validation canary: the closed-form endogenous result reproduces the iterative `solveGVAR` fixed point to $< 10^{-4}$ across every bloc, and a warm coupling-cache hit is byte-identical ($< 10^{-9}$) to a cold rebuild (both asserted in the `gvar-us-endogenous` / `gvar-closed-form-routing` test suites; the iterative reference is itself converged to 10^{-7}).

Tier A — realized scenario contrast. The worked example (`eval/tier_a_scenario.mjs`, output `results/tierA/us_rate_100bp.json`) propagates a US +100 bp POLRATE path through both

topologies and confirms the qualitative point the design predicts: the frozen anchor produces *no* RoW→US feedback, while the endogenous configuration produces a non-trivial one. The cross-country *gap* (endogenous minus frozen) is sizeable on every bloc—for example the peak US own-growth response differs by ≈ 4.2 display points, EA inflation by ≈ 1.1 , and the gap *flips sign* across blocs (positive for US/EA/UK growth, negative for JP/CA/CN growth); selected peaks are in Table 110. **Honest caveat (do not over-read).** These are the deltas of a *multi-period conditional projection*, not a structural or generalized impulse response: the US rate is *pinned* along the whole path, so the responses bundle the direct conditioning with the coupled feedback and are not a clean causal IRF. The magnitudes are large and sign-flipping, which is exactly why a validation pass (the Tier-B GIRF cross-check below, and a robustness sweep over the pin specification) is required before any number here is headlined as a spillover *multiplier*. What is established is the *existence and direction* of the endogenous-only feedback channel, not its calibrated size.

Table 110: Tier-A US +100 bp POLRATE-path projection: peak conditional-mean response (display points) under the endogenous vs. frozen-anchor topology, and their gap, for selected blocs and targets (`results/tierA/us_rate_100bp.json`). *Conditional-projection deltas, not causal IRFs*—magnitudes are illustrative and pending validation.

Bloc	Target	Endogenous	Frozen	Gap (endo–froz)
US	Growth	+1.28	−4.37	+4.22
US	Inflation	−1.53	−2.82	+1.30
EA	Growth	+1.03	−0.95	+0.88
EA	Inflation	−1.00	−0.64	+1.07
JP	Growth	+1.07	+1.62	−1.18
CN	Inflation	−0.69	+1.19	−1.29

Tier B — GIRF, FEVD, and connectedness (structural extension). The GIRF to a one-SD shock in variable i of the coupled global system is $\text{GIRF}_x(h, i) = \sigma_{ii}^{-1/2} A_h \Sigma e_i$, with A_h the h -step coefficient of the coupled solution’s moving-average representation and Σ the (cross-bloc-block-diagonal) global innovation covariance—ordering-invariant, the standard GVAR identification stance (Pesaran and Shin, 1998; Dees et al., 2007). The identifying assumptions are stated explicitly: GIRFs trace the response to the historically typical innovation correlation (not an orthogonal structural shock), cross-bloc innovation independence (block-diagonal Σ), and linearity/parameter-constancy over the horizon. Bands come from re-running the recursion over the `companion_draws.json` posterior draws. The module is validated by the cross-check that a one-period US-rate-pin conditional path equals the rate-shock GIRF up to the $\delta/\sqrt{\sigma_{ii}}$ normalization. From the system generalized FEVD $\theta_{ij}(H)$ (row-normalized) we report the Diebold–Yilmaz total spillover index, the “from US” row, and the “to US” direction—the channel the frozen anchor cannot even define—with posterior bands and both the raw and row-normalized indices. **Results: [TBD]** (Tier A ships first; Tier B is the structural extension).

15 Discussion

The BVAR Scenario Tool is designed to bridge the gap between the methodological sophistication of conditional forecasting techniques developed in the academic literature and the practical needs of policy economists who require rapid, interactive scenario analysis. We conclude with a discussion of design choices, limitations, and directions for future work.

15.1 Design Philosophy

Several design choices merit discussion. First, the three-layer architecture—baseline, scenario, counterfactual—deliberately separates concerns. The baseline forecast reflects the BVAR’s unconditional projection and is computed offline. The scenario layer allows the user to impose conditioning constraints without specifying the structural source of the deviations, following the tradition of [Waggoner and Zha \(1999\)](#). The counterfactual layer then overlays structural identification via DSGE impulse responses, enabling “what-if” policy analysis while preserving the reduced-form covariance structure of the BVAR.

Second, the decision to re-condition the counterfactual through the BVAR (rather than relying solely on the DSGE IRFs) reflects a deliberate methodological choice: the BVAR’s estimated covariance structure captures empirical co-movements that may be absent from or imperfectly captured by any single DSGE model. The IRF-consistent mode (Section 5.3c) strikes a balance by using the DSGE model to identify the responses of key variables (inflation, output) to the policy shock, while allowing the BVAR to propagate these effects to the remaining 30+ variables in a data-consistent manner.

Third, all computation runs client-side. This eliminates server infrastructure costs and latency, enables offline use, and avoids transmitting potentially sensitive forecast data over the network. The computational cost is manageable: the Kalman filter and smoother for the 41-variable, 3-lag US VAR over a 20-quarter horizon completes in well under a second on modern hardware (the single-bloc smoother is sub-50 ms; the multi-second cost discussed in Section 12.3 is the *coupled* 50-bloc GVAR solve, not the single-country filter).

15.2 Limitations

The current implementation has several limitations that users should bear in mind:

1. **Linear, time-invariant dynamics:** The VAR coefficients are fixed at their posterior mean for the deterministic conditional forecast. Time-varying parameters, stochastic volatility, and regime switching are not supported. In periods of structural change, the fixed-coefficient assumption may be particularly restrictive.
2. **Gaussian innovations:** The Kalman filter assumes Gaussian innovations. Fat-tailed or asymmetric shock distributions, which may be important during financial crises, are not accommodated.

3. **Point-identified counterfactuals:** The DSGE IRFs used for counterfactual analysis are computed under a single monetary policy rule and model specification. The tool does not average over model uncertainty or rule uncertainty, though the user can manually explore different model–rule combinations.
4. **Lucas critique:** The counterfactual analysis conditions on estimated reduced-form dynamics that may themselves change under the alternative policy. The IRF-consistent mode partially addresses this by using structurally identified responses for the key transmission variables, but the remaining variables’ responses are governed by the estimated BVAR covariance structure, which is held fixed.
5. **Data vintage:** The BVAR is estimated on a fixed data vintage. Real-time data revisions and the publication lag of macroeconomic series are not handled.

15.3 Known issues flagged for fix

The limitations above are intrinsic to the modelling choices. The following are *specific, identified defects* in the current deployment—known and flagged, but *not yet fixed*—documented here in the interest of honesty for a live research tool. Each cites the source file where it lives.

1. **China fiscal debt double-counts inflation.** The fiscal debt-snowball roll-forward (`fiscalAccounting.ts`) builds the quarterly nominal GDP growth as $\Delta(\text{idx.gdp}) + \Delta(\text{idx.deflator})$, i.e. real growth plus inflation. For China the resolved `idx.gdp` points at `GDP_CN`, which is *nominal* GDP, while `idx.deflator` points at `CPI_CN`—so inflation is added *twice* and the CN debt-to-GDP path is biased. The fix requires a proper real/nominal split for the CN bloc (the other fiscal blocs carry a real GDP variable, so they are correct).
2. **Entropic tilt does not filter non-finite long-run moments.** `longRunTilt.ts` returns NaN long-run moments for posterior draws with a unit-root/near-singular companion (the steady-state moment is undefined), but `solveEntropicTilt` consumes the moment vector without filtering: a single NaN propagates through the softmax ($\exp(\text{NaN})$) and turns *every* weight NaN, silently emptying the anchor band. Until a finite-mask is added, the user-defined long-run anchor should be used with the (finite) forecast moment, which is the default, and not the raw steady-state moment.
3. **No divergence guard on the closed-form GVAR solve.** The live solver (`gvarSolve.ts`, Neumann iteration and direct inverse) does not check $\rho(M) < 1$ before solving. The deployed coupling is contractive ($\rho(M) \approx 0.924$), but a future roster or a degenerate pin set that pushed $\rho(M) \geq 1$ would return a *confidently wrong* forecast from a diverged fixed point with no warning. A spectral-radius backstop is needed.
4. **Conditional band uses the filtered, not smoothed, covariance.** The predictive-band variances are extracted from the Kalman *filter* covariance P_t^{filt} (`kalman.ts`), not the Durbin–Koopman *smoother* covariance. For quarters *preceding* a future pin, the smoother would

tighten the covariance (a later observation reduces uncertainty about an earlier state), so the displayed band is *overstated* on the run-up to a future conditioning point. The central path is unaffected; only the band width is conservative.

5. **The porter does not regenerate `fiscal_meta.json`.** The fiscal block is addressed by *positional* indices stored in each bloc’s `fiscal_meta.json` (`idx.gdp`, `idx.deflator`, `idx.revenue`, ...). The model export does not rebuild this file, so any change to a bloc’s variable order leaves the indices pointing at the wrong columns—the cause of a recent EA/UK debt-path explosion—and they must be repaired manually (`pipeline/fix_fiscal_meta_indices.mjs`). Folding the meta regeneration into the porter is the durable fix.

15.4 Directions for Future Work

Several extensions are planned or under consideration:

- **Time-varying parameters:** Incorporating stochastic volatility or time-varying coefficients (e.g., [Sims and Zha, 1998](#)) to capture evolving dynamics.
- **Structural identification:** Extending the counterfactual layer to support sign restrictions, narrative restrictions, or scenario priors ([Farkas and Jakab, 2026](#)) for identification within the BVAR itself, rather than relying on external DSGE models.
- **Mixed-frequency data:** Supporting monthly–quarterly mixed-frequency estimation to incorporate higher-frequency indicators.
- **Model averaging:** Bayesian model averaging over DSGE specifications and policy rules for the counterfactual layer.

A Glossary of terms

This glossary defines the named concepts used throughout the manual; the mathematical symbols are collected in Table 1.

BVAR (Bayesian vector autoregression)

A reduced-form VAR whose coefficients are estimated under a prior, here the hierarchical Minnesota/GLP prior of [Giannone et al. \(2015\)](#) with data-driven hyperparameter selection (Section 2).

Minnesota / GLP prior

The shrinkage prior that centers each equation on a univariate own-lag model and selects its tightness λ and sum-of-coefficients μ by marginal-likelihood maximization. “GLP” refers to the [Giannone et al. \(2015\)](#) hierarchical implementation.

Own-lag prior mean (δ): RW vs. WN

The prior mean on a variable’s own first lag: $\delta = 1$ (random walk, RW, the default—best forecast is

the current level) or $\delta = 0$ (stationary white noise, WN—best forecast is the mean). Implemented as `diagb(pos) = 0` for the WN index set `pos` (Section 2.5). It shifts the prior *center*, not a hard constraint.

GVAR (Global VAR)

A collection of per-economy (“bloc”) BVARs linked through trade-weighted foreign variables; here the linkage is imposed at solve time, not in estimation (Sections 6, 6.4).

Trade-weighted “star” variable

A bloc’s foreign aggregate (ROWDEM demand, ROWINF inflation, ROWFIN financial), built as the trade-weight w_{cj} -weighted average of its partners’ corresponding series, renormalized over the partners present each quarter (Section 6.2).

Endogenous dominant unit

The configuration in which the US is a full bloc inside the fixed point—with its own ROW*_US stars and a US weights row—so shocks abroad feed *back* into the US (the RoW→US channel a frozen anchor cannot represent).

Frozen anchor

The legacy configuration in which the US is estimated once as a closed, exogenous driver: it enters the coupling only through the forcing d , with no equation for the world to feed back into.

Global commons

The four US-determined common series (oil, the US 10-year yield, the US BAA corporate yield, US equity) pinned 1:1 onto every RoW bloc (Section 6.6).

COVID-as-missing

The treatment of the fixed 2020Q2–Q4 window as *missing data*: the macro columns and the two macro stars are NaN-masked while financials and ROWFIN are kept observed, and the Durbin–Koopman simulation smoother imputes the holes inside a data-augmentation Gibbs sampler (Section 9.4).

EM loop

The block-coordinate iteration over (foreign stars, completed-data states, VAR coefficients): an E-step that completes the panels and rebuilds the stars and an M-step that re-estimates every bloc, iterated to a fixed point in the completed stars (Algorithm 3).

Durbin–Koopman simulation smoother

The forward-filter / backward-draw recursion (Durbin and Koopman, 2002) that produces draws of the conditioned/missing states by simulating an unconditional path, smoothing it, and adding back the smoothed discrepancy (Algorithm 1).

Coupling operator M , fixed point $x = Mx + d$

The affine solve-time operator that propagates a scenario across blocs; assembled by Kalman-probing

each pinned cell, solved by Gauss–Seidel/Neumann sweeps or the direct $(I - M)^{-1}$ (Algorithm 2). $\rho(M) < 1$ makes it contractive.

SFA (stock-flow adjustment)

The deficit–debt discrepancy: the part of the observed change in gross debt that the flow side (deficit plus the interest–growth snowball) does not explain; an estimated WN-prior BVAR variable (Section 7.1).

EFFI (effective interest rate)

The average rate the government actually pays (net interest over lagged gross debt, annualized); a persistent RW-prior variable distinct from the policy rate and market yields.

Entropic tilt

An after-estimation reweighting of the predictive draws that imposes a user long-run anchor by minimum relative-entropy (Kullback–Leibler) distortion, monitored by the effective sample size (Section 8.2).

CRPS (continuous ranked probability score)

A strictly proper density score (Gneiting and Raftery, 2007) computed from the draw ensemble; the primary density metric of the evaluation, robust at modest draw counts.

PIT (probability integral transform)

The rank of each realization in its predictive ensemble; uniform PITs indicate good calibration (Berkowitz, 2001).

Giacomini–White (GW) test

The conditional predictive-ability test (Giacomini and White, 2006) used as the headline inferential test: valid under nesting and for the recursive/expanding-window scheme, and loss-agnostic (applies to point, CRPS, and log-score differentials).

Clark–West (CW) test

The nested-model point/MSE test (Clark and West, 2007) that adjusts the loss differential for the larger model’s extra-parameter estimation noise.

Diebold–Mariano (DM) test

The equal-predictive-accuracy test (Diebold and Mariano, 1995), used here only for *non-nested* pairs, with the Harvey et al. (1997) small-sample correction.

Romano–Wolf / Model Confidence Set

Multiplicity controls: the Romano–Wolf step-down (Romano and Wolf, 2005) for FWER across the test family, and the MCS (Hansen et al., 2011) reporting the models statistically indistinguishable from the best.

GIRF / Diebold–Yilmaz index

Generalized (ordering-invariant) impulse responses (Pesaran and Shin, 1998) and the connectedness

index ([Diebold and Yilmaz, 2009, 2012](#)) of the Tier-B spillover module (Section [14.3](#)).

Disclaimer

The views expressed herein are those of the author and should not be attributed to the International Monetary Fund, its Executive Board, or its management. This tool is provided “as is” without warranty of any kind, express or implied. The author and the IMF disclaim all liability for any errors, omissions, or for any loss or damage arising from its use. Users are solely responsible for any decisions made based on the output of this tool.

References

- Marta Bańbura, Domenico Giannone, and Michele Lenza. Conditional forecasts and scenario analysis with vector autoregressions for large cross-sections. *International Journal of Forecasting*, 31(3):739–756, 2015.
- Regis Barnichon and Geert Mesters. A sufficient statistics approach for macro policy evaluation. *American Economic Review*, 113(11):2809–2845, 2023.
- Jeremy Berkowitz. Testing density forecasts, with applications to risk management. *Journal of Business & Economic Statistics*, 19(4):465–474, 2001.
- Henning Bohn. The behavior of U.S. public debt and deficits. *The Quarterly Journal of Economics*, 113(3):949–963, 1998.
- Lawrence J. Christiano, Martin Eichenbaum, and Charles L. Evans. Nominal rigidities and the dynamic effects of a shock to monetary policy. *Journal of Political Economy*, 113(1):1–45, 2005.
- Alexander Chudik and M. Hashem Pesaran. Theory and practice of GVAR modelling. *Journal of Economic Surveys*, 30(1):165–197, 2016.
- Todd E. Clark and Kenneth D. West. Approximately normal tests for equal predictive accuracy in nested models. *Journal of Econometrics*, 138(1):291–311, 2007.
- Günter Coenen, Christopher J. Erceg, Charles Freedman, Davide Furceri, Michael Kumhof, René Lalonde, Douglas Laxton, Jesper Lindé, Annabelle Mourougane, Dirk Muir, Susanna Mursula, Carlos de Resende, John Roberts, Werner Roeger, Stephen Snudden, Mathias Trabandt, and Jan in’t Veld. Effects of fiscal stimulus in structural models. *American Economic Journal: Macroeconomics*, 4(1):22–68, 2012.
- Stephane Dees, Filippo di Mauro, M. Hashem Pesaran, and L. Vanessa Smith. Exploring the international linkages of the euro area: a global VAR analysis. *Journal of Applied Econometrics*, 22(1):1–38, 2007.
- Francis X. Diebold and Roberto S. Mariano. Comparing predictive accuracy. *Journal of Business & Economic Statistics*, 13(3):253–263, 1995.
- Francis X. Diebold and Kamil Yilmaz. Measuring financial asset return and volatility spillovers, with application to global equity markets. *The Economic Journal*, 119(534):158–171, 2009.
- Francis X. Diebold and Kamil Yilmaz. Better to give than to receive: Predictive directional measurement of volatility spillovers. *International Journal of Forecasting*, 28(1):57–66, 2012.
- Thomas Doan, Robert Litterman, and Christopher Sims. Forecasting and conditional projection using realistic prior distributions. *Econometric Reviews*, 3:1–100, 1984.

- James Durbin and Siem Jan Koopman. A simple and efficient simulation smoother for state space time series analysis. *Biometrika*, 89(3):603–616, 2002.
- James Durbin and Siem Jan Koopman. *Time Series Analysis by State Space Methods*. Oxford University Press, Oxford, 2nd edition, 2012.
- Julio Escolano. A practical guide to public debt dynamics, fiscal sustainability, and cyclical adjustment of budgetary aggregates. IMF Technical Notes and Manuals 2010/02, International Monetary Fund, 2010.
- Mátyás Farkas and Zoltán Jakab. Scenario priors for Bayesian SVARs. *Working Paper*, 2026.
- Raffaella Giacomini and Halbert White. Tests of conditional predictive ability. *Econometrica*, 74(6):1545–1578, 2006.
- Domenico Giannone, Michele Lenza, and Giorgio E. Primiceri. Prior selection for vector autoregressions. *The Review of Economics and Statistics*, 97(2):436–451, 2015.
- Domenico Giannone, Michele Lenza, and Giorgio E. Primiceri. Priors for the long run. *Journal of the American Statistical Association*, 114(526):565–580, 2019.
- Tilmann Gneiting and Adrian E. Raftery. Strictly proper scoring rules, prediction, and estimation. *Journal of the American Statistical Association*, 102(477):359–378, 2007.
- Peter R. Hansen, Asger Lunde, and James M. Nason. The model confidence set. *Econometrica*, 79(2):453–497, 2011.
- David Harvey, Stephen Leybourne, and Paul Newbold. Testing the equality of prediction mean squared errors. *International Journal of Forecasting*, 13(2):281–291, 1997.
- International Monetary Fund. Staff guidance note for public debt sustainability analysis in market-access countries. Policy paper, International Monetary Fund, 2013.
- S. Kullback and R. A. Leibler. On information and sufficiency. *The Annals of Mathematical Statistics*, 22(1):79–86, 1951.
- Robert B. Litterman. Forecasting with Bayesian vector autoregressions — five years of experience. *Journal of Business & Economic Statistics*, 4(1):25–38, 1986.
- Massimiliano Marcellino, James H. Stock, and Mark W. Watson. A comparison of direct and iterated multistep ar methods for forecasting macroeconomic time series. *Journal of Econometrics*, 135(1–2):499–526, 2006.
- H. Hashem Pesaran and Yongcheol Shin. Generalized impulse response analysis in linear multivariate models. *Economics Letters*, 58(1):17–29, 1998.

- M. Hashem Pesaran, Til Schuermann, and Scott M. Weiner. Modeling regional interdependencies using a global error-correcting macroeconometric model. *Journal of Business & Economic Statistics*, 22(2):129–162, 2004.
- John C. Robertson, Ellis W. Tallman, and Charles H. Whiteman. Forecasting using relative entropy. *Journal of Money, Credit and Banking*, 37(3):383–401, 2005.
- Joseph P. Romano and Michael Wolf. Stepwise multiple testing as formalized data snooping. *Econometrica*, 73(4):1237–1282, 2005.
- Christopher A. Sims and Tao Zha. Bayesian methods for dynamic multivariate models. *International Economic Review*, 39(4):949–968, 1998.
- Frank Smets and Rafael Wouters. Shocks and frictions in US business cycles: A Bayesian DSGE approach. *American Economic Review*, 97(3):586–606, 2007.
- John B. Taylor. Discretion versus policy rules in practice. *Carnegie-Rochester Conference Series on Public Policy*, 39:195–214, 1993.
- Daniel F. Waggoner and Tao Zha. Conditional forecasts in dynamic multivariate models. *Review of Economics and Statistics*, 81(4):639–651, 1999.
- Volker Wieland, Tobias Cwik, Gernot J. Müller, Sebastian Schmidt, and Maik Wolters. A new comparative approach to macroeconomic modeling and policy analysis. *Journal of Economic Behavior & Organization*, 83(3):523–541, 2012.
- Volker Wieland, Elena Afanasyeva, Meguy Kuete, and Jinhyuk Yoo. New methods for macro-financial model comparison and policy analysis. In John B. Taylor and Harald Uhlig, editors, *Handbook of Macroeconomics*, volume 2. North Holland/Elsevier, 2016.